A Guided Introduction to Quantum Mechanics

Concepts, Derivations, and Applications

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Preface

It would be foolish to imagine that students, simply by encountering the experiments that led to quantum mechanics, could deduce its principles for themselves. The subject itself is *deeply nonintuitive*, and more generally, experimental results do not uniquely dictate the theory we use to describe them. Nature does not tend to hand us a formalism along with the data! Still, there is value in guiding students from observation toward theoretical structure. By tracing the questions raised by experiment and the ideas introduced to answer them, students can see quantum mechanics not as a collection of rules handed down from nowhere, but as a framework built to make sense of what we observe. Experience may not reveal the theory on its own, but it is the root from which understanding grows.

In that spirit, this book develops quantum mechanics through explicit derivations and concrete examples. Students analyze Stern–Gerlach experiments, study spin precession and magnetic resonance, use wave functions to derive bound states and understand tunneling phenomena, and apply perturbation theory to explain shifts in hydrogen's spectrum. Later chapters explore identical particles, Pauli exclusion, electronic structure, and energy bands. The quantized electromagnetic field is introduced and applied to light–matter interactions and spontaneous emission. The aim is to show how to use the mathematical formalism and how its assumptions connect to the physical phenomena it describes.

This textbook is a work in progress. You may find typos, mistakes, and places where the discussion could be improved. Homework problems are not included here; they are available for students on the course website. If you spot an error or have a suggestion, please let me know. I welcome your feedback and will use it to improve future versions.

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Chapter 1

Stern-Gerlach Experiments

1.1 Introduction

1.1.1 Welcome to Quantum Mechanics

This course marks a turning point in your study of physics. Up to now, you have mostly been working within the framework of classical mechanics. Developed over centuries by Newton, Lagrange, Hamilton, and others, classical mechanics is a triumph of human imagination and reasoning. We assume that particles have definite positions and momenta, regardless of how accurately we can measure them, and that the state of a closed system deterministically evolves over time. These are powerful ideas, and they remain useful. In many systems, *classical mechanics is still the best tool we have.* But it's not the final word.

In the early 20th century, physicists encountered phenomena that classical mechanics couldn't explain, as you may be well-aware. Blackbody radiation, the photoelectric effect, atomic spectra, etc. all pointed to a deeper structure underlying matter and energy. Quantum mechanics was born from the minds of people working hard to solve these puzzles, and what emerged was not just a set of ad hoc rules, but a radically different theoretical framework, grounded in a strange but consistent mathematical formalism. Many classical concepts like energy, momentum, and conservation laws remain central to quantum mechanics, but they are *redefined and reinterpreted.* Classical mechanics becomes an approximation–a limiting case of quantum mechanics–and this continuity is crucial. Quantum mechanics reveals where and how the assumptions of classical mechanics break down and what stands in their place.

1.1.2 Disclaimer

These notes are intended for pedagogical purposes and are designed to guide students through the core concepts of quantum mechanics in a way that emphasizes understanding and intuition. They make assumptions about what students already know in order to focus attention on the new ideas being introduced and make the transition to these ideas as comfortable as possible (a fool's errand).

The presentation favors conceptual development over formal rigor, and as such, the notes are not meant to serve as a mathematical treatise. Rather than functioning as a comprehensive reference, they are structured as an interactive guide—something to be worked through actively, not merely consulted. Students seeking more formal or detailed treatments are encouraged to supplement these notes with standard textbooks and references (see the recommendations on Brightspace).

We will introduce quantum mechanics using the *Stern-Gerlach experiment* as a concrete entry point. It's important to understand, however, that quantum mechanics is not merely induced from experimental observations, nor is it derived from deeper theoretical principles. It is built on a set of postulates, which are statements informed by experiments but not proven by them. Postulates are creative, conceptual starting points that organize and give meaning to observations and allow us to explain and predict a wide range of phenomena. If the postulates of quantum theory follow from a deeper framework, that framework is not currently known or widely accepted.

1.2 Stern-Gerlach Experiment

The Stern-Gerlach experiment, published in 1922 by Otto Stern and Walther Gerlach, demonstrated that particles possess quantized intrinsic angular momentum, or spin [1]. The experiment consisted of an oven that produced a beam of neutral atoms, a region of space with an inhomogeneous magnetic field, and a photographic plate to serve as a detector. When a beam of silver atoms passed through the magnetic field, it split into two distinct beams; one deflected upwards and the other deflected downwards, relative to the direction of the magnetic field gradient.

Let's analyze the experiment classically to understand why this was a surprising result. A neutral particle will interact with a magnetic field if it possesses a magnetic dipole moment[1]. The potential energy of this interaction is $U = -\boldsymbol{\mu}\cdot\mathbf{B}$ which results in a force given by $\mathbf{F} = \nabla(\boldsymbol{\mu}\cdot\mathbf{B})$. The magnetic field must have a nonzero gradient in some direction to exert a force in that direction, and this was accomplished with an asymmetrical arrangement of permanent magnets–one sharp-pointed pole (like a wedge or cone) and one broad, flat pole. Let's assume the magnetic field is

$$\mathbf{B} = -\alpha x\hat{\mathbf{x}} + (B_0 + \alpha z)\hat{\mathbf{z}},$$

where $\alpha = \partial B_z/\partial z$ and B_0 are constants. The force on the magnetic dipole is

$$\mathbf{F} = -\alpha\mu_x\hat{\mathbf{x}} + \alpha\mu_z\hat{\mathbf{z}}$$

According to Maxwell's equations, $\mathbf{B}$ must be divergence-less, so we end up with a force having both x and z components. It turns out that we can ignore the x component as long as the constant B_0 is very large–so large that μ_x will oscillate due to Larmor precession and average to zero (we'll investigate this phenomenon later in the context of quantum mechanics, but it's a well-established result of classical electromagnetism).

[1] See Griffiths, *Introduction to Electrodynamics*

A beam of silver atoms, initially traveling in the y direction, feels a net force in the z direction given by

$$F_z = \alpha\mu_z = \mu_z \frac{\partial B_z}{\partial z}$$

Every atom consists of charged particles which can produce loops of current that give rise to a magnetic moment. A loop of area A and current I produces a magnetic moment $\mu = IA$. For instance, if this loop of current arises from a particle with charge q and mass m traveling in a circle with radius r at speed v, then

$$\mu = \frac{q}{T}\pi r^2 = \frac{q}{2\pi r/v}\pi r^2 = \frac{qrv}{2}.$$

On the other hand, the angular momentum of such a particle about the center of the circle is $L = mvr$, so

$$\mu = \frac{q}{2m}L$$

In the same way that the earth revolves around the sun and rotates around its own axis, we can also imagine a charged particle in an atom having orbital angular momentum $\mathbf{L}$ and a new property, the intrinsic angular momentum, which we label $\mathbf{S}$ and call spin. The relationship between $\boldsymbol{\mu}$ and $\mathbf{S}$ is very similar to that between $\boldsymbol{\mu}$ and $\mathbf{L}$, except an integration over current loops must be performed. This contributes an overall constant factor g called the g-factor that depends on the geometry of the particle. Hence,

$$\boldsymbol{\mu} = g\frac{q}{2m}\mathbf{S}$$

It turns out that the magnetic moment of a neutral silver atom is almost entirely due to the spin of its outermost, unpaired electron. The magnetic moment of this electron, and therefore of the entire neutral silver atom, is

$$\boldsymbol{\mu} = -g\frac{e}{2m_e}\mathbf{S}$$

where e is the magnitude of the electron's charge. The classical force on the atom is

$$F_z = -g\frac{e}{2m_e}S_z\frac{\partial B_z}{\partial z}$$

Hence, the deflection of the beam in the Stern-Gerlach experiment is a measure of the z-component of spin, which is the orientation of the magnetic field gradient. In the thermal environment of the oven, we expect a random distribution of spin directions, so the z-component of spin should be a continuous distribution between $-|\mathbf{S}|$ and $+|\mathbf{S}|$, which would yield a continuous spread in deflections of the silver atomic beam.

As stated earlier, Stern and Gerlach observed two deflections, indicating only two values for the z-component of the electron spin. The magnitudes of deflection are consistent with

$$S_z = \pm\frac{\hbar}{2}$$

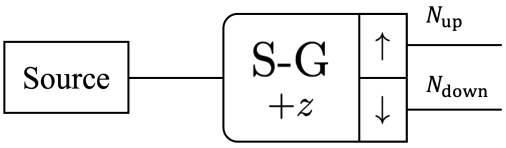


Figure 1.1: A simplified diagram of the Stern-Gerlach experiment. N_{up} is the number of atoms that are spin up, and N_{down} is the number of atoms that are spin down, as recorded by the counter.

where $\hbar$ is Planck's constant h divided by 2π and has the numerical value

$$\boxed{\hbar = 1.054\ 571\ 817 \times 10^{-34}\ \text{J} \cdot \text{s} = 6.582\ 119\ 569 \times 10^{-16}\ \text{eV} \cdot \text{s}}$$

The factor of $1/2$ in the values of S_z is why we say the electron is a spin-1/2 particle. When $S_z = +\hbar/2$, we say the electron is spin up, and when $S_z = -\hbar/2$ the electron is spin down.

1.3 Preparing a Quantum State

As it turns out, the concept of intrinsic spin can be fully described within the framework of classical field theory [2]. The *quantization of spin* is entirely quantum mechanical and can be understood in the context of quantum field theory. While the magnitude of the electron's spin is always a fixed value, *there is no way to predict which component of spin you will observe for a single electron* in a Stern-Gerlach experiment. Quantization of spin is certainly interesting, but this is our first key insight into quantum mechanics.

Your physics education has mostly dealt with the certainty and exactness of classical mechanics. The predictions of quantum mechanics, on the other hand, are intrinsically probabilistic. In most areas of science, probabilities reflect ignorance about the exact state of a system. For example, in statistical mechanics, we assign probabilities because we cannot realistically keep track of the overwhelming number of the microstates. In quantum mechanics, probabilities are intrinsic to the theory; even with complete knowledge of the state of a system, the outcomes are probabilistic. Philosophers make a distinction between *epistemic* probability and *ontological* probability; the former has to do with one's knowledge whereas the latter has to do with existence.

When a radioactive nucleus decays, it does so spontaneously; there simply is no rule that will tell you when it decays or in what direction the particles will travel afterwards. When a particle of light strikes a half-silvered mirror, it either reflects off it or passes through, and the best we can do is assign probabilities for each occurrence. Truly random numbers have been generated using Geiger counters and quantum optics, and in 2025, a team at NIST built the first random number generator that uses quantum entanglement to produce verifiable random numbers [3]. We'll talk more about entanglement later, but the idea is that quantum systems can exhibit correlations that are much stronger than fluctuating classical systems. We can verify a system is "truly quantum" by measuring these strong correlations.

Figure 1.1 illustrates a simplified version of the Stern-Gerlach experiment consisting of an oven

that produces the beam of atoms, a Stern-Gerlach device with two output ports for the possible values of the spin component, and two counters to detect the atoms leaving the output ports. Analyzing the atoms that come directly from the source involves a longer story than we're ready to tell at the moment (involving the density matrix, a topic reserved for Physics 438B).

We can simplify the situation by blocking one of the output ports and keeping track of the number of atoms that exit the other port. This is called **state preparation** because the quantum state of the atoms that exit the free port can be easily determined. The output beams are labeled with symbols called **kets**. We use the ket $|+z\rangle$ to represent the state of atoms that exit with spin up and $|-z\rangle$ to represent the state of atoms that exit with spin down. The ket notation we will use in this course was developed by Paul Dirac. We will discuss the mathematics of kets in full detail later. **Informally, the first postulate of quantum mechanics states that kets describe the quantum state–they contain all the information that we can know about the system.** Kets are elements of a vector space, which is a concept from linear algebra.

1.4 Ideal S-G Experiments

Now let's consider a series of simple Stern-Gerlach experiments with slight variations that help to illustrate important features of quantum mechanics. The experiments are summarized in Figure 1.2. We first describe the experiments and their results and draw some qualitative and general conclusions about the nature of quantum mechanics. Then we introduce the formal mathematics show how it can be used to describe and predict the results of each experiment.

1.4.1 Experiment 1

The first experiment is depicted in Figure 1.2(a). Atoms from the source enter a Stern-Gerlach analyzer with a magnetic field gradient in the $+z$ direction. The lower port is blocked and the upper port is free. Atoms that exit the upper port are fed into a second analyzer oriented in the same direction as the first. For every 100 atoms that exit the upper port of the first analyzer, 100 atoms will exit the upper port of the second. This might not be surprising, but it sets the stage for the following experiments, and it illustrates the concept of state preparation. We find a similar result if we block the upper port of the first analyzer and leave the lower port free–all atoms that exit the second analyzer will exit the lower port.

1.4.2 Experiment 2

Figure 1.2(b) shows the second experiment, which is similar to the first except for the fact that the second analyzer is rotated 90° relative to the first. The magnetic field gradient of the second analyzer is aligned with the $+x$-axis. We find that atoms appear from both output ports of the second analyzer this time, and in equal proportion. For every 100 atoms that exit the upper port of the first analyzer, 50 will exit the upper port of the second and 50 will exit the lower port. We

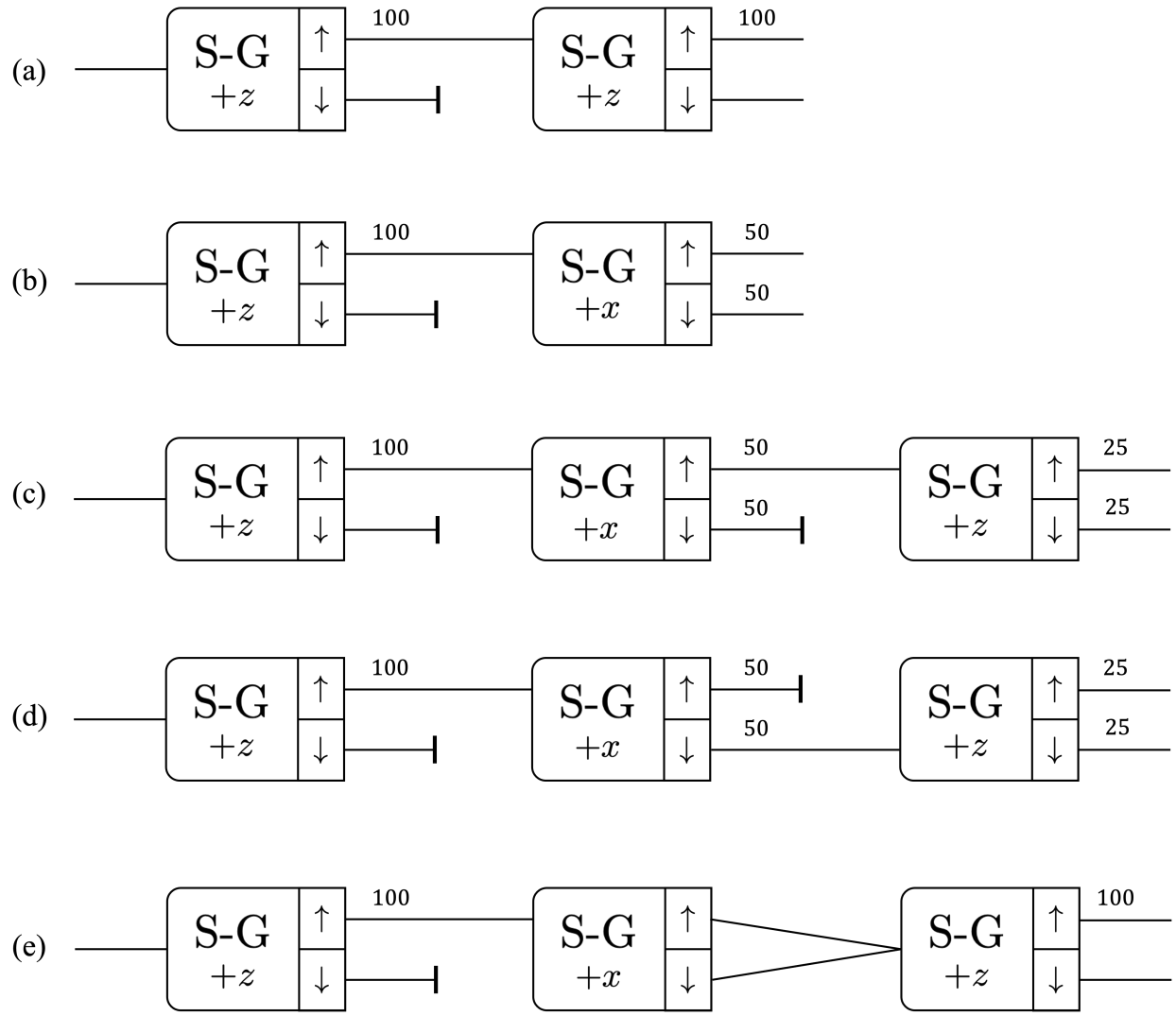


Figure 1.2: Ideal Stern-Gerlach thought experiments.

find the same result if we block the upper port of the first analyzer–for every 100 that exit the lower port of the first, 50 exit the upper port of the second and 50 exit the lower port.

The most important feature of this experiment is revealed when we look at the data collected by the counters. Assuming the atoms go through one at a time, there is no way to predict which port of the second analyzer any particular atom will exit. It is only by letting the experiment unfold after many repetitions that a pattern begins to emerge, namely that atoms will exit both ports in roughly equal amounts. In what sense can we claim to know the state of the system if we can't determine exactly what happens to it? One might be tempted to say we simply don't know enough about the system to predict which port an atom will exit. Perhaps one day we'll have a fully deterministic theory, but all signs indicate that nature is fundamentally probabilistic. We'll see that knowing the state of the atom after the first analyzer, which is $|+z\rangle$, we can predict the statistical distribution of outcomes from the second analyzer–that is the sense in which we claim

to know the state of the system.

1.4.3 Experiment 3

The experiments depicted in Figures 1.2(c) and 1.2(d) extend experiment 2 by adding a third analyzer aligned along the z-axis. With the lower ports of the first and second analyzers blocked, for every 100 atoms that exit the upper port of the first, 50 will exit the upper port of the second, 25 will exit the upper of the third, and 25 will exit the lower port of the third. Naively, we might expect all the atoms to exit the upper port of the third analyzer due to the state preparation of the first, but that is not what's observed. The system behaves as if the second analyzer "resets" the quantum state. By analogy with the first analyzer, the state of the atoms that exit the upper port of the second analyzer is $|+x\rangle$, and the state of the atoms that exit the lower port is $|-x\rangle$.

This result demonstrates another key feature of quantum mechanics: a measurement disturbs the system. The second analyzer has disturbed the system in such a way to make the spin component along the z-axis unknown, even though we measured it with the first analyzer. There is a fundamental incompatibility in trying to measure the spin component of the atom along two different directions. We say that S_x and S_z are "incompatible observables."

1.4.4 Experiment 4

Given the results of experiment 3, it would be interesting to study what happens if the two beams exiting the second analyzer are combined and then fed into the third analyzer, as illustrated in Figure 1.2(e). The results depend on exactly how that's done. If we keep the counters in place and record which atoms exit which port of the second analyzer, then we get a 50/50 split just like experiment 3 (except this time 50 atoms exit the upper port of the third analyzer and 50 exit the lower port). If instead we remove the counters and simply combine the beams with a suitable arrangement of magnets[2], all the atoms exit the upper port of the third analyzer and *none will exit the lower port*. The atoms appear to "remember" that they were initially measured to have spin up along the z-axis. They behave as if the second analyzer isn't even there.

This result demonstrates *quantum mechanical interference*–the chance that an atom will exit the lower port of the third analyzer is zero despite giving the atom more routes through the experiment. As long as there is no record of which path the atoms take through the second analyzer, the two paths seemingly interfere with one another in a way that is reminiscent of wave interference. The upper port experiences constructive interference while the lower port shows destructive interference. It's important to understand that there is no literal wave interference necessarily happening, and this is entirely due to way states and probabilities work in quantum mechanics. Early attempts to understand quantum mechanical interference fell for this trap, and the terminology developed during this period persists even today.

The existence of quantum mechanical interference makes it difficult to say that kets are simply

[2]See Feynman's *Lectures on Physics*, Volume 3.

about our knowledge of a physical system (meaning they are epistemic quantities). Kets encode possibilities, and different possible outcomes can apparently "interfere with one another" like classical waves as long as nothing disturbs them. Any attempt to determine which outcome will occur spoils the interference. Surprisingly, the postulates of quantum mechanics explain interference as a generic feature of physical systems. Interference effects have been observed for photons, electrons, atoms, and even molecules as large as carbon-60 and carbon-70. As systems grow larger, so does the chance that something will spoil interference, and this partially explains why you don't diffract through a doorway...

1.5 Quantum State Vectors

The kets we introduced earlier are abstract objects belonging to a vector space and they obey many of the familiar rules you know about ordinary geometric vectors. Kets are related to **quantum state vectors** and they are elements of a complex vector space with important properties (we'll talk about the properties as they become relevant). The dimension of the vector space depends on the mathematical model of the physical system and, most importantly, on the number of measurement outcomes. In the Stern-Gerlach experiments we discussed, there are only two possible results for a spin component measurement, so the vector space is two dimensional.

In the two-dimensional vector space of a spin-1/2 system, the two kets $|\pm z\rangle$ form a basis, just like the Cartesian unit vectors $\{\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}\}$ form a basis for vectors in three-dimensional Euclidean space. In particular, the Cartesian unit vectors form an orthonormal basis with the follow properties:

$$\begin{aligned} &\text{normalization:} && \hat{\mathbf{x}} \cdot \hat{\mathbf{x}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{z}} \cdot \hat{\mathbf{z}} = 1 \\ &\text{orthogonality:} && \hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = \hat{\mathbf{x}} \cdot \hat{\mathbf{z}} = \hat{\mathbf{y}} \cdot \hat{\mathbf{z}} = 0 \\ &\text{completeness:} && \mathbf{A} = a_x \hat{\mathbf{x}} + a_y \hat{\mathbf{y}} + a_z \hat{\mathbf{z}} \end{aligned}$$

Notice that the scalar product (dot product) is central to the description of these properties. In light of normalization and orthogonality, we can express completeness as

$$\mathbf{A} = (\hat{\mathbf{x}} \cdot \mathbf{A})\hat{\mathbf{x}} + (\hat{\mathbf{y}} \cdot \mathbf{A})\hat{\mathbf{y}} + (\hat{\mathbf{z}} \cdot \mathbf{A})\hat{\mathbf{z}}$$

since each component can be found by simply taking the scalar product of $\mathbf{A}$ with the corresponding unit vector, i.e. $a_x = (\hat{\mathbf{x}} \cdot \mathbf{A})$, etc.

By analogy with spatial vectors, the kets $|\pm z\rangle$ form a complete orthonormal basis. Completeness implies that a general vector $|\psi\rangle$ is a linear combination of the two basis kets:

$$|\psi\rangle = \alpha\,|+z\rangle + \beta\,|-z\rangle \tag{1.1}$$

where α and β are complex scalar numbers multiplying each basis ket. The sense in which $|\pm z\rangle$ are

normalized and orthogonal depends on the definition of the inner product, which is a generalization of the familiar scalar product that applies to geometric vectors.

An **inner product** is a map that takes any two elements from the vector space and returns a complex number. For kets $|\psi\rangle$ and $|\phi\rangle$, the inner product is written as $\langle\psi|\phi\rangle$ and it has the following properties:

1. **Linearity** (in the second argument):

$$\langle u| \Big(a\,|v\rangle + b\,|w\rangle \Big) = a\,\langle u|v\rangle + b\,\langle u|w\rangle \tag{1.2}$$

for complex numbers a, b and kets $|u\rangle, |v\rangle, |w\rangle$.

2. **Conjugate symmetry**:

$$\langle v|w\rangle = \langle w|v\rangle^* \tag{1.3}$$

where * denotes the complex conjugate. Since $a = a^*$ if and only if a is a real number, $\langle\psi|\psi\rangle$ is always a real number.

3. **Positive semi-definiteness**:

$$\langle\psi|\psi\rangle \geq 0 \tag{1.4}$$

and $\langle\psi|\psi\rangle = 0$ if and only if $|\psi\rangle = 0$.

1.5.1 Bra Vectors and the Induced Norm

The inner product allows us to define a new object called a **bra vector**, which is an element of a dual vector space. Each ket $|\psi\rangle$ can be identified with a bra vector $\langle\psi|$, which is a linear functional that takes a single ket and returns a complex number. Anytime you see the notation $\langle\psi|$, you should see it as a linear map awaiting a ket to produce a complex number in a manner that is consistent with the inner product defined above. When a bra and ket are combined to form an inner product, we get a bracket or bra-ket (cute huh?). More precisely, for every ket $|\psi\rangle$ in the vector space, there exists a unique bra $\langle\psi|$ such that

$$\langle\psi| \big(|\phi\rangle \big) = \langle\psi|\phi\rangle \qquad \text{(bra vector induced by inner product)}$$

for all kets $|\phi\rangle$. The inner product also allows us to define the **norm** of a ket, which is a generalization of length. The norm of $|\psi\rangle$ is defined as

$$\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle} \qquad \text{(norm induced by inner product)}$$

All quantum state vectors are normalized kets in the sense that $\||\psi\rangle\| = \sqrt{\langle\psi|\psi\rangle} = 1$. This is crucial for the probabilistic interpretation that we'll explain soon.

Given the state $|\psi\rangle$ in Eq. (1.1) we can construct the corresponding bra vector $\langle\psi|$ using the properties of the inner product. Let $|\phi\rangle$ be an arbitrary ket. Then, $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle^*$ by conjugate symmetry, and by linearity,

$$\langle\phi| \Big(\alpha \, |+z\rangle + \beta \, |-z\rangle \Big) = \alpha \, \langle\phi|+z\rangle + \beta \, \langle\phi|-z\rangle \, .$$

Complex conjugation is itself a linear operation on complex numbers, so the complex conjugate of the preceding expression is

$$\alpha^* \, \langle\phi|+z\rangle^* + \beta^* \, \langle\phi|-z\rangle^* = \alpha^* \, \langle+z|\phi\rangle + \beta^* \, \langle-z|\phi\rangle$$

Hence, $\langle\psi| = \alpha^* \, \langle+z| + \beta^* \, \langle-z|$ for all kets in the vector space, since $|\phi\rangle$ was arbitrary.

1.5.2 Probabilistic Interpretation

The properties of normalization, orthogonality, and completeness can be expressed in the case of a two-state spin-1/2 quantum system as:

$$\begin{aligned} &\text{normalization:} && \langle+z|+z\rangle = \langle-z|-z\rangle = 1 \\ &\text{orthogonality:} && \langle+z|-z\rangle = \langle-z|+z\rangle = 0 \\ &\text{completeness:} && |\psi\rangle = \alpha \, |+z\rangle + \beta \, |-z\rangle \end{aligned}$$

The inner product can be used to find the coefficients of a ket in the $|\pm z\rangle$ basis. For instance,

$$\langle+z|\psi\rangle = \alpha \, \langle+z|+z\rangle + \beta \, \langle+z|-z\rangle = \alpha$$

$$\langle-z|\psi\rangle = \alpha \, \langle-z|+z\rangle + \beta \, \langle-z|-z\rangle = \beta$$

If we apply normalization to $|\psi\rangle$, we obtain a constraint on the coefficients:

$$\langle\psi|\psi\rangle = \Big(\alpha^* \, \langle+z| + \beta^* \, \langle-z| \Big) \Big(\alpha \, |+z\rangle + \beta \, |-z\rangle \Big) = 1$$

$$\alpha^* \alpha \, \langle+z|+z\rangle + \beta^* \beta \, \langle-z|-z\rangle = 1$$

$$|\alpha|^2 + |\beta|^2 = 1$$

or in terms of the expressions above,

$$|\, \langle+z|\psi\rangle \,|^2 + |\, \langle-z|\psi\rangle \,|^2 = 1 \tag{1.5}$$

In other words, the sum of the absolute squares is equal to 1. Since each term in the sum is greater than or equal to zero, the absolute squares can be used to define a discrete probability distribution.

Now comes a crucial element of quantum mechanics. **We postulate that each term in the sum is equal to the probability that the quantum state described by the normalized ket $|\psi\rangle$ is measured to be in the corresponding basis state**. The quantity

$$p(+z) = |\langle +z|\psi\rangle|^2 \tag{1.6}$$

is the probability that atoms in the state $|\psi\rangle$ are found in the state $|+z\rangle$ when a measurement of S_z is made, meaning $S_z = +\hbar/2$, and

$$p(-z) = |\langle -z|\psi\rangle|^2 \tag{1.7}$$

is the probability that atoms in the state $|\psi\rangle$ are found in the state $|-z\rangle$ when a measurement of S_z is made, meaning $S_z = -\hbar/2$ for those atoms.

Because the quantum mechanical probability is found by squaring an inner product, we refer to the inner product $\langle +z|\psi\rangle$, for instance, as a **probability amplitude** or just an amplitude–much like a classical wave intensity is found by squaring the wave amplitude. The probabilistic interpretation gives us physical insight into the inner product. For instance, the inner product $\langle +z|+z\rangle = 1$ means that atoms in the state $|+z\rangle$ have a 100% chance of being measured in the $|+z\rangle$ state, and $\langle -z|+z\rangle = 0$ means that atoms in the state $|+z\rangle$ have a 0% chance of being measured in the $|-z\rangle$ state. The possible outcomes are *mutually exclusive*. By convention, the input or initial state appears on the right and the output or final state goes on the left, although $|\langle \text{out}|\text{in}\rangle|^2 = |\langle \text{in}|\text{out}\rangle|^2$, so it doesn't really matter.

Example 1.1: Measurement of S_z

A measurement of S_z is carried out on a particle in the state

$$|\psi\rangle = \frac{1}{2}|+z\rangle + \frac{i\sqrt{3}}{2}|-z\rangle$$

What are the possible results of this measurement and with what probability do these results occur?

Solution: Start by computing the inner products with $|\pm z\rangle$.

$$\langle +z|\psi\rangle = \frac{1}{2} \quad \text{and} \quad \langle -z|\psi\rangle = \frac{i\sqrt{3}}{2}$$

The probability to obtain $S_z = \hbar/2$ is

$$\text{prob}(S_z = \hbar/2) = |\langle +z|\psi\rangle|^2 = \frac{1}{4}$$

and the probability to obtain $S_z = -\hbar/2$ is

$$\text{prob}(S_z = -\hbar/2) = |\langle +z|\psi\rangle|^2 = \langle -z|\psi\rangle^* \langle -z|\psi\rangle = \left(\frac{-i\sqrt{3}}{2}\right)\left(\frac{i\sqrt{3}}{2}\right) = \frac{3}{4}$$

Since the state $|\psi\rangle$ is normalized, the probabilities sum to one.

Example 1.2: Normalizing a Ket

Normalize the ket $|\psi\rangle = C(|+z\rangle + 2i\,|-z\rangle)$ by finding an appropriate value for the constant C, often referred to as a normalization constant.

Solution: Compute the inner product of $|\psi\rangle$ with itself and set it equal to one.

$$\begin{aligned}
1 &= \langle\psi|\psi\rangle \\
&= C^*\Big(\langle +z| - 2i\langle -z|\Big)\cdot C\Big(|+z\rangle + 2i\,|-z\rangle\Big) \\
&= |C|^2\Big(\underbrace{\langle +z|+z\rangle}_{=1} + 2i\underbrace{\langle -z|+z\rangle}_{=0} - 2i\underbrace{\langle -z|+z\rangle}_{=0} - 4i^2\underbrace{\langle -z|-z\rangle}_{=1}\Big) \\
&= 5|C|^2
\end{aligned}$$

Hence, $|C|^2 = 1/5$, and a properly normalized ket would be $|\psi\rangle = \frac{1}{\sqrt{5}}(|+z\rangle + 2i\,|-z\rangle)$.

1.6 Observables and Measurements

The Stern-Gerlach experiment is understood within a quantum mechanical framework where we no longer assume that all physical quantities have definite values at all times. Instead, we focus on how those quantities manifest as a result of measurement. In quantum theory, an **observable** is any measurable physical quantity like position, velocity, or spin, and we use linear operators to describe observables and model the impact of measurement on a system.

A **linear operator** is a map between vectors that preserves addition and scalar multiplication. For example,

$$A\left[c_1\,|\psi\rangle + c_2\,|\phi\rangle\right] = c_1 A\,|\psi\rangle + c_2 A\,|\phi\rangle \tag{1.8}$$

where A is an operator (usually clear from context, but sometimes not), c_1 and c_2 are scalars (complex numbers), and $|\psi\rangle$ and $|\phi\rangle$ are kets. Additionally, $A\,|\psi\rangle$ and $A\,|\phi\rangle$ are both elements of the same vector space as $|\psi\rangle$ and $|\phi\rangle$.

1.6.1 Matrix Elements

In the same way that vectors can be written as a linear combination of basis vectors, linear operators can also be written as a linear combination of a basis formed by *outer products*.

The **outer product** of vectors $|\phi\rangle$ and $|\psi\rangle$, denoted $|\phi\rangle\langle\psi|$, is a linear operator on vectors that takes any vector $|\varphi\rangle$, "applies" the bra vector $\langle\psi|$ to compute the inner product $\langle\psi|\varphi\rangle$, and outputs the vector $\langle\psi|\varphi\rangle|\phi\rangle$. In other words,

$$|\phi\rangle\langle\psi|\big(|\varphi\rangle\big) = \langle\psi|\varphi\rangle|\phi\rangle = |\phi\rangle\langle\psi|\varphi\rangle \tag{1.9}$$

For example, any linear operator acting on a two-dimensional vector space can be written as

$$A = a|+z\rangle\langle+z| + b|-z\rangle\langle+z| + c|+z\rangle\langle-z| + d|-z\rangle\langle-z| \tag{1.10}$$

where a, b, c, d are all complex numbers. Since any vector is a linear combination of basis states, an operator is uniquely specified by its action on basis states. In other words, if we know how an operator acts on basis states, we know how it acts on any vector. By linearity and orthonormality,

$$A|+z\rangle = a|+z\rangle + b|-z\rangle$$

$$A|-z\rangle = c|+z\rangle + d|-z\rangle$$

and it follows that

$$\langle+z|A|+z\rangle = a$$

$$\langle-z|A|+z\rangle = b$$

$$\langle+z|A|-z\rangle = c$$

$$\langle-z|A|-z\rangle = d$$

These are called the matrix elements of the operator A, even though we haven't introduce matrix notation at this point. Notice the matrix elements are basis dependent.

1.6.2 Eigenvalues and Projections

There are special kets that are not changed by the operation of a particular operator, except for a multiplicative constant. Such kets are known as eigenvectors of the operator and the multiplicative constants are the eigenvalues. **We postulate that the only possible result of measuring a physical observable is one of the eigenvalues of the corresponding operator. After the measurement, the state of the system is the eigenstate corresponding to the measured**

eigenvalue. The relationship between an observable, a possible measurement result, and the ket corresponding to that result is called an eigenvalue equation. For example, the eigenvalue equations for the S_z operator are

$$\begin{aligned} S_z \left|+z\right\rangle &= \tfrac{\hbar}{2} \left|+z\right\rangle \\ S_z \left|-z\right\rangle &= -\tfrac{\hbar}{2} \left|+z\right\rangle \end{aligned} \tag{1.11}$$

and the matrix elements are $\langle +z| S_z |+z\rangle = \hbar/2$, $\langle -z| S_z |-z\rangle = -\hbar/2$, and the rest are zero. Hence,

$$S_z = \tfrac{\hbar}{2} \left|+z\right\rangle \langle +z| - \tfrac{\hbar}{2} \left|-z\right\rangle \langle -z| \tag{1.12}$$

Observe that S_z is a linear combination of operators that square to themselves; for instance $(|+z\rangle \langle +z|)^2 = |+z\rangle \langle +z|$. This is analogous to finding the z-component of a Euclidean vector. Applying the projection twice gives you the same result as applying it once.

A **projection operator** P is a linear operator P such that $P^2 = P$ and $P^\dagger = P$, where $P^\dagger$ denotes the *Hermitian adjoint* (defined in the next section).

In case it isn't clear that $|+z\rangle \langle +z|$ square to itself, let $|\phi\rangle$ be an arbitrary vector:

$$\begin{aligned} (\left|+z\right\rangle \langle +z|)^2 \left|\phi\right\rangle &= \left|+z\right\rangle \langle +z| \Big[\left|+z\right\rangle \langle +z| \left(\left|\phi\right\rangle \right) \Big] \\ &= \left|+z\right\rangle \langle +z| \Big[\langle +z|\phi\rangle \left|+z\right\rangle \Big] \\ &= \langle +z|\phi\rangle \left|+z\right\rangle \\ &= \left|+z\right\rangle \langle +z| \left(\left|\phi\right\rangle \right) \end{aligned}$$

Hence, $\left(\left|+z\right\rangle \langle +z| \right)^2 = \left|+z\right\rangle \langle +z|$ since $|\phi\rangle$ was arbitrary. Notice that you get the same result by naively multiplying the two operators and "pulling apart" the outer product notation:

$$\left(\left|+z\right\rangle \langle +z| \right)^2 = \left|+z\right\rangle \ \langle +z|+z\rangle \langle +z| = \left|+z\right\rangle \langle +z| \,.$$

Although this "proof" is much shorter, it invokes unjustified assumptions about how the outer product works, and you should be careful when manipulating new mathematical objects. When in doubt, go back to the definitions, but in this case it works just fine.

Informally, a projection operator projects a vector onto a single vector or combination of vectors. If $P_+ = |+z\rangle \langle +z|$ and $P_- = |-z\rangle \langle -z|$, then

$$S_z = \frac{\hbar}{2} P_+ - \frac{\hbar}{2} P_-$$

and

$$P_+ + P_- = |+z\rangle\langle+z| + |-z\rangle\langle-z| = \mathbb{1} \tag{1.13}$$

where $\mathbb{1}$ is the identity operator. Eq. (1.13) is called a completeness relation, closure, or resolution of the identity. To see why, consider an arbitrary vector $|\psi\rangle$:

$$\begin{aligned} |\psi\rangle &= |+z\rangle\langle+z|\psi\rangle + |-z\rangle\langle-z|\psi\rangle \\ &= \Big(|+z\rangle\langle+z| + |-z\rangle\langle-z| \Big) |\psi\rangle \end{aligned}$$

The quantity in parentheses is a sum of the projection operators $P_+ + P_-$ and clearly the sum is equal to the identity by comparing the left- and right-hand sides of the last equation. Notice that P_+ and P_- project onto *orthogonal subspaces* since P_+ always returns a vector proportional to $|+z\rangle$ and P_- always returns a vector proportional to $|-z\rangle$, i.e., $P_+P_- = 0$.

Recall that quantum measurements disturb the state of the system. Projection operators allow us to model measurements mathematically. For instance, if an atom with a normalized input state $|\psi\rangle$ is measured to have $S_z = \hbar/2$, then the output state is $|+z\rangle$. The post-measurement state is determined by a *non-linear operation* on the pre-measurement state:

$$|\psi'\rangle = \frac{P_+|\psi\rangle}{\sqrt{\langle\psi|P_+|\psi\rangle}} = |+z\rangle \tag{1.14}$$

The probability that $S_z = \hbar/2$ can be written in terms of the corresponding projector as

$$|\langle+z|\psi\rangle|^2 = \langle\psi|+z\rangle\langle+z|\psi\rangle = \langle\psi|P_+|\psi\rangle \tag{1.15}$$

The "projection postulate" is often referred to as *state vector collapse*, and it clearly states that quantum measurements cannot be made without disturbing the system in an irreversible way. We do not really know what is going on in the measurement process, so we cannot explain the mechanism of the collapse of the quantum state vector. This lack of understanding makes some people uncomfortable with quantum mechanics and has been the source of much controversy. Trying to better understand the measurement process is an ongoing research problem.

1.6.3 Hermitian Adjoint

An operator A acts on kets as well as bras, but we have to be careful. For instance, the ket $|\phi\rangle = A|\psi\rangle$ does not correspond to the bra found by computing $\langle\psi|A$, where A acts from the right. The adjoint of an operator is defined precisely for this reason.

The **Hermitian adjoint** (or just adjoint for brevity) of a linear operator A is the unique operator denoted $A^\dagger$ such that

$$\langle \phi | A\psi \rangle = \left\langle A^\dagger \phi \middle| \psi \right\rangle \tag{1.16}$$

for all kets $|\psi\rangle$ and $|\phi\rangle$ in a complex vector space.

(i) An operator is called **self-adjoint** if it is equal to its adjoint, i.e., $A = A^\dagger$, and the domain of A is the same as the domain of $A^\dagger$.

(ii) An operator is **Hermitian** if $A^\dagger = A$ and **anti-Hermitian** if $A^\dagger = -A$.

By conjugate symmetry of the inner product,

$$\left\langle A^\dagger \phi \middle| \psi \right\rangle^* = \langle \psi | \, A^\dagger \, | \phi \rangle = \langle \phi | \, A \, | \psi \rangle^*$$

This relates the matrix elements of A to the matrix elements of $A^\dagger$. We can also find the bra vector corresponding to $|\phi\rangle = A\,|\psi\rangle$. If $|\varphi\rangle$ is an arbitrary ket, then

$$\langle \phi | \varphi \rangle = \langle \varphi | \phi \rangle^* = \langle \varphi | \, A \, | \psi \rangle^* = \langle \psi | \, A^\dagger \, | \varphi \rangle$$

Hence, $\langle \psi | \, A^\dagger$ is the bra corresponding to the ket $A\,|\psi\rangle$. Here are more properties of $A^\dagger$:

(a) The adjoint is antilinear with respect to scalar multiplication: $(\alpha A)^\dagger = \alpha^* A^\dagger$

(b) Reversal of products: $(AB)^\dagger = B^\dagger A^\dagger$

(c) The adjoint of the adjoint is the original operator: $(A^\dagger)^\dagger = A$

(d) The adjoint of the identity is the identity: $\mathbb{1}^\dagger = \mathbb{1}$

(e) The adjoint of a sum is the sum of the adjoints: $(A + B)^\dagger = A^\dagger + B^\dagger$

(f) Technically, the adjoint is only defined over linear operators, but we often write the relationship between a bra and its ket as if the bra is the "adjoint" of its ket

$$\langle \psi | = (|\psi\rangle)^\dagger \quad \text{and} \quad |\psi\rangle = (\langle \psi |)^\dagger$$

This abuse of notation makes more sense in the context of matrices, where the adjoint corresponds to the conjugate transpose of an operator's matrix representation.

1.7 Analysis of S-G Experiments

Armed with new mathematical rules and tools, we will analyze the ideal Stern-Gerlach experiments discussed earlier. In the process, we'll learn more about the mathematics of kets, observables, and projectors. We'll also discuss their physical significance.

1.7.1 Analysis of Experiment 1

The atoms coming from the upper port of the first analyzer are in the state $|+z\rangle$, and the projector corresponding to the upper port of the second analyzer is $P_+ = |+z\rangle\langle+z|$. Hence, the probability that atoms will exit the upper port of the second analyzer is

$$\langle\psi| P_+ |\psi\rangle = |\langle+z|+z\rangle|^2 = 1$$

and the post-measurement state for atoms that exit the upper port is

$$\frac{P_+ |+z\rangle}{\sqrt{\langle+z| P_+ |+z\rangle}} = |+z\rangle$$

No atoms exit the lower port because $\langle+z| P_- |+z\rangle = |\langle-z|+z\rangle|^2 = 0$.

1.7.2 Analysis of Experiment 2

The second analyzer effectively performs a measurement of the S_x operator. Following the same reasoning as for the S_z operator, it follows that

$$S_x = \tfrac{\hbar}{2} |+x\rangle\langle+x| - \tfrac{\hbar}{2} |+x\rangle\langle+x| \tag{1.17}$$

where $|\pm x\rangle$ also form an orthonormal basis for the vector space. In other words, there was nothing special about the z-axis in experiment 1, meaning we could have rotated the entire apparatus by $90°$ and obtained the same results along the x-axis.

The results of experiment 2 would be the same if we blocked the upper port instead. Including orthonormality, we have the following results:

$$\begin{aligned} |\langle+x|+z\rangle|^2 = |\langle-x|+z\rangle|^2 &= \tfrac{1}{2} \\ |\langle+x|-z\rangle|^2 = |\langle-x|-z\rangle|^2 &= \tfrac{1}{2} \\ \langle+x|+x\rangle = \langle-x|-x\rangle &= 1 \\ \langle+x|-x\rangle = \langle-x|+x\rangle &= 0 \end{aligned} \tag{1.18}$$

This is almost enough information to determine an expression for $|\pm x\rangle$ as linear combinations of $|\pm z\rangle$. With a, b, c, d all complex numbers,

$$|+x\rangle = a\,|+z\rangle + b\,|-z\rangle$$

$$|-x\rangle = c\,|+z\rangle + d\,|-z\rangle$$

Each complex number has a real and imaginary part, so there are actually 8 numbers to determine. The experimental results provide us with 6 independent conditions, so we'll need to make a couple

arbitrary choices at some point, and for that we use the following somewhat cryptic convention: the states $|+x\rangle$, $|+y\rangle$, $|+z\rangle$ form a "right-handed triplet." We'll leave that discussion for later, but for now it is straightforward to verify that the following conventional expressions match the experimental results:

$$|+x\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + |-z\rangle \Big) \tag{1.19}$$

$$|-x\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle - |-z\rangle \Big) \tag{1.20}$$

Consider Experiment 2 in the case where the second Stern-Gerlach analyzer is aligned along the y-axis. The experimental results are identical to when the second analyzer was aligned along the x-axis, and we can use that information to determine the $|\pm y\rangle$ states in terms of $|\pm z\rangle$ as

$$|+y\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + e^{i\alpha}\,|-z\rangle \Big)$$

$$|-y\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle - e^{i\alpha}\,|-z\rangle \Big)$$

where α is a real number yet to be determined. Consider a further modification to experiment 2 where the first Stern-Gerlach analyzer is aligned along the x-axis and the second analyzer is aligned along the y-axis. As you might expect, we find

$$|\,\langle +x|+y\rangle\,|^2 = \tfrac{1}{2},$$

and after expanding in the $|\pm z\rangle$ basis,

$$|\,\langle +x|+y\rangle\,|^2 = \tfrac{1}{4}\left|1 + e^{i\alpha}\right|^2 = \tfrac{1}{2}\left(1 + \cos\alpha\right) = \tfrac{1}{2}$$

This requires $\alpha = \pm\pi/2$ (or any odd multiple of $\pm\pi/2$). By convention, we choose $\alpha = \pi/2$ such that

$$|+y\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + i\,|-z\rangle \Big) \tag{1.21}$$

$$|-y\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle - i\,|-z\rangle \Big) \tag{1.22}$$

Notice that the first coefficient, the one in front of $|+z\rangle$, is a real positive constant in each of the final expressions for $|\pm x\rangle\,, |\pm y\rangle\,, |\pm z\rangle$. This was not by accident, and it turns out that the state vector has an extra degree of freedom we needed to account for with another convention. Any normalized ket $|\psi\rangle$ is *physically indistinguishable* from $e^{i\alpha}\,|\psi\rangle$ for any real number α. We simply fix the first coefficient to be real and positive.

1.7.3 Analysis of Experiment 3

With the lower port of the second analyzer blocked, the state of the atoms coming from the upper port that go into the third analyzer is

$$|+x\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + |-z\rangle \Big)$$

Hence, the probability an atom from the upper port of the second will exit the upper port of the third is $|\langle +z|+x\rangle|^2 = 1/2$. Recall that the probability that an atom from the upper port of the first analyzer will exit the upper port of the second is $|\langle +x|+z\rangle|^2 = 1/2$. Overall, the probability that an atom from the upper port of the first analyzer will exit the upper port of the second and the upper port of the third is

$$\begin{aligned}\mathcal{P}_{\text{upper},+} &= \langle +x| P_+ |+x\rangle \langle +z| P_{+x} |+z\rangle \\ &= |\langle +z|+x\rangle|^2 |\langle +x|+z\rangle|^2 = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}\end{aligned}$$

Likewise, the probability that an atom from the upper port of the first analyzer will exit the upper port of the second and the lower port of the third is

$$\begin{aligned}\mathcal{P}_{\text{upper},-} &= \langle +x| P_- |+x\rangle \langle +z| P_{+x} |+z\rangle \\ &= |\langle -z|+x\rangle|^2 |\langle +x|+z\rangle|^2 = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}\end{aligned}$$

We get the same results if we block the upper port of the second analyzer and measure the state of the atoms to be $|-x\rangle$ from the lower port before passing them off to the third analyzer. Hence,

$$\begin{aligned}\mathcal{P}_{\text{lower},+} &= \langle -x| P_+ |-x\rangle \langle +z| P_{-x} |+z\rangle \\ &= |\langle +z|-x\rangle|^2 |\langle -x|+z\rangle|^2 = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}\end{aligned}$$

$$\begin{aligned}\mathcal{P}_{\text{lower},-} &= \langle -x| P_- |-x\rangle \langle +z| P_{-x} |+z\rangle \\ &= |\langle -z|-x\rangle|^2 |\langle -x|+z\rangle|^2 = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}\end{aligned}$$

1.7.4 Analysis of Experiment 4

Experiment #4 is a much simpler example of quantum interference than the famous single-electron (or single-photon) double-slit experiment that you may have learned in a previous course. In my opinion, the simplicity of this experimental design does a much better job at exposing some of the most fascinating aspects of quantum theory and the fundamental nature of reality itself.

Let's consider two experimental designs–one that demonstrates quantum interference and one

that doesn't. In the design that *does not* demonstrate interference, we record which atoms exit the output ports of each analyzer. It doesn't matter if we actually keep a record, only that some interaction is taking place between the atom and its environment that would in principle allow us to distinguish its path. In this problem, we discuss the probability that an atom leaving the first analyzer in the $|+z\rangle$ state is detected in one of the counters connected to the output ports of the third analyzer. Such a probability involves two measurements at the second and third analyzers. The total probability is the product of the individual probabilities of each measurement.

For each output port of the third analyzer, there are two mutually exclusive paths that an atom could take. There is a probability that an atom from the upper port of the first analyzer exits the upper port of the second and the upper port of the third, and there is also a probability an atom from the upper port of the first analyzer exits the lower port of the second and the upper port of the third. The rules of probability (axiomatic in fact) dictate that the probabilities of mutually exclusive events must add. In other words, the probability that an atom will exit the upper port of the third analyzer is

$$p(+z) = \mathcal{P}_{\text{upper},+} + \mathcal{P}_{\text{lower},+} = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}$$

Now suppose we do not keep track of the atoms exiting the two ports of the second analyzer and simply arrange magnets to carefully recombine the beams. Classically it seems like there's no difference, but the rules of quantum mechanics say otherwise. After the second analyzer, it is possible to measure each atom in the state $|+x\rangle$ or $|-x\rangle$, so the state vector after the second analyzer must be the linear combination

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\,|+x\rangle + \tfrac{1}{\sqrt{2}}\,|-x\rangle$$

since this state would give equal probability of measuring $|\pm x\rangle$. In fact, since the initial state of the atoms coming from the upper port of the first analyzer is $|+z\rangle$, we can also write the state as

$$|\psi\rangle = \langle +x|+z\rangle\,|+x\rangle + \langle -x|+z\rangle\,|-x\rangle$$

Depending on how we combine the beams, there could be some relative phase between the coefficients but we'll ignore that for now. The magnets are arranged to combine the beams and introduce no other effects. Eqs. (1.19) & (1.20) can be used to show that $|\psi\rangle = |+z\rangle$ and the analysis is pretty straightforward, but let's pretend we don't know this shortcut and proceed anyways.

The probability that an atom from the first analyzer will exit the upper port of the third analyzer, given that it could go through either port of the second is

$$\begin{aligned} p(+z) &= \langle\psi|\,P_+\,|\psi\rangle \\ &= \Big|\,\langle +x|+z\rangle\,\langle +z|+x\rangle + \langle -x|+z\rangle\,\langle +z|-x\rangle\,\Big|^2 \end{aligned}$$

Letting $c_1 = \langle +x|+z\rangle \langle +z|+x\rangle$ and $c_2 = \langle -x|+z\rangle \langle +z|-x\rangle$ denote the amplitudes,

$$\begin{aligned} p(+z) &= |c_1|^2 + |c_2|^2 + 2\mathrm{Re}\left[c_1 c_2\right] \\ &= \mathcal{P}_{\mathrm{upper},+} + \mathcal{P}_{\mathrm{lower},+} + \text{interference terms} \end{aligned}$$

In this case, $p(+z) = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$, so we have *constructive quantum mechanical interference.* The probability that an atom from the first analyzer will exit the lower port of the third, given that it could go through either port of the second is $p(-z) = 0$, so there is *destructive interference.*

We can summarize our conclusions by saying that if no measurement is made on the intermediate state, then we add amplitudes and then square to find the probability, while if an intermediate measurement is performed, we square the amplitudes first and then add to find the probability. One is the square of a sum and the other is the sum of squares, and only the former exhibits interference.

1.8 Matrix Notation

So far we have mathematically defined kets in terms of their inner products with other kets. If we choose a basis such as $|\pm z\rangle$, then any ket may be written as

$$|\psi\rangle = a\,|+z\rangle + b\,|-z\rangle \tag{1.23}$$

and the coefficients $a = \langle +z|\psi\rangle$ and $b = \langle -z|\psi\rangle$ uniquely specify the state. We can simplify the notation by representing the ket as a column vector. In order to do this, we first need to establish the column vectors corresponding to the basis kets, and for that we choose the "standard basis" by convention. In other words,

$$|+z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |-z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1.24}$$

An arbitrary state may now be written as a column vector of the form

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \tag{1.25}$$

For instance,

$$|+x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \tag{1.26}$$

$$|+y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |-y\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{1.27}$$

Let $|\psi\rangle = a\,|+z\rangle + b\,|-z\rangle$ and $|\phi\rangle = c\,|+z\rangle + d\,|-z\rangle$. Their inner product is

$$\langle\psi|\phi\rangle = (a^*\,\langle+z| + b^*\,\langle-z|)\,(c\,\langle+z| + d\,\langle-z|) = a^*c + b^*d. \tag{1.28}$$

In order for this result to be consistent with the rules of matrix multiplication, we use row vectors to represent bras. For instance,

$$\langle\psi| = \begin{pmatrix} a^* & b^* \end{pmatrix} \tag{1.29}$$

which is the conjugate transpose of the corresponding column vector. In matrix notation,

$$\langle\psi|\phi\rangle = \begin{pmatrix} a^* & b^* \end{pmatrix} \begin{pmatrix} c \\ d \end{pmatrix} = a^*c + b^*d \tag{1.30}$$

The simplest way to understand the matrix representation of operators is to use the standard basis where $|i\rangle$ is represented by a column vector with zeros everywhere except where it has a one in the ith row:

$$|1\rangle = \begin{pmatrix} 1 \\ 0 \\ \vdots \end{pmatrix}, \quad |2\rangle = \begin{pmatrix} 0 \\ 1 \\ \vdots \end{pmatrix}, \quad |i\rangle = \begin{pmatrix} \vdots \\ 1 \\ \vdots \end{pmatrix}$$

The standard orthonormal basis obeys a convenient inner product relation

$$\langle j|i\rangle = \delta_{ij} \tag{1.31}$$

where $|i\rangle$ and $|j\rangle$ belong to the standard basis and δ_{ij} is the Kronecker delta symbol. The standard basis allows us to easily compute the matrix elements of any linear operator. Suppose

$$\left|\psi'\right\rangle = A\,|\psi\rangle\,.$$

We can expand the kets as linear combinations of the standard orthonormal basis vectors as

$$\left|\psi'\right\rangle = \sum_i c'_i\,|i\rangle\,, \quad |\psi\rangle = \sum_i c_i\,|i\rangle$$

where c_i and c'_i are the components of the vectors in matrix representation. Since A is linear,

$$\sum_i c'_i\,|i\rangle = A\sum_i c_i\,|i\rangle = \sum_i c_i A\,|i\rangle\,.$$

Using Eq. 1.31, we can solve for each component of $|\psi'\rangle$:

$$\left\langle j\middle|\psi'\right\rangle = \sum_i c'_i\,\langle j|i\rangle = \sum_i c'_i\delta_{ij} = c'_j \tag{1.32}$$

$$\Rightarrow c'_j = \sum_i c_i \langle j| A |i\rangle = \sum_i A_{ji} c_i \tag{1.33}$$

In other words, the jth component of $|\psi'\rangle$ is a linear combination of the components of $|\psi\rangle$ in a manner that is consistent with matrix multiplication if we define **the matrix elements of the operator** A as

$$A_{ji} = \langle j| A |i\rangle \tag{1.34}$$

Eq. 1.33 can be written in matrix form as

$$\begin{pmatrix} c'_1 \\ c'_2 \\ \vdots \end{pmatrix} = \begin{pmatrix} \langle 1| A |1\rangle & \langle 1| A |2\rangle & \cdots \\ \langle 2| A |1\rangle & \langle 2| A |2\rangle & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix} \tag{1.35}$$

Even in cases where the bra and ket are not elements of the standard basis, we still refer to an expression like $\langle \phi| A |\psi\rangle$ as a matrix element.

Let's compute the matrix elements of the operator S_z acting on spin-1/2 states, with $|+z\rangle = |1\rangle$ and $|-z\rangle = |2\rangle$ as our basis vectors:

$$S_z = \begin{pmatrix} \langle +z| S_z |+z\rangle & \langle +z| S_z |-z\rangle \\ \langle -z| S_z |+z\rangle & \langle -z| S_z |-z\rangle \end{pmatrix} = \begin{pmatrix} \frac{\hbar}{2} & 0 \\ 0 & -\frac{\hbar}{2} \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{1.36}$$

Similarly, we find that

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

Alternatively, we could have used $|\pm x\rangle$ as our basis, in which case

$$S_x = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is why it's important to think of operators as more abstract quantities than their matrix representations–the matrix is basis dependent but the operator is not. Changing the basis does not affect any measurement outcomes.

Let's now compute the matrix elements of the tensor product operator $|i\rangle \langle j|$ when $|i\rangle$, $|j\rangle$ are from the standard basis.

$$\left(|i\rangle \langle j| \right)_{kl} = \langle k| \left(|i\rangle \langle j| \right) |l\rangle = \langle k| \left(\langle j|l\rangle |i\rangle \right) = \langle k|i\rangle \langle j|l\rangle = \delta_{ki}\delta_{jl}$$

This matrix therefore has zero elements everywhere except when $k = i$ and $l = j$. In other words,

the matrix $|i\rangle\langle j|$ is zero everywhere except for a 1 in the ith row and jth column:

$$|1\rangle\langle 1| = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 0 & \\ \vdots & & \ddots \end{pmatrix}, \quad |1\rangle\langle 2| = \begin{pmatrix} 0 & 1 & \cdots \\ 0 & 0 & \\ \vdots & & \ddots \end{pmatrix}.$$

Hence, any linear operator A can be written as

$$A = \sum_{i,j} A_{ij} |i\rangle\langle j| \tag{1.37}$$

In particular, when $i = j$, we have an elementary projection operators $|i\rangle\langle i|$. The completeness relation can be written as

$$\sum_i |i\rangle\langle i| = \mathbb{1} \tag{1.38}$$

The completeness relation is very useful in many calculations.

Given the matrix elements of an operator, we can easily compute the matrix elements of its adjoint:

$$A^\dagger_{ij} = \langle i| A^\dagger |j\rangle = \langle j| A |i\rangle^* = A^*_{ji}$$

In other words, the adjoint matrix is the conjugate transpose of the operator matrix: compute the transpose and apply complex conjugation to every matrix element. For example,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^\dagger = \begin{pmatrix} a^* & c^* \\ b^* & d^* \end{pmatrix}$$

Additionally, every bra vector is the conjugate transpose of the corresponding ket, and we often use the same symbol to denote the adjoint even though this is technically an abuse of notation:

$$\begin{pmatrix} a \\ b \end{pmatrix}^\dagger = \begin{pmatrix} a^* & b^* \end{pmatrix}$$

1.9 Statistical Analysis

In experiment 2, we perform measurements of S_x given the input state $|+z\rangle$ and collect statistical data on the measurement outcomes. We calculate the **expectation value** of S_x as a weighted average of the measurement outcomes, weighted by the measurement-outcome probability:

$$\langle S_x \rangle = \tfrac{\hbar}{2} \cdot \tfrac{1}{2} + \left(-\tfrac{\hbar}{2}\right) \cdot \tfrac{1}{2} = 0 \tag{1.39}$$

Notice that the term "expectation value" is a bit misleading, since we do not expect nor do we ever observe $S_x = 0$ as a measurement outcome. A better name for the quantity might be "expected

mean value" because it's the average we expect of a large number of experiments. It is not a time average, but an average over many identical experiments.

There is clearly **uncertainty** in the measurements, since they do not all yield the same value, and we quantify the uncertainty by computing the standard deviation:

$$\begin{aligned}(\Delta S_x)^2 &= \langle (S_x - \langle S_x \rangle)^2 \rangle \\ &= \langle S_x^2 - 2S_x\langle S_x\rangle + S_x\rangle^2\rangle \\ &= \langle S_x^2\rangle - 2\langle S_x\rangle\langle S_x\rangle + \langle S_x\rangle^2 \\ &= \langle S_x^2\rangle - \langle S_x\rangle^2 \end{aligned} \tag{1.40}$$

The expectation value of S_x^2 is again a weighted average of each value of S_x^2 measured:

$$\langle S_x^2\rangle = \left(\tfrac{\hbar}{2}\right)^2 \cdot \tfrac{1}{2} + \left(-\tfrac{\hbar}{2}\right)^2 \cdot \tfrac{1}{2} = \tfrac{\hbar^2}{4} \tag{1.41}$$

Hence, the uncertainty of S_x given the state $|+z\rangle$ is $\Delta S_x = \hbar/2$.

We often write the expectation value of an operator A as

$$\langle A\rangle = \langle\psi|\, A\, |\psi\rangle \tag{1.42}$$

where $|\psi\rangle$ is a normalized ket representing the quantum state that A is being measured on. For instance, consider the expectation value of S_z in some arbitrary state $|\psi\rangle$:

$$\begin{aligned}\langle S_z\rangle &= \tfrac{\hbar}{2}|\, \langle +z|\psi\rangle\, |^2 - \tfrac{\hbar}{2}|\, \langle -z|\psi\rangle\, |^2 \\ &= \tfrac{\hbar}{2}\, \langle\psi|+z\rangle\, \langle +z|\psi\rangle - \tfrac{\hbar}{2}\, \langle\psi|-z\rangle\, \langle -z|\psi\rangle \\ &= \langle\psi|\left(\tfrac{\hbar}{2}\, |+z\rangle\, \langle +z| - \tfrac{\hbar}{2}\, |-z\rangle\, \langle -z|\right)|\psi\rangle \\ &= \langle\psi|\, S_z\, |\psi\rangle \end{aligned}$$

The uncertainty, or standard deviation, of an operator A is

$$\Delta A = \sqrt{\langle A^2\rangle - \langle A\rangle^2} \tag{1.43}$$

Notice that the uncertainty only makes sense mathematically if $\langle A^2\rangle - \langle A\rangle^2 \geq 0$.

Example 1.3: Expectation Value and Uncertainty

A spin-1/2 particle is in the state

$$|\psi\rangle = \frac{1}{\sqrt{5}}(|+z\rangle + 2i\,|-z\rangle)$$

What are the expectation value $\langle S_z\rangle$ and uncertainty ΔS_z for this state?

Solution: First we calculate the expectation value of S_z as

$$\begin{aligned}\langle S_z\rangle &= \langle\psi|\,S_z\,|\psi\rangle \\ &= \left[\frac{1}{\sqrt{5}}(\langle+z| - 2i\,\langle-z|)\right] S_z \left[\frac{1}{\sqrt{5}}(|+z\rangle + 2i\,|-z\rangle)\right] \\ &= \frac{1}{5}\Big(\langle+z|\,S_z\,|+z\rangle + 2i\,\langle+z|\,S_z\,|-z\rangle - 2i\,\langle-z|\,S_z\,|+z\rangle + 4\,\langle-z|\,S_z\,|-z\rangle\Big) \\ &= \frac{1}{5}\left(\frac{\hbar}{2} - \frac{4\hbar}{2}\right) = -\frac{3\hbar}{10}\end{aligned}$$

Now we calculate the expectation value of S_z^2.

$$\langle S_z^2\rangle = \langle\psi|\,S_z^2\,|\psi\rangle = \frac{1}{5}\left(\frac{\hbar^2}{4} + \frac{4\hbar^2}{4}\right) = \frac{\hbar^2}{4}$$

The uncertainty is then

$$\Delta S_z = \sqrt{\langle S_z^2\rangle - \langle S_z\rangle^2} = \sqrt{\frac{25\hbar^2}{100} - \frac{9\hbar^2}{100}} = \frac{4\hbar}{10}$$

Each measurement of S_z will produce either $\hbar/2$ or $-\hbar/2$, but after repeated measurements of S_z on identically prepared systems in the state $|\psi\rangle$, we find that the expectation value of S_z is $-0.3\hbar$ and the uncertainty is $0.4\hbar$.

1.10 Commuting Observables

In experiment 3, we learned the two observables S_x and S_z could not be known or simultaneously measured, and we called them "incompatible." Let's explore what it means for two observables to be incompatible and how incompatibility affects measurement results.

Measuring S_x projects the input state to an eigenstate of S_x, and a subsequent measurement of the observable S_z projects an eigenstate of S_x to an eigenstate of S_z. In the end, we have an eigenstate of S_z which is a superposition of S_x eigenstates. If we measure S_z and then S_x, we end up with an eigenstate of S_x which is a superpositon of S_z eigenstates. Notice that the order of operations matters with respect to the final state, and the eigenstates of S_x are different than the eigenstates of S_z. We use a mathematical operation called the commutator to measure the degree

to which two operators fail to commute.

The **commutator** of two operators is defined as the difference between the products of the two operators taken in alternate orders:

$$[A, B] = AB - BA \tag{1.44}$$

If the commutator is equal to zero, we say that the operators commute.

Let A be diagonalizable with a set of non-degenerate eigenvalues. If a is an eigenvalue of A, there exists an eigenvector $|a\rangle$ such that $A|a\rangle = a|a\rangle$. If B is an operator that commutes with A,

$$A(B|a\rangle) = BA|a\rangle = B(a|a\rangle) = a(B|a\rangle).$$

Since A is non-degenerate, the vector $B|a\rangle$ must be some scalar multiple of $|a\rangle$. Let's call the scalar b such that $B|a\rangle = b|a\rangle$; this is an eigenvalue equation for B with eigenvalue a and eigenvector $|a\rangle$. Even when there is degeneracy, it turns out that **if A and B commute and are both diagonalizable, then they can be simultaneously diagonalized**, i.e., there exists a basis of common eigenvectors for both operators. Hence, we can know the eigenvalues of any two commuting observables simultaneously. It is common to extend this language and say that these two observables can be measured simultaneously. What we mean is that we can measure one observable without erasing our knowledge of the previous results of the other observable. Observables A and B are said to be **compatible**.

If two operators do not commute, then they are incompatible observables and cannot be measured or known simultaneously. For example,

$$[S_z, S_x] = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} - \frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\frac{\hbar}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} = i\hbar S_y$$

In fact, none of the spin component operators commute and the complete set of *commutation relations* is

$$[S_x, S_y] = i\hbar S_z, \quad [S_z, S_x] = i\hbar S_y, \quad [S_y, S_z] = i\hbar S_x \tag{1.45}$$

We'll learn the origins of these commutation relations in chapter #4.

1.11 Uncertainty Principle

The **uncertainty principle** of quantum mechanics states that if A and B are *self-adjoint linear operators* on a complex vector space, then

$$\Delta A \Delta B \geq \tfrac{1}{2}\left|\langle [A,B] \rangle\right| \tag{1.46}$$

where $\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$ is the uncertainty (standard deviation) and

$$\langle [A,B] \rangle = \langle \psi | \, [A,B] \, | \psi \rangle$$

for all states $|\psi\rangle$ in the vector space.

We will prove this theorem after we've learned more about self-adjoint operators (also called Hermitian operators), which play a significant role in quantum theory. For now, let's consider what it says about a simple Stern-Gerlach experiment. For example,

$$\begin{aligned} \Delta S_x \Delta S_y &\geq \tfrac{1}{2}\left|\langle [S_x, S_y] \rangle\right| \\ &\geq \tfrac{1}{2}\left|i\hbar \langle S_z \rangle\right| \\ &\geq \tfrac{\hbar}{2}\left|\langle S_z \rangle\right| \end{aligned} \tag{1.47}$$

If the quantum state is an eigenstate of either S_x or S_y, then either ΔS_x or ΔS_y will vanish, and it's easy to show that the expectation value of S_z vanishes as well. Anytime S_z has a nonzero expectation value, the uncertainty principle tells us that neither ΔS_x nor ΔS_y can be zero. For instance, suppose we know that $S_z = \hbar/2$ such that the state of the system is $|+z\rangle$. Then,

$$\Delta S_x \Delta S_y \geq \left(\tfrac{\hbar}{2}\right)^2$$

The uncertainties can be calculated and it turns out that, in this case, $\Delta S_x = \hbar/2$ and $\Delta S_y = \hbar/2$ such that

$$\Delta S_x \Delta S_y = \left(\tfrac{\hbar}{2}\right)^2$$

and the uncertainty principle is satisfied. We say that the spin-up state $|+z\rangle$ is a **minimum uncertainty state** for the observables S_x and S_y, meaning the uncertainty relation is saturated in this state. Although S_x and S_y are individually uncertain, their uncertainties are balanced in such a way that their product attains the minimum allowed by quantum mechanics.

Example 1.4: Uncertainty Principle

A spin-1/2 particle is in the state

$$|\psi\rangle = \frac{1}{\sqrt{5}}(|+z\rangle + 2i\,|-z\rangle)$$

Repeated measurements of S_x and S_y are made on the particle prepared in this state. Verify the uncertainty relation $\Delta S_x \Delta S_y \geq \frac{\hbar}{2}|\langle S_z\rangle|$.

Solution: Let's use matrix notation to speed this along.

$$\langle S_x\rangle = \langle\psi|\, S_x\, |\psi\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix}1 & -2i\end{pmatrix}\cdot\frac{\hbar}{2}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\cdot\frac{1}{\sqrt{5}}\begin{pmatrix}1\\ 2i\end{pmatrix}$$

$$= \frac{\hbar}{10}\begin{pmatrix}1 & -2i\end{pmatrix}\begin{pmatrix}2i\\ 1\end{pmatrix} = 0$$

$$\langle S_x^2\rangle = \langle\psi|\, S_x^2\, |\psi\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix}1 & -2i\end{pmatrix}\cdot\frac{\hbar^2}{4}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\cdot\frac{1}{\sqrt{5}}\begin{pmatrix}1\\ 2i\end{pmatrix}$$

$$= \frac{\hbar^2}{20}\begin{pmatrix}1 & -2i\end{pmatrix}\begin{pmatrix}1\\ 2i\end{pmatrix} = \frac{\hbar^2}{4}$$

The uncertainty $\Delta S_x = 0.5\hbar$.

$$\langle S_y\rangle = \langle\psi|\, S_y\, |\psi\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix}1 & -2i\end{pmatrix}\cdot\frac{\hbar}{2}\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix}\cdot\frac{1}{\sqrt{5}}\begin{pmatrix}1\\ 2i\end{pmatrix}$$

$$= \frac{\hbar}{10}\begin{pmatrix}1 & -2i\end{pmatrix}\begin{pmatrix}2\\ i\end{pmatrix} = \frac{4\hbar}{10}$$

$$\langle S_y^2\rangle = \langle\psi|\, S_y^2\, |\psi\rangle = \frac{1}{\sqrt{5}}\begin{pmatrix}1 & -2i\end{pmatrix}\cdot\frac{\hbar^2}{4}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\cdot\frac{1}{\sqrt{5}}\begin{pmatrix}1\\ 2i\end{pmatrix}$$

$$= \frac{\hbar^2}{20}\begin{pmatrix}1 & -2i\end{pmatrix}\begin{pmatrix}1\\ 2i\end{pmatrix} = \frac{\hbar^2}{4}$$

The uncertainty $\Delta S_y = 0.3\hbar$. In an earlier example, we found $\langle S_z\rangle = -0.3\hbar$.

$$\Delta S_x \Delta S_y = (0.5\hbar)(0.3\hbar) = 0.15\hbar^2 \quad \text{and} \quad \frac{\hbar}{2}|\langle S_z\rangle| = 0.15\hbar^2$$

Evidently, $|\psi\rangle$ is a "minimum uncertainty state" for S_x and S_y. **Your turn**. Verify that the following uncertainty relation is satisfied: $\Delta S_y \Delta S_z \geq \frac{\hbar}{2}|\langle S_x \rangle|$.

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Chapter 2

Postulates of Quantum Mechanics

2.1 Brief Historical Digression

We'll zoom out from the Stern-Gerlach experiment and discuss the postulates of quantum mechanics more generally, although mostly within the context of finite-dimensional vector spaces. We'll revise the postulates later when we need to work with *wave functions* and infinite-dimensional vector spaces. Much of the intuition we gain from the finite-dimensional case will carry over to infinite dimensional spaces anyways. Let's begin with a brief historical digression–not because the historical details matter but because quantum theory is abstract and it can be hard to imagine where these principles came from. You may find quantum theory to be more palatable if you treat it like a game with simple but arbitrary rules.

The rules were first formulated in 1925 by Heisenberg, Born, and Jordan. Heisenberg in particular took a radical step by abandoning particle trajectories and formulating a theory only in terms of measurable quantities (observables). He found that the relationships between observables could be expressed using arrays of numbers (matrices), leading to a new algebraic, non-visual formulation of quantum mechanics called matrix mechanics. Heisenberg's work is significant because it marked a departure from classical intuition and set the stage for the more abstract nature of quantum mechanics.

Schrödinger developed wave mechanics in 1926 as an alternative formulation of quantum mechanics, offering a more intuitive and visualizable approach compared to Heisenberg's matrix mechanics. Inspired by Louis de Broglie's matter wave hypothesis, Schrödinger proposed that particles, like electrons, could be described by wave functions evolving in space and time according to a differential equation (the Schrödinger equation). Unfortunately, as elucidated by Max Born (the maternal grandfather of famed actress Olivia Newton-John), wave functions are ultimately used to compute probabilities, and they do not exist in three-dimensional physical space like ordinary waves, so their interpretation is not as intuitive as Schrödinger initially hoped.

In 1928, Dirac merged Schrödinger's wave mechanics and Heisenberg's matrix mechanics into a

single, abstract framework (and also introduced the bra-ket notation)[1]. Finally, in 1932, John von Neumann provided mathematical rigor, turning quantum mechanics into a well-defined axiomatic system and inventing a large portion of a field known as *functional analysis* (beyond the scope of our class). The postulates of quantum mechanics dictate how to treat a quantum mechanical system mathematically and how to interpret the mathematics to learn about the physical system in question. These postulates cannot be proven, but they have been successfully tested by many experiments, so we accept them as an accurate way to describe quantum mechanical systems. Quantum mechanics has five basic rules, and I've come up with a silly mnemonic to help you remember them:

Small **O**bjects **B**reak **C**lassical **T**heories

2.2 Kinematical Postulates

Postulate 1

The **state** of a quantum system is represented by a normalized vector $|\psi\rangle$ belonging to a complex vector space with an inner product.

The quantum state vector is a normalized vector (ket) and it contains complex components corresponding to all possible measurement configurations. Kets can be multiplied by complex numbers and added together. Any linear combination of kets is also a ket, but this does not mean that the system "exists in multiple states at once" (this is a very common phrase but it's unnecessary and vague). Two states are said to be *physically equivalent* if they differ only by a multiplicative phase factor, i.e. $e^{i\alpha}$ where α is a real-valued constant.

Mathematicians call the vector space of quantum states a *Hilbert space*. A Hilbert space is a real or complex vector space equipped with an inner product, and it is also a *complete metric space* with respect to the distance induced by that inner product. The inner product allows us to define lengths and angles between states, while the associated metric gives a precise notion of distance between vectors.

Completeness is a crucial mathematical property. Informally, it ensures that the space contains all of the limits it ought to contain: if a sequence of vectors becomes arbitrarily close to one another, then there exists a vector in the space to which the sequence converges. This allows us to use the tools of calculus; limits, infinite sums, and continuous transformations.

In quantum mechanics, completeness is especially important because many physically relevant states arise as limits of infinite superpositions. When we work with infinite-dimensional vector spaces, completeness ensures that infinite superpositions of states correspond to well-defined vectors in the Hilbert space. For example, a general *wave function* may be expressed as an infinite sum of

[1]It was Schrödinger who demonstrated equivalence between matrix mechanics and wave mechanics.

energy eigenstates, and completeness guarantees that such sums converge to legitimate quantum states.

Because of this property, many of the techniques and intuitions developed for finite-dimensional systems, such as expansions in orthonormal bases and the use of linear operators, carry over to infinite-dimensional systems with minimal modification. This mathematical stability is one reason Hilbert spaces provide the natural setting for quantum mechanics.

A detailed study of the mathematics of Hilbert spaces lies beyond the scope of this course. In what follows, we will take many of these mathematical structures and results for granted and focus instead on their physical meaning and applications.

Postulate 2

Every measurable physical quantity, called an **observable**, is described by a self-adjoint linear operator. The result of measuring an observable is always an eigenvalue of the corresponding operator, and the operator's eigenvectors form a basis for the vector space.

Recall that a self-adjoint operator has the property that it is equal to its adjoint, i.e. $A^\dagger = A$. While postulate #2 states unambiguously (and without proof) that observables are represented by self-adjoint linear operators, we gain confidence and intuition by examining the plausibility of this postulate. We need to establish a few properties of self-adjoint operators before we can remark on their ability to represent observables.

(a) The expectation value of a self-adjoint operator $A = A^\dagger$ is a real number for all vectors.

$$\langle A\rangle^* = \langle\psi|\, A\,|\psi\rangle^* = \langle\psi|\, A^\dagger\,|\psi\rangle = \langle A\rangle$$

(b) The eigenvalues of a self-adjoint operator $A = A^\dagger$ are real. Let $|a_n\rangle$ be a normalized eigenvector of A, i.e. $A\,|a_n\rangle = a_n\,|a_n\rangle$ and $\langle a_n|a_n\rangle = 1$. From (a), it follows immediately that $a_n = \langle a_n|\, A\,|a_n\rangle$ is real.

Interpretation of (a) and (b): All measurement outcomes in quantum mechanics will be real numbers. We don't have to worry about computing a complex value for position or spin just because we're working with vectors over complex numbers.

(c) Eigenvectors corresponding to distinct eigenvalues are orthogonal. Let $|a_1\rangle$ and $|a_2\rangle$ be two normalized eigenvectors of A such that $A\,|a_1\rangle = a_1\,|a_1\rangle$ and $A\,|a_2\rangle = a_2\,|a_2\rangle$. It follows that

$$\langle a_2|\, A\,|a_1\rangle = a_1\,\langle a_2|a_1\rangle \;\text{ and }\; \langle a_1|\, A\,|a_2\rangle = a_2\,\langle a_1|a_2\rangle$$

Since A is self-adjoint, $\langle a_1|\, A\,|a_2\rangle^* = \langle a_2|\, A\,|a_1\rangle$, and since the eigenvalues are real,

$$a_1\,\langle a_2|a_1\rangle = a_2^*\,\langle a_1|a_2\rangle^* = a_2\,\langle a_2|a_1\rangle$$

$$(a_2 - a_1)\langle a_2|a_1\rangle = 0$$

If the eigenvalues are distinct, namely $a_2 \neq a_1$, then $\langle a_2|a_1\rangle = 0$.

Interpretation of (c): Two measurement outcomes that are distinct should correspond to the system being in two different states. Otherwise the states would not be well-defined. The measurement outcomes are mutually exclusive, i.e. if one measurement outcome occurs, a different outcome certainly does not occur. Note that if two states share an eigenvalue, their inner product may not necessarily be zero (the eigenvalues may be *degenerate*).

In finite-dimensional vector spaces, the eigenvectors of a self-adjoint operator provably span the vector space. Under precise conditions, the spectral theorem from functional analysis states that self-adjoint linear operators can be diagonalized with a complete set of "generalized eigenstates," even in infinite dimensions. In this course, we'll simply take completeness as an axiom.

2.2.1 Spectral theorem (finite dimensional case)

Let A be a self-adjoint linear operator on a finite-dimensional complex vector space. The following statements are equivalent:

(i) There exists an orthonormal basis of the vector space consisting of eigenvectors of A.

(ii) A is diagonalizable by a unitary transformation $A = UDU^\dagger$ where U is is a unitary matrix whose columns are orthonormal eigenvectors of A and D is a real diagonal matrix whose entries are the eigenvalues of A.

The **spectral decomposition** of a self-adjoint operator is a weighted sum of projectors given by

$$A = \sum_n a_n A_n = \sum_n a_n |a_n\rangle\langle a_n| \tag{2.1}$$

where $\{a_n\}$ are the eigenvalues (the weights) and $\{|a_n\rangle\}$ is an orthonormal set of eigenvectors. We will accept the spectral theorem without proof in this class, although you are welcome to consult supplementary resources.

Postulate 3

The probability of obtaining an eigenvalue a_n in the measurement of an observable A on a quantum system in the state $|\psi\rangle$ is given by the **Born rule**:

$$p(a_n) = \langle\psi| P_n |\psi\rangle \tag{2.2}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n.

It is possible for an observable to have multiple orthogonal eigenstates corresponding to the same eigenvalue. In this case, we say the eigenvalue is degenerate. When measuring the observable, there is no way to distinguish between the eigenstates corresponding to a degenerate eigenvalue.

(i) When the eigenvalue a_n is non-degenerate, then $P_n = |a_n\rangle \langle a_n|$ for a unique normalized vector $|a_n\rangle$ and the probability becomes

$$p(a_n) = |\langle a_n|\psi\rangle|^2 \tag{2.3}$$

This can also be viewed as the probability that the state collapses to $|a_n\rangle$, even when the associated eigenvalue is degenerate.

(ii) When the eigenvalue a_n is degenerate, there are multiple eigenvectors with the same eigenvalue, and the projection operator is a sum over projectors

$$P_n = \sum_{i=1}^{k} |a_i\rangle \langle a_i| \tag{2.4}$$

where k is the degeneracy of the eigenvalue and the probability becomes

$$p(a_n) = \sum_{i=1}^{k} |\langle a_i|\psi\rangle|^2 \tag{2.5}$$

From postulate #2, we can express any vector from the vector space in terms of an observable's eigenvectors, which we can assume are orthonormal without loss of generality:

$$|\psi\rangle = \sum_i \langle a_i|\psi\rangle |a_i\rangle$$

$$\langle\psi|\psi\rangle = \sum_i |\langle a_i|\psi\rangle|^2 = 1$$

The quantities $|\langle a_i|\psi\rangle|^2 \geq 0$ sum to 1, so it seems plausible they define a discrete probability distribution over the measurement outcomes in a finite dimensional setting.

2.2.2 Proof of the Uncertainty Principle

Recall the **uncertainty principle** states that for observables A and B,

$$\Delta A \Delta B \geq \tfrac{1}{2}\big|\langle [A, B]\rangle\big| \tag{2.6}$$

The uncertainty principle alone does not mean that any single measurement is necessarily imprecise. In fact, each time you perform a single measurement of an observable, you may get an infinitely precise value. Instead, the uncertainty principle is a statistical statement about repeated measure-

ments of different observables on identically prepared quantum states. It is possible to derive the uncertainty principle from the postulates of quantum mechanics.

We begin by proving the **Schwarz inequality**

$$\langle\alpha|\alpha\rangle\,\langle\beta|\beta\rangle \geq |\,\langle\alpha|\beta\rangle\,|^2 \tag{2.7}$$

which is analogous to $|\mathbf{a}|^2|\mathbf{b}|^2 \geq |\mathbf{a}\cdot\mathbf{b}|^2$ for real Euclidean space. For kets $|\alpha\rangle\,, |\beta\rangle$,

$$(\langle\alpha| + \lambda^*\,\langle\beta|) \cdot (|\alpha\rangle + \lambda\,|\beta\rangle) \geq 0$$

since the inner product of a ket with itself is nonnegative. The relation above holds for any complex number λ, even when

$$\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle} \quad \text{where } |\beta\rangle \neq 0$$

Expanding the inner product and multiplying the relation by $\langle\beta|\beta\rangle$, we find

$$\langle\alpha|\alpha\rangle\,\langle\beta|\beta\rangle - |\,\langle\alpha|\beta\rangle\,|^2 \geq 0$$

Consider $|\alpha\rangle = \tilde{A}\,|\psi\rangle$ and $|\beta\rangle = \tilde{B}\,|\psi\rangle$ for an arbitrary ket $|\psi\rangle$ where $\tilde{A} = A - \langle A\rangle$ and $\tilde{B} = B - \langle B\rangle$. Since $\tilde{A}$ and $\tilde{B}$ are Hermitian,

$$(\Delta A)^2(\Delta B)^2 \geq |\langle\tilde{A}\tilde{B}\rangle|^2$$

where $(\Delta A)^2 = \langle\tilde{A}^2\rangle$ and $(\Delta A)^2 = \langle\tilde{B}^2\rangle$ are the uncertainties in A and B respectively. Note

$$\tilde{A}\tilde{B} = \tfrac{1}{2}[\tilde{A},\tilde{B}] + \tfrac{1}{2}\{\tilde{A},\tilde{B}\}$$

where $\{\tilde{A},\tilde{B}\} = \tilde{A}\tilde{B} + \tilde{B}\tilde{A}$ denotes the anti-commutator. The anti-commutator of two Hermitian operators is obviously Hermitian, but the commutator is anti-Hermitian:

$$[\tilde{A},\tilde{B}]^\dagger = [A,B]^\dagger = (AB - BA)^\dagger = -[A,B]$$

Taking the expectation value, we find

$$\langle\tilde{A}\tilde{B}\rangle = \tfrac{1}{2}\langle[\tilde{A},\tilde{B}]\rangle + \tfrac{1}{2}\langle\{\tilde{A},\tilde{B}\}\rangle$$

where $\langle[\tilde{A},\tilde{B}]\rangle$ is pure imaginary and $\langle\{\tilde{A},\tilde{B}\}\rangle$ is pure real. Hence,

$$|\langle\tilde{A}\tilde{B}\rangle|^2 = \tfrac{1}{4}|\langle[\tilde{A},\tilde{B}]\rangle|^2 + \tfrac{1}{4}|\langle\{\tilde{A},\tilde{B}\}\rangle|^2 \geq \tfrac{1}{4}|\langle[\tilde{A},\tilde{B}]\rangle|^2 = \tfrac{1}{4}|\langle[A,B]\rangle|^2$$

By omitting the anti-commutator term, we make the resulting inequality even stronger. The uncertainty relation immediately follows:

$$\Delta A\Delta B \geq \tfrac{1}{2}|\langle[A,B]\rangle|$$

2.3 Dynamical Postulates

Postulate 4

If a measurement of an observable A on a quantum system in the state $|\psi\rangle$ yields the result a_n, the quantum system irreversibly **collapses** to a new state

$$|\psi'\rangle = \frac{P_n |\psi\rangle}{\sqrt{\langle\psi| P_n |\psi\rangle}} \tag{2.8}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n. The new state is the normalized projection of $|\psi\rangle$ onto the ket (or kets) corresponding to the result of the measurement.

Allow me to emphasize just how *annoying* this postulate is in light of postulate #3. Not only can you not predict with 100% certainty which outcome you will get, but the state irreversibly changes depending on the outcome!

(i) When the eigenvalue a_n is non-degenerate, then the state vector collapses to the single normalized ket $|a_n\rangle$ that corresponds to the eigenvalue a_n.

(ii) When the eigenvalue is degenerate, then the state collapses to the eigensubspace that corresponds to the eigenvalue a_n. It is a common misconception that the state will collapse to an equal superposition of eigenstates with the same eigenvalue obtained in the measurement, but it depends on the state of the system right before measurement.

We know how to describe a quantum system at any given time, and we know how to model the effect of a measurement, but we haven't yet discussed what quantum systems are doing "when nobody's watching."

Postulate 5

The **time evolution** of a closed quantum system is described by a unitary transformation. That is, the state of the system at time t_0 is related to the state of the system at time t by a unitary operator U which depends on t and t_0 such that

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle \tag{2.9}$$

For continuous evolution, the state vector obeys a differential equation called the Schrödinger equation given by

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H(t) |\psi(t)\rangle \tag{2.10}$$

where the Hamiltonian $H(t)$ is the observable associated with the total energy of the

system and $\hbar$ is the reduced Planck's constant (a universal constant).

Julian Schwinger once remarked, when lecturing on quantum mechanics, "for fundamental properties we will borrow only names from classical physics." You may recognize the terms "Hamiltonian" and "energy" in the fifth postulate, but I would kindly ask you to remember that we are describing a theory for which classical mechanics is merely an approximation. It may be the case that we have to abandon certain ideas attached to classical energy in order to fully capture the behavior of quantum systems.

We'll cover postulate #5 in more detail in the next section, but for now I'd like to discuss unitary transformations and the role they play in time evolution. The fact that one state vector is related to another by a linear operator is not too surprising. A unitary operator has the additional property that its left or right inverse is equal to its corresponding adjoint operator. In other words, if U is a unitary operator, then $U^\dagger U = UU^\dagger = \mathbb{1}$.

Claim: If U is a linear operator, and $|\psi\rangle$ is a normalized vector, then $U|\psi\rangle$ is normalized if and only if U is a unitary operator.

Proof: If U is unitary, then the norm squared of $U|\psi\rangle$ is $\langle\psi|U^\dagger U|\psi\rangle = \langle\psi|\mathbb{1}|\psi\rangle = \langle\psi|\psi\rangle = 1$. On the other hand, if $U|\psi\rangle$ is normalized, then

$$\langle\psi|U^\dagger U|\psi\rangle = 1 = \langle\psi|\mathbb{1}|\psi\rangle$$

It might be tempting to immediately conclude that $U^\dagger U = \mathbb{1}$ just because $|\psi\rangle$ is arbitrary, but there's no obvious sense of cancellation that applies here. Rearranging the expression, we find

$$\langle\psi|\left(U^\dagger U - \mathbb{1}\right)|\psi\rangle = \langle\psi|A|\psi\rangle = 0$$

where $A = U^\dagger U - \mathbb{1}$. Again, it might be tempting to immediately conclude that $A = 0$ and $U^\dagger U = \mathbb{1}$, but this isn't obvious, and in fact, it simply isn't true for an arbitrary linear operator A! In case you don't buy it, here's a simple counterexample; consider the matrix

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For any real vector $|\psi\rangle = \begin{pmatrix} x \\ y \end{pmatrix}$, we see that $\langle\psi|A|\psi\rangle = xy - yx = 0$ even though $A \neq 0$.

Here's where it gets a bit creative–if you can think of a better proof, please let me know! Consider the linear combination $|\phi + \alpha\psi\rangle = |\phi\rangle + \alpha|\psi\rangle$ where α is an arbitrary scalar. Since this is guaranteed to be a ket in the vector space,

$$0 = \langle\psi + \alpha\phi|A|\psi + \alpha\phi\rangle$$

$$= \langle\psi| A |\psi\rangle + \alpha^* \langle\phi| A |\psi\rangle + \alpha \langle\psi| A |\phi\rangle + \alpha^*\alpha \langle\phi| A |\phi\rangle$$

$$= \alpha^* \langle\phi| A |\psi\rangle + \alpha \langle\psi| A |\phi\rangle$$

Let $c = \langle\psi| A |\phi\rangle$. Since A is Hermitian, $c^* = \langle\phi| A |\psi\rangle$ and it follows that

$$\alpha c + (\alpha c)^* = 0$$

This expression must hold for all complex numbers α. When $\alpha = 1$, $c + c^* = 0$, and we conclude the real part of c must vanish. On the other hand, when $\alpha = i$, $c - c^* = 0$, which means the imaginary part of c must vanish. If both the real and imaginary parts are zero, then $c = \langle\psi| A |\phi\rangle = 0$. In particular, for all $|\psi\rangle = A |\phi\rangle$, we find that $\langle\psi|\psi\rangle = 0$, and it follows from the definition of the inner product that $|\psi\rangle = A |\phi\rangle = 0$. By definition of the zero operator, $A = 0$, and we finally conclude that $U^\dagger U = \mathbb{1}$.

The point of this exercise was to demonstrate that unitary operators are the only operators that map normalized kets to normalized kets, and since a normalized ket contains all the information of a quantum system, a unitary operator is "information conserving" in a very informal sense. It is crucial for the statistical interpretation that unitary operators preserve the norm of the quantum state vector as it evolves over time in a closed system.

The argument above assumed that time evolution is represented by a *linear operator*. **Wigner's theorem** shows that this assumption is not needed: any transformation of pure states that preserves the transition probabilities between every pair of states is represented by either a unitary or an antiunitary operator. Since time evolution varies continuously and reduces to the identity when $t = t_0$, it must be unitary. Thus, unitarity follows from preserving transition probabilities and continuous time evolution, without assuming linearity at the outset. The proof of Wigner's theorem is well beyond the scope of this class, but the theorem itself is worth mentioning!

2.4 Generator of Time Translations

In the "Schrödinger picture" of quantum mechanics, the state vector evolves over time between measurements. The law governing time evolution is the unitary transformation

$$|\psi(t)\rangle = U(t, t_0) |\psi(t_0)\rangle \tag{2.11}$$

where $U(t, t) = \mathbb{1}$ for any time t. It's easy to demonstrate that the time evolution operators obey a composition rule. Given three times t_0, t_1, and t_2, the state can evolve unitarily from $t_0 \to t_1$ and then again from $t_1 \to t_2$. The composition rule is

$$U(t_2, t_0) = U(t_2, t_1)U(t_1, t_0) \tag{2.12}$$

where the operator $U(t_2, t_1)$ can be understood as $U(t_2, t_0)U^\dagger(t_1, t_0)$ in case there happens to be something "special" about t_0. Assuming time is a continuous parameter, the composition rule implies that time evolution divides into infinitesimal unitary transformations. We imagine the state vector is evolving smoothly just like a particle's trajectory in classical mechanics.

If $U(t', t)$ is differentiable, we may consider an infinitesimal time evolution from time t to time $t' = t + \epsilon$ where ϵ is small. Expanding $U(t + \epsilon, t)$ to first order in ϵ,

$$U(t + \epsilon, t) = U(t, t) + \epsilon \left.\frac{\partial U(t', t)}{\partial t'}\right|_{t'=t} + O(\epsilon^2)$$

or

$$U(t + \epsilon, t) = \mathbb{1} - i\epsilon\Omega(t) + O(\epsilon^2) \tag{2.13}$$

where we've defined a new linear operator $\Omega(t)$ with units of $[\text{time}]^{-1}$ as

$$\Omega(t) = i\left.\frac{\partial U(t', t)}{\partial t'}\right|_{t'=t} \tag{2.14}$$

In defining this operator, we've included the factor of i to ensure $\Omega^\dagger(t) = \Omega(t)$, which follows from unitarity of $U(t', t)$. The adjoint is given by

$$U^\dagger(t + \epsilon, t) = \mathbb{1} + i\epsilon\Omega^\dagger(t)$$

and it follows that

$$\begin{aligned} U^\dagger(t + \epsilon, t)U(t + \epsilon, t) &= \Big(\mathbb{1} + i\epsilon\Omega^\dagger(t)\Big)\Big(\mathbb{1} - i\epsilon\Omega(t)\Big) \\ &= \mathbb{1} + i\epsilon\Big(\Omega^\dagger(t) - \Omega(t)\Big) + O(\epsilon^2) \end{aligned}$$

The linear term must vanish, so $\Omega^\dagger(t) = \Omega(t)$, and we've shown that $\Omega(t)$ is Hermitian. Presumably, it corresponds to some physical quantity that we can measure. Formally, $\Omega(t)$ is the *generator of time translations* in quantum mechanics, because it gives the small correction needed to advance a state from time t to $t + \epsilon$. Evolution over a finite time interval can be regarded as the result of a large number of small displacements in time, so knowing $\Omega(t)$ allows us to evolve the system over any time interval.

2.4.1 The Schrödinger Equation

We can obtain a differential equation for $U(t, t_0)$ by differentiating with respect to the final time t and expanding the definition of the derivative:

$$\frac{\partial U(t, t_0)}{\partial t} = \lim_{\epsilon\to 0} \frac{U(t + \epsilon, t_0) - U(t, t_0)}{\epsilon} \tag{2.15}$$

Using the composition rule, $U(t+\epsilon, t_0) = U(t+\epsilon, t)U(t, t_0)$, and we have

$$\frac{\partial U(t, t_0)}{\partial t} = \left[\lim_{\epsilon \to 0} \frac{U(t+\epsilon, t) - \mathbb{1}}{\epsilon}\right] U(t, t_0) \tag{2.16}$$

where the term in square brackets is $-i\Omega(t)$. A differential equation for $U(t, t_0)$ is then

$$i\frac{\partial U(t, t_0)}{\partial t} = \Omega(t) U(t, t_0) \tag{2.17}$$

We can also find a differential equation for the quantum state by differentiating Eq. (2.11) with respect to the final time t and substituting Eq. (2.17)

$$\frac{d}{dt}|\psi(t)\rangle = \frac{\partial U(t, t_0)}{\partial t}|\psi(t_0)\rangle = -i\Omega(t)U(t, t_0)|\psi(t_0)\rangle$$

or

$$i\frac{d}{dt}|\psi(t)\rangle = \Omega(t)|\psi(t)\rangle \tag{2.18}$$

So far, there has been no mention of energies or Hamiltonians. In other words, up until this point, there has been very little (recognizable) physics. A straightforward application of the product rule reveals an important identity concerning the expectation value of $\Omega(t)$:

$$\frac{d}{dt}\langle\psi(t)|\,\Omega(t)\,|\psi(t)\rangle = \left[\frac{d}{dt}\langle\psi(t)|\right] \underbrace{\Omega(t)|\psi(t)\rangle}_{i\frac{d}{dt}|\psi(t)\rangle} + \underbrace{\langle\psi(t)|\,\Omega(t)}_{-i\frac{d}{dt}\langle\psi(t)|} \left[\frac{d}{dt}|\psi(t)\rangle\right] + \left\langle \frac{\partial\Omega(t)}{\partial t} \right\rangle$$

$$\therefore \quad \frac{d}{dt}\langle\Omega(t)\rangle = \left\langle \frac{\partial\Omega(t)}{\partial t} \right\rangle \tag{2.19}$$

Hence, the expectation value of $\Omega(t)$ is conserved when the operator itself does not depend on time. In classical mechanics, the Hamiltonian is (i) used to take the state of the system at one time and produce the state of the system at an infinitesimal time later, and (ii) conserved when it does not depend explicitly on time. The analogy could be taken further, but ultimately, we simply define the quantum mechanical Hamiltonian as

$$H(t) = \hbar\Omega(t) = i\hbar\frac{\partial U(t', t)}{\partial t'}\bigg|_{t'=t} \tag{2.20}$$

where $\hbar$ is a universal constant with units of (energy) $\times$ (time) so that $H(t)$ has units of energy, and the precise numerical value of $\hbar$ must be obtained from experiments. We then arrive at the **Schrödinger equation**, a first-order differential equation for the state vector, as a consequence of smooth unitary time evolution:

$$i\hbar\frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle \tag{2.21}$$

The differential equation for $U(t, t_0)$ can be written in terms of $H(t)$ as

$$i\hbar \frac{\partial U(t, t_0)}{\partial t} = H(t)U(t, t_0) \tag{2.22}$$

The Hamiltonian differs from the generator of time translations by an overall multiplicative constant, so we might as well regard the Hamiltonian itself as the generator. When $H(t)$ doesn't depend explicitly on time, its expectation value is guaranteed to be conserved for any time-dependent state vector. This gives us confidence that the Hamiltonian corresponds to the energy observable. In fact, by the postulates of quantum mechanics, the eigenvalues of the Hamiltonian define the possible energies of the system.

2.4.2 Time-Translation Invariance

In classical mechanics, you learn that the Hamiltonian is conserved when the laws governing the evolution of a system are time-translation invariant. There is a similar and even more elegant example of that same idea in quantum mechanics. We say that a quantum system is time-translation invariant when $U(t, t_0)$ depends only on the time difference $t - t_0$ such that

$$U(t, t_0) = U(t - t_0) \tag{2.23}$$

The composition law for a time-translation invariant system becomes

$$U(t_2 - t_0) = U(t_2 - t_1)U(t_1 - t_0) \tag{2.24}$$

Suppose $t_1 = t_0 + s_1$ and $t_2 = t_1 + s_2$, then

$$U(s_1 + s_2) = U(s_2)U(s_1) = U(s_1)U(s_2) \tag{2.25}$$

If we assume that $U(0) = \mathbb{1}$, then $U(t)$ applied to $|\psi(0)\rangle$ gives the state of the system at any later time. We now show that time-translation invariance implies the quantum mechanical Hamiltonian is time-independent, and hence conserved. Using a change of variables $\tau = t' - t$,

$$H(t) = i\hbar \frac{\partial U(t', t)}{\partial t'}\bigg|_{t'=t} = i\hbar \frac{\partial U(\tau)}{\partial \tau}\bigg|_{\tau=0}$$

The expression on the far right is the derivative of U with respect to τ evaluated at $\tau = 0$. In other words, it is independent of time. Thus, $H(t)$ must be independent of time, i.e. $H(t) = H$. Additionally, the operator $U(t)$ itself obeys the differential equation

$$i\hbar \frac{\partial U(t)}{\partial t} = HU(t) \tag{2.26}$$

In many situations, the Hamiltonian will be supplied by the model, and when the Hamiltonian

is independent of time, we know the evolution will be time-translation invariant, and we can easily solve for the unitary time operators by invoking Stone's theorem. We can understand the result intuitively by taking a finite interval t and dividing it many times into N intervals of length δt such that $t = N\delta t$. Then,

$$U(t) = U(N\delta t) = \big[U(\delta t)\big]^N \approx \left[1 + \frac{\partial U(t)}{\partial t}\bigg|_{t=0} \delta t\right]^N$$

or

$$U(t) = \lim_{N\to\infty} \left[1 + \frac{\partial U(t)}{\partial t}\bigg|_{t=0} \delta t\right]^N = \lim_{N\to\infty} \left[1 - \frac{iHt/\hbar}{N}\right]^N$$

Recall that, as a function over real numbers, the exponential e^x can be written as

$$e^x = \lim_{N\to\infty} \left(1 + \frac{x}{n}\right)^n$$

It is therefore tempting to write the solution for $U(t)$ as

$$U(t) = e^{-iHt/\hbar} \tag{2.27}$$

and this actually turns out to be correct, but our "derivation" only works for a limited class of operators and the exponential of a self-adjoint operator is properly understood in the context of functional analysis, and in particular, the spectral theorem.

In a finite-dimensional vector space, the exponential of an operator A can be defined via the Taylor series as

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!} \tag{2.28}$$

If A is self-adjoint and finite dimensional, it has a spectral decomposition $UDU^\dagger$ such that D is diagonal and U is a unitary matrix consisting of an orthonormal set of eigenvectors of A. Hence,

$$e^A = \sum_{n=0}^{\infty} \frac{(UDU^\dagger)^n}{n!} = Ue^D U^\dagger \tag{2.29}$$

Since D is diagonal, e^D can be represented by a diagonal matrix consisting of entries like e^λ where λ is an eigenvalue of A. In fact, many functions (those with convergent series expansions) applied to self-adjoint linear operators may be written as

$$f(A) = Uf(D)U^\dagger. \tag{2.30}$$

In terms of the spectral decomposition,

$$f(A) = \sum_n f(a_n) \left|a_n\right\rangle \left\langle a_n\right| \tag{2.31}$$

and furthermore, $f(A)\left|a_n\right\rangle = f(a_n)\left|a_n\right\rangle$ where we are using an abuse of notation where f can act on the space of operators and real numbers. For instance, $\cos(A)\left|a_n\right\rangle = \cos(a_n)\left|a_n\right\rangle$.

Example 2.1: Spectral Theorem

Use the spectral theorem to compute $e^{i\omega S_x t/\hbar}$ in the basis of S_z where ω is a constant with units of frequency and t is a time parameter.

Solution: Start by writing S_x in terms of its eigenvalues and eigenprojectors:

$$S_x = \frac{\hbar}{2}\left|+x\right\rangle\left\langle +x\right| - \frac{\hbar}{2}\left|-x\right\rangle\left\langle -x\right|$$

The exponential is then

$$e^{i\omega S_x t/\hbar} = e^{i\omega t/2}\left|+x\right\rangle\left\langle +x\right| + e^{-i\omega t/2}\left|-x\right\rangle\left\langle -x\right|$$

In the standard S_z basis,

$$\begin{aligned} e^{i\omega S_x t/\hbar} &= \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} e^{i\omega t/2} + \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} e^{-i\omega t/2} \\ &= \begin{pmatrix} \cos(\omega t/2) & i\sin(\omega t/2) \\ i\sin(\omega t/2) & \cos(\omega t/2) \end{pmatrix} \end{aligned}$$

2.5 Interpretations

We have laid out the basic postulates of quantum mechanics: states as vectors in a complex vector space, observables as operators, unitary time evolution, measurement with probabilistic outcomes, and the projection postulate (state collapse). Together, these rules form a remarkably successful framework for predicting the results of experiments. From atomic spectra to semiconductors to lasers to magnetic resonance, quantum mechanics works extraordinarily well *for all practical purposes*.

For the majority of this course, we'll adopt an *instrumentalist* view of quantum mechanics. For the applications we're interested in, it will be clear when a measurement occurs and there will be little-to-know ambiguity in the formalism. We will simply use the postulates without worrying about the underlying physical reality they may or may not be describing.

In this section, and later in the course if we have time, we'll address *realism*, the attitude that a

successful physical theory is not just a tool for predicting observations, but a description of a world that exists (independently of us). There have been many attempts to understand the physical reality implied by quantum mechanics, and each has a different take on the issues of probability, measurement, and state collapse, to name a few. We'll briefly address each attempt in the context of a thought experiment devised by Eugene Wigner.

2.5.1 Wigner's Friend

The primary purpose of the Wigner's friend thought experiment is to convince you that the measurement problem is indeed a problem! Wigner asks his friend to perform a Stern-Gerlach experiment inside a large container that can be perfectly isolated from the environment. The initial spin state of the atoms is $|+x\rangle$, a superposition of $|\pm z\rangle$, and this state is known to both Wigner and his friend. Suppose Wigner's friend performs a measurement of the z-component of spin angular momentum. The friend measures and records a definite outcome, for example "spin up," in a notebook. From the friend's point of view, the story is straightforward: before the measurement, the system was in a superposition; after the measurement, it has a definite value. The state vector has collapsed.

Now consider Wigner, outside the sealed lab. He treats everything inside—the atom, the apparatus, the friend, and the notebook—as one big *isolated quantum system*. Quantum mechanics tells him to assign a state that evolves unitarily. For Wigner, the experiment puts the lab into a superposition of two outcomes, meaning no definite outcome has occurred. Both descriptions are allowed by the quantum postulates, depending on where we draw the line between "system" and "observer." But they cannot both be true if collapse is supposed to be an objective physical process. Did the state collapse when the friend looked? Or only later, when Wigner opens the lab? Or never?

Suppose you aren't convinced of the ambiguity. Wigner may have no knowledge of the outcome, but Wigner's friend certainly obtains one. What if Wigner's friend was actually a highly trained capuchin monkey? A very smart dog? What if Wigner's friend was a tardigrade? Can tardigrades collapse quantum states? The magnetic field alone doesn't collapse the state; in the next chapter we'll learn that a spin state evolves *unitarily* inside a magnetic field. When does the collapse actually occur at the level of atoms and interactions between atoms (or subatomic particles)? In the words of John Bell, "...does not any analysis of measurement require concepts more fundamental than measurement?"

2.5.2 The Flow Chart

We'll briefly discuss various attempts to better understand, solve, or subvert the measurement problem by navigating the "Wigner's friend flow chart" in Figure 2.1. Almost everyone ends up somewhere on the chart, even if you think quantum mechanics is completely wrong and not just missing some subtle details about nature. Assuming you agree that quantum mechanics is mostly correct at some level, you likely fall into one of several camps:

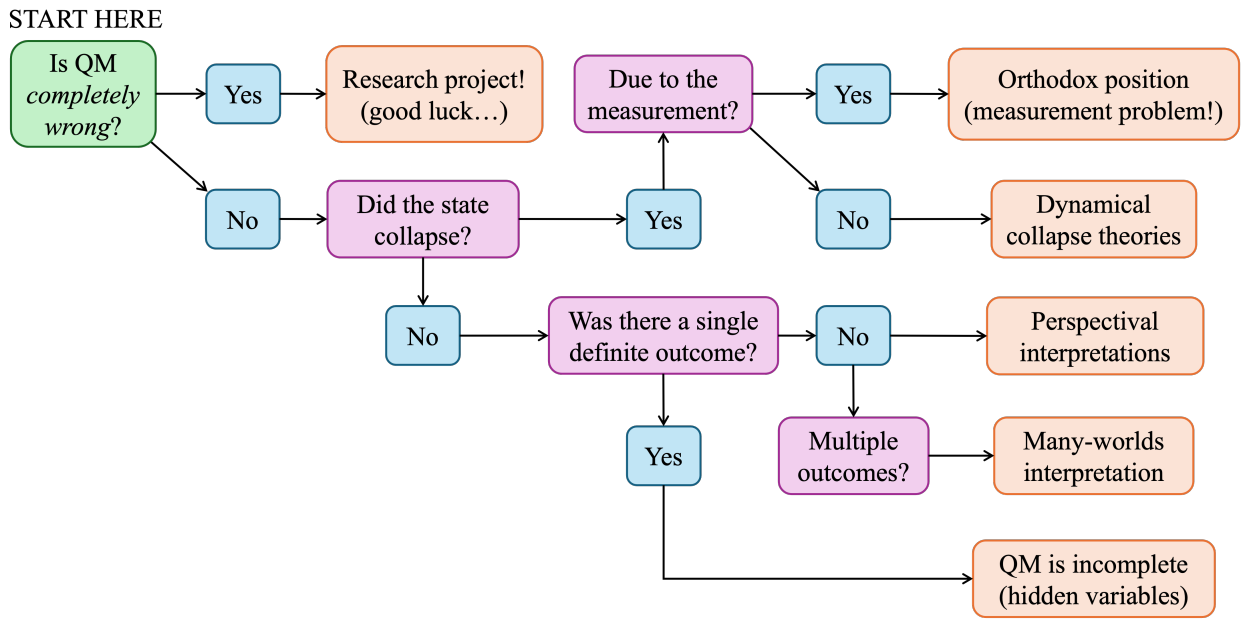


Figure 2.1: The "Wigner's friend flow chart" maps out various interpretations of quantum mechanics (credit: Dr. Jacob Barandes, Havard).

(a) The **orthodox position** remains largely agnostic about physical reality. The postulates are a set of tools, so "shut up and calculate." The measurement problem is not a *practical problem*, but it might be worthy of study. This position is often conflated with the **Copenhagen interpretation**, which crucially involves a distinction between the "classical world" and the "quantum world." Read Heisenberg's book, *Physics and Philosophy*, if you're interested in learning more.

(b) **Dynamical collapse theories**, like Ghirardi–Rimini–Weber (GRW), assert that state vector collapse is an objective physical process and try to describe how it works.

(c) In a **perspectival or relational interpretation** of quantum mechanics, the properties of a system, including its quantum state, are not absolute, but always defined relative to another system with which it interacts. Look up Carlo Rovelli's *relational quantum mechanics* if you want to learn more.

(d) In the **many-worlds interpretation** (by Everett), the state vector is taken to be a real, physical object that always evolves according to the Schrödinger equation and never collapses. What we call a measurement is just an ordinary interaction that *entangles* the system with the apparatus and observer, causing the total state to branch into non-interacting components in which each possible outcome is realized. In each branch, the observer experiences a single definite result, even though all outcomes exist in the global state vector (or wave function). The apparent randomness of measurement reflects which branch an observer finds themselves in, not any fundamental indeterminism, so the theory is fully deterministic at the level of the

universal state.

(e) In a **hidden-variable theory**, the quantum state does not give a *complete description* of an individual physical system. Instead, there exist additional, unobserved parameters called *hidden variables* that determine the properties and outcomes of measurements. In deterministic hidden-variable theories (e.g., pilot wave theory), the apparent randomness of quantum mechanics reflects our ignorance of the hidden variables like in classical statistical mechanics. In stochastic hidden-variable theories, indeterminism is a fundamental feature of the rules by which the hidden variables change over time. Whether deterministic or indeterministic, the system always possesses at least some well-defined properties, and measurements are physical interactions that can either reveal those properties or create new, emergent properties. Contrary to what you may have read or heard, Bell's theorem does not rule out hidden variables.

Some extreme positions suggest the state vector never collapses and no definite outcome occurs; these are largely self-undermining and rarely discussed seriously.

Chapter 3

Quantum Dynamics

3.1 Time Evolution in the Energy Eigenbasis

The Schrödinger equation of quantum mechanics is analagous to Newton's second law in classical mechanics—formally true but physically empty without additional information. For any given physical system, you must build a model, and that includes specifying a suitable Hamiltonian[1]. Whether or not you have the *right* Hamiltonian is a serious question, and addressing it requires physical insight. Once you have a Hamiltonian, it is often convenient to express the state vector in the energy eigenbasis, where the Hamiltonian takes a diagonal form. We'll consider finite-dimensional quantum systems in this chapter, but many of the concepts apply equally well to infinite-dimensional systems with discrete energy spectra (such as bound particle system). Questions related to the convergence of sequences naturally arise in such contexts, but we won't be concerned with them in this class.

If we label the allowed energies as E_n, then the energy eigenvalue equation is

$$H\left|E_n\right\rangle = E_n\left|E_n\right\rangle \tag{3.1}$$

where $\left|E_n\right\rangle$ is the energy eigenstate corresponding to the eigenvalue E_n. In our present discussion, we'll assume the eigenvalues are non-degenerate and that the Hamiltonian is time-independent for simplicity. The spectral decomposition of H is

$$H = \sum_n E_n\left|E_n\right\rangle\left\langle E_n\right| \tag{3.2}$$

and the unitary time evolution operator is

$$U(t) = e^{-iHt/\hbar} = \sum_n e^{-iE_nt/\hbar}\left|E_n\right\rangle\left\langle E_n\right| \tag{3.3}$$

[1] In almost any scientific paper on a quantum system, the first or second equation is usually a Hamiltonian!

If the initial state vector is

$$|\psi(0)\rangle = \sum_n c_n |E_n\rangle \tag{3.4}$$

where the coefficients are $c_n = \langle E_n|\psi(0)\rangle$, then it follows that

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle \tag{3.5}$$

We can also expand the final state vector in the energy eigenbasis using time-dependent coefficients such that

$$|\psi(t)\rangle = \sum_n c_n(t) |E_n\rangle \tag{3.6}$$

where $c_n(0) = c_n$ are the initial coefficients defined above. The energy eigenstates are orthonormal, meaning $\langle E_k|E_n\rangle = \delta_{kn}$ where δ_{kn} is the Kronecker delta symbol. We can use this fact to solve for the coefficients of the state vector in the energy eigenbasis:

$$\langle E_k|\psi(t)\rangle = \sum_n c_n(t) \langle E_k|E_n\rangle = \sum_n c_n(t)\delta_{kn} = c_k(t) \tag{3.7}$$

From Eq. (3.5), it follows that

$$\begin{aligned} \langle E_k|\psi(t)\rangle &= \sum_n c_n e^{-iE_n t/\hbar} \langle E_k|E_n\rangle \\ &= \sum_n c_n e^{-iE_n t/\hbar}\delta_{kn} \\ &= c_k e^{-iE_k t/\hbar} \end{aligned} \tag{3.8}$$

Hence,

$$c_n(t) = c_n e^{-iE_n t/\hbar} \tag{3.9}$$

If we measure an observable A on a quantum system in the state $|\psi(t)\rangle$, there are general statements we can make regarding the probability to measure an eigenvalue a_n of the observable A that depend on depend on properties of A and $|\psi(t)\rangle$, which we will outline:

(i) If $|\psi(0)\rangle$ is an energy eigenstate, then the probability to measure a_n is time independent and is equal to the probability at the initial time. For this reason, the energy eigenstates are called **stationary states**–there is no *measurable* time evolution.

(ii) The probability to measure any particular energy of a quantum system will not change over time (assuming H is time-independent and the system is closed).

(iii) If $[H, A] = 0$, then A and H will share common eigenstates, and the probability to measure any particular eigenvalue of A will not change over time.

(iv) If $[H, A] \neq 0$, and $|\psi(0)\rangle$ is a superposition of eigenstates, then the probability to measure any particular eigenvalue of A will change over time.

Proof of (i). Let $|\psi(0)\rangle = |E_k\rangle$ such that $|\psi(t)\rangle = e^{-iE_k t/\hbar}|E_k\rangle$. The probability to measure a_n is

$$|\langle a_n|\psi(t)\rangle|^2 = |\langle a_n| e^{-iE_k t/\hbar}|E_k\rangle|^2 = |\langle a_n|E_k\rangle|^2 \tag{3.10}$$

which is independent of time. The claim is also true when a_n is degenerate.

Proof of (ii). The probability to measure E_n is

$$\begin{aligned} |\langle E_n|\psi(t)\rangle|^2 &= \Big|\sum_k c_k e^{-iE_k t/\hbar}\langle E_n|E_k\rangle\Big|^2 \\ &= |c_n|^2 \end{aligned} \tag{3.11}$$

Proof of (iii). When the observable A commutes with H, it has a spectral decomposition in terms of the energy eigenstates–it is diagonal in the energy eigenbasis. Hence,

$$A = \sum_n a_n |E_n\rangle\langle E_n| \tag{3.12}$$

In other words, the probability to measure a_n is the same as the probability to measure E_n, assuming they both correspond to the same eigenvector. The probability to measure a_n is therefore $|c_n|^2$ as we found in the proof of statement (ii).

Proof of (iv). Here's where things get interesting. I'll do this in a rather technical way and leave the simpler, more illuminating examples for you to work through on your own. Since the energy eigenstates form a basis, we can expand the eigenstates of A in terms of energy eigenstates as

$$|a_n\rangle = \sum_k \alpha_{nk}|E_k\rangle$$

The probability of measuring the eigenvalue a_n is

$$\begin{aligned} |\langle a_n|\psi(t)\rangle|^2 &= \Big|\sum_{k\ell}\alpha^*_{nk}c_\ell e^{-iE_\ell t/\hbar}\langle E_k|E_\ell\rangle\Big|^2 \\ &= \Big|\sum_k \alpha^*_{nk}c_k e^{-iE_k t/\hbar}\Big|^2 \\ &= \sum_{k\ell}\alpha^*_{nk}\alpha_{n\ell}c_k c^*_\ell e^{-i(E_k-E_\ell)t/\hbar} \\ &= \sum_k |\alpha_{nk}|^2|c_k|^2 + \sum_{k\neq\ell}\alpha^*_{nk}\alpha_{n\ell}c_k c^*_\ell e^{-i(E_k-E_\ell)t/\hbar} \end{aligned}$$

$$= \sum_k |\alpha_{nk}|^2 |c_k|^2 + \sum_{k<\ell} 2\mathrm{Re}\Big[\alpha_{nk}^* \alpha_{n\ell} c_k c_\ell^* e^{-i(E_k - E_\ell)t/\hbar}\Big]$$

We see that the probability separates into a part that is time independent and a term that oscillates with a combination of **Bohr frequencies** defined as

$$\omega_{k\ell} = \frac{E_k - E_\ell}{\hbar} \tag{3.13}$$

It's reassuring that a measurable aspect of time evolution depends on *energy differences* rather than energies themselves. In classical mechanics, energy is only well defined up to an overall constant and dynamical evolution is tied to changes in energy that occur over time.

3.2 Spin Precession

Consider a single magnetic dipole (e.g., a silver atom) in a uniform magnetic field. From classical electromagnetism, the potential energy of interaction between the dipole and the magnetic field is

$$U = -\boldsymbol{\mu} \cdot \mathbf{B} \tag{3.14}$$

Recall that the magnetic dipole of an electron is given by

$$\boldsymbol{\mu} = g\frac{q}{2m_e}\mathbf{S} \tag{3.15}$$

where $q = -e$ and we'll take $g = 2$, even though the g-factor of the electron famously differs slightly from 2. We'll *define* our quantum system by the Hamiltonian

$$H = \frac{e}{m_e}\mathbf{S} \cdot \mathbf{B} \tag{3.16}$$

where $\mathbf{S}$ is not a unit vector, or even an ordinary vector–it's an *operator-valued vector* with components S_x, S_y, and S_z. The magnetic field is an ordinary vector (we are treating the magnetic field as a classical vector field), so it's actually unclear what the dot product means in this context. In fact, it's just the usual dot product in terms of Cartesian coordinates:

$$\mathbf{S} \cdot \mathbf{B} = B_x S_x + B_y S_y + B_z S_z$$

While the Hamiltonian operator represents the total energy of the system, only energy differences are relevant. We may consider only energy terms that differentiate between the two possible spin states in the system.

3.2.1 Spin Component in a General Direction

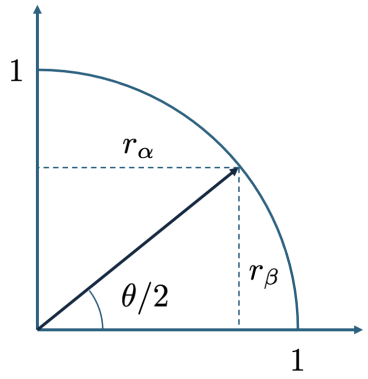
We'll see where the spin operators and their eigenstates come from in chapter #4, but for now I'd like to discuss the spin component in an arbitrary direction. Let's adopt a mathematical perspective given that state vectors are notoriously abstract and physical intuition takes time and effort to build. An arbitrary two-dimensional state vector may be written as a linear combination of $|+z\rangle$ and $|-z\rangle$ in the following manner:

$$|\psi\rangle = \alpha\,|+z\rangle + \beta\,|-z\rangle\,, \quad \text{where } |\alpha|^2 + |\beta|^2 = 1$$

Starting with a general spin state in the z basis, let

$$\alpha = r_\alpha e^{i\phi_\alpha}, \quad \beta = r_\beta e^{i\phi_\beta} \quad \text{(complex numbers in polar form)}$$

where $r_\alpha, r_\beta > 0$ are real numbers (amplitudes) and ϕ_α, ϕ_β are phases (also real numbers). From the condition, $|\alpha|^2 + |\beta|^2 = 1$, it follows that $r_\alpha^2 + r_\beta^2 = 1$. This is the equation for a quarter circle since $r_\alpha, r_\beta > 0$ (see diagram below).



Reparameterization of the state vector:

$$r_\alpha = \cos(\theta/2), \quad r_\beta = \sin(\theta/2), \quad \theta \in [0, \pi]$$

$$|\psi\rangle = \cos(\theta/2)e^{i\phi_\alpha}\,|+z\rangle + \sin(\theta/2)e^{i\phi_\beta}\,|-z\rangle$$

$$e^{-i\phi_\alpha}\,|\psi\rangle = \cos(\theta/2)\,|+z\rangle + \sin(\theta/2)e^{i(\phi_\beta - \phi_\alpha)}\,|-z\rangle$$

Let $\phi = (\phi_\beta - \phi_\alpha) \in [0, 2\pi)$, and note that the state $e^{-i\phi_\alpha}\,|\psi\rangle$ is *physically equivalent* to $|\psi\rangle$. The most general spin-1/2 state can be written as

$$|\psi\rangle = \cos(\theta/2)\,|+z\rangle + \sin(\theta/2)e^{i\phi}\,|-z\rangle \tag{3.17}$$

where $0 \leq \theta \leq \pi$, and $0 \leq \phi < 2\pi$. Hence, any point on the surface of a unit sphere represents a

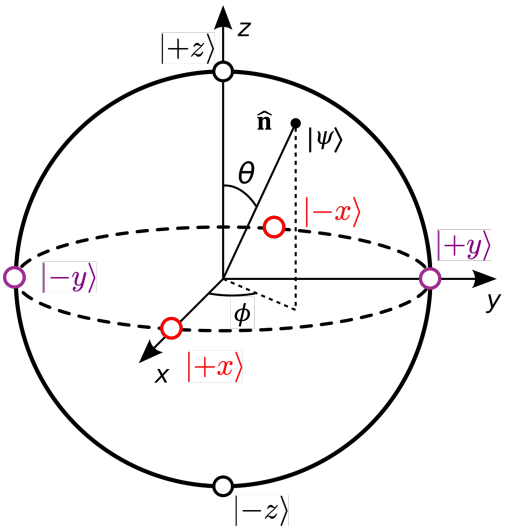


Figure 3.1: The Bloch sphere–a convenient way to visualize spin-1/2 systems. Any pure state can be mapped to some point on the Bloch sphere.

valid spin-1/2 state–we call the unit sphere the **Bloch sphere** in this context.

For every spin-1/2 state, there corresponds a unit vector in 3D Euclidean space called the **Bloch vector** given by

$$\hat{\mathbf{n}} = \hat{\mathbf{x}} \sin\theta \cos\phi + \hat{\mathbf{y}} \sin\theta \sin\phi + \hat{\mathbf{z}} \cos\theta \tag{3.18}$$

Importantly, do not confuse the unit vector $\hat{\mathbf{n}}$ with the state vector $|+n\rangle$. There is only a mathematical correspondence, a mapping, and we'll explore the origins and significance of this mapping in chapter #4. The spin component along this direction is obtained by projecting the spin vector $\mathbf{S}$ onto the Bloch vector:

$$S_n = \mathbf{S} \cdot \hat{\mathbf{n}} = S_x \sin\theta \cos\phi + S_y \sin\theta \sin\phi + S_z \cos\theta$$

In matrix form,

$$S_n = \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta\ e^{-i\phi} \\ \sin\theta\ e^{i\phi} & -\cos\theta \end{pmatrix}$$

It is relatively straightforward to show that the general spin-1/2 state given by Eq. (3.17) is an eigenvector of S_n when the parameters θ and ϕ match–that is, when the Bloch vector is aligned with the spin component. The way I've set this up guarantees alignment... For notational convenience, let

$$|+n\rangle = \cos(\theta/2)\,|+z\rangle + \sin(\theta/2)e^{i\phi}\,|-z\rangle \tag{3.19}$$

such that

$$S_n\,|+n\rangle = \frac{\hbar}{2}\,|+n\rangle$$

To find the other eigenvector (there must be two after all), we use the guess-and-check method,

informed by intuition. If $|+n\rangle$ is the eigenvector aligned with $\hat{\mathbf{n}}$, then $|-n\rangle$ must be an eigenvector aligned with $-\hat{\mathbf{n}}$. The vector $\hat{\mathbf{n}}$ is mapped to $-\hat{\mathbf{n}}$ under the transformation $(\theta, \phi) \to (\pi - \theta, \phi + \pi)$. Hence,

$$|-n\rangle = \sin(\theta/2)\,|+z\rangle - \cos(\theta/2)e^{i\phi}\,|-z\rangle \tag{3.20}$$

and indeed,

$$S_n\,|-n\rangle = -\frac{\hbar}{2}\,|-n\rangle$$

3.2.2 Uniform Magnetic Field

A uniform magnetic field points in some direction, so we might as well define that to be the z direction such that $\mathbf{B} = B_0\hat{\mathbf{z}}$. The Hamiltonian in Eq. (3.16) simplifies to

$$H = \frac{eB_0}{m_e}S_z = \omega_0 S_z \tag{3.21}$$

where the constant $\omega_0 = eB_0/m_e$ has units of frequency. Since the Hamiltonian commutes with S_z, it shares the same eigenstates. Let's write the Hamiltonian as a matrix in S_z representation:

$$H = \frac{\hbar\omega_0}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{3.22}$$

The unitary time evolution operator is therefore

$$U(t) = e^{-iHt/\hbar} = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix} = e^{-i\omega_0 t/2}\begin{pmatrix} 1 & 0 \\ 0 & e^{i\omega_0 t} \end{pmatrix} \tag{3.23}$$

We consider the most general state vector written in the Bloch sphere representation as

$$|\psi(0)\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix} \tag{3.24}$$

Hence, the final state vector is

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle = e^{-i\omega_0 t/2}\begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i(\phi+\omega_0 t)} \end{pmatrix} \tag{3.25}$$

The overall phase does not have a measurable effect here, so we can interpret the evolved state as a spin up eigenstate along a direction that maintains a constant polar angle θ as it revolves around the z axis with an angular frequency of ω_0.

We can test a few of the statements we made in the opening section. For instance, the probability of measuring the spin component along the z-axis is

$$|\,\langle +z|\psi(t)\rangle\,|^2 = \cos^2(\theta/2) \tag{3.26}$$

which is time independent because the S_z eigenstates are also energy eigenstates for this problem (H commutes with S_z). Alternatively, the probability for measuring spin up along the x-axis is

$$|\langle +x|\psi(t)\rangle|^2 = \tfrac{1}{2}\big[1 + \sin\theta\cos(\phi + \omega_0 t)\big] \tag{3.27}$$

This probability is time dependent because the S_x eigenstates are not energy eigenstates (H does not commute with S_x).

While it may be helpful to think of the state vector in the Bloch sphere representation as revolving around the z axis, one should be cautious when ascribing any physical picture to the state vector itself. A better approach to understanding spin precession would be to calculate the expectation values of the spin components. One may verify

$$\langle S_z\rangle = \frac{\hbar}{2}\cos\theta \tag{3.28}$$

$$\langle S_x\rangle = \frac{\hbar}{2}\sin\theta\cos(\phi + \omega_0 t) \tag{3.29}$$

$$\langle S_y\rangle = \frac{\hbar}{2}\sin\theta\sin(\phi + \omega_0 t) \tag{3.30}$$

After many measurements of the spin components performed on identical systems, we expect the average results to be consistent with the classical picture in which the spin vector precesses around the magnetic field direction with frequency ω_0. The precession of the spin vector is *Larmor precession* and the frequency of precession is the *Larmor frequency*. This precession of the spin vector suggests the system has angular momentum, rather than simply having a magnetic dipole moment. The equivalence of the classical Larmor precession and the expectation value of the quantum mechanical spin vector is one example of *Ehrenfest's theorem*, which states that quantum mechanical expectation values obey classical laws.

Example 3.1: Spin Precession

Regarding degrees of freedom associated with spin, the Hamiltonian for an electron in a magnetic field $\mathbf{B} = B_0\hat{\mathbf{x}}$ is

$$H = \omega_0 S_x$$

where $\omega_0 \approx eB_0/m_e$ is the Larmor frequency. If the particle is initially in the state $|\psi(0)\rangle = |+z\rangle$, determine $|\psi(t)\rangle$, the state of the electron at time t.

Solution: The Hamiltonian is time independent, so the unitary time evolution operator is

$$U(t) = e^{-\omega_0 S_x t/\hbar} = \begin{pmatrix} \cos(\omega_0 t/2) & -i\sin(\omega_0 t/2) \\ -i\sin(\omega_0 t/2) & \cos(\omega_0 t/2) \end{pmatrix}$$

where this result was obtained in a previous example. The state of the electron is

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle = \begin{pmatrix} \cos(\omega_0 t/2) \\ -i\sin(\omega_0 t/2) \end{pmatrix} = \cos(\omega_0 t/2)\,|+z\rangle - i\sin(\omega_0 t/2)\,|-z\rangle$$

3.3 Magnetic Resonance

Here we study a simple, yet rich example to understand the phenomena of *magnetic resonance*. By applying a transverse oscillating magnetic field with a frequency close to the Larmor frequency, we can induce transitions between energy (spin) states. This same model of the interaction between a two-level system and an applied time-dependent field is also used to explain how atoms absorb and emit light.

The Hamiltonian we'll consider in this example is time-dependent and given by

$$\begin{aligned} H(t) &= -\boldsymbol{\mu}\cdot\mathbf{B} \\ &= \frac{e}{m_e}\mathbf{S}\cdot\Big[B_0\hat{\mathbf{z}} + B_1\cos(\omega t)\hat{\mathbf{x}} + B_1\sin(\omega t)\hat{\mathbf{y}}\Big] \\ &= \omega_0 S_z + \omega_1\Big[\cos(\omega t)S_x + \sin(\omega t)S_y\Big] \end{aligned} \tag{3.31}$$

where ω is the "driving frequency" of the magnetic field and the two Larmor frequencies are

$$\omega_0 = \frac{eB_0}{m_e} \quad \text{and} \quad \omega_1 = \frac{eB_1}{m_e} \tag{3.32}$$

The matrix representation of the Hamiltonian in the S_z basis is

$$H(t) = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix} \tag{3.33}$$

We can't use Stone's theorem because the Hamiltonian is time dependent. Instead, we'll write out the components of the Schrödinger equation:

$$i\hbar\frac{d}{dt}\,|\psi(t)\rangle = H(t)\,|\psi(t)\rangle \tag{3.34}$$

Here we'll use the S_z eigenstates as our basis, even though they aren't the energy eigenstates anymore. The state vector is

$$|\psi(t)\rangle = c_+(t)\,|+z\rangle + c_-(t)\,|-z\rangle = \begin{pmatrix} c_+(t) \\ c_-(t) \end{pmatrix} \tag{3.35}$$

and the components of the Schrödinger equation are

$$i\dot{c}_+(t) = \frac{\omega_0}{2}c_+(t) + \frac{\omega_1}{2}e^{-i\omega t}c_-(t) \tag{3.36}$$

$$i\dot{c}_-(t) = \frac{\omega_1}{2}e^{i\omega t}c_+(t) - \frac{\omega_0}{2}c_-(t) \tag{3.37}$$

where $\dot{c}_\pm(t)$ denotes the time derivative. Coupled differential equations are notoriously difficult to work with, but we can solve these with a clever *gauge transformation*.

3.3.1 Dynamical Gauge Transformations

The Schrödinger equation and nearly all empirical predictions of quantum mechanics are invariant under transformation by an *arbitrary time-dependent unitary operator* $V(t)$:

$$|\psi(t)\rangle \mapsto V(t)\,|\psi(t)\rangle$$

$$A(t) \mapsto V(t)A(t)V^\dagger(t)$$

$$H(t) \mapsto V(t)H(t)V^\dagger(t) + i\hbar\dot{V}(t)V^\dagger(t)$$

where $A(t)$ is a time-dependent observable and $\dot{V}(t)$ is the time derivative of $V(t)$. This set of transformations was discovered by Foldy and Wouthuysen in 1950 [1].

In order to establish the validity of these transformations, we must consider the *gauge invariant quantities* of quantum mechanics. These are the quantities that do not depend on our choice of $V(t)$. For instance, the expectation value of an observable should not depend on $V(t)$, and indeed

$$\langle\tilde{A}(t)\rangle = \langle\tilde{\psi}(t)|\tilde{A}(t)\left|\tilde{\psi}(t)\right\rangle = \langle\psi|\,V^\dagger V A V^\dagger V\,|\psi\rangle = \langle\psi|\,A\,|\psi\rangle = \langle A(t)\rangle$$

Probabilities generated via the Born rule will also be unaffected. Given that $V(t)$ is time-dependent, if we insist on using the Schrödinger equation for any choice of $V(t)$, then the Hamiltonian must be modified:

$$\begin{aligned}
\frac{d}{dt}\left|\tilde{\psi}(t)\right\rangle &= \dot{V}(t)\,|\psi(t)\rangle + V(t)\frac{d}{dt}\,|\psi(t)\rangle \\
&= \dot{V}(t)\,|\psi(t)\rangle - \frac{i}{\hbar}V(t)H(t)\,|\psi(t)\rangle \\
&= \dot{V}(t)V^\dagger(t)\left|\tilde{\psi}(t)\right\rangle - \frac{i}{\hbar}V(t)H(t)V^\dagger(t)\left|\tilde{\psi}(t)\right\rangle \\
i\hbar\frac{d}{dt}\left|\tilde{\psi}(t)\right\rangle &= \left[V(t)H(t)V^\dagger(t) + i\hbar\dot{V}(t)V^\dagger(t)\right]\left|\tilde{\psi}(t)\right\rangle = \tilde{H}(t)\left|\tilde{\psi}(t)\right\rangle
\end{aligned}$$

Hence, the Schrödinger equation is preserved under this transformation.

3.3.2 Rotating Frame

Recall from Section 3.2.2 that the unitary operator

$$U(t) = \begin{pmatrix} e^{-i\omega_0 t/2} & 0 \\ 0 & e^{i\omega_0 t/2} \end{pmatrix}$$

corresponds to continuous rotation around the z axis with constant frequency ω_0. If we transform to a frame that rotates around the z axis at the same rate as the oscillating component of the magnetic field, then that component will appear stationary and it might be easier to solve the problem.

Define the time-dependent rotation operator in the S_z representation as

$$V(t) = \begin{pmatrix} e^{-i\omega t/2} & 0 \\ 0 & e^{i\omega t/2} \end{pmatrix} \tag{3.38}$$

and the transformed state vector in the rotating frame is

$$\left|\tilde{\psi}(t)\right\rangle = V^\dagger(t)\left|\psi(t)\right\rangle \tag{3.39}$$

where the idea is that if $|\psi(t)\rangle$ appears to be rotating in the "lab frame," then $\left|\tilde{\psi}(t)\right\rangle$ is stationary in the rotating frame... Ultimately, we're simply using a gauge transformation of the type discussed in the previous section where $V \to V^\dagger$, so we won't strain ourselves to understand the physical significance of Eq. (3.39).

The transformed Hamiltonian in the rotating frame is given by

$$\tilde{H}(t) = V^\dagger(t)H(t)V(t) + i\hbar\dot{V}^\dagger(t)V(t) \tag{3.40}$$

After working through some calculations, the transformed Hamiltonian can be written as

$$\tilde{H} = \frac{\hbar}{2}\begin{pmatrix} \Delta\omega & \omega_1 \\ \omega_1 & -\Delta\omega \end{pmatrix} \tag{3.41}$$

where $\Delta\omega = \omega_0 - \omega$. The transformed Hamiltonian is time-independent! Furthermore,

$$\tilde{H} = \omega_1 S_x + \Delta\omega S_z = \mathbf{S}\cdot\left(\omega_1\hat{\mathbf{x}} + \Delta\omega\hat{\mathbf{z}}\right)$$

The vector in parentheses can be written in unit-vector notation as

$$\left(\omega_1\hat{\mathbf{x}} + \Delta\omega\hat{\mathbf{z}}\right) = \sqrt{\Delta\omega^2 + \omega_1^2}\ \hat{\mathbf{n}}$$

where $\hat{\mathbf{n}} = \sin\theta\hat{\mathbf{x}} + \cos\theta\hat{\mathbf{z}}$, and θ is the polar angle of a spherical coordinate system. The sine and

cosine of the angle θ are given by

$$\sin\theta = \frac{\omega_1}{\sqrt{\Delta\omega^2 + \omega_1^2}}, \qquad \cos\theta = \frac{\Delta\omega}{\sqrt{\Delta\omega^2 + \omega_1^2}}$$

We can interpret this parameterization as the result of an effective magnetic field in the rotating frame. Nevertheless, it follows that

$$\tilde{H} = \sqrt{\Delta\omega^2 + \omega_1^2}\; S_n \tag{3.42}$$

where S_n is the spin component along the unit vector $\hat{\mathbf{n}}$ in the xz-plane (the azimuthal angle is zero in this case). It follows that the eigenvalues of $\tilde{H}$, denoted $E_\pm$, are

$$E_\pm = \pm\frac{\hbar}{2}\sqrt{\Delta\omega^2 + \omega_1^2} \tag{3.43}$$

and the corresponding eigenvectors are $|\pm n\rangle$ with $\phi = 0$ (on the xz-plane). The unitary time evolution operator in the rotating frame is therefore

$$\tilde{U}(t) = e^{-iE_+t/\hbar}\,|+n\rangle\langle+n| + e^{-iE_-t/\hbar}\,|-n\rangle\langle-n| \tag{3.44}$$

3.3.3 Spin Flip

Let's consider a specific scenario known as a **spin flip** where we start with a spin up eigenstate of S_z and compute the probability for it to flip to a spin down state. Starting with the spin up eigenstate of S_z, the probability to measure $S_z = -\hbar/2$ after some time t is

$$\mathcal{P}_{+\to-} = |\,\langle -z|\psi(t)\rangle\,|^2 \tag{3.45}$$

The dynamical gauge transformation, by design, does not change the probability, so

$$\mathcal{P}_{+\to-} = |\left\langle -\tilde{z}\middle|\tilde{\psi}(t)\right\rangle|^2$$

and since $|-\tilde{z}\rangle = V^\dagger(t)\,|-z\rangle = e^{-i\omega t/2}\,|-z\rangle$,

$$\mathcal{P} = |\left\langle -\tilde{z}\middle|\tilde{\psi}(t)\right\rangle|^2 = |\left\langle -z\middle|\tilde{\psi}(t)\right\rangle|^2$$

After some tedious calculations, it follows that

$$\mathcal{P}_{+\to-} = \frac{\omega_1^2}{(\omega_0-\omega)^2 + \omega_1^2}\sin^2\left[\frac{\sqrt{(\omega_0-\omega)^2+\omega_1^2}}{2}\;t\right] \tag{3.46}$$

The static magnetic field case ($\omega = 0$) is generally referred to as spin precession, while the rotating field case is referred to as **Rabi flopping** (or Rabi oscillations), and the equation above is a gen-

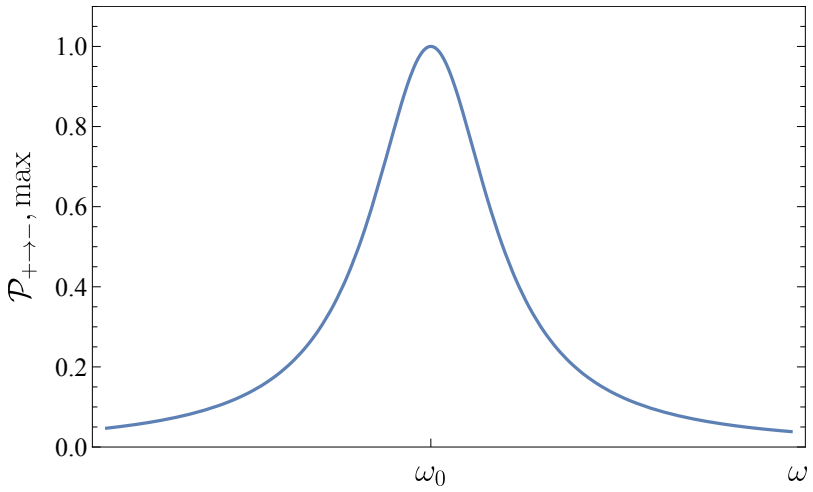


Figure 3.2: Magnetic resonance curve showing the maximum probability of a spin flip as a function of the applied frequency.

eralization of **Rabi's formula**. Rabi oscillations represent transitions between energy eigenstates; there is exchange of energy between the system and the applied field. The energy exchange occurs because the Hamiltonian is time dependent.

The probability of a Rabi spin flip oscillates with an angular frequency given by

$$\Omega = \sqrt{(\omega_0 - \omega)^2 + \omega_1^2} \tag{3.47}$$

that is typically referred to as the **generalized Rabi frequency**. The term **Rabi frequency** generally refers to the frequency ω_1, which is the value of the generalized Rabi frequency when the frequency ω of the rotating field is on **resonance** (i.e., ω is set equal to the Larmor precession frequency ω_0 of the system in the presence of the magnetic field B_0 alone). For this choice of $\omega = \omega_0$, the probability of a spin flip becomes

$$\mathcal{P} = \sin^2\left(\frac{\omega_1 t}{2}\right) \tag{3.48}$$

which implies that the spin is flipped with 100% probability at times $t = \pi/\omega_1, 3\pi/\omega_1$, etc., *even if the transverse field is weak*. For other off-resonance choices of the frequency ω, the probability of a spin flip oscillates with an amplitude smaller than one. A graph of $\mathcal{P}$ as a function of ω shows a curve called a Lorentzian with a peak at $\omega = \omega_0$ and a full width at half-maximum of $2\omega_1$.

3.4 Light-Matter Interactions

The same model we used to study spin flipping is used to explain how atoms absorb and emit light. In the interaction of atoms with light, the oscillating electric field of the light wave interacts with the electric dipole of the atom, and energy exchange between the field and the atom corresponds

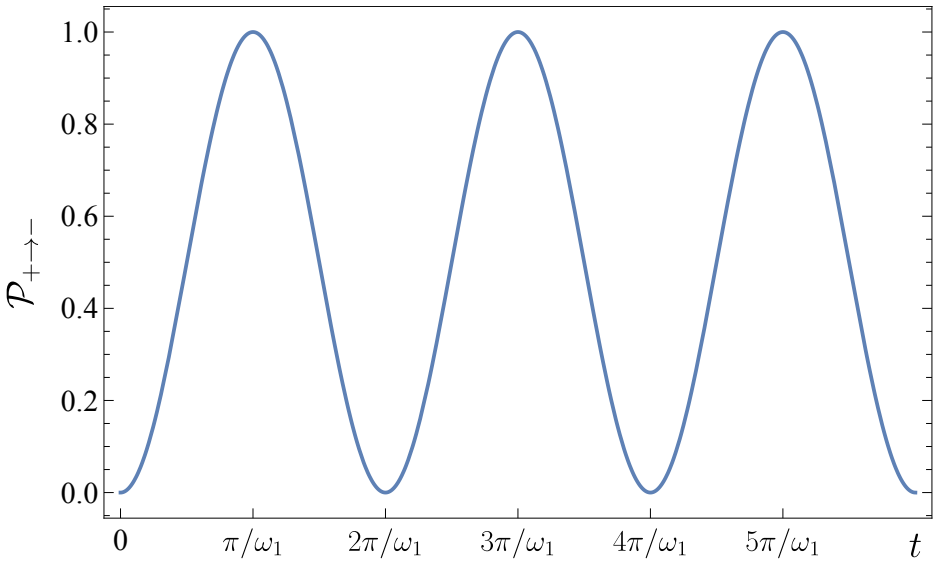


Figure 3.3: Rabi oscillations: probability for a spin-flip to occur over time when the driving field is resonant with the Larmor precession of the dipole.

to absorption and emission of photons. Though atoms have more than two energy levels, we can reduce the problem to a two-level system if the frequency ω of the applied light field is close to just one of the Bohr frequencies of the atom. Label the lower state $|g\rangle$ (for ground state) and the upper state $|e\rangle$ (for excited state). The energy difference between the two levels is defined to be

$$E_e - E_g = \hbar\omega_0 \tag{3.49}$$

The applied light field (e.g., laser beam) has a frequency ω that is close to, but not necessarily equal to, the atomic Bohr frequency ω_0. By analogy with the spin problem, we can express the Hamiltonian for this atom-light system in two parts

$$H = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & -\omega_0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} \omega_0 & 0 \\ 0 & -\omega_0 \end{pmatrix} + \frac{\hbar}{2}\begin{pmatrix} 0 & \omega_1 e^{-i\omega t} \\ \omega_1 e^{i\omega t} & 0 \end{pmatrix} \tag{3.50}$$

where the first term as the atomic Hamiltonian and the second term as the interaction Hamiltonian. We can interpret the parameter ω_1 as an off-diagonal matrix element of the interaction Hamiltonian that connects the two states. The Rabi formula then gives the probability for the light field to cause transitions between the two atomic energy states. Studying induced transitions is the most powerful tool we have for discovering what the energy levels of a system are and ultimately for determining the Hamiltonian of the system. This tool is known as **spectroscopy** and has played a pivotal role in relating experiments and theory in quantum mechanics.

3.4.1 Atomic Clocks

Atomic clocks are among the most precise measurement devices ever created, and at their core lies the quantum physics of driven two-level systems. These clocks operate by exploiting transitions between two discrete energy levels in atoms–typically hyperfine levels of an alkali metal like cesium or rubidium. The transition frequency between the energy levels can be used to define a unit of time: one second is *defined* as a specific number of oscillations (e.g., 9,192,631,770 for cesium-133) of radiation corresponding to this transition.

Physically, the clock works by preparing the atoms in one hyperfine state and then subjecting them to an oscillating electromagnetic field (usually in the microwave or optical range) whose frequency corresponds to the energy difference between the two levels. The interaction of this field with the atom is an example of a *driven two-level system*, governed by the physics of *Rabi oscillations*. When the driving field is resonant with the energy splitting, the population of the atom oscillates between the two levels with a frequency known as the *Rabi frequency*, which depends on the strength of the field and the dipole moment of the transition.

By tuning the driving field and observing the population of the final state, experimenters can determine the frequency at which the transition is maximally excited. That is, the frequency of the field is exactly resonant with the atomic transition. This frequency becomes the reference for the clock. The narrower the transition linewidth and the more stable the oscillation, the more precise the clock. Advanced atomic clocks (like optical lattice clocks) push this technique to its extreme, using ultra-stable lasers and atoms trapped in standing light waves to suppress motion and environmental noise, yielding timing precision at the level of 1 part in 10^{18}.

3.5 The Heisenberg Picture

In the "Heisenberg picture," the state vector is time-independent and the operators evolve in time. Starting from the Schrödinger picture, consider a time-independent operator A_S in a quantum system where the Hamiltonian is also time-independent. If we know the initial state of the system, we can write the expectation value of A_S at a later time as

$$\langle\psi(t)|\, A_S\, |\psi(t)\rangle = \langle\psi(0)|\, U^\dagger(t) A_S U(t)\, |\psi(0)\rangle\,, \tag{3.51}$$

and we define the **Heisenberg picture operator** as

$$A_H(t) \equiv U^\dagger(t) A_S U(t) \tag{3.52}$$

Evidently, the same expectation value can be obtained in either picture. Furthermore, since

$$\frac{d}{dt}U(t) = -\frac{i}{\hbar} H U(t) \tag{3.53}$$

it follows that

$$\begin{aligned}\frac{d}{dt}A_H(t) &= \left(\frac{d}{dt}U^\dagger(t)\right)A_S U(t) + U^\dagger(t)A_S\left(\frac{d}{dt}U(t)\right) \\ &= \frac{i}{\hbar}U^\dagger(t)HA_S U(t) - \frac{i}{\hbar}U^\dagger(t)A_S HU(t) \\ &= \frac{i}{\hbar}HU^\dagger(t)A_S U(t) - \frac{i}{\hbar}U^\dagger(t)A_S U(t)H \\ &= \frac{i}{\hbar}[H, A_H(t)]\end{aligned}$$

This is called Heisenberg's equation of motion in quantum mechanics. We get a similar equation for the expectation value of A for a quantum system in the state $|\psi\rangle$, also called Heisenberg's equation of motion, given by

$$\frac{d\langle A\rangle}{dt} = \frac{i}{\hbar}\Big\langle [H, A]\Big\rangle \tag{3.54}$$

This equation holds in both the Heisenberg picture and the Schrödinger picture.

3.6 Energy-Time Uncertainty Relation

Consider a time-independent observable A and a Hamiltonian H for some quantum system. The product of their uncertainties must satisfy

$$\Delta H \Delta A \geq \tfrac{1}{2}\big|\big\langle [H, A]\big\rangle\big| \tag{3.55}$$

where ΔH denotes the uncertainty in H. The Heisenberg equation of motion for A is

$$\frac{d\langle A\rangle}{dt} = \frac{i}{\hbar}\langle [H, A]\rangle \tag{3.56}$$

so it follows that

$$\Delta H \Delta A \geq \frac{\hbar}{2}\left|\frac{d\langle A\rangle}{dt}\right| \tag{3.57}$$

Define δt_A as the time for $\langle A\rangle$ to change by an amount on the order of ΔA. Informally, it is the timescale over which a "significant" change in $\langle A\rangle$ will occur. More precisely, if

$$\delta t_A = \frac{\Delta A}{|d\langle A\rangle/dt|} \tag{3.58}$$

then

$$\Delta H \ \delta t_A \geq \frac{\hbar}{2} \tag{3.59}$$

This relation **does not** say that time is uncertain in the same way that observables are uncertain. There is no time operator in quantum theory! The energy-time uncertainty relation expresses a

tradeoff between the uncertainty in energy and the speed of evolution. A state with well-defined energy ($\Delta H = 0$) doesn't evolve in time, so $d\langle A\rangle/dt = 0$ for any observable A. A state that changes rapidly must involve a superposition of energy eigenstates and hence $\Delta H > 0$.

References

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Chapter 4

Rotations & Angular Momentum

4.1 Rotations in 3D Space

Rotations play a central role in quantum mechanics and serve as a gentle introduction to the *group-theoretic framework* that underlies modern physics. Group theory provides the mathematical language for describing symmetries in a precise way. Informally, a group is a set of transformations that can be combined and inverted, capturing the structure of symmetry operations. By studying the group of rotations, we will uncover an underlying algebraic structure that allows us to derive quantum states systematically, understand their behavior under rotations, and describe phenomena such as spin.

We begin by studying the group of *proper rotations*, denoted SO(3). Given a unit vector $\hat{\mathbf{u}} = (u_x, u_y, u_z)$, a proper rotation R is a linear map from $\hat{\mathbf{u}}$ to another unit vector

$$\hat{\mathbf{u}}' = R\hat{\mathbf{u}}$$

and it preserves the orientation of a right-handed triplet. Namely,

$$R\hat{\mathbf{z}} \cdot (R\hat{\mathbf{x}} \times R\hat{\mathbf{y}}) = \det(R) \det \begin{bmatrix} \hat{\mathbf{z}} & \hat{\mathbf{x}} & \hat{\mathbf{y}} \end{bmatrix} = \det(R) = 1$$

Since R is a linear map between unit vectors, it must be orthogonal:

$$\hat{\mathbf{u}}' \cdot \hat{\mathbf{u}}' = (R\hat{\mathbf{u}}) \cdot (R\hat{\mathbf{u}}) = \hat{\mathbf{u}}^T R^T R \hat{\mathbf{u}} = 1 \quad \Rightarrow \quad R^T R = \mathbb{1}$$

Hence, the elements of SO(3) are represented by matrices with the following properties:

$$R^T R = R R^T = \mathbb{1} \quad \text{and} \quad \det(R) = 1 \tag{4.1}$$

4.1.1 Rotations About a Fixed Axis

Consider a rotation about some axis $\hat{\mathbf{n}}$ by an angle θ in which the sense is determined by the right hand rule. Denote this rotation by $R(\hat{\mathbf{n}}, \theta)$. Rotations about the same axis commute and the angles

add under matrix multiplication:

$$R(\hat{\mathbf{n}}, \theta_1)R(\hat{\mathbf{n}}, \theta_2) = R(\hat{\mathbf{n}}, \theta_2)R(\hat{\mathbf{n}}, \theta_1) = R(\hat{\mathbf{n}}, \theta_1 + \theta_2)$$

Consecutive rotations about different axes do not commute; the group of rotations is called "non-abelian" for this reason. The rotations about the three coordinate axes are

$$R(\hat{\mathbf{x}}, \theta) = R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \tag{4.2}$$

$$R(\hat{\mathbf{y}}, \theta) = R_y(\theta) = \begin{pmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{pmatrix} \tag{4.3}$$

$$R(\hat{\mathbf{z}}, \theta) = R_z(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4.4}$$

and you can work out for yourself simple examples where $R_z(\theta_1)R_x(\theta_2) \neq R_z(\theta_2)R_x(\theta_1)$ for instance. I'll probably ask you to perform rotations on your phone in class...

One can show that any proper rotation can be represented as $R(\hat{\mathbf{n}}, \theta)$, for some axis $\hat{\mathbf{n}}$ and some angle θ. Thus, there is no loss of generality in writing a proper rotation in this form. We will call this the *axis-angle parameterization* of the rotations.

4.1.2 Infinitesimal Rotations

Consider an infinitesimal rotation of the vector $\mathbf{u}$ about the axis $\hat{\mathbf{n}}$ by a small angle $\delta\theta$ (see Figure 4.1 for a geometric argument):

$$R(\hat{\mathbf{n}}, \delta\theta)\mathbf{u} = \mathbf{u} + \delta\theta\hat{\mathbf{n}} \times \mathbf{u} + \mathcal{O}(\delta\theta^2) \tag{4.5}$$

The cross product can be written as a matrix equation using the following identity:

$$\mathbf{a} \times \mathbf{u} = A\mathbf{u} = \begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix} \begin{pmatrix} u_x \\ u_y \\ u_z \end{pmatrix} \tag{4.6}$$

The matrix A is antisymmetric ($A^T = -A$) and admits the following decomposition:

$$A = \mathbf{a} \cdot \boldsymbol{\mathcal{J}} = a_x\mathcal{J}_x + a_y\mathcal{J}_y + a_z\mathcal{J}_z \tag{4.7}$$

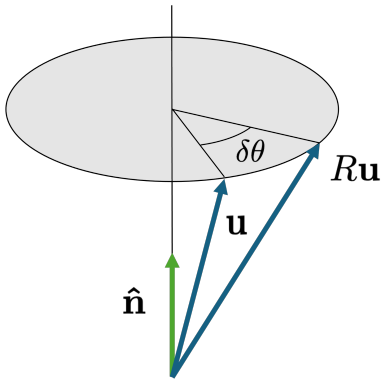


Figure 4.1: Infinitesimal rotation of the vector $\mathbf{u}$ about the axis specified by $\hat{\mathbf{n}}$. The component of $\mathbf{u}$ parallel to $\hat{\mathbf{n}}$ remains constant while the perpendicular component rotates in a positive sense (given by the right hand rule) around the axis defined by $\hat{\mathbf{n}}$.

where $\boldsymbol{\mathcal{J}} = (\mathcal{J}_x, \mathcal{J}_y, \mathcal{J}_z)$ is a "vector" of the matrices

$$\mathcal{J}_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathcal{J}_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \mathcal{J}_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{4.8}$$

The matrices $\mathcal{J}_x, \mathcal{J}_y$ and $\mathcal{J}_z$ encode the geometric structure of rotations (and ultimately the structure of 3D Euclidean space). The non-commutativity of rotations is fully captured by the commutators, which you can verify by direct calculation:

$$[\mathcal{J}_x, \mathcal{J}_y] = \mathcal{J}_z, \quad [\mathcal{J}_z, \mathcal{J}_x] = \mathcal{J}_y, \quad [\mathcal{J}_y, \mathcal{J}_z] = \mathcal{J}_x \tag{4.9}$$

Any infinitesimal rotation can be written as

$$R(\hat{\mathbf{n}}, \delta\theta) = 1 + \delta\theta \hat{\mathbf{n}} \cdot \boldsymbol{\mathcal{J}} \tag{4.10}$$

and it follows that any finite rotation can then be written as the matrix exponential

$$R(\hat{\mathbf{n}}, \theta) = \exp\left(\theta \hat{\mathbf{n}} \cdot \boldsymbol{\mathcal{J}}\right) \tag{4.11}$$

It turns out that knowledge of the commutation relations in Eq. (4.9) is enough to compute the multiplication of any two finite rotations in terms of axis-angle parameters.

4.2 Rotations in Quantum Mechanics

Our goal for this section is to find operators that act on quantum states and represent rotations at the classical level–rotations in 3D Euclidean space. We saw in chapter #3 there was a sense in which a spin-1/2 system precesses in a magnetic field. Specifically, the expectation value of the

spin vector rotates over time around the direction of the magnetic field. In the context of spin precession, a unitary operator evolves the state over time in direct correspondence with a rotating classical vector **S**. We postulate for each proper rotation matrix R in 3D Euclidean space, there corresponds a unitary operator $U(R)$ acting on state vectors. The specific form of $U(R)$ depends on the system, but the essential properties of quantum rotations follow from the properties of classical rotations.

4.2.1 Angular Momentum

Recall that Eq. (4.11) for a classical rotation depends on the product $\theta\hat{\mathbf{n}}$. Define $\boldsymbol{\theta} = \theta\hat{\mathbf{n}}$ such that for an infinitesimal rotation

$$R(\boldsymbol{\theta}) = \mathbb{1} + \boldsymbol{\theta} \cdot \boldsymbol{\mathcal{J}} + \cdots \tag{4.12}$$

which implies

$$\mathcal{J}_x = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_x}\right|_{\boldsymbol{\theta}=0}, \quad \mathcal{J}_y = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_y}\right|_{\boldsymbol{\theta}=0}, \quad \mathcal{J}_z = \left.\frac{\partial R(\boldsymbol{\theta})}{\partial \theta_z}\right|_{\boldsymbol{\theta}=0} \tag{4.13}$$

The unitary operator that corresponds to $R(\boldsymbol{\theta})$ is ultimately a function of $\boldsymbol{\theta}$, and for an infinitesimal rotation, we can approximate $U(\boldsymbol{\theta})$ by the leading terms of its Taylor series

$$U(\boldsymbol{\theta}) = \mathbb{1} - \frac{i}{\hbar}\boldsymbol{\theta} \cdot \mathbf{J} + \cdots \tag{4.14}$$

where (by analogy with the classical case)

$$J_x = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_x}\right|_{\boldsymbol{\theta}=0}, \quad J_y = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_y}\right|_{\boldsymbol{\theta}=0}, \quad J_z = i\hbar \left.\frac{\partial U(\boldsymbol{\theta})}{\partial \theta_z}\right|_{\boldsymbol{\theta}=0} \tag{4.15}$$

Formally, components of **J** are the *generators of rotation* in quantum mechanics. When a classical system is invariant under rotation about some axis, the angular momentum about that axis is conserved. Likewise, when the Hamiltonian for a quantum system commutes with the generator of rotation about some axis, the expectation value of the generator is conserved. We define the **angular momentum** of a quantum system as the vector operator $\mathbf{J} = (J_x, J_y, J_z)$. The factor $\hbar$ ensures (i) **J** has the units of angular momentum and (ii) **J** corresponds to classical angular momentum in the appropriate limit. The factor of i ensures the generators are self-adjoint, i.e. $J_x^\dagger = J_x$, etc. Any finite rotation corresponds to the unitary operator

$$U(\boldsymbol{\theta}) = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta} \cdot \mathbf{J}\right) = \exp\left(-\frac{i}{\hbar}\theta\hat{\mathbf{n}} \cdot \mathbf{J}\right) \tag{4.16}$$

4.2.2 The Adjoint Formula

Up to this point, we have constructed quantum rotation operators by analogy with classical rotations. We began with the geometric structure of SO(3), extracted its generators $\boldsymbol{\mathcal{J}}$, and then introduced self-adjoint operators **J** whose exponentials generate unitary transformations on Hilbert

space. We have not, however, explicitly related $R \to U(R)$. The assignment must be compatible with how rotations act on ordinary vectors.

Consider an operator-valued vector, or vector operator, denoted $\mathbf{A} = (A_x, A_y, A_z)$, where each component represents a Hermitian operator ($A_i^\dagger = A_i$). In other words, $\mathbf{A}$ is a vector whose components are observables. If $\mathbf{A}$ was an ordinary vector in 3D Euclidean space, then a rotation would be a linear mapping $\mathbf{A} \to R\mathbf{A}$. Now consider a unitary transformation $U(R)$ corresponding to the 'classical rotation' R. On a quantum system in the state $|\psi\rangle$,

$$\left|\psi'\right\rangle = U(R)\left|\psi\right\rangle \tag{4.17}$$

If $U(R)$ truly represents the classical rotation R, then the expectation values must transform like ordinary vectors. More precisely, when the quantum state is rotated, the expectation value of the vector operator rotates exactly like a classical vector in $\mathbb{R}^3$. Mathematically,

$$\left\langle\psi'\right| \mathbf{A} \left|\psi'\right\rangle = R\left\langle\psi\right| \mathbf{A} \left|\psi\right\rangle, \tag{4.18}$$

Since $|\psi'\rangle = U(R)|\psi\rangle$, it follows that

$$\left\langle\psi\right| U^\dagger \mathbf{A} U \left|\psi\right\rangle = R\left\langle\psi\right| \mathbf{A} \left|\psi\right\rangle, \tag{4.19}$$

On the right-hand side, we make use of the Einstein summation convention to write

$$\Big[R\langle\mathbf{A}\rangle\Big]_i = R_{ij}\langle A_j\rangle = \langle R_{ij} A_j\rangle \tag{4.20}$$

Since the expression above holds for each component, it follows that

$$\left\langle\psi\right| U^\dagger \mathbf{A} U \left|\psi\right\rangle = \left\langle\psi\right| R\mathbf{A} \left|\psi\right\rangle \tag{4.21}$$

If A and B are Hermitian operators and $\langle\psi|A|\psi\rangle = \langle\psi|B|\psi\rangle$ for all states $|\psi\rangle$, then $A = B$ (you proved this statement in homework #2; it's not as obvious as it seems). We then arrive at the following adjoint formula for the vector operator $\mathbf{A}$:

$$\boxed{U^\dagger \mathbf{A} U = R\mathbf{A}} \tag{4.22}$$

The adjoint formula for rotations defines the relationship between a unitary rotation operator $U(R)$ and its corresponding rotation R. Notice the mapping is not one-to-one; the rotation R can be mapped to the unitary operators $U(R)$ and $-U(R)$. As it turns out, we've encountered an example of this equation during our study of spin precession for the vector operator $\mathbf{S}$. Let's investigate the adjoint formula in more detail.

Example 4.1: Rotations in Quantum Mechanics

Consider a spin-1/2 system. Verify Eq. (4.22) for rotation around the z axis by an angle φ by explicit computation. Use the matrix representations in the S_z basis.

Solution: According to Eq. (4.22),

$$U^\dagger(\hat{\mathbf{z}},\varphi)\mathbf{S}U(\hat{\mathbf{z}},\varphi) = R(\hat{\mathbf{z}},\varphi)\mathbf{S}$$

$$\begin{pmatrix} \exp(i\varphi S_z/\hbar)S_x\exp(-i\varphi S_z/\hbar) \\ \exp(i\varphi S_z/\hbar)S_y\exp(-i\varphi S_z/\hbar) \\ \exp(i\varphi S_z/\hbar)S_z\exp(-i\varphi S_z/\hbar) \end{pmatrix} = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix}$$

Since S_z commutes with any analytic function of itself, the z component is trivially satisfied. Let's check the x component by explicit calculation. On the left-hand side,

$$\exp(i\varphi S_z/\hbar)S_x\exp(-i\varphi S_z/\hbar) = \begin{pmatrix} e^{i\varphi/2} & 0 \\ 0 & e^{-i\varphi/2} \end{pmatrix} \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{i\varphi/2} \end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

And on the right-hand side,

$$S_x\cos\varphi - S_y\sin\varphi = \frac{\hbar}{2}\begin{pmatrix} 0 & \cos\varphi \\ \cos\varphi & 0 \end{pmatrix} - \frac{\hbar}{2}\begin{pmatrix} 0 & -i\sin\varphi \\ i\sin\varphi & 0 \end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix} 0 & e^{i\varphi} \\ e^{-i\varphi} & 0 \end{pmatrix}$$

As for the y component, work out the calculations for yourself and verify that both sides equal

$$\frac{\hbar}{2}\begin{pmatrix} 0 & -ie^{i\varphi} \\ ie^{-i\varphi} & 0 \end{pmatrix}$$

It's good to see that the expression works when we're given the observables for a physical system, but ultimately Eq. (4.22) is a foundational relationship that connects unitary rotation operators to their corresponding rotations in 3D Euclidean space. In the next section, we'll see how the observables corresponding to the components of $\mathbf{S}$ in a spin-1/2 system (and others) naturally arise from the quantum theory of rotations.

Example 4.2: Spin Precession

Let's revisit spin precession from chapter #3 in the context of quantum rotations. Consider a uniform magnetic field along the z axis such that the Hamiltonian takes the form $H = \omega_0 S_z$ and the unitary time evolution operator is

$$U(t) = \exp(-i\omega_0 S_z t/\hbar)$$

We recognize this as a unitary rotation operator related to rotations about the z axis, since S_z is the generator of rotations about the z axis. Given the initial state

$$|\psi(0)\rangle = \cos(\theta/2)\,|+z\rangle + e^{-i\phi}\sin(\theta/2)\,|-z\rangle\,,$$

we know the expectation value of $\mathbf{S}$ is

$$\langle \mathbf{S}\rangle = \frac{\hbar}{2}(\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$$

From the adjoint formula, we can easily compute the time-evolved (rotated) vector:

$$\langle S_x(t)\rangle = \langle S_x\rangle\cos\omega_0 t - \langle S_y\rangle\sin\omega_0 t = \frac{\hbar}{2}\sin\theta\cos(\phi+\omega_0 t)$$

$$\langle S_y(t)\rangle = \langle S_x\rangle\sin\omega_0 t + \langle S_y\rangle\cos\omega_0 t = \frac{\hbar}{2}\sin\theta\sin(\phi+\omega_0 t)$$

and of course,

$$\langle S_z(t)\rangle = \langle S_z\rangle = \frac{\hbar}{2}\cos\theta$$

We have obtained the same results using the adjoint formula, as expected. Note, once again, that our results relied on having the correct spin component operators, so ultimately this result is circular. We determined the spin operators as far as we could from experimental considerations alone. If you remember, we had to adopt a few "phase conventions," and we can ultimately trace those to the theory of rotations in quantum mechanics, and in particular, the adjoint formula.

4.2.3 Commutation Relations

Consider a finite rotation $R_i(\theta) = R(\hat{\mathbf{e}}_i, \theta)$ about the i-th Cartesian coordinate axis, where $i = 1, 2, 3$ corresponds to x, y, z respectively. From the adjoint formula,

$$U_i^\dagger(\theta) A_j U_i(\theta) = R_i(\theta)_{jk} A_k, \tag{4.23}$$

where $U_i(\theta) = U(\hat{\mathbf{e}}_i, \theta)$ is the unitary operator associated with $R_i(\theta)$ and $R_i(\theta)_{jk}$ denotes the components of R_i. Note, once again, we're using the summation convention with an implied sum

over the repeated index k. From Eq. (4.23), it follows that

$$\frac{d}{d\theta}\Big[U_i^\dagger(\theta)A_jU_i(\theta)\Big]_{\theta=0} = \frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} \tag{4.24}$$

On the left-hand side,

$$\begin{aligned}\frac{d}{d\theta}\Big[U_i^\dagger(\theta)A_jU_i(\theta)\Big]_{\theta=0} &= \frac{d}{d\theta}\Big[U_i^\dagger(\theta)\Big]_{\theta=0}A_jU_i(0) + U_i^\dagger(0)A_j\frac{d}{d\theta}\Big[U_i(\theta)\Big]_{\theta=0} \\ &= \frac{i}{\hbar}J_iA_j - \frac{i}{\hbar}A_jJ_i \\ &= \frac{i}{\hbar}[J_i, A_j]\end{aligned}$$

And on the right-hand side,

$$\frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} = \frac{d}{d\theta}\Big[R_i(\theta)_{jk}\Big]_{\theta=0}A_k = \big[\mathcal{J}_i\big]_{jk}A_k$$

where $\big[\mathcal{J}_i\big]_{jk}$ denotes the components of the classical rotation generator $\mathcal{J}_i$. At this point, we introduce the three-dimensional Levi-Civita symbol given by

$$\varepsilon_{ijk} = \begin{cases} +1 & \text{if } (i,j,k) \text{ is } (1,2,3),(3,1,2),(2,3,1) \\ -1 & \text{if } (i,j,k) \text{ is } (3,2,1),(2,1,3),(1,3,2) \\ 0 & \text{if any two indices are equal} \end{cases} \tag{4.25}$$

An easy way to remember the nonzero components of the Levi-Civita is to remember that it's $+1$ when (i,j,k) is an even permutation of (1,2,3), meaning the permutation is formed by an even number of transpositions, and it's -1 when (i,j,k) is an odd permutation, meaning the permutation is formed by an odd number of transpositions. In 3D only, the cyclic permutations of $(1,2,3)$ are all even permutations, and the anticyclic permutations are all odd permutations. Incidentally, $\big[\mathcal{J}_i\big]_{jk} = -\varepsilon_{ijk}$, as you can easily verify[1], and it follows that

$$\frac{d}{d\theta}\Big[R_i(\theta)_{jk}A_k\Big]_{\theta=0} = -\varepsilon_{ijk}A_k$$

Putting together the left- and right-hand sides, we find

$$[J_i, A_j] = i\hbar\varepsilon_{ijk}A_k \tag{4.26}$$

On its own, this result is noteworthy, but we are primarily interested in the case $\mathbf{A} = \mathbf{J}$. We finally arrive at the commutation relations for the angular momentum operators:

$$[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k \tag{4.27}$$

[1]Note the convention here is that with i fixed, j indicates the row and k indicates the column.

where, as another reminder, we are using the summation convention. In particular,

$$[J_x, J_y] = i\hbar J_z, \quad [J_z, J_x] = i\hbar J_y, \quad [J_y, J_z] = i\hbar J_x \tag{4.28}$$

We have achieved something rather remarkable; starting only from the geometric properties of rotations in ordinary three-dimensional space and the requirement that quantum expectation values transform in the same way as classical vectors under these rotations, we have deduced—without ever referring to any specific physical system—that the generators of rotations in quantum mechanics must satisfy the commutation relations in Eq. (4.28).

The commutation relations of angular momentum not only encode the structure of rotations but also determine the allowed values, or spectrum, of angular momentum in quantum systems. Because the components of angular momentum do not commute, it is impossible to know all three simultaneously. Instead, we can diagonalize J^2 and one component, usually J_z, and we find discrete, quantized values for both. The commutation relations define a Lie algebra—specifically, the algebra of infinitesimal rotations—and it is the representation theory of this algebra that constrains the possible measurement outcomes and leads to the quantized spectrum of angular momentum.

4.3 Representations of Angular Momentum

We will now explore the consequences of the angular momentum commutation relations for the construction of angular momentum eigenstates. Consider the operator

$$J^2 = J_x^2 + J_y^2 + J_z^2 \tag{4.29}$$

and note that $[J^2, J_x] = [J^2, J_y] = [J^2, J_z] = 0$. Since J^2 commutes with the components of $\mathbf{J}$, we can construct simultaneous eigenstates of J^2 and any one of the components. However, since these components do not commute with each other, we cannot find simultaneous eigenstates of more than one component of $\mathbf{J}$. By convention we look for simultaneous eigenstates of J^2 and J_z. Every basis vector is labeled by two numbers (the eigenvalues)

$$J^2 \left|\alpha, \beta\right\rangle = \hbar^2 \alpha \left|\alpha, \beta\right\rangle \tag{4.30}$$

$$J_z \left|\alpha, \beta\right\rangle = \hbar \beta \left|\alpha, \beta\right\rangle \tag{4.31}$$

We'll assume without loss of generality that the eigenstates are orthonormal. Note that

$$J^2 - J_z^2 = J_x^2 + J_y^2 \geq 0$$

since J_x and J_y are self-adjoint. It follows that

$$\alpha \geq \beta^2$$

Since $0 \leq \alpha < \infty$, β is bounded above and below for a given value of α. In other words,

$$-\sqrt{\alpha} \leq \beta \leq \sqrt{\alpha}$$

Therefore, β has an upper bound $\beta_{\max}$ and a lower bound $\beta_{\min}$.

Whenever eigenvalues are bounded, it is useful to search for *raising* and *lowering* operators that can be used to construct eigenvectors by repeated application. The only generators that can map an eigenstate to a different vector are J_x and J_y, and after careful consideration, we define the following raising and lowering operators (ladder operators)

$$J_+ = J_x + iJ_y, \quad J_- = J_x - iJ_y \tag{4.32}$$

I realize this is the least motivated step in the whole process, but it really is the key that unlocks the eigenstates. This process of considering a complex linear combination, instead of the original generators, is called a *complexification*, and it's a common technique. The commutation relations for the ladder operators are

$$[J_z, J_\pm] = \pm\hbar J_\pm, \quad [J_+, J_-] = 2\hbar J_z \tag{4.33}$$

Notice that complexification breaks the cyclical symmetry of the original commutation relations. Not only that, but we can use the commutation relations to justify the claim that $J_\pm$ are the raising and lowering operators. For starters, $J_\pm$ does not alter the eigenvalue of J^2 since $[J^2, J_\pm] = 0$.

$$J^2 \left(J_\pm \left|\alpha, \beta\right\rangle\right) = J_\pm J^2 \left|\alpha, \beta\right\rangle = \hbar^2 \alpha \left(J_\pm \left|\alpha, \beta\right\rangle\right)$$

From the commutator of L_z and $L_\pm$, it follows that

$$\begin{aligned} J_z \left(J_\pm \left|\alpha, \beta\right\rangle\right) &= J_\pm J_z \left|\alpha, \beta\right\rangle + \left[J_z, J_\pm\right] \left|\alpha, \beta\right\rangle \\ &= \hbar\beta J_\pm \left|\alpha, \beta\right\rangle \pm \hbar J_\pm \left|\alpha, \beta\right\rangle \\ &= \hbar(\beta \pm 1) \left(J_\pm \left|\alpha, \beta\right\rangle\right) \end{aligned}$$

Hence, $J_\pm \left|\alpha, \beta\right\rangle$ is proportional to $\left|\alpha, \beta \pm 1\right\rangle$. We'll find the constant of proportionality later by requiring that $\left|\alpha, \beta\right\rangle$ is an orthonormal set. Since β is bounded above and below,

$$J_+ \left|\alpha, \beta_{\max}\right\rangle = 0 \quad \text{and} \quad J_- \left|\alpha, \beta_{\min}\right\rangle = 0 \tag{4.34}$$

Furthermore,

$$\begin{aligned} 0 = J_- J_+ \left|\alpha, \beta_{\max}\right\rangle &= \left(J_x^2 + J_y^2 - \hbar J_z\right) \left|\alpha, \beta_{\max}\right\rangle \\ &= \left(J^2 - J_z^2 - \hbar J_z\right) \left|\alpha, \beta_{\max}\right\rangle \\ &= \left(\hbar^2 \alpha - \hbar^2 \beta_{\max}^2 - \hbar^2 \beta_{\max}\right) \left|\alpha, \beta_{\max}\right\rangle \end{aligned}$$

$$\alpha = \beta_{\max} \left(\beta_{\max} + 1\right)$$

Similarly, using $J_- \left|\alpha, \beta_{\min}\right\rangle = 0$ and $J_+ J_- = J^2 - J_z^2 + \hbar J_z$, we find

$$\alpha = \beta_{\min} \left(\beta_{\min} - 1\right)$$

We now introduce the standard notation $j = \beta_{\max}$ and $\beta = m$ and relabel the eigenstates

$$J^2 \left|j, m\right\rangle = \hbar^2 j(j+1) \left|j, m\right\rangle \quad \text{and} \quad J_z \left|j, m\right\rangle = \hbar m \left|j, m\right\rangle$$

where

$$-j \leq m \leq j \tag{4.35}$$

Starting from the vector $|j, j\rangle$, we obtain the vector $|j, -j\rangle$ by repeated application of J_-. After each application, the eigenvalue is lowered by one, so the difference between the highest and lowest eigenvalue must be an integer n. It follows that

$$j - (-j) = 2j = n \quad \rightarrow \quad j = \frac{n}{2} \tag{4.36}$$

Hence, the only values of j allowed are non-negative integers or half-integers:

$$j = 0,\ \frac{1}{2},\ 1,\ \frac{3}{2},\ \ldots$$

When we say that j belongs to the indicated set of values (integers and half-integers), we mean that the commutation relations alone tell us that j can take on only these values. They do not tell us which of these values actually occur in a specific application.

Let's look at the eigenvalues for the first few representations:

$$\begin{array}{lll} j = 0 & j(j+1) = 0 & m = 0 \\ j = 1/2 & j(j+1) = 3/4 & m = -1/2, 1/2 \\ j = 1 & j(j+1) = 2 & m = -1, 0, 1 \\ \ldots & & \end{array}$$

The dimension of the vector space is $2j + 1$ for a given representation, and we can construct the

eigenvectors of J_z using the ladder operators (the raising the lowering operators). Recall

$$J_\pm |j,m\rangle = C_\pm |j,m\pm 1\rangle$$

for constants $C_\pm$ that depend on j and m. The norm squared of $J_+ |j,m\rangle$ is

$$\begin{aligned}\|J_+ |j,m\rangle\|^2 = |C_+|^2 &= \langle j,m| J_+^\dagger J_+ |j,m\rangle \\ &= \langle j,m| J_- J_+ |j,m\rangle \\ &= \langle j,m| \left(J^2 - J_z^2 - \hbar J_z\right) |j,m\rangle \\ &= \hbar^2 \left(j(j+1) - m^2 - m\right)\end{aligned}$$

A similar calculation gives $|C_-|^2$. There is ambiguity in the overall phase factor, so we choose

$$C_\pm = \hbar\sqrt{j(j+1) - m(m\pm 1)} \tag{4.37}$$

Given an eigenstate $|j,m\rangle$, a ladder operator will produce a higher or lower eigenstate:

$$J_\pm |j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\, |j,m\pm 1\rangle \tag{4.38}$$

4.3.1 Spin 1/2 and SU(2)

The smallest non-trivial value of j is $1/2$. The generator J_z has eigenvalues $\hbar/2$ and $-\hbar/2$, and because J_z is conventionally chosen to be diagonal,

$$J_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The corresponding eigenvectors are

$$\left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{4.39}$$

As for the ladder operators,

$$J_+ \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle = \hbar \left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle \quad \text{and} \quad J_- \left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \hbar \left|\tfrac{1}{2},-\tfrac{1}{2}\right\rangle$$

and their corresponding matrix representations are

$$J_+ = \hbar\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad J_- = \hbar\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{4.40}$$

From $J_+ = J_x + iJ_y$ and $J_- = J_x - iJ_y$, we can invert the expressions to find

$$J_x = \frac{1}{2}(J_+ + J_-) = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$J_y = \frac{1}{2i}(J_+ - J_-) = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

These are the spin-1/2 angular momentum operators we inferred from experiments earlier in the course! Keep in mind there is no guarantee that any physical system will align with this mathematical description. As it turns out, if you combine quantum mechanics, special relativity, and rotational invariance, *spin is unavoidable*. You would learn more about this in a course on quantum field theory.

There's a topic I've been putting off for a while, but now seems like a good time to discuss the group SU(2), the group of special unitary 2×2 complex matrices. In the same way that complex numbers can be used to represent 2D rotations (rotation around a single, fixed axis), there is a way to use 2×2 special unitary matrices to represent rotations in 3D Euclidean space. I'd like to show you how that works.

The unitary rotation operators in a spin-1/2 system belong to SU(2). They are 2×2 complex, unitary matrices, and they are "special" because $\det U = 1$. In the axis-angle parameterization, any unitary rotation operator in a spin-1/2 system can be written as

$$U(\hat{\mathbf{n}}, \theta) = \exp\left(-\frac{i}{\hbar}\theta\hat{\mathbf{n}} \cdot \mathbf{S}\right) = \exp\left(-\frac{i}{2}\theta\hat{\mathbf{n}} \cdot \boldsymbol{\sigma}\right)$$

where we use $\mathbf{S}$ to denote angular momentum, and $\boldsymbol{\sigma}$ denotes the **Pauli matrices**:

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z) \quad \text{where} \quad S_x = \frac{\hbar}{2}\sigma_x,$$

and for reference,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Any 2×2 Hermitian matrix can be written as a linear combination of the identity operator and the Pauli matrices. To see why, consider an arbitrary 2×2 Hermitian matrix

$$A = \begin{pmatrix} a_0 + a_z & a_x + ia_y \\ a_x - ia_y & a_0 - a_z \end{pmatrix}$$

where c and the components of $\mathbf{a} = (a_x, a_y, a_z)$ are real numbers. It follows that

$$A = a_0 \mathbb{1} + \mathbf{a} \cdot \boldsymbol{\sigma}$$

This result suggests that any 2×2 Hermitian matrix encodes four parameters, and three of them correspond to a vector in 3D Euclidean space.

Just as in the Heisenberg picture, we regard state vectors to be stationary while the operators evolve unitarily over time, we can do the same thing for rotations. Start by considering the expectation value of A under unitary rotation of the state $|\psi\rangle$:

$$\langle \psi' | A | \psi' \rangle = \langle \psi | U^\dagger A U | \psi \rangle$$

The unitary transformation of A is then

$$A' = U^\dagger A U$$

which has the same structure as a time-evolving Heisenberg operator. It follows that

$$A' = U^\dagger (a_0 \mathbb{1} + \mathbf{a} \cdot \boldsymbol{\sigma}) U = a_0 \mathbb{1} + U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U$$

Using Eq. (4.22), we can expand the second term as

$$U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U = \sum_i a_i U^\dagger \sigma_i U = \sum_i a_i \sum_j R_{ij} \sigma_j$$

Note that $R_{ji} = R^T_{ji}$ such that

$$U^\dagger (\mathbf{a} \cdot \boldsymbol{\sigma}) U = \sum_j \left(\sum_i R^T_{ji} a_i \right) \sigma_j = \left(R^T \mathbf{a} \right) \cdot \boldsymbol{\sigma}$$

Altogether, we find

$$A' = U^\dagger A U = a_0 \mathbb{1} + \left(R^T \mathbf{a} \right) \cdot \boldsymbol{\sigma}$$

In the *Schrödinger picture of rotations*, the state vectors unitarily rotate in correspondence with the classical rotation R of a Bloch vector. In the *Heisenberg picture of rotations*, state vectors are fixed and the operators are unitarily rotating in the opposite sense since $R^T = R^{-1}$. This is certainly an interesting result, but let's revisit that second term. Let $R^T \to R$ and note that $U(R^T) = U^\dagger(R)$. Hence,

$$U(R) (\mathbf{a} \cdot \boldsymbol{\sigma}) U^\dagger(R) = (R\mathbf{a}) \cdot \boldsymbol{\sigma}$$

This result, also known as an adjoint formula for rotations, constitutes a mapping from elements of SU(2) to elements of SO(3). Notice the mapping is two-to-one because both $U(R)$ and $-U(R)$ map to the same rotation matrix R.

4.3.2 Spin 1

In the case $j = 1$,

$$J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \tag{4.41}$$

and the basis vectors are

$$|1,1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad |1,0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.42}$$

Example 4.3: Stern-Gerlach experiment with spin-1 particles

Consider a Stern-Gerlach experiment with a neutral beam of spin-1 particles. What fraction of particles with $S_y = \hbar$ will be found to have $S_z = \hbar$ with the appropriate arrangement of analyzers?

Solution: The quantitative answer to this equation is the square of an inner product:

$$|\langle 1,1|1,1\rangle_y|^2$$

where $|1,1\rangle$ is the eigenvector of S_z with eigenvalue $\hbar$ and $|1,1\rangle_y$ is the eigenvector of S_y with eigenvalue $\hbar$. Find $|1,1\rangle_y$ in the basis of S_z. The eigenvalue equation is

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \hbar \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

From which it follows that

$$-ib = \sqrt{2}a, \quad i(a-c) = \sqrt{2}b, \quad ib = \sqrt{2}c$$

Hence, $b = i\sqrt{2}a$ and $c = -a$. After normalizing the state,

$$|1,1\rangle_y = \frac{1}{2} \begin{pmatrix} 1 \\ i\sqrt{2} \\ 1 \end{pmatrix}$$

and finally,

$$|\langle 1,1|1,1\rangle_y|^2 = \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{4}$$

Furthermore, 1/2 of the particles will have $S_z = 0$ and 1/4 of them will have $S_z = -\hbar$. **Your turn.** What fraction of particles with $S_y = 0$ with be found to have $S_z = \hbar, 0$ and $-\hbar$ with the appropriate arrangement of analyzers?

Chapter 5

Wave Mechanics for 1D Motion

5.1 Introduction

In this section, we will begin studying the quantum physics of a non-relativistic spinless particle with mass m whose environmental interactions can be modeled by a potential energy function. This model applies to a variety of physical systems whenever particle velocities are much smaller than the speed of light and spin degrees of freedom either do not exist or do not play a significant role. Physically, it describes a structureless particle with no relevant internal dynamics moving in a potential that does not couple to spin. This situation arises in several important contexts across atomic, nuclear, and condensed matter physics.

One of the clearest realizations is in the physics of ultracold neutral atoms trapped in optical potentials. In such systems, atomic velocities are extremely small, and often the internal electronic structure is effectively "frozen out." If magnetic fields are negligible or if one focuses purely on motional degrees of freedom, spin can be ignored, and the single-particle Hamiltonian with kinetic plus potential energy accurately describes the system. This framework is the standard starting point for describing atoms in optical traps and optical lattices, as well as the single-particle picture underlying Bose–Einstein condensation.

Another particularly clean example is the alpha particle in nuclear physics. An alpha particle (a helium-4 nucleus) has total spin zero and is frequently non-relativistic in nuclear processes such as alpha decay. In tunneling models of alpha decay and in low-energy scattering problems, it is treated as a structureless point particle governed by the non-relativistic Schrödinger equation. Similarly, helium-4 atoms, which also have total spin zero, behave as spinless bosons at low temperatures. Their center-of-mass motion in superfluid helium or dilute Bose gases is well described by the non-relativistic Schrödinger equation.

Spinless approximations also appear in effective descriptions of quasiparticles in condensed matter systems. Certain collective excitations, such as phonons or other bosonic modes, can behave like particles with an effective mass moving in an effective potential. In these cases, the same Hamiltonian applies, with the physical mass replaced by an effective mass. Additionally, the center-of-mass motion of composite objects—such as cold molecules in traps—can be modeled as that of

a spinless particle if magnetic and spin-dependent interactions are negligible.

At low energies, even fundamentally relativistic spin-0 particles such as pions can be approximately described by the non-relativistic Hamiltonian, since their relativistic energy reduces to

$$E \approx mc^2 + \frac{p^2}{2m}$$

when momenta are small compared to mc. However, this is only an approximation valid in the low-momentum regime.

The model ultimately breaks down when relativistic effects become important or when spin-dependent interactions play a significant role, as in electrons subject to magnetic fields, fine structure, spin–orbit coupling, or Pauli exclusion effects. In such cases, one must use either the Pauli or Dirac equations. Nevertheless, as a universal low-energy limit for massive particles without relevant spin dynamics, the non-relativistic spinless Hamiltonian remains one of the most broadly applicable and physically realized models in quantum physics.

5.2 Position-Space Wave Functions

The observables considered so far have all been assumed to exhibit discrete eigenvalue spectra. In quantum mechanics, however, there are observables with continuous eigenvalues. Furthermore, observables can have portions of their spectra that are discrete and portions that are continuous. When the spectrum is continuous, there are infinitely many eigenvectors, and the associated vector space has infinite dimension. Thankfully, much of the intuition we've gained by working with finite-dimensional vector spaces will carry over in a relatively straightforward way. I'll be sure to point out the important differences as we go.

Consider a non-relativistic particle moving in one dimension. We assume that we can measure its position anywhere along the x axis. It is therefore natural to introduce an orthonormal set of position eigenstates $|\tilde{x}\rangle$ such that

$$\hat{x}\,|\tilde{x}\rangle = x\,|\tilde{x}\rangle \tag{5.1}$$

where $\hat{x}$ is the position operator and the value of x runs over all possible values of the position of the particle, that is, from $-\infty$ to ∞. From postulate #3, we may write a general state vector describing the particle as a superposition of position states such that

$$|\psi\rangle = \sum_i \langle \tilde{x}_i|\psi\rangle\,|\tilde{x}_i\rangle$$

and it follows that the probability of finding the particle at a particular location x_i is

$$p(x_i) = |\,\langle \tilde{x}_i|\psi\rangle\,|^2.$$

Unfortunately, since x lies in a continuum, there are infinitely many positions the particle could

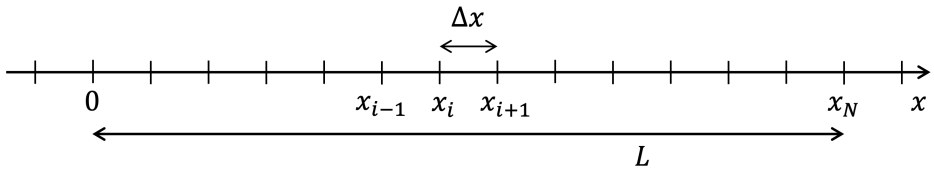


Figure 5.1: A particle moves on a line with a nonzero probability of being found within a region of length L. We discretize the allowed positions and then take a continuum limit.

have, so the probability of obtaining exactly one position out of infinitely many is zero. This is easier to illustrate if we imagine that a quantum particle can occupy only a discrete set of positions x_i, each separated by Δx such that $\Delta x = x_{i+1} - x_i = x_i - x_{i-1}$, etc. Within some range of length $L \gg \Delta x$, we expect the total probability of finding the particle to be nonzero. Let N be the number of discrete positions in the range of length L such that $N\Delta x = L$. The probability the particle will be found at a particular position is proportional to $1/N$. In the continuum limit, $N \to \infty$ and $\Delta x \to 0$, so the probability of finding the particle at just one location vanishes.

Assuming the total probability to find the particle in the range $(0, L)$ is some nonzero constant that is independent of our course-graining procedure, then

$$\sum_{i=1}^{N} |\langle \tilde{x}_i|\psi\rangle|^2 = \sum_{i=1}^{N} \Delta x \left[\frac{|\langle \tilde{x}_i|\psi\rangle|^2}{\Delta x}\right] = \text{constant}$$

Look closely at the term in brackets. In both the numerator and the denominator, we have quantities that scale like $1/N$, so in the limit $N \to \infty$, we're left with a constant value. Furthermore, the summation above becomes an integral in the continuum limit.

In order to make use of this result in a mathematically consistent way, we need to redefine our position eigenstates. They will still be orthogonal, but no longer normalized in the familiar sense. We simultaneously require

$$|x\rangle = \lim_{\Delta x \to 0} \frac{|\tilde{x}\rangle}{\sqrt{\Delta x}}$$

and for the completeness relation to remain valid. In other words,

$$\sum_i |\tilde{x}_i\rangle\langle\tilde{x}_i| = \sum_i \Delta x \;\; |x_i\rangle\langle x_i| = \mathbb{1}$$

and in the limit $\Delta x \to 0$ (or $N \to \infty$),

$$\int_{-\infty}^{\infty} dx \;\; |x\rangle\langle x| = \mathbb{1} \tag{5.2}$$

So far so good, but now it looks like the inner product $\langle x|x\rangle \to \infty$ when $\Delta x \to 0$. This might be a problem for the finite-dimensional vector spaces we've been working with, but it actually works in the vector space of *functions*. Consider "inserting the completeness relation" into the inner product

of $|x\rangle$ with an arbitrary ket $|\psi\rangle$:

$$\langle x|\psi\rangle = \langle x| \left(\int_{-\infty}^{\infty} dx' \ \big|x'\big\rangle \big\langle x'\big| \right) |\psi\rangle$$

$$= \int_{-\infty}^{\infty} dx' \ \big\langle x\big|x'\big\rangle \big\langle x'\big|\psi\big\rangle \tag{5.3}$$

The quantity $\langle x|\psi\rangle$ is the projection of $|\psi\rangle$ onto the basis ket $|x\rangle$. For a fixed state $|\psi\rangle$, the quantity $\langle x|\psi\rangle$ depends on the value of x much like a function. In fact, we define

$$\psi(x) = \langle x|\psi\rangle \tag{5.4}$$

and call it the **wave function** for purely historical reasons. You'll have to excuse the abuse of notation taking place, where ψ appears as a label inside the ket and as a function. It will be clear from context which object you are working with.

Eq. 5.3 can now be written as

$$\psi(x) = \int_{-\infty}^{\infty} dx' \ \big\langle x\big|x'\big\rangle \, \psi(x') \tag{5.5}$$

which is (nearly) the defining property of a mathematical object called the **Dirac delta function**. Hence,

$$\big\langle x\big|x'\big\rangle = \delta(x' - x) \tag{5.6}$$

where $\delta(x' - x)$ is the delta function defined as the generalized function (a distribution) on the real numbers, whose value is zero everywhere except where $x = x'$, and whose integral over the entire real line is equal to one:

$$\int_{-\infty}^{\infty} dx' \ \delta(x' - x) = 1 \tag{5.7}$$

Eq. 5.6 is a generalized form of orthonormality suited for continuous variables. It reflects the fact that states like $|x\rangle$ are non-normalizable limits, yet they form a complete basis in the sense of distributions, allowing us to expand and reconstruct functions using integrals.

We can now define the inner product over wave functions by integrating over x. Beginning with the inner product of $|\phi\rangle$ with $|\psi\rangle$, we insert the completeness relation:

$$\langle \phi|\psi\rangle = \int_{-\infty}^{\infty} dx \ \langle \phi|x\rangle \langle x|\psi\rangle = \int_{-\infty}^{\infty} dx \ \phi^*(x)\psi(x) \tag{5.8}$$

and in particular,

$$\langle \psi|\psi\rangle = \int_{-\infty}^{\infty} dx \ \psi^*(x)\psi(x) = \int_{-\infty}^{\infty} dx \ |\psi(x)|^2 \tag{5.9}$$

A *normalized wave function* has the property

$$\int_{-\infty}^{\infty} dx\ |\psi(x)|^2 = 1 \tag{5.10}$$

We identify the quantity $dx\ |\psi(x)|^2$ as the probability of finding the particle between x and $x+dx$. The probability to find the particle in the range $[a,b]$ is given by

$$P(a<x<b) = \int_a^b dx\ |\psi(x)|^2 \tag{5.11}$$

Normalization therefore reflects the statement that the probability of finding the particle *anywhere* on the x axis is 1, which makes sense given the fundamental assumption that we are indeed dealing with a single particle.

Example 5.1: Position-space wave function

At time $t=0$, a particle has an associated wavefunction

$$\psi(x) = \begin{cases} A(x/a) & 0 \le x < a \\ A(b-x)/(b-a) & a \le x \le b \\ 0 & \text{otherwise} \end{cases}$$

where A, a, and b are all positive constants. (a) Normalize ψ by finding an appropriate value for A. (b) Where is the particle most likely to be found at $t=0$? (c) What is the probability of finding the particle to the left of a? (d) What is the expectation value of x?

Solution: (a) Normalize by taking the square-modulus of the wavefunction and setting the integral equal to one.

$$1 = \int_0^b dx\ |\psi(x)|^2 = \int_0^a \frac{A^2}{a^2}x^2 dx + \int_a^b A^2\frac{(b-x)^2}{(b-a)^2}dx$$

$$1 = A^2\cdot\frac{a}{3} + A^2\cdot\frac{(b-a)}{3} = A^2\cdot\frac{b}{3}$$

Hence, $A=\sqrt{3/b}$ is a suitable choice for the constant A. (b) The particle is most likely to be found wherever $|\psi(x)|^2$ has a maximum. In this case, that corresponds to $x=a$ where $|\psi(a)|^2 = 3/b$. (c) The probability to find the particle to the left of a is

$$P(x<a) = \int_0^a |\psi(x)|^2 dx = \int_0^a \frac{3}{b}\cdot\frac{x^2}{a^2}dx = \frac{a}{b}$$

Notice if $a=b$, then the probability of finding the particle to the left of a is 1 since the wavefunction is zero for $x>a$. If $b=2a$, then the probability of finding the particle to the

left of a is 0.5; the wavefunction is symmetric about $x = a$. (d) To find the expectation value of $\hat{x}$, apply the definition and use the completeness relation:

$$\begin{aligned}\langle \hat{x} \rangle &= \langle \psi | \, \hat{x} \, | \psi \rangle = \langle \psi | \, \hat{x} \left(\int_{-\infty}^{\infty} dx \;\; |x\rangle \, \langle x| \right) |\psi\rangle \\ &= \int_{-\infty}^{\infty} dx \;\, \psi^*(x) x \psi(x) \\ &= \int_0^a \frac{3}{b} \cdot \frac{x^3}{a^2} dx + \int_a^b \frac{3}{b} \cdot \frac{(b-x)^2 x}{(b-a)^2} dx \\ &= \frac{3a^2}{4b} + \frac{a}{2} - \frac{3a^2}{4b} + \frac{b}{4} \\ &= \frac{a}{2} + \frac{b}{4}\end{aligned}$$

When $b = 2a$, the wavefunction (and corresponding probability distribution) is symmetric about $x = a$, which is the most probable location to find the particle. It therefore makes sense that $x = a$ would also be the expectation value of $\hat{x}$. Note, however, that the most probable location and the expectation value do not necessarily coincide. For instance, when $a = b$, the most likely place to find the electron is still $x = a$, but the expectation value of $\hat{x}$ is $3a/4$ since the wavefunction is asymmetric (we would say it is left-skewed).

5.3 Translations in Quantum Mechanics

Mirroring our study of rotations in chapter #4, we postulate the existence of a unitary operator T acting on state vectors that corresponds to a translation along one dimension of 3D Euclidean space. If we denote the operator that translates a state by an amount a as $T(a)$, then its action on a position eigenstate is defined by

$$T(a) \, |x\rangle = |x + a\rangle$$

Applying two translations in opposite directions should bring us back to the same point:

$$T(a)T(-a) \, |x\rangle = T(a) \, |x - a\rangle = |x\rangle$$

Thus, $T(-a)T(a) = I$, and since $T^\dagger(a)T(a) = \mathbb{1}$,

$$T^\dagger(a) = T(-a)$$

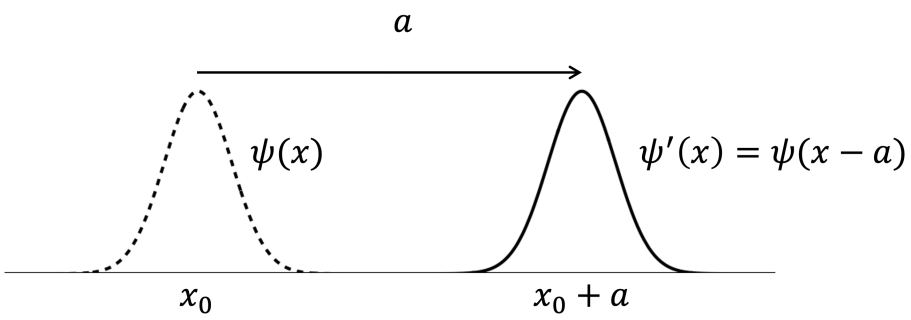


Figure 5.2: For $a > 0$, the translation operator shifts the wave function to the right such that the new wave function $\psi'(x) = T(a)\psi(x) = \psi(x-a)$, at position x, takes the values of the old wave function $\psi(x)$ at position $x - a$.

Because performing a translation by a followed by translation by b is equivalent to a translation by $a + b$, the translation operators satisfy

$$T(a+b) = T(a)T(b) = T(b)T(a)$$

We can now examine how the position operator transforms under translation. Acting with $T(a)$ shifts the position eigenvalues by a

$$T^\dagger(a)\hat{x}T(a) = \hat{x} + a\mathbb{1} = \hat{x} + a$$

where we have used a shorthand notation suppressing the identity operator.

Let us now see how the translation operator acts on wave functions. The position representation of a state is $\langle x|\psi\rangle$. Acting with $T(a)$ gives

$$\langle x|\, T(a)\, |\psi\rangle = \psi(x-a)$$

showing explicitly that $T(a)$ shifts the wave function to the right by a. We can also consider a translation operator acting directly on the wave function such that

$$T(a)\psi(x) \equiv \langle x|\, T(a)\, |\psi\rangle = \psi(x-a)$$

5.3.1 Momentum Operator

Consider an infinitesimal translation by δx

$$T(\delta x) = \mathbb{1} - \frac{i}{\hbar}\hat{p}_x \delta x \tag{5.12}$$

where the operator $\hat{p}_x$ is called the generator of translations such that any finite translation may be written as

$$T(a) = \exp\left(-\frac{i}{\hbar}\hat{p}_x a\right) \tag{5.13}$$

When a classical system is invariant under translations along some axis, the linear momentum about that axis is conserved. Likewise, when the Hamiltonian for a quantum system commutes with the generator of translation along some axis, the expectation value of the generator is conserved. We define the x-component of **linear momentum** of a quantum particle as the generator of translation $\hat{p}_x$ along the x axis. In other words,

$$\hat{p}_x = i\hbar \left.\frac{\partial T(a)}{\partial a}\right|_{a=0} \tag{5.14}$$

where the factor of i ensures that $\hat{p}_x$ is self-adjoint[1] and the factor of $\hbar$ is included so that $\hat{p}_x$ has the correct units and corresponds to classical momentum in the appropriate limit.

So far we've been working with operators on the vector space of kets. It will often be convenient to work directly with operators on the space of wave functions. For instance, the position operator can act directly on the wave function $\psi(x)$ as

$$\hat{x}\psi(x) \equiv \langle x|\,\hat{x}\,|\psi\rangle = x\psi(x) \tag{5.15}$$

The momentum operator can also act directly on the space of wave functions. Start by writing out the action of the translation operator on the wave function:

$$T(a)\psi(x) = \psi(x-a) = \psi(u(x,a))$$

where $u(x,a) = x - a$. Differentiating with respect to the translation parameter a while taking care to implement the chain rule on the right-hand side, we find

$$\left.\frac{\partial T(a)}{\partial a}\right|_{a=0}\psi(x) = \frac{\partial\psi(u)}{\partial u}\left.\frac{\partial u}{\partial a}\right|_{a=0} = -\frac{\partial\psi(x)}{\partial x}$$

After multiplying both sides by $i\hbar$, we find the following identity for the momentum operator on the space of position-space wave functions:

$$i\hbar\left.\frac{\partial T(a)}{\partial a}\right|_{a=0}\psi(x) = \hat{p}_x\psi(x) = -i\hbar\frac{\partial\psi(x)}{\partial x}$$

Hence, the momentum operator satisfies

$$\hat{p}_x\psi(x) = \langle x|\,\hat{p}_x\,|\psi\rangle = -i\hbar\frac{\partial\psi(x)}{\partial x} \tag{5.16}$$

Please excuse the abuse of notation, but it's extremely common and it should be clear from context

[1] We'll see very soon that a mere factor of i is actually not enough to ensure that momentum is self-adjoint.

whether the operator is acting on state vectors or wave functions directly.

5.3.2 Physical Wave Functions

In the quantum mechanics of a non-relativistic particle, certain mathematical conditions must be satisfied to define position $\hat{x}$ and momentum $\hat{p}_x$ operators as self-adjoint operators on a suitable space of wave functions.

1. Wave functions must be square-integrable:

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \tag{5.17}$$

   This ensures inner products are well-defined and wave functions can be normalized.

2. Wave functions must vanish at infinity, i.e. $\psi(x) \to 0$ as $x \to \pm\infty$.

Condition #1 is pretty clear, but the reason we impose condition #2 mostly has to do with the status of momentum as a valid observable in quantum theory. If $\hat{p}_x$ is self-adjoint, then $\langle f|\,\hat{p}_x\,|g\rangle^* = \langle g|\,\hat{p}_x\,|f\rangle$ for any two functions f and g. Let's see if this holds:

$$\langle g|\,\hat{p}_x\,|f\rangle = \int_{-\infty}^{\infty} dx\ \langle g|\,\hat{p}_x\,|x\rangle\,\langle x|f\rangle = \int_{-\infty}^{\infty} dx\ \left(i\hbar\frac{dg^*(x)}{dx}\right) f(x)$$

$$\langle f|\,\hat{p}_x\,|g\rangle^* = \int_{-\infty}^{\infty} dx\ \left(-i\hbar\frac{df(x)}{dx}\right) g^*(x) = -i\hbar f(x)g^*(x)\Big|_{-\infty}^{\infty} + i\hbar\int_{-\infty}^{\infty} dx\ \frac{dg^*(x)}{dx} f(x)$$

For arbitrary functions f and g, $\langle f|\,\hat{p}_x\,|g\rangle^* \neq \langle g|\,\hat{p}_x\,|f\rangle$, so the momentum operator is not self-adjoint! Condition #2 ensures the boundary term vanishes so that momentum is an observable for all physical wave functions–those that may correspond to a particle with position and momentum.

5.3.3 Canonical Commutation Relation

Given a wave function $\psi(x)$, we can easily find the commutator of $\hat{x}$ and $\hat{p}_x$:

$$\begin{aligned}
[\hat{x},\hat{p}_x]\psi(x) &= \hat{x}\hat{p}_x\left[\psi(x)\right] - \hat{p}_x\hat{x}\left[\psi(x)\right] \\
&= -i\hbar x\frac{\partial\psi(x)}{\partial x} + i\hbar\frac{\partial(x\psi(x))}{\partial x} \\
&= -i\hbar x\frac{\partial\psi(x)}{\partial x} + i\hbar\frac{\partial\psi(x)}{\partial x} + i\hbar\psi(x) \\
&= i\hbar\psi(x)
\end{aligned}$$

Hence,

$$[\hat{x},\hat{p}_x] = i\hbar\mathbb{1} = i\hbar \tag{5.18}$$

where it's common to drop the identity operator. Any two observables that satisfy the commutation relation $[A, B] = i\hbar$ are said to be conjugate to one another.

5.3.4 Ehrenfest's Theorem

A non-relativistic particle constrained to move along one dimension of space is essentially a model defined by the Hamiltonian

$$H = \frac{\hat{p}_x^2}{2m} + V(\hat{x}) \tag{5.19}$$

where the first term is the kinetic energy operator and $V(\hat{x})$ is the potential energy operator that models a particle's interaction with its environment—a very large subsystem whose motion/dynamics can be neglected. The potential energy operator $V(\hat{x})$ is a function of the position operator $\hat{x}$ and it multiplies the position-space wave function $\psi(x)$ by the corresponding scalar potential energy function evaluated at position x. In other words,

$$V(\hat{x})\psi(x) = V(x)\psi(x) \tag{5.20}$$

For example, $V(\hat{x}) = \frac{1}{2}k\hat{x}^2$ is the potential energy operator for a Hooke's law force; $V(\hat{x})$ can generally be written as a linear combination of monomials $\hat{x}^n$ where n is an integer.

The expectation value of the position operator changes over time in a way that matches the definition of momentum in classical mechanics. Observe that

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{i}{\hbar}\langle[H, \hat{x}]\rangle = \frac{i}{2m\hbar}\langle[\hat{p}_x^2, x]\rangle$$

The commutator is found by expanding in terms of the canonical commutation relation:

$$\begin{aligned}
[\hat{p}_x^2, x] &= \hat{p}_x\hat{p}_x\hat{x} - \hat{x}\hat{p}_x\hat{p}_x \\
&= \hat{p}_x\hat{p}_x\hat{x} - \hat{p}_x\hat{x}\hat{p}_x + \hat{p}_x\hat{x}\hat{p}_x - \hat{x}\hat{p}_x\hat{p}_x \\
&= \hat{p}_x[\hat{p}_x, \hat{x}] + [\hat{p}_x, \hat{x}]\hat{p}_x \\
&= -2i\hbar\hat{p}_x
\end{aligned}$$

Hence,

$$\frac{d\langle\hat{x}\rangle}{dt} = \frac{\langle\hat{p}_x\rangle}{m} \tag{5.21}$$

which is the result of classical mechanics. This is yet another example of Ehrenfest's theorem, which states that the expectation values of quantum mechanical operators for position and momentum obey equations of motion analogous to classical mechanics.

The equation of motion for the expectation value of momentum is given by

$$\frac{d\langle\hat{p}_x\rangle}{dt} = \frac{i}{\hbar}\langle[H,\hat{p}_x]\rangle = \frac{i}{\hbar}\langle[V(\hat{x}),\hat{p}_x]\rangle$$

where the commutator $[V(\hat{x},\hat{p}_x)]$ can be evaluated with an arbitrary wave function $\psi(x)$:

$$\Big(V(\hat{x})\hat{p}_x\Big)\psi(x) = -i\hbar V(x)\psi'(x)$$

$$\Big(\hat{p}_x V(\hat{x})\Big)\psi(x) = \hat{p}_x\Big(V(x)\psi(x)\Big) = -i\hbar V'(x)\psi(x) - i\hbar V(x)\psi'(x)$$

$$[V(\hat{x}),\hat{p}_x]\psi(x) = i\hbar V'(x)\psi(x)$$

where we've used prime notation to denote derivatives with respect to x. Hence,

$$\frac{d\langle\hat{p}_x\rangle}{dt} = \frac{i}{\hbar}\left\langle i\hbar\frac{\partial V}{\partial x}\right\rangle = \left\langle -\frac{\partial V(\hat{x})}{\partial x}\right\rangle \tag{5.22}$$

which is the analog of Newton's second law in quantum mechanics. Note that to evaluate the commutator leading to Eq. (5.21), I used an "algebraic method" consisting of algebraic manipulations involving commutators and the canonical commutation relation. On the other hand, I used a trial wave function to evaluate $[V(\hat{x}),\hat{p}_x]$ directly. The point is that there's more than one way to skin a cat, whether it's dead or alive...

While Ehrenfest's theorem could be used to justify Eq. (5.19), a better argument comes from symmetry considerations. The Hamiltonian for a non-relativistic spinless particle can be motivated from spacetime symmetry rather than assumed by analogy with classical mechanics. Non-relativistic physics is invariant under the Galilean group (spatial translations, time translations, rotations, and boosts), and in quantum mechanics these symmetries are implemented by unitary operators. The generators of these transformations satisfy the *Galilean Lie algebra*, in which mass appears as a parameter in the commutator between boosts and momentum. Requiring consistency of this algebra forces the free-particle Hamiltonian to be quadratic in momentum, yielding $H = \hat{p}_x^2/2m$, up to an additive constant. With an external potential, the most general compatible form is $H = \hat{p}_x^2/2m + V(\hat{x})$. If you're interested in learning more, Ballentine provides an accessible derivation in [1] based on ideas originally developed by V. Bargmann in [2].

5.3.5 Definite-Energy States

Substituting Eq. (5.19) into the Schrödinger equation (postulate #5),

$$i\hbar\frac{\partial}{\partial t}|\psi(t)\rangle = \left(\frac{\hat{p}_x^2}{2m} + V(\hat{x})\right)|\psi(t)\rangle$$

We can turn this into a partial differential equation for the wave function $\psi(x,t)$ which is the projection of $|\psi\rangle$ in the position basis. The Schrödinger equation for $\psi(x,t)$ is

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x)\psi(x,t) \tag{5.23}$$

Historically, this was the first "Schrödinger equation" and people will often call it the time-dependent Schrödinger equation for obvious reasons. It is a linear, first-order in time, second-order in space partial differential equation (PDE).

In the next few chapters, we're going to solve for the time evolution of wave functions describing a single particle in various potentials. We found in our first introduction to quantum dynamics that it was convenient to express the state vector in the energy eigenbasis, where the Hamiltonian takes a diagonal form. Let E be the energy eigenvalue corresponding to an energy eigenfunction $\psi_E(x,t)$. The eigenvalue equation for the Hamiltonian is

$$H(x)\psi_E(x,t) = i\hbar\frac{\partial\psi_E(x,t)}{\partial t} = E\psi_E(x,t) \tag{5.24}$$

This first-order differential equation has the general solution

$$\psi_E(x,t) = \psi(x)e^{-iEt/\hbar} \tag{5.25}$$

where $\psi(x)$ is a normalized wave function of x only, so we'll refer to it as the "time-independent wave function." Eq. (5.25) describes a **definite-energy state**. Substituting the definite energy state into the time-dependent Schrödinger equation, we obtain the so-called time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x) \tag{5.26}$$

Comparing with Eq. (5.23), you'll see that our notation is getting sloppy, but it's very common so we might as well get used to it. It will be clear from context whether we're dealing with a time-dependent or time-independent wave function.

5.4 Momentum-Space Wave Functions

In position space, we describe the quantum state of a particle using a wave function $\psi(x)$ whose squared magnitude $|\psi(x)|^2$ gives the probability density for finding the particle near position x. It can often be useful to consider the *momentum representation*, where the wave function is written in terms of eigenvalues of the momentum operator. Given the position-space wave function $\psi(x)$, we can obtain the momentum-space wave function by a change of basis between the eigenstates of position and those of momentum.

5.4.1 Momentum Eigenfunctions

Let $\psi_{x'}(x)$ denote a position eigenstate wave function $\langle x|x'\rangle$ (eigenfunction for short). We've already seen that position eigenstates are in fact delta functions:

$$\hat{x}\psi_{x'}(x) = x'\psi_{x'}(x) = x' \langle x|\psi_{x'}\rangle = x'\delta(x'-x) \Rightarrow \psi_{x'}(x) = \delta(x'-x) \tag{5.27}$$

Now we search for position-space momentum eigenfunctions

$$\hat{p}_x\psi_{p_x}(x) = p_x\psi_{p_x}(x) \tag{5.28}$$

where

$$\psi_{p_x}(x) = \langle x|p_x\rangle \quad \text{and} \quad \hat{p}_x |p_x\rangle = p_x |p_x\rangle$$

Eq. (5.28) leads to a first-order differential equation for $\psi_{p_x}(x)$

$$-i\hbar\frac{\partial\psi_{p_x}(x)}{\partial x} = p_x\psi_{p_x}(x)$$

which has the general solution

$$\psi_{p_x}(x) = Ae^{ip_x x/\hbar} \tag{5.29}$$

where A is a constant to be determined. Neither the position eigenfunctions nor the momentum eigenfunctions are normalizable–they are not square integrable and the momentum eigenfunctions do not vanish at infinity. We can assume that momentum eigenstates form a complete basis, so for any state vector $|\psi\rangle$ corresponding to a physical wave function,

$$\langle p_x|\psi\rangle = \langle p_x| \left(\int_{-\infty}^{\infty} dp'_x \;\; |p'_x\rangle \langle p'_x|\right) |\psi\rangle$$

$$= \int_{-\infty}^{\infty} dp'_x \;\; \langle p_x|p'_x\rangle \langle p'_x|\psi\rangle$$

$$\psi(p_x) = \int_{-\infty}^{\infty} dp'_x \;\; \langle p_x|p'_x\rangle \psi(p'_x)$$

It follows that momentum eigenstates must be Dirac orthonormal in the sense that

$$\langle p_x|p'_x\rangle = \delta(p'_x - p_x) \tag{5.30}$$

This constraint allows us to determine the constant A in Eq. (5.29). Using completeness of the position eigenstates,

$$\langle p_x|p'_x\rangle = \int_{-\infty}^{\infty} dx \;\; \langle p_x|x\rangle \langle x|p'_x\rangle$$

$$= \int_{-\infty}^{\infty} dx\ \psi_{p_x}(x)\psi_{p'_x}(x)$$

$$= \int_{-\infty}^{\infty} dx\ |A|^2 e^{i(p_x - p'_x)x/\hbar}$$

At this point, we must make use of a mathematical identity from complex analysis:

$$\int_{-\infty}^{\infty} dx\ e^{ix(k'-k)} = 2\pi\delta(k'-k) \tag{5.31}$$

We'll call this the "integral representation of the Dirac delta function." Hence,

$$\langle p_x | p'_x \rangle = 2\pi\hbar\ |A|^2\ \delta(p'_x - p_x)$$

There is technically ambiguity in the phase of A, but nonetheless if we choose

$$A = \frac{1}{\sqrt{2\pi\hbar}}$$

then our momentum eigenstates will be Dirac-orthornormal. Finally, we have

$$\psi_{p_x}(x) = \langle x | p_x \rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ip_x x/\hbar} \tag{5.32}$$

Momentum eigenfunctions do not belong to the space of "physical wave functions" that we discussed earlier. They are not square-integrable, and hence they are non-normalizable. They do however form a complete basis for wave functions in the sense that any wave function may be written as a linear superposition of the momentum eigenfunctions. In the space of physical wave functions, the momentumm operator is self-adjoint and has a complete set of orthonormal eigenstates which are normalized in a distributional sense by the delta function.

5.4.2 Fourier Transform

Position and momentum eigenstates satisfy integral completeness

$$\int_{-\infty}^{\infty} dp_x\ \ |p_x\rangle \langle p_x| = \mathbb{1} \qquad \text{and} \qquad \int_{-\infty}^{\infty} dx\ \ |x\rangle \langle x| = \mathbb{1}$$

Given a quantum system in the state $|\psi\rangle$, we can determine the position-space wave function or the momentum-space wave function as

$$\psi(x) = \langle x|\psi\rangle \qquad \text{and} \qquad \tilde{\psi}(p_x) = \langle p_x|\psi\rangle \tag{5.33}$$

where a tilde is added over the momentum-space wave function for notational convenience. It will often be useful or necessary to go from one representation directly to the other. Given the

completeness of momentum states,

$$\psi(x) = \int_{-\infty}^{\infty} dp_x \, \langle x|p_x\rangle \, \langle p_x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp_x \; \tilde{\psi}(p_x) e^{ip_x x/\hbar} \tag{5.34}$$

and from completeness of position states,

$$\tilde{\psi}(p_x) = \int_{-\infty}^{\infty} dx \, \langle p_x|x\rangle \, \langle x|\psi\rangle = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx \; \psi(x) e^{-ip_x x/\hbar} \tag{5.35}$$

The latter equation is called the *Fourier transform* of the function $\psi(x)$, and we interpret this as a change of basis from "position space" to "momentum space." The first expression is the inverse Fourier transform in the sense that substituting $\tilde{\psi}(p_x)$ from the second expression should result in the identity operation:

$$\begin{aligned}
\psi(x) &= \int \frac{dp_x}{\sqrt{2\pi\hbar}} e^{-ip_x x/\hbar} \left\{ \int \frac{dx'}{\sqrt{2\pi\hbar}} e^{-ip_x x'/\hbar} \psi(x') \right\} \\
&= \int dx' \int \frac{dp_x}{2\pi\hbar} e^{ip_x(x-x')/\hbar} \psi(x') \\
&= \int dx' \delta(x'-x)\psi(x') \\
&= \psi(x)
\end{aligned}$$

For example, the Fourier transform of the momentum eigenfuction in position space should give the momentum eigenfunction in momentum space:

$$\tilde{\psi}_{p'_x}(p_x) = \int \frac{dx}{2\pi\hbar} e^{i(p'_x - p_x)x/\hbar} = \delta(p'_x - p_x)$$

5.4.3 Operators in Momentum Space

Nearly all observables we care about will be expressible in terms of position and momentum. We should know how those operators work in momentum space. For instance, we cannot apply $\hat{p}_x = -i\hbar\partial/\partial x$ to the wave function $\tilde{\psi}(p_x)$ since there simply is no x dependence.

Momentum Operator in Momentum Space

Dirac notation and completeness are especially helpful here.

$$\hat{p}_x \tilde{\psi}(p_x) = \langle p_x| \, \hat{p}_x \, |\psi\rangle = \int_{-\infty}^{\infty} dx \;\; \langle p_x|x\rangle \, \langle x| \, \hat{p}_x \, |\psi\rangle$$

This equation is basically the Fourier transform of the position-space wave function $\hat{p}_x\psi(x)$. Using the position-space momentum operator, we apply integration by parts and eliminate the boundary

term since $\psi(x) \to 0$ at $x \to \pm\infty$:

$$\begin{aligned}
\hat{p}_x\tilde{\psi}(p_x) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx \left(-i\hbar\frac{\partial\psi(x)}{\partial x}\right) e^{-ip_x x/\hbar} \\
&= \underbrace{\frac{-i\hbar}{\sqrt{2\pi\hbar}}\psi(x)e^{-ip_x x/\hbar}\Big|_{-\infty}^{\infty}}_{\text{boundary term} \to 0} - (-i\hbar)(-ip_x/\hbar)\underbrace{\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx\ \psi(x)e^{-ip_x x/\hbar}}_{\tilde{\psi}(p_x)} \\
&= p_x\tilde{\psi}(p_x)
\end{aligned}$$

Hence, in the momentum representation, the momentum operator returns the value of momentum, just like the position operator in position space.

Position Operator in Momentum Space

Now consider the position operator in position space. Using Dirac notation,

$$\hat{x}\tilde{\psi}(p_x) = \langle p_x|\,\hat{x}\,|\psi\rangle = \int_{-\infty}^{\infty} dx\ \langle p_x|x\rangle\,\langle x|\,\hat{x}\,|\psi\rangle$$

which says the position operator in momentum space is the Fourier transform of the position operator in position space. Use the position operator in position space and observe the integrand is a total derivative with respect to p_x:

$$\begin{aligned}
\hat{x}\tilde{\psi}(p_x) &= \frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx\ x\psi(x)e^{-ip_x x/\hbar} \\
&= i\hbar\frac{\partial}{\partial p_x}\left\{\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty} dx\ \psi(x)e^{-ip_x x/\hbar}\right\} \\
&= i\hbar\frac{\partial\tilde{\psi}(p_x)}{\partial p_x}
\end{aligned}$$

The position operator in momentum space acts almost like the momentum operator in position space, but with an overall minus sign.

Example 5.2: Momentum-Space Wave Function

At a particular moment in time, a localized particle is described by the position-space wave function

$$\psi(x) = \begin{cases} A, & -a < x < a \\ 0, & \text{otherwise} \end{cases}$$

where A and a are positive real constants. Find the momentum-space wave function.

Solution: First we need to normalize the wave function:

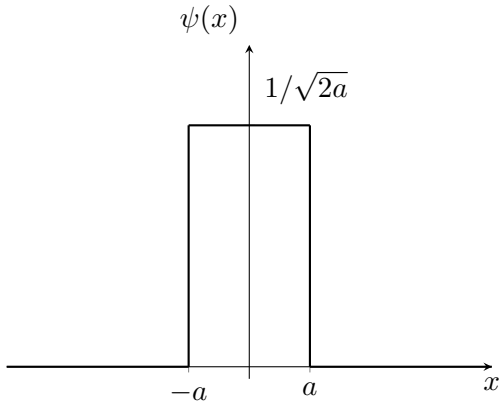
$$1 = \int_{-\infty}^{\infty} dx\ |\psi(x)|^2 = \int_{-a}^{a} dx\ A^2 = (2a)A^2 \Rightarrow A = \frac{1}{\sqrt{2a}}$$

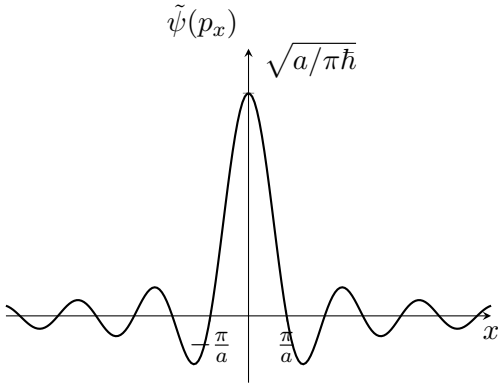
Now we can calculate the momentum-space wave function:

$$\tilde{\psi}(p_x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dx\ \psi(x) e^{-ip_x x/\hbar} = \frac{1}{\sqrt{4\pi\hbar a}} \int_{-a}^{a} dx\ e^{-ip_x x/\hbar}$$

$$\tilde{\psi}(p_x) = \frac{1}{\sqrt{4\pi\hbar a}} \frac{2\hbar \sin(p_x a/\hbar)}{p_x} = \sqrt{\frac{\hbar}{\pi a}} \frac{\sin(p_x a/\hbar)}{p_x}$$

The position-space wave funtion and it's Fourier transform are plotted below.





5.4.4 Position-Momentum Uncertainty

According to the Heisenberg uncertainty relation, it follows from the canonical commutation relation for position and momentum that

$$\Delta x \Delta p_x \geq \frac{\hbar}{2} \tag{5.36}$$

It's important to understand what this relation is saying (and what it's not saying). In standard (non-relativistic) quantum mechanics, there is no principle that forbids an arbitrarily precise measurement of position or momentum. What is "forbidden" is preparing a state with both arbitrarily precise position and arbitrarily precise momentum. Recall that the uncertainty relation is a statement about the statistical spreads of outcomes in a given quantum state. It is not a statement that you cannot measure $\hat{x}$ or $\hat{p}_x$ precisely. If the system is in an idealized position eigenstate, $\psi(x) = \delta(x - x_0)$, then $\Delta x = 0$ and $\Delta p_x = \infty$, which is completely consistent with the uncertainty relation. So mathematically, quantum mechanics absolutely allows an arbitrarily sharp position measurement at the price of an arbitrarily broad momentum distribution (and vice versa).

There are, of course, a few important caveats. The first is that position eigenstates are not physical states as defined earlier, so while you can approach arbitrarily small Δx, an exact delta function entails an infinite spread in momentum, suggesting a practical limitation on reliable local-

ization. In relativistic quantum field theory, localizing a particle to within its so-called 'Compton wavelength' requires energies high enough to create particle-antiparticle pairs. If you try to localize too sharply, you no longer have a single particle.

5.5 Quantum Dynamics of a Free Particle

Newton's first law states that a free particle–one that is absent of any applied forces–will travel in a straight line at constant speed. The probabilistic nature of quantum dynamics immediately calls for revision; if we can't know for certain where a free particle will move in a short time interval, the velocity is simply undefined.

Consider a particle moving in a constant potential, and for the sake of simplicity, let $V = 0$. The Hamiltonian is then

$$H = \frac{\hat{p}_x^2}{2m}$$

and since $\hat{p}_x$ is Hermitian, it follows that the energy of a free particle must be non-negative:

$$\langle\psi| H |\psi\rangle = \frac{1}{2m} \langle\psi| \hat{p}_x^2 |\psi\rangle = \frac{1}{2m} \|\hat{p}_x\psi\|^2 \geq 0$$

The time-independent Schrödinger equation for the definite-energy wave function $\psi(x)$ is

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi = -k^2\psi \tag{5.37}$$

where $k = \pm\sqrt{2mE}/\hbar$. Clearly $\psi(x)$ is an eigenstate of $\hat{p}_x$ since $[H, \hat{p}_x] = 0$. Hence,

$$\psi(x) = \frac{1}{\sqrt{2\pi\hbar}} e^{ip_x x/\hbar} \tag{5.38}$$

where $p_x = \hbar k = \pm\sqrt{2mE}$. We know these functions are not square-integrable and don't represent physical wave functions on their own, but they can be combined to produce physical wave functions. In fact, the most general solution to the time-dependent Schrödinger equation for a free particle is

$$\psi(x,t) = \int_{-\infty}^{\infty} \frac{dp_x}{\sqrt{2\pi\hbar}} \tilde{\psi}(p_x) e^{i(kx-\omega(k)t)} \tag{5.39}$$

where $\omega(k) = E/\hbar = \hbar k^2/2m$. This is a superposition of complex waves, each of the form $e^{i(kx-\omega t)}$, weighted by the amplitude $\tilde{\psi}(p_x)$. Each wave has a well-defined **wavevector** k and **angular frequency** ω, which are related to the momentum and energy of the particle via de Broglie's relations:

$$p_x = \hbar k \quad \text{and} \quad E = \hbar\omega \tag{5.40}$$

There are two ways to justify the general solution given by Eq. (5.39). We could substitute directly into the Schrödinger equation, or we can build up the superposition using orthonormality

and completeness. The momentum eigenstates are also the eigenstates of the Hamiltonian. Hence,

$$H\,|p_x\rangle = \frac{\hbar^2 k^2}{2m}\,|p_x\rangle \quad \text{where } \hbar k = \sqrt{2mE}$$

Since the momentum eigenstates are complete, we can write any initial state vector as

$$|\psi(0)\rangle = \int_{-\infty}^{\infty} dp_x\ \ |p_x\rangle\,\langle p_x|\psi(0)\rangle = \int_{-\infty}^{\infty} dp_x\ \tilde{\psi}(p_x)\,|p_x\rangle$$

The time-evolved state vector is

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle = e^{-iHt/\hbar}\,|\psi(0)\rangle = \int_{-\infty}^{\infty} dp_x\ \tilde{\psi}(p_x)e^{-i\omega(k)t}\,|p_x\rangle$$

where $\omega(k)$ is defined above. The time-dependent wave function in the position basis is

$$\psi(x,t) = \langle x|\psi(t)\rangle = \int_{-\infty}^{\infty} dp_x\ \tilde{\psi}(p_x)e^{-i\omega(k)t}\,\langle x|p_x\rangle = \int_{-\infty}^{\infty} \frac{dp_x}{\sqrt{2\pi\hbar}}\tilde{\psi}(p_x)e^{i(kx-\omega(k)t)}$$

Here's another argument in terms of wave functions–in case that appeals to you. Since the Hamiltonian is diagonal in the momentum basis, it is useful to consider its momentum representation. For a momentum-space wave function $\tilde{\psi}(p_x,t)$,

$$H\tilde{\psi}(p_x,t) = \frac{\hat{p}_x^2}{2m}\tilde{\psi}(p_x,t) = \frac{p_x^2}{2m}\tilde{\psi}(p_x,t)$$

The time-dependent Schrödinger equation for $\tilde{\psi}(p_x)$ is

$$i\hbar\frac{\partial\tilde{\psi}(p_x,t)}{\partial t} = \frac{p_x^2}{2m}\tilde{\psi}(p_x,t)$$

and the solution is

$$\tilde{\psi}(p_x,t) = \tilde{\psi}(p_x,0)e^{-i\omega t}$$

where $\omega = p_x^2/2m\hbar$. Hence, the momentum-space probability distribution remains constant:

$$|\tilde{\psi}(p_x,t)|^2 = |\tilde{\psi}(p_x,0)|^2$$

The position-space wave function is then given by the inverse Fourier transform of $\tilde{\psi}(p_x)$.

5.5.1 Phase and Group Velocity

Consider a single complex wave $e^{i(kx-\omega t)} = e^{i\phi}$. The **phase** $\phi = kx - \omega t$ is constant along points were $x/t = \omega/k$. Thus, we define the **phase velocity** as

$$v_{\text{phase}} = \frac{\omega}{k} \tag{5.41}$$

The phase velocity of a non-relativistic particle with definite energy and momentum is

$$v_{\text{phase}} = \frac{\omega}{k} = \frac{E}{p_x} = \frac{p_x}{2m} \tag{5.42}$$

This is not the classical particle velocity. Instead, it's the speed at which the wave crests move, but not necessarily the speed at which the particle or its probability density moves.

A more physically meaningful quantity is the group velocity, which tells us how the envelope of a wave packet (and thus the probability distribution) propagates. This is particularly relevant when $\tilde{\psi}(p_x)$ is peaked sharply around some value of $p_x = \hbar k$. As we'll explain soon, the **group velocity** is defined as

$$v_{\text{group}} = \frac{d\omega}{dk} \tag{5.43}$$

The group velocity of a non-relativistic particle with definite energy and momentum is

$$v_{\text{group}} = \frac{d\omega}{dk} = \frac{dE}{dp_x} = \frac{p_x}{m} \tag{5.44}$$

This is exactly the classical velocity of a particle with momentum p_x and mass m. We say it's the rate at which the wave packet moves. To understand this result, let's assume the wave packet is nearly sinusoidal in the sense that $\tilde{\psi}(p_x)$ is sharply peaked around some value $p_{x0} = \hbar k_0$. The angular frequency depends on the wavenumber, and we call this *dispersion*. The dispersion relation is $\omega(k)$. When the wave packet is nearly sinusoidal, the angular frequency will vary slightly from $\omega_0 = \omega(k_0)$, and we can Taylor expand to first order since $\Delta k = k - k_0$ is small. Hence,

$$\omega(k) \approx \omega(k_0) + \left.\frac{d\omega}{dk}\right|_{k_0} (k - k_0) + \mathcal{O}(\Delta k^2) = \omega_0 + v_g \Delta k + \mathcal{O}(\Delta k^2)$$

Changing variables to Δk, the wave packet is

$$\begin{aligned}\psi(x,t) &\approx e^{i(k_0 x - \omega_0 t)} \int \frac{d(\Delta p_x)}{\sqrt{2\pi\hbar}} \tilde{\psi}(p_{x0} + \Delta p_x) e^{i\Delta p_x (x - v_g t)/\hbar} \\ &= e^{i(k_0 x - \omega_0 t)}\ f(x - v_g t)\end{aligned}$$

where the first term is a complex wave with phase velocity ω_0/k and the second term is a function with a fixed shape moving with the group velocity v_g. Generally speaking, the dispersion relation describes the spreading of a wave packet over time, since different k components will move at different speeds. We can analyze this effect by keeping the second order term in the series expansion for the angular frequency $\omega(k)$:

$$\begin{aligned}\omega(k) &\approx \omega(k_0) + \left.\frac{d\omega}{dk}\right|_{k_0} (k - k_0) + \frac{1}{2}\left.\frac{d^2\omega}{dk^2}\right|_{k_0} (k - k_0)^2 + \mathcal{O}(\Delta k^3) \\ &= \omega_0 + v_g \Delta k + \alpha \Delta k^2\end{aligned}$$

where $\alpha = \frac{1}{2}\frac{d^2\omega}{dk^2}\big|_{k_0}$. The resulting wave packet is

$$\psi(x,t) \approx e^{i(k_0 x - \omega_0 t)} \int \frac{d(\Delta p_x)}{\sqrt{2\pi\hbar}} \tilde{\psi}(p_{x0} + \Delta p_x) e^{i\Delta p_x [x - (v_g + \alpha\Delta k)t]/\hbar}$$

Since different k components move at different group speeds, we can no longer think of the integral as a function describing a fixed shape moving at speed v_g. Instead, the wave packet spreads out over time. By dimensional analysis, the spreading is proportional to

$$\Delta x \propto \alpha \Delta k t = \frac{1}{2}\frac{d^2\omega}{dk^2}\bigg|_{k_0} \Delta k t = \frac{\Delta p_x t}{m}$$

The spreading increases over time, is greater for k components further from k_0, and it depends on the mass–the rate of spreading is evidently slower for an object with larger mass.

The "quantum revision" to Newton's first law might be something like: in the absence of external forces, the momentum-space distribution of a quantum particle remains constant, while its position-space distribution spreads over time. In other words, a free particle moves with constant *average momentum*, but its position becomes increasingly uncertain as its wave function spreads. As soon as a position measurement is made on the particle, it will spontaneously "localize," but we'll lose knowledge of the momentum.

5.5.2 Free Gaussian Wave Packet

Consider a free particle with a normalized initial wave function given by

$$\psi(x,0) = \left(\frac{1}{2\pi\sigma_x^2}\right)^{1/4} \exp\left(-\frac{x^2}{4\sigma_x^2}\right) \tag{5.45}$$

where σ_x is a positive constant. The probability distribution $|\psi(x)|^2$ is called a Gaussian or normal distribution. In order to verify that the wave function is normalized, we need to use the following integral identity, which I'll state without proof but you can find many online:

$$\frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} dx\ e^{-x^2} = 1 \tag{5.46}$$

Normalization can be demonstrated by applying a change of variables where $u = x/\sqrt{2}\sigma_x$.

$$\begin{aligned}
\int_{-\infty}^{\infty} dx\ |\psi(x)|^2 &= \left(\frac{1}{2\pi\sigma_x^2}\right)^{1/2} \int_{-\infty}^{\infty} dx\ \exp\left(-\frac{x^2}{2\sigma_x^2}\right) \\
&= \left(\frac{1}{2\pi\sigma_x^2}\right)^{1/2} \sqrt{2}\sigma_x \int_{-\infty}^{\infty} du\ e^{-u^2} \\
&= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} du\ e^{-u^2}
\end{aligned}$$

$$= 1$$

Uncertainty Principle

According to the uncertainty principle, the product of uncertainties in position and momentum for a quantum particle in any state must satisfy the inequality

$$\Delta p_x \Delta x \geq \frac{1}{2}|\langle[\hat{x}, \hat{p}_x]\rangle| = \frac{\hbar}{2} \tag{5.47}$$

As it turns out, the Gaussian wave function is a *minimum uncertainty state*. The expectation value of position in the state $\psi(x)$ is

$$\langle\hat{x}\rangle = \int_{-\infty}^{\infty} dx\ x|\psi(x)|^2 = 0$$

since x is an odd function about $x = 0$ and the Gaussian distribution is even. To compute the uncertainty in position, we need to compute the expectation value of $\hat{x}^2$, which is

$$\langle\hat{x}^2\rangle = \int_{-\infty}^{\infty} dx\ x^2|\psi(x)|^2 = \sigma_x^2 \tag{5.48}$$

and you can verify this with integration by parts or using a fun little trick. First apply the same change of variables as before with $u = x/\sqrt{2}\sigma_x$:

$$\int_{-\infty}^{\infty} dx\ x^2|\psi(x)|^2 = \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} du\ u^2 e^{-u^2}$$

Observe that the integral may be written in a strange but strategic form

$$\int_{-\infty}^{\infty} du\ u^2 e^{-u^2} = \int_{-\infty}^{\infty} du\ \left(-\frac{\partial}{\partial z} e^{-u^2 z}\right)_{z=1}$$

In this form, we can make use of Eq. (5.46) to compute the integral above:

$$\begin{aligned}
\int_{-\infty}^{\infty} du\ u^2 e^{-u^2} &= \int_{-\infty}^{\infty} du\ \left(-\frac{\partial}{\partial z} e^{-u^2 z}\right)_{z=1} \\
&= -\frac{d}{dz}\left[\int du\ e^{-u^2 z}\right]_{z=1} \\
&= -\frac{d}{dz}\left(\sqrt{\frac{\pi}{z}}\right)_{z=1} \\
&= \frac{\sqrt{\pi}}{2}
\end{aligned}$$

Now that we've verified the expectation value of $\hat{x}^2$, it follows that the initial uncertainty in the position is

$$\Delta x = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2} = \sigma_x \tag{5.49}$$

In order to compute the uncertainty in momentum, it will be easier to compute the momentum-space wave function $\tilde{\psi}(p_x)$. As an added benefit, this will allow us to solve for the time-evolved wave function $\psi(x,t)$ later on. Applying the Fourier transform,

$$\tilde{\psi}(p_x) = \left(\frac{1}{2\pi(\hbar/2\sigma_x)^2} \right)^{1/4} \exp\left(-\frac{p_x^2}{4(\hbar/2\sigma_x)^2} \right) \tag{5.50}$$

After some calculations, we learn that the Fourier transform of a Gaussian function is itself a Gaussian function. In this form, we can immediately identify the expectation values of $\hat{p}_x$ and $\hat{p}_x^2$ by analogy with the position-space wave function. The expectation value of $\hat{p}_x$ is zero, and the expectation value of $\hat{p}_x^2$ is $(\hbar/2\sigma_x)^2$. Hence, the uncertainty in momentum is

$$\Delta p_x = \frac{\hbar}{2\sigma_x} \tag{5.51}$$

and the product of uncertainties in position and momentum is

$$\Delta p_x \Delta x = \frac{\hbar}{2} \tag{5.52}$$

Note that the uncertainty in momentum is inversely related to the uncertainty in the position. An initial wave function with a large spread in position will have a narrow spread in momentum. As time goes on, we expect the spread in position to grow while the spread in momentum remains constant (the momentum-space distribution is stationary).

Time-Dependent Wave Function

Given the initial wave function in Eq. (5.45), the time-dependent wave function for the free particle is given by the general solution

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} dp_x \; \tilde{\psi}(p_x) e^{i(kx - \omega t)}$$

where $k = p_x/\hbar$ and $\omega = E/\hbar = p_x^2/2m\hbar$. Inserting $\tilde{\psi}(p_x)$, we have

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \left(\frac{1}{2\pi(\hbar/2\sigma_x)^2} \right)^{1/4} \int_{-\infty}^{\infty} dp_x \; \exp\left(-\frac{\sigma_x^2 p_x^2}{\hbar^2} + \frac{i p_x x}{\hbar} - \frac{i p_x^2 t}{2m\hbar} \right)$$

The integral may look intimidating, but we can easily extend the identity in Eq. (5.46) to handle any quadratic function by simply completing the square in the argument of the exponential. This

leads to a more generally applicable integral identity:

$$\int_{-\infty}^{\infty} dx\ e^{-(ax^2+bx+c)} = \sqrt{\frac{\pi}{a}} e^{(b^2/4a)-c} \tag{5.53}$$

For the situation at hand, the argument of the exponential can be written as

$$-\frac{\sigma_x^2 p_x^2}{\hbar^2} + \frac{i p_x x}{\hbar} - \frac{i p_x^2 t}{2m\hbar} = -\sigma_x^2 \left(1 + \frac{i\hbar t}{2m\sigma_x^2}\right) \frac{p_x^2}{\hbar^2} + ix\frac{p_x}{\hbar}$$

Hence, the time-dependent wave function is

$$\psi(x,t) = \left(\frac{1}{2\pi\sigma_x^2(1+it/\tau)}\right)^{1/4} \exp\left(-\frac{x^2}{4\sigma_x^2(1+it/\tau)}\right) \tag{5.54}$$

where we've defined a time scale τ over which the wave function changes significantly:

$$\tau = \frac{2m\sigma_x^2}{\hbar} \tag{5.55}$$

We can more easily obtain a physical interpretation of the wave function by computing the probability distribution:

$$|\psi(x,t)|^2 = \frac{1}{\sqrt{2\pi\sigma_x^2(1+t^2/\tau^2)}} \exp\left(-\frac{x^2}{2\sigma_x^2(1+t^2/\tau^2)}\right) \tag{5.56}$$

The probability distribution spreads out over time in a way that can be quantified by the uncertainty in position:

$$\Delta x(t) = \sigma_x \sqrt{1 + t^2/\tau^2} \tag{5.57}$$

Indeed, τ gives us the time scale over which the probability distribution changes significantly. Notice the spreading will happen faster for less massive particles that are highly localized in their initial state. After a time interval much larger than τ, i.e. $t \gg \tau$, the uncertainty in $\hat{x}$ is approximately

$$\Delta x_{t\gg\tau} \approx \frac{\sigma_x t}{\tau} = \frac{\hbar t}{2m\sigma_x} = \frac{\Delta p_x t}{m}$$

which matches the result of our dimensional argument in Section 5.5.1.

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[2] V. Bargmann. "On Unitary Ray Representations of Continuous Groups". In: *Annals of Mathematics* 59.1 (1954), pp. 1–46. ISSN: 0003486X, 19398980. URL: `http://www.jstor.org/stable/1969831`.

Chapter 6

Bound States and Scattering

6.1 Energy Diagrams & Qualitative Solutions

6.1.1 Energy Diagrams

In the previous chapter, we introduced the wave function and solved for the dynamics of a free particle–one for which the potential energy was a constant value at all positions. We will now consider potential energy functions that vary with position as in Figure 6.1.

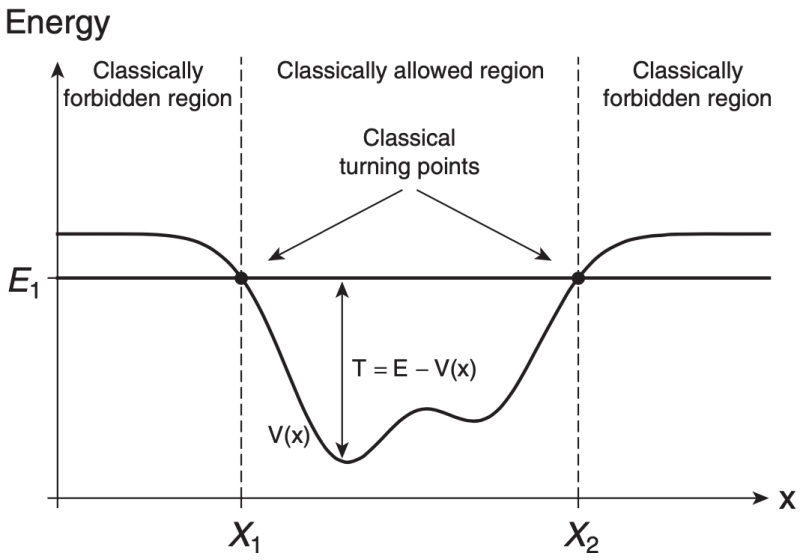


Figure 6.1: A generic potential energy well (McIntyre, 2022).

For a classical particle, the kinetic energy cannot be negative, so a classical particle with the energy E_1 chosen in Figure 6.1 has its motion constrained to the region between x_1 and x_2. These extreme points of the classical motion are called **classical turning points** and the region within the turning points is called the **classically allowed region**, while the regions beyond are called **classically forbidden regions**. Particles that have their motion constrained by the potential

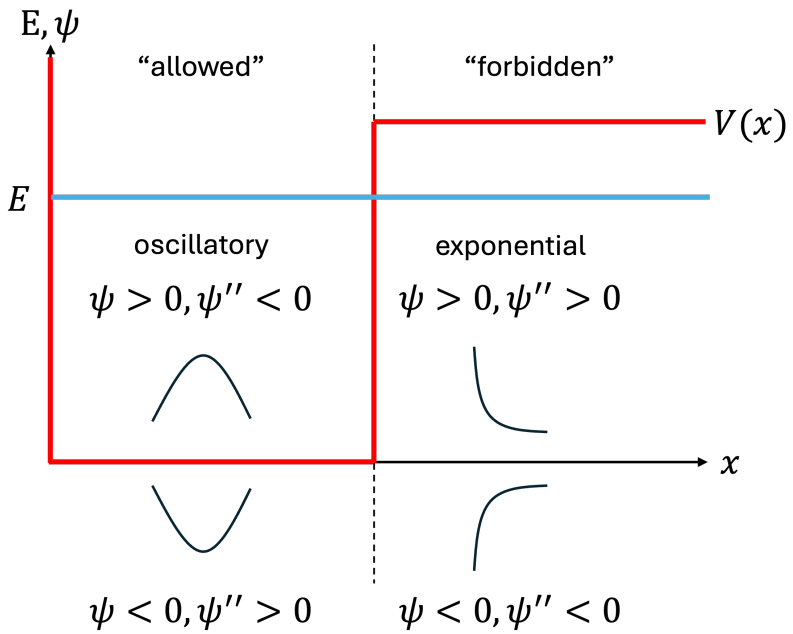


Figure 6.2: Curvature of the energy eigenstate wave functions in the allowed and forbidden regions.

well are said to be in **bound states**. Particles with energies above the top of the potential well do not have their motion constrained and so are in **unbound states**. Note that the extent of the classically forbidden and allowed regions depends on the specific value of the energy, E_1, for a particular bound state.

6.1.2 Qualitative Solutions

Recall that a state with definite energy satisfies the energy eigenvalue equation

$$H\psi = E\psi$$

where H is the Hamiltonian, E is the energy eigenvalue, and ψ is the position-dependent part of a wave function describing a system with definite energy. The full time-dependent definite-energy wave function is given by

$$\psi_E(x,t) = \psi(x)e^{-iEt/\hbar}$$

just as we found in the last chapter. For a bound system, the first step is to solve the energy eigenvalue equation. The general solution is then constructed as a superposition of the corresponding eigenstates.

The energy eigenvalue equation can be written in the position basis as

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

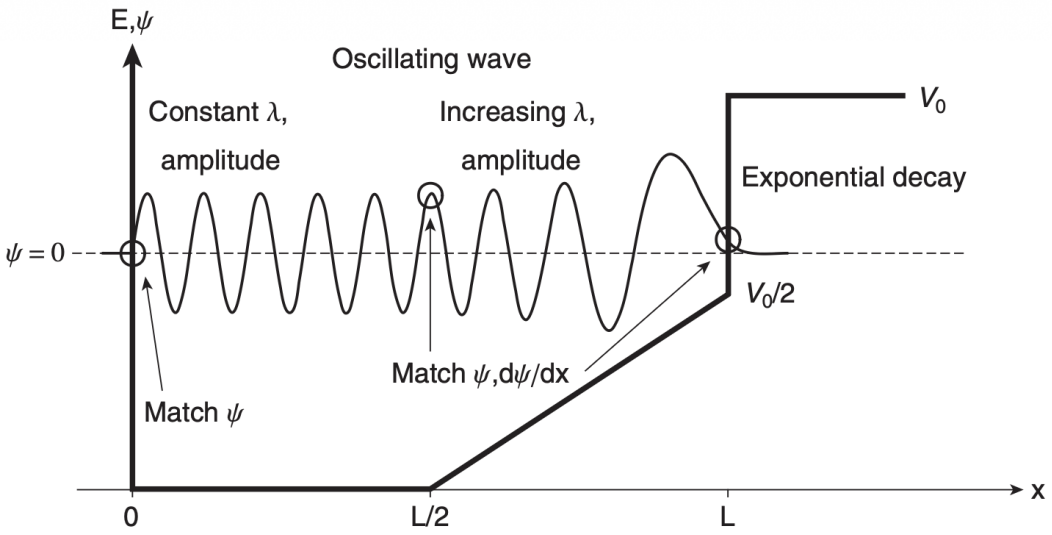


Figure 6.3: Drawing approximate energy eigenstate solutions. (McIntyre, 2022)

or

$$\psi''(x) = \frac{2m(V(x) - E)}{\hbar^2}\psi(x)$$

For smoothly varying potentials, we can make a few general statements about the behavior of solutions based on the energy eigenvalue equation:

1(a) In the classically allowed region, $E > V$, solutions are oscillatory

1(b) with wavelength proportional to $1/\sqrt{E - V(x)}$,

1(c) and the amplitude is large for large wavelengths and small for small wavelengths.

2(a) In the classically forbidden region, $E < V$, and solutions are real exponential functions

2(b) with decay/growth length proportional to $1/\sqrt{V(x) - E}$.

3(a) The wave function must be continuous everywhere.

3(b) The first derivative of the wave function is continuous everywhere V is finite.

Rules 3(a) and 3(b) are called **matching conditions**, and they are necessary for the existence of the second derivative at all points. We'll explain rule 1(c) in more detail later (or in class). Using these rules of thumb, we can get a very good idea of the wave function before we tackle the differential equation that gives us the exact solution.

6.2 Infinite Square Well

Consider a "super ball" bouncing between two perfectly elastic walls. We call this system a **particle in a box**. We observe three important characteristics of this classical system: (1) the ball flies

freely between the walls, (2) the ball is reflected perfectly at each bounce, and (3) the ball remains in the box no matter how large its energy. These three observations are consistent with (1) zero force on the ball when it is between the walls, (2) infinite force on the ball at the walls, and (3) infinite potential energy outside the box.

The potential energy function for the classical super ball is a piecewise function

$$V(x) = \begin{cases} 0, & 0 < x < a \\ \infty, & \text{otherwise} \end{cases} \tag{6.1}$$

and we posit that the quantum mechanical potential energy operator is simply $V(\hat{x})$ where x is "promoted" to an operator. We're interested in the definite-energy states of this system, and for a particle with finite energy E, $\psi(x) = 0$ outside the box, where $\psi(x)$ is the time-independent portion of the definite-energy state wave function. Inside the box,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x)$$

which is the same differential equation we encountered for the free particle. Define

$$k^2 = 2mE/\hbar^2 \qquad \text{(wave vector squared)}$$

and with foresight, let's write the general solution in terms of real functions

$$\psi(x) = A\sin(kx) + B\cos(kx) \tag{6.2}$$

Since $\psi(x)$ must vanish at the two boundaries, we have $B = 0$ and

$$\sin(ka) = 0 \tag{6.3}$$

This is a transcendental equation that places limitations on the allowed values of the wave vector k. The wave vectors that satisfy this equation are

$$k_n = \frac{n\pi}{a}, \quad \text{where } n = 1, 2, \ldots \tag{6.4}$$

where the negative n values can be absorbed into the constant A since $\sin(-\theta) = -\sin(\theta)$ and $n = 0$ is excluded because $\psi(x) = 0$ everywhere is not physical (normalizable). Only discrete wave vectors are allowed, so this is called a **quantization condition**. The index n is the **quantum number**, which we use to label the quantized states and energies. The wave vector quantization condition leads directly to energy quantization:

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2\pi^2\hbar^2}{2ma^2}, \quad \text{where } n = 1, 2, \ldots \tag{6.5}$$

Any time the energy of this system is measured, the only possible values are given by Eq. (6.5), and the wave function will collapse to the energy eigenstate corresponding to whatever value of energy is actually obtained in the measurement.

The normalized energy eigenstates, labeled by the quantum number n, are given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad \text{where } n = 1, 2, \ldots \tag{6.6}$$

Each solution looks just like the standing waves on a string of length a. We call the solution with lowest energy, ψ_1, the ground state. All solutions with higher energy are excited states. All together, the functions $\psi_n(x)$ constitute a countably infinite set of eigenfunctions.

Recall our discussion of quantum dynamics in chapter #3. Even though we assumed the system was finite-dimensional, the same analysis applies to discrete systems which may be infinite dimensional. Due to the quantization of energy eigenvalues, the Hamiltonian for a particle in a 1D box admits a spectral decomposition

$$H = \sum_{n=1}^{\infty} E_n \left|E_n\right\rangle \left\langle E_n\right|$$

where the upper bound now goes to infinity, E_n are the energy eigenvalues, and $|E_n\rangle$ are the energy eigenstates in Dirac notation. Given an initial state vector

$$\left|\psi(0)\right\rangle = \sum_n c_n \left|E_n\right\rangle,$$

the most general time-dependent solution to the Schrödinger equation is

$$\left|\psi(t)\right\rangle = \sum_n c_n e^{-iE_n t/\hbar} \left|E_n\right\rangle$$

In some sense, the energy eigenstates form the "preferred basis" in which to expand a general quantum state vector, with the time evolution determined by phase factors dependent on the energy of each component state.

We can now translate everything into the position basis. The energy eigenfunctions are

$$\psi_n(x) = \langle x|E_n\rangle$$

and the initial wave function is given by

$$\psi(x,0) = \langle x|\psi(0)\rangle = \sum_n c_n \psi_n(x)$$

We can solve for the coefficients c_n using orthonormality of the energy eigenstates and the com-

pleteness relation for the position eigenstates:

$$c_n = \langle E_n|\psi(0)\rangle = \langle E_n| \int_{-\infty}^{\infty} dx\, |x\rangle\, \langle x|\psi(0)\rangle = \int_{-\infty}^{\infty} dx\; \psi_n^*(x)\psi(x,0)$$

Physically, the coefficients are related to the probabilities for measuring the system to have one of the energy eigenvalues. For instance, the probability of measuring E_n is

$$\mathcal{P}(E_n) = |c_n|^2$$

and the expectation value of the Hamiltonian can be calculated as

$$\langle H\rangle = \sum_n |c_n|^2 E_n$$

Once we have the wave function expansion coefficients in the energy basis, we can predict the future time evolution of the system. Then we can calculate any physical quantities we need to, such as probabilities and expectation values.

Example 6.1: Infinite Square Well

A particle in an infinite square well has the initial wave function

$$\psi(x,0) = Ax(a-x), \qquad (0 \le x \le a)$$

for some constant A. Outside the well, of course, $\psi = 0$. Find $\psi(x,t)$.

Solution: First we need to determine a suitable value for A that ensures ψ is a normalized wave function. Normalizing the wave function, we find

$$1 = \int_0^a dx\; |\psi(x,0)|^2 = |A|^2 \int_0^a dx\; x^2(a-x)^2 = |A|^2 \frac{a^5}{30} \to A = \sqrt{\frac{30}{a^5}}$$

The nth coefficient in the expansion is

$$\begin{aligned} c_n &= \sqrt{\frac{2}{a}} \int_0^a dx\; \sin\left(\frac{n\pi x}{a}\right) \sqrt{\frac{30}{a^5}} x(a-x) \\ &= \frac{2\sqrt{15}}{a^3}\left[a\int_0^a dx\; x \sin\left(\frac{n\pi x}{a}\right) - \int_0^a dx\; x^2 \sin\left(\frac{n\pi x}{a}\right)\right] \\ &= \begin{cases} 0, & n \text{ even} \\ 8\sqrt{15}/(n\pi)^3, & n \text{ odd} \end{cases} \end{aligned}$$

Thus,

$$\psi(x,t) = \sqrt{\frac{30}{a^5}}\left(\frac{2}{\pi}\right)^3 \sum_{n=1,3,5,\ldots} \frac{1}{n^3}\sin\left(\frac{n\pi x}{a}\right)\exp\left(-\frac{in^2\pi^2\hbar t}{2ma^2}\right) \tag{6.7}$$

As an interesting side-quest, let's consider the sum of the absolute squares of the coefficients. Since the initial state is normalized, it must be the case that

$$\sum_{n=1,3,5,\ldots} |c_n|^2 = 1$$

Hence,

$$\frac{960}{\pi^6}\left(\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots\right) = 1$$

and indeed, one can show from purely a mathematical argument that

$$\frac{1}{1^6} + \frac{1}{3^6} + \frac{1}{5^6} + \cdots = \frac{\pi^6}{960}$$

The expectation value of the Hamiltonian is

$$\langle H\rangle = \sum_n E_n|c_n|^2 = \sum_n \frac{n^2\pi^2\hbar^2}{2ma^2}\left(\frac{8\sqrt{15}}{n^3\pi^3}\right)^2 = \frac{480\hbar^2}{\pi^4 ma^2}\sum_{n=1,3,5,\ldots}\frac{1}{n^4} = \frac{5\hbar^2}{ma^2}$$

since

$$\sum_{n=1,3,5,\ldots}\frac{1}{n^4} = \frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \cdots = \frac{\pi^4}{96}$$

The value of $\langle H\rangle$ is very close to E_1, as one might expect given the similarity of the initial state to that of the ground state.

6.3 Finite Square Well

Consider the finite square well potential energy function given by

$$V(x) = \begin{cases} 0, & x < -a, \\ -V_0, & -a < x < a, \\ 0, & x > a \end{cases} \tag{6.8}$$

For now we will look for bound state solutions; we'll save the unbound states for the next section. In this case, bound state solutions correspond to a particle with energy $-V_0 < E < 0$, the maximum of the potential energy. Since there is a discontinuity in $V(x)$ at $x = \pm a$, we split the wave function into three regions and enforce the following boundary conditions:

1) The wave function $\psi(x)$ is everywhere continuous.

2) The first derivative $\frac{d}{dx}\psi(x)$ is continuous except where $V = \infty$.

The energy eigenvalue equation for the region $x < -a$ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi \qquad (E < 0) \tag{6.9}$$

and the general solution to this differential equation can be written as

$$\psi(x) = Ae^{-qx} + Be^{qx}, \quad \text{where } q = \frac{\sqrt{2m|E|}}{\hbar} > 0 \tag{6.10}$$

which corresponds to a superposition of exponential growth and decay. The first term grows without bound as $x \to -\infty$, so we must choose $A = 0$ to have a physical wave function.

In the region $-a < x < a$, the energy eigenvalue equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - V_0\psi = E\psi \qquad (E < 0)$$

or

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad \text{where } k = \frac{\sqrt{2m(V_0 - |E|)}}{\hbar} > 0$$

The general solution is oscillatory, and it can be written in a form that makes it easier to implement the boundary conditions:

$$\psi(x) = C\sin(kx) + D\cos(kx) \tag{6.11}$$

The last region, $x > a$, is identical to the first region where $x < -a$, expect we want the wave function to vanish when $x \to \infty$. The energy eigenstate must be constructed by connecting solutions in the three regions. We write the general solution as

$$\psi(x) = \begin{cases} Be^{qx}, & x < -a \\ C\sin(kx) + D\cos(kx), & -a < x < a \\ Fe^{-qx}, & x > a \end{cases} \tag{6.12}$$

6.3.1 Boundary at the left wall

The boundary conditions at the left wall where $x = -a$ give

$$\begin{aligned} \psi(-a): &\quad Be^{-qa} = -C\sin(ka) + D\cos(ka) \\ \psi'(-a): &\quad qBe^{-qa} = kC\cos(ka) + kD\sin(ka) \end{aligned} \tag{6.13}$$

6.3.2 Boundary at the right wall

The boundary conditions at the right wall where $x = a$ give

$$\begin{aligned} \psi(a): &\quad Fe^{-qa} = C\sin(ka) + D\cos(ka) \\ \psi'(a): &\quad -qFe^{-qa} = kC\cos(ka) - kD\sin(ka) \end{aligned} \tag{6.14}$$

Divide derivative equations by k, then add and subtract the equations to get

$$\begin{aligned} 2D\cos(ka) &= (B+F)e^{-qa} \\ 2C\sin(ka) &= (F-B)e^{-qa} \\ 2C\cos(ka) &= \frac{q}{k}(B-F)e^{-qa} \\ 2D\sin(ka) &= \frac{q}{k}(B+F)e^{-qa} \end{aligned}$$

Eliminate B and F from the equations to find

$$\begin{aligned} 2C\left[\sin(ka) + \frac{k}{q}\cos(ka)\right] &= 0 \\ 2D\left[\cos(ka) - \frac{k}{q}\sin(ka)\right] &= 0 \end{aligned}$$

So either $C = 0$ or $D = 0$ (both cannot be zero, otherwise the wave function would be zero). This means we have either cosine or sine solutions; the energy eigenstates have definite parity (even or odd) just like the infinite square well. For the even cosine solutions ($C = 0$),

$$k\tan(ka) = q \tag{6.15}$$

and for the odd sine solutions ($D = 0$),

$$-k\cot(ka) = q \tag{6.16}$$

Equations (6.15) and (6.16) constitute quantization conditions; the solutions for the allowed energies come in discrete values which must be solved for numerically. This will easier if we work with dimensionless variables. To find a natural set of units, let's write the eigenvalue equation inside the well as

$$\frac{\hbar^2}{2ma^2}\left(a^2\frac{d^2\psi}{dx^2}\right) = -(V_0 - |E|)\psi$$

We define units such that

$$[\text{energy}] = \frac{\hbar^2}{2ma^2}, \qquad [\text{momentum}] = \frac{\hbar}{a}, \qquad [\text{length}] = a$$

which is equivalent to setting $\hbar = 2m = a = 1$. In dimensionless units,

$$k^2 + q^2 = V_0, \quad \text{or} \quad q = \sqrt{V_0 - k^2}$$

The quantization conditions can be written in terms of dimensionless variables as

$$\begin{aligned} k \tan k &= \sqrt{V_0 - k^2} \\ -k \cot k &= \sqrt{V_0 - k^2} \end{aligned} \tag{6.17}$$

which are transcendental equations which must be solved numerically/graphically. Once the value of k is found, we can then find the dimensionless energy E using

$$E = -q^2 = k^2 - V_0,$$

and then simply multiply by $\hbar^2/2ma^2$ to recover ordinary units. For $V_0 = 36$, we find four points of intersection between the left- and right-hand sides of Eq. (6.17), as shown in Figure 6.4. The corresponding energy eigenvalues are, in ordinary units,

$$k_1 a = 1.34475 : \; E_1 = -34.1916 \frac{\hbar^2}{2ma^2}$$

$$k_2 a = 2.67878 : \; E_2 = -28.8241 \frac{\hbar^2}{2ma^2}$$

$$k_3 a = 3.98583 : \; E_3 = -20.1132 \frac{\hbar^2}{2ma^2}$$

$$k_4 a = 5.22596 : \; E_4 = -8.68931 \frac{\hbar^2}{2ma^2}$$

The limit of an infinitely deep well corresponds to the case that $V_0 \to \infty$. In other words, the radius of the quarter circle in Figure 6.4 tends to infinity and the allowed values of k become the asymptotes of $\tan k$ and $\cot k$. In this limit, the allowed values are given by

$$k_n a = \frac{n\pi}{2} \Rightarrow k_n = \frac{n\pi}{2a}$$

which is exactly the same quantization condition for the infinite well. We can redefine the zero of potential energy by simply adding V_0 to the energy values, and we find

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$$

In other words, we recover the energy eigenvalues of the infinite energy well. On the other hand, when V_0 is very small, there is always at least one intersection that occurs for $ka < \pi/2$.

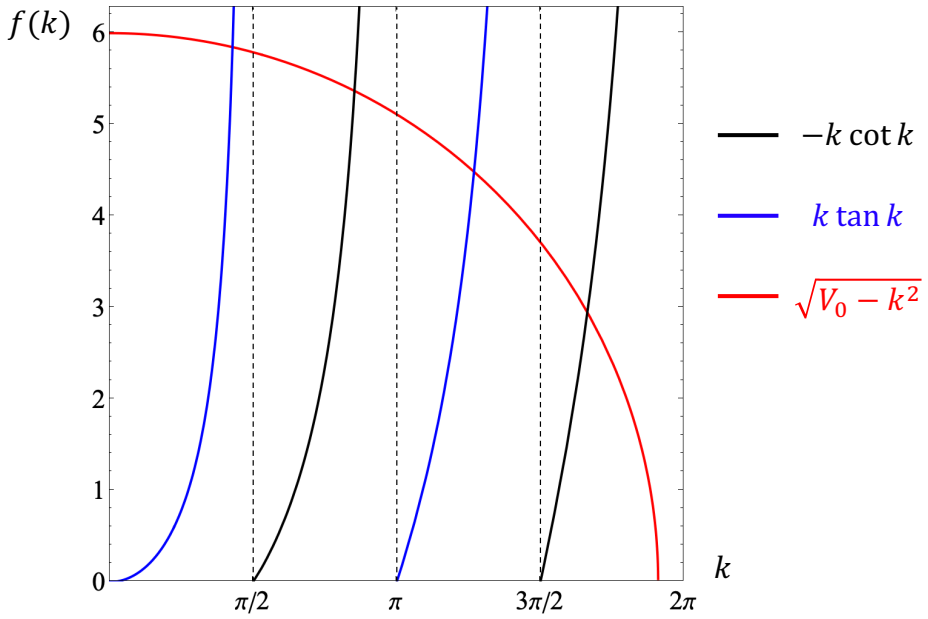


Figure 6.4: Graphical solution of the transcendental equations for the allowed energies of a finite square well for $V_0 = 36$ in dimensionless units.

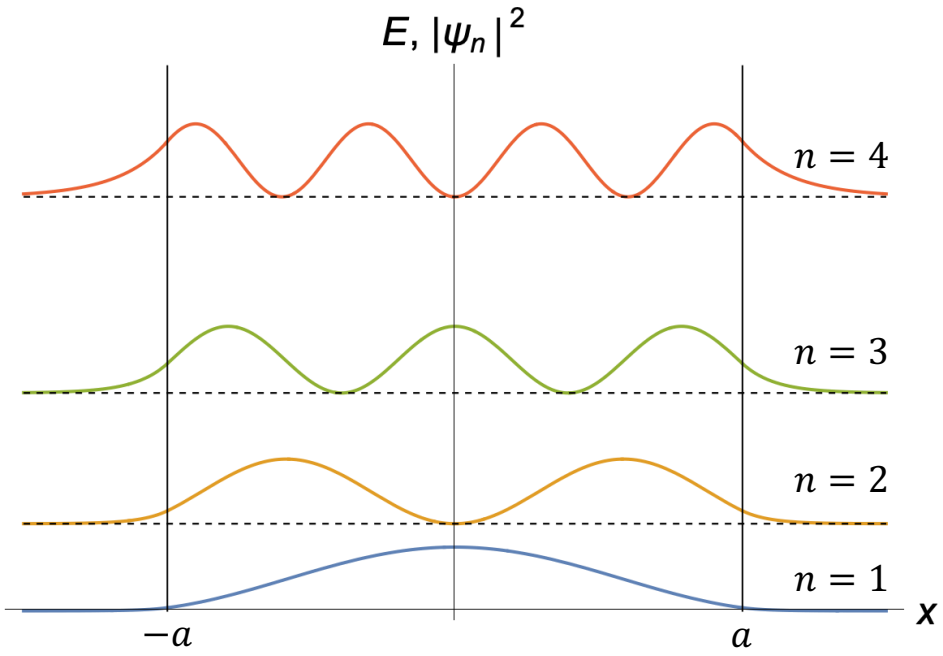


Figure 6.5: Square magnitude of the energy eigenstates for a finite square well with $V_0 = 36$ in dimensionless units.

The energy eigenfunctions are constrained by the boundary conditions and normalization. After some calculations, the even solutions are

$$\psi_{\text{even}}(x) = \frac{1}{\sqrt{a + \frac{1}{q}}} \begin{cases} e^{qa}\cos(ka)e^{-q|x|}, & |x| > a, \\ \cos(kx), & |x| < a \end{cases} \tag{6.18}$$

and the odd solutions are

$$\psi_{\text{odd}}(x) = \frac{1}{\sqrt{a + \frac{1}{q}}} \begin{cases} \text{sgn}(x)e^{qa}\sin(ka)e^{-q|x|}, & |x| > a, \\ \sin(kx), & |x| < a \end{cases} \tag{6.19}$$

where $\text{sgn}(x)$ is the "sign" function which is equivalent to $x/|x|$ when $x \neq 0$. The energy eigenfunctions are plotted in Figure 6.5 for the case $V_0 = 36$ in dimensionless units.

6.4 Unbound States and Scattering

Unbound states have an energy that is greater than the potential energy at infinity, in contrast to bound states, which have an energy that is less than the potential energy at infinity. Bound states must "fit" into the potential well, which leads to energy quantization, while unbound states lie above the well and extend to infinity. A free particle is characterized by a wave packet, which is a continuous superposition of plane wave solutions.

Consider a scenario in which a distant free particle travels to a region with a localized potential energy function. The interaction will generally produce nonzero amplitudes for the wave packet to both transmit and reflect. If the wave packet is sharply peaked in momentum space so that its width in position space is large compared to that of the potential, we can treat the wave packet as a plane wave to good approximation. However, plane waves are stationary states for the free-particle Hamiltonian, so how can we calculate the probabilities for reflection and transmission? Think of the particles being continually projected onto a target; some incident flux is reflected, and some is transmitted. The probability of reflection will be the ratio of reflected flux to incident flux, and the probability of transmission will be the ratio of transmitted to incident flux. We can formalize this intuition by studying the *probability current*, which is related to a conservation law implied by the Schrödinger equation for a non-relativistic particle with mass m.

6.4.1 Probability Current

Define the **probability density** as

$$\rho(x,t) \equiv |\psi(x,t)|^2 \tag{6.20}$$

where $\psi(x,t)$ is a position-space wave function satisfying

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi$$

For a non-stationary state, ψ (and thus ρ) changes with time. By the chain rule,

$$\frac{\partial\rho}{\partial t} = \psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}$$

and by the Schrödinger equation,

$$-i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*, \quad (V \text{ is a real function})$$

Putting it all together, we find

$$\begin{aligned}\frac{\partial\rho}{\partial t} &= \frac{\psi^*}{i\hbar}\left[-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi\right] - \frac{\psi}{i\hbar}\left[-\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*\right] \\ &= -\frac{i\hbar}{2m}\left(\psi\frac{\partial^2\psi}{\partial x^2} - \psi^*\frac{\partial^2\psi}{\partial x^2}\right) \\ &= -\frac{\partial j_x}{\partial x}\end{aligned}$$

where j_x is the **probability current** defined as

$$j_x \equiv \frac{i\hbar}{2m}\left(\psi\frac{\partial\psi^*}{\partial x} - \psi^*\frac{\partial\psi}{\partial x}\right) \tag{6.21}$$

We arrive at the following one-dimensional continuity equation:

$$\frac{\partial\rho}{\partial t} + \frac{\partial j_x}{\partial x} = 0 \tag{6.22}$$

The continuity equation expresses a local conservation principle: if the probability density changes at some location, it must be consistent with a "flow of probability" to or from that location. Integrating over some region from $x = a$ to $x = b > a$, we find

$$\frac{d}{dt}\int_a^b dx\ |\psi|^2 = -j_x(b,t) + j_x(a,t)$$

where $j_x(b,t) < 0$ and $j_x(a,t) > 0$ both correspond to *inward flow* of probability.

The probability current is especially useful in analyzing scattering phenomena. Consider the scattering from a localized potential as discussed earlier. The unbound state in front of the target is a superposition of incident and reflected waves, whereas the unbound state on the far side of the target is a single transmitted wave. By conservation of probability, the probability flux entering a

region must equal the flux exiting the region. In other words

$$|j_{\text{ref}}| + |j_{\text{tran}}| = |j_{\text{inc}}|$$

where each term represents the probability current for the reflected, transmitted, and incident waves respectively (reading from left to right). We assume that no probability is lost (or that no particles are absorbed). Rearranging the expression above, we find,

$$\frac{|j_{\text{ref}}|}{|j_{\text{inc}}|} + \frac{|j_{\text{tran}}|}{|j_{\text{inc}}|} = 1$$

which resembles a discrete probability distribution! We interpret the first term as the probability of reflection and the second term as the probability of transmission. Define the **reflection and transmission coefficients** as

$$R \equiv \frac{|j_{\text{ref}}|}{|j_{\text{inc}}|}, \qquad T \equiv \frac{|j_{\text{tran}}|}{|j_{\text{inc}}|} \tag{6.23}$$

For momentum eigenstates (plane waves), $\psi(x) = Ae^{ikx}$, and the corresponding probability current is given by

$$j_x = \frac{i\hbar}{2m}\left[|A^2|(-ik) - |A|^2(ik)\right] = \frac{\hbar k}{m}|A|^2$$

where the wave (and probability current) is moving right (in the $+x$ direction) when $k > 0$ and left when $k < 0$. We can interpret this result by recognizing that $\hbar k/m$ is the group velocity (the classical particle velocity) and $|A|^2$ is the probability density for the momentum eigenstate ψ. In a typical scattering problem, we solve for the amplitudes of reflected and transmitted waves using boundary conditions.

6.4.2 Scattering from a Potential Well

Let's return to the finite square well, but this time the particle has energy $E > 0$, and thus occupies an unbound state. The energy eigenvalue equations are

$$\frac{d^2\psi}{dx^2} = -k_1^2\psi \qquad \text{for } |x| > a$$
$$\frac{d^2\psi}{dx^2} = -k_2^2\psi \qquad \text{for } |x| < a$$

where $k_1 = \sqrt{2mE}/\hbar > 0$ and $k_2 = \sqrt{2m(E+V_0)}/\hbar > 0$. Hence, in all regions we have sinusoidal solutions which must satisfy the boundary conditions at $x = \pm a$. Let

$$\psi(x) = \begin{cases} Ae^{ik_1x} + Be^{-ik_1x}, & x < -a \\ Ce^{ik_2x} + De^{-ik_2x}, & -a < x < a \\ Fe^{ik_1x} + Ge^{-ik_1x}, & x > a \end{cases} \tag{6.24}$$

When solving a scattering problem, we treat the energy E as an initial condition; we can prepare particles with energy E by accelerating them through a fixed potential energy difference. Let's imagine we prepare particles with energy E and send them as projectiles into the potential well from $x = -\infty$. The potential well represents an interaction with some other, much larger particle (a target). In this context, since e^{ikx} represents a traveling wave moving to the right and e^{-ikx} travels to the left, we can set $G = 0$; there is no probability to measure a projectile coming from $x = \infty$ due to the experimental design.

Notice that we have five unknowns, A, B, C, D and F, but the boundary conditions give us only four constraints. We can eliminate A by effectively normalizing our solutions to the amplitude of the incoming wave. Letting $A = 1$, the constant B is the relative amplitude of *reflected waves* and F is the relative amplitude of *transmitted waves*. Altogether,

$$\psi(x) = \begin{cases} e^{ik_1 x} + Be^{-ik_1 x}, & x < -a \\ Ce^{ik_2 x} + De^{-ik_2 x}, & -a < x < a \\ Fe^{ik_1 x}, & x > a \end{cases} \tag{6.25}$$

6.4.3 Boundary at the left wall

The boundary conditions at the left wall where $x = -a$ give

$$\begin{aligned} \psi(-a): &\quad e^{-ik_1 a} + Be^{ik_1 a} = Ce^{-ik_2 a} + De^{ik_2 a} \\ \psi'(-a): &\quad ik_1 e^{-ik_1 a} - ik_1 Be^{ik_1 a} = ik_2 Ce^{-ik_2 a} - ik_2 De^{ik_2 a} \end{aligned} \tag{6.26}$$

6.4.4 Boundary at the right wall

The boundary conditions at the right wall where $x = a$ give

$$\begin{aligned} \psi(a): &\quad Ce^{ik_2 a} + De^{-ik_2 a} = Fe^{ik_1 a} \\ \psi'(a): &\quad ik_2 Ce^{ik_2 a} - ik_2 De^{-ik_2 a} = ik_1 Fe^{ik_1 a} \end{aligned} \tag{6.27}$$

After some tedious calculations, this system of equations can be solved for F and B to give

$$F = \frac{e^{-2ik_1 a}}{\cos(2k_2 a) - i\dfrac{k_1^2 + k_2^2}{2k_1 k_2}\sin(2k_2 a)}$$

$$B = -iF\frac{k_1^2 - k_2^2}{2k_1 k_2}\sin(2k_2 a)$$

The absolute square of F gives the relative probability T that an incident particle is transmitted through the potential well, which we call the **transmission coefficient**. The transmission

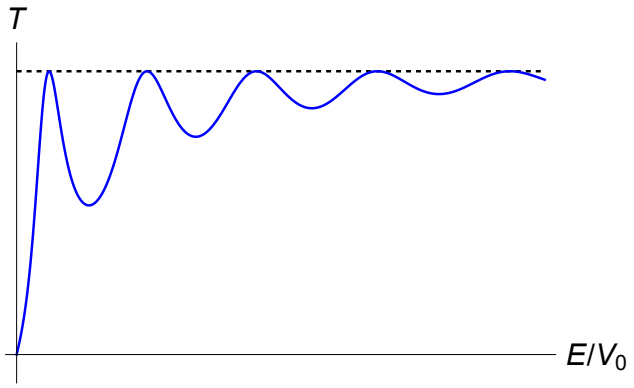


Figure 6.6: Transmission coefficient vs. energy for $V_0 = 12$ in dimensionless units.

coefficient for a finite square well is

$$T = \frac{1}{1 + \dfrac{(k_1^2 - k_2^2)^2}{4k_1^2 k_2^2} \sin^2(2k_2 a)} \tag{6.28}$$

Expressed in terms of the energy E and potential well depth V_0,

$$T = \frac{1}{1 + \dfrac{V_0^2}{4E(E+V_0)} \sin^2\left(\dfrac{2a}{\hbar}\sqrt{2m(E+V_0)}\right)} \tag{6.29}$$

The **reflection coefficient** $R = |B|^2$ is given by

$$R = \frac{1}{1 + \dfrac{4k_1^2 k_2^2}{(k_1^2 - k_2^2)^2 \sin^2(2k_2 a)}} \tag{6.30}$$

In this finite square well problem, there is no absorption of particles by the well, so the reflection and transmission coefficients add up to unity:

$$R + T = 1$$

In contrast to quantum mechanical particles, classical particles do not reflect from potential wells. They merely speed up and then slow down as they traverse the well. The reflection of quantum mechanical particles is thus further evidence of the wave nature of particle motion.

The transmission coefficient equals one when either the energy of the projectiles overwhelms the potential energy well, or when the wavelength of the wave function matches a **resonance condition**:

$$2k_2 a = n\pi$$

$$2\left(\frac{2\pi}{\lambda_2}\right) a = n\pi \Rightarrow 2a = n\frac{\lambda_2}{2}$$

When this condition is met, the wave function will have nodes on either side of the well, so there it effectively "hops the gap" so to speak. In terms of the energy, the resonance condition is

$$\left(\frac{2a}{\hbar}\right)^2 2m(E+V_0) = n^2\pi^2$$

$$E = -V_0 + \frac{n^2\pi^2\hbar^2}{2ma^2}$$

Thus, the energies of the transmission resonances (with respect to the bottom of the well) correspond to the bound-state energies of the infinite well. A similar effect is seen in atomic physics, where it is called the Ramsauer-Townsend effect.

6.4.5 Tunneling Through a Barrier

If the energy of the particle is below the barrier height, then the barrier region is classically forbidden and a classical particle reflects perfectly from the barrier. In the quantum mechanical treatment there is a possibility that the particle can penetrate the barrier and come out on the other side! This is because the quantum mechanical wave function penetrates into the classically forbidden region. This phenomenon is called **quantum mechanical tunneling**. We can study the tunneling phenomenon by inverting the potential well into a potential barrier with $-V_0 \to +V_0$. The "incident particles" have energy E greater than zero but less than V_0. The energy eigenvalue equations in each region are

$$\frac{d^2\psi}{dx^2} = -k^2\psi \qquad \text{for } |x| > a$$
$$\frac{d^2\psi}{dx^2} = q^2\psi \qquad \text{for } |x| < a$$

where $k = \sqrt{2mE}/\hbar > 0$ and $q = \sqrt{2m(V_0 - E)}/\hbar > 0$. Using the same experimental design as in the last section, we can write the wave function as

$$\psi(x) = \begin{cases} e^{ikx} + Be^{-ikx}, & x < -a \\ Ce^{qx} + De^{-qx}, & -a < x < a \\ Fe^{ikx}, & x > a \end{cases} \tag{6.31}$$

After implementing the boundary conditions, we can solve for the **transmission coefficient** $T = |F|^2$, and we find

$$T = \frac{1}{1 + \dfrac{V_0^2}{4E(V_0 - E)} \sinh^2\left(\dfrac{2a}{\hbar}\sqrt{2m(V_0 - E)}\right)} \tag{6.32}$$

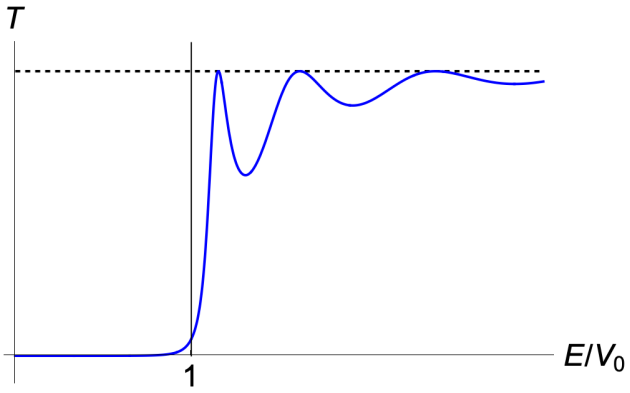


Figure 6.7: Transmission coefficient for scattering from a square barrier for $V_0 = 12$ in dimensionless units. Nonzero T for $E < V_0$ corresponds to quantum mechanical tunneling.

As for the reflection coefficient,

$$R = |B|^2 = 1 - T = \frac{1}{1 + \dfrac{4E(V_0 - E)}{V_0^2 \sinh^2\left(\frac{2a}{\hbar}\sqrt{2m(V_0 - E)}\right)}} \tag{6.33}$$

The transmission coefficient is plotted in Figure 6.7. In the tunneling case, the transmission is nearly zero except near the top of the barrier, where the tunneling probability increases exponentially. As the energy of the incident particle exceeds the barrier height, the transmission becomes large and exhibits the same resonances seen in the finite well problem. For large energy, the transmission goes to unity, which is to be expected because the potential barrier becomes insignificant.

Chapter 7

Quantum Harmonic Oscillator

7.1 Introduction

The harmonic oscillator plays a central role in quantum mechanics and appears, in one form or another, across nearly every area of physics. In atomic and molecular physics, the harmonic oscillator describes small oscillations of atoms about equilibrium configurations, governing vibrational spectra and setting the energy scales relevant for infrared absorption and emission. In condensed matter physics, collective excitations such as lattice vibrations (phonons) and charge-density fluctuations can be modeled as quantized harmonic oscillators, providing the foundation for understanding thermal properties, electrical transport, and superconductivity. Similarly, electromagnetic modes in cavities and waveguides are described by harmonic oscillators, a fact that underlies much of quantum optics and quantum information science.

The harmonic oscillator also appears naturally in modern experimental platforms. The motion of trapped ions and neutral atoms in optical traps is well approximated by harmonic potentials, making the oscillator an essential ingredient in precision metrology and quantum simulation. In superconducting circuits, effective harmonic oscillator modes form the starting point for the design of qubits and resonators used in quantum computing. Even in high-energy physics and quantum field theory, each mode of a free field behaves as an independent harmonic oscillator, with particle excitations emerging from the quantization of these modes.

In many of these contexts the harmonic oscillator description is only approximate, but it provides a controlled starting point for systematic improvements. Perturbation theory, semiclassical methods, and numerical techniques often rely on the harmonic oscillator as a reference system around which more realistic interactions are introduced. For this reason, a thorough understanding of the quantum harmonic oscillator is indispensable, both as an exactly solvable model and as a conceptual framework that extends far beyond the specific system it was originally introduced to describe. In these notes we focus primarily on the quantum mechanics of a single harmonic oscillator.

7.2 The Classical Harmonic Oscillator

A particle of mass m, constrained to move along the x axis, is interacting with some larger system that is essentially fixed. The force of interaction is conservative, and the system is in stable equilibrium at the point $x = 0$. For small displacements, $|x| \ll 1$, and we can Taylor expand the potential energy function about the equilibrium point:

$$V(x) = V(0) + \left(\frac{dV}{dx}\Big|_{x=0}\right) x + \frac{1}{2}\left(\frac{d^2V}{dx^2}\Big|_{x=0}\right) x^2 + O(x^3)$$

The leading term is quadratic because $V(0)$ can be defined to be zero without loss of generality and the linear term is zero because the first derivative of the potential energy function vanishes at an equilibrium point. Furthermore, since the equilibrium point is stable,

$$k \equiv \frac{d^2V}{dx^2}\Big|_{x=0} > 0$$

By Newton's second law,

$$m\frac{d^2x}{dt^2} = -kx$$

where the net force is $F = -dV/dx$. We can rewrite the equation of motion as

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

which describes a simple harmonic oscillator with angular frequency $\omega = \sqrt{k/m}$. One valid way to write the solution to this differential equation is

$$x(t) = A\cos(\omega t + \phi)$$

where the amplitude of oscillations A and the phase constant ϕ are both determined by initial conditions. A prototypical example would be a simple mass-on-spring system.

In this situation, the total energy of the particle is conserved and corresponds to the system's classical Hamiltonian (covered in PHYS 304). The total energy is

$$H = T + V = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

and since $p = mv$ and $k = m\omega^2$,

$$H = \frac{p_x^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

7.3 Quantum Harmonic Oscillator

We define the 1D quantum harmonic oscillator by the Hamiltonian

$$H = \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2$$

where $x \to \hat{x}$ and $p_x \to \hat{p}_x$ are "promoted" to operators. This procedure–constructing a quantum Hamiltonian by replacing classical observables with operators–is known as **canonical quantization**. It should be emphasized that quantization is not a well-defined procedure: there is no general algorithm that takes a classical system and produces its quantum counterpart without ambiguity. Canonical quantization is a physically motivated heuristic that works remarkably well for simple systems like the harmonic oscillator, while more complicated systems often require additional principles or more sophisticated arguments.

The goal for this section is to find the energy eigenstates of the Hamiltonian, i.e. to find the states $|E\rangle$ such that $H|E\rangle = E|E\rangle$, where $E > 0$ (the minimum of the potential energy function) to have normalizable wave functions. This lower bound on the energy eigenvalues is our cue to find ladder operators similar to our treatment of angular momentum in chapter #4. Before we get started, the presentation will be much cleaner if we use dimensionless variables. Since $\hbar\omega$ has units of energy, we can write the Hamiltonian as

$$H = \frac{\hbar\omega}{2}\left(\frac{\hat{p}_x^2}{m\hbar\omega} + \frac{\hat{x}^2}{\hbar/m\omega}\right)$$

The quantity in parentheses must be dimensionless, so we can redefine units such that

$$[\text{energy}] = \hbar\omega, \qquad [\text{momentum}] = \sqrt{m\hbar\omega}, \qquad [\text{length}] = \sqrt{\hbar/m\omega}$$

In these units, the Hamiltonian becomes

$$H = \frac{1}{2}\left(\hat{p}_x^2 + \hat{x}^2\right),$$

which is the same as if we simply set $m = \omega = \hbar = 1$. Likewise, the commutator between $\hat{x}$ and $\hat{p}_x$ is now given by $[\hat{x}, \hat{p}_x] = i$. Notice the *classical Hamiltonian* can be easily factored as

$$H = \frac{1}{2}(x - ip_x)(x + ip_x)$$

since x and p_x "commute" in classical mechanics. Applying the same factorization to the quantum operators, we find

$$\frac{1}{2}(\hat{x} - i\hat{p}_x)(\hat{x} + i\hat{p}_x) = \frac{1}{2}\left(\hat{x}^2 + \hat{p}_x^2 + i[\hat{x}, \hat{p}_x]\right) = H - \frac{1}{2}$$

Hence, due to the non-commutativity of the position and momentum operators,

$$H = \frac{1}{2}(\hat{x} - i\hat{p}_x)(\hat{x} + i\hat{p}_x) + \frac{1}{2}$$

We now define the operator a, called the "lowering operator," such that

$$a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p}_x)$$

and its Hermitian conjugate, $a^\dagger$, is the "raising operator" given by

$$a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p}_x)$$

Together, a and $a^\dagger$ are called *ladder operators* for reasons that will be clear very soon. They have the commutation relation

$$[a, a^\dagger] = 1$$

which follows from $[\hat{x}, \hat{p}_x] = i$. The dimensionless Hamiltonian can be written as

$$H = a^\dagger a + \tfrac{1}{2} = N + \tfrac{1}{2}$$

where $N = a^\dagger a$ is the "number operator." Let $|n\rangle$ be a normalized eigenvector of N with eigenvalue n, i.e. $N|n\rangle = n|n\rangle$. Given that H and N commute, it follows that $|n\rangle$ is also an eigenvector of H. Since N is Hermitian, we know that n must be real. Also, we can use properties of the inner product to demonstrate that n must be positive:

$$n = \langle n| N |n\rangle = \|a|n\rangle\|^2 \geq 0$$

The number operator has following commutation relations:

$$[N, a] = -a, \quad \text{and} \quad [N, a^\dagger] = a^\dagger$$

from which it follows that

$$N(a|n\rangle) = aN|n\rangle + [N, a]|n\rangle = (n-1)(a|n\rangle)$$

$$N\left(a^\dagger|n\rangle\right) = a^\dagger N|n\rangle + [N, a^\dagger]|n\rangle = (n+1)(a^\dagger|n\rangle)$$

Together, these equations imply that if $a|n\rangle$ and $a^\dagger|n\rangle$ do not vanish, they are eigenstates of the number operator with eigenvalues $n-1$ and $n+1$ respectively. If they do vanish, then they reduce to the trivial case $0 = 0$. We normally require eigenstates to be nonzero. Consider the inner product of $a|n\rangle$ with itself:

$$\|a|n\rangle\|^2 = \langle n| a^\dagger a |n\rangle = n \geq 0$$

Hence, $a|n\rangle$ vanishes if and only if $n = 0$. On the other hand,

$$\left\| a^\dagger |n\rangle \right\|^2 = \langle n| aa^\dagger |n\rangle = \langle n| (N+1) |n\rangle = n + 1$$

which never vanishes since $n \geq 0$, i.e., the state $a^\dagger |n\rangle$ never vanishes. Since the raising and lowering operators change the eigenvalue by ± 1, it follows that n must be an integer. Suppose n was not an integer, for example, suppose $n = 1/2$. Then $a\left|\frac{1}{2}\right\rangle$ is a nonzero ket that is an eigenstate of N with eigenvalue $-1/2$, which is impossible. On the other hand, if n is an integer, then we obtain no contradiction, since the eigenvalue can be lowered in integer steps by applying a until we reach $|0\rangle$, and then the next application of a gives $a|0\rangle = 0$. The spectrum of N therefore consists of all nonnegative integers, and since the Hamiltonian has the same eigenstates as N,

$$H|n\rangle = E_n |n\rangle$$

where, in ordinary units,

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right)$$

Note that the ground state does not have zero energy, in contrast to the classical harmonic oscillator. Rather, the quantum mechanical ground state has a **zero-point energy** of $\hbar\omega/2$. The structure of the quantum harmonic oscillator–particularly the commutation relations, the ladder operators, and the existence of a lowest energy state–forces the energy spectrum to be discrete. Furthermore, the energy levels are equally spaced:

$$E_{n+1} - E_n = \hbar\omega$$

This is Planck's famous hypothesis–that the energy of the radiation emitted by the *oscillating charges* must come only in bundles (quanta) that are integral multiples of $\hbar\omega$.

7.3.1 Ladder Operators

In this section we'll work out the raising and lowering operators in more detail. Since $a|n\rangle$ is an eigenstate with eigenvalue $n-1$ for $n > 0$, it must be proportional to the state $|n-1\rangle$. Let c_n be the constant of proportionality such that

$$a|n\rangle = c_n |n-1\rangle$$

Computing the norm squared, we find

$$\|a|n\rangle\|^2 = n = |c_n|^2$$

since the eigenstates are normalized. Setting the overal phase equal to 1, we have

$$a\,|n\rangle = \sqrt{n}\,|n-1\rangle$$

Likewise, $a^\dagger\,|n\rangle$ is an eigenstate with eigenvalue $n+1$, so

$$a^\dagger\,|n\rangle = d_n\,|n+1\rangle$$

where d_n is a complex constant. Computing the norm squared,

$$\left\|a^\dagger\,|n\rangle\right\|^2 = n+1 = |d_n|^2$$

Again, setting the overall phase equal to 1,

$$a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

In summary, we have

$$a\,|n\rangle = \sqrt{n}\,|n-1\rangle$$

$$a^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

Starting from the lowest energy state $n=0$, we can systematically determine all energy eigenstates by repeated application of the raising operator:

$$\begin{aligned}
|1\rangle &= a^\dagger\,|0\rangle \\
|2\rangle &= \frac{1}{\sqrt{2}}a^\dagger\,|1\rangle = \frac{1}{\sqrt{2}}(a^\dagger)^2\,|0\rangle \\
|3\rangle &= \frac{1}{\sqrt{3}}a^\dagger\,|2\rangle = \frac{1}{\sqrt{3!}}(a^\dagger)^3\,|0\rangle \\
&\vdots \\
|n\rangle &= \frac{1}{\sqrt{n!}}(a^\dagger)^n\,|0\rangle
\end{aligned}$$

7.3.2 Energy Eigenfunctions

We can determine any eigenstate by successive applications of the operator $a^\dagger$ on the ground state, so we start by solving for the position space wave function of the ground state. Since the operator

a "annihilates" the ground state,

$$\langle x|\, a\, |0\rangle = \langle x|\, \frac{1}{\sqrt{2}}\left(\hat{x} + i\hat{p}_x\right)|0\rangle = 0$$

or

$$\left(x + \frac{d}{dx}\right)\psi_0(x) = 0$$

where $\psi_0(x) = \langle x|0\rangle$ is the position-space wave function for the ground state. We seek a function whose derivative gives $-x$ times the original function. Clearly $e^{-x^2/2}$ satisfies that criterion, so the normalized energy eigenfunction in dimensionless units is given by

$$\psi_0(x) = \frac{1}{\pi^{1/4}} e^{-x^2/2}$$

In ordinary units, $\psi_0(x)$ has units of $[\text{length}]^{-1/2}$, so

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar^2}$$

The nth excited state in dimensionless units is given by repeated application of $a^\dagger$,

$$\psi_n(x) = \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{2^n n!}} \left(x - \frac{d}{dx}\right)^n e^{-x^2/2}$$

We can "simplify" this equation by using a set of special functions called Hermite polynomials, which are a set of polynomials generated by the Rodriguez formula

$$H_n(x) = (-1)^n e^{x^2} \left(\frac{d}{dx}\right)^n e^{-x^2}$$

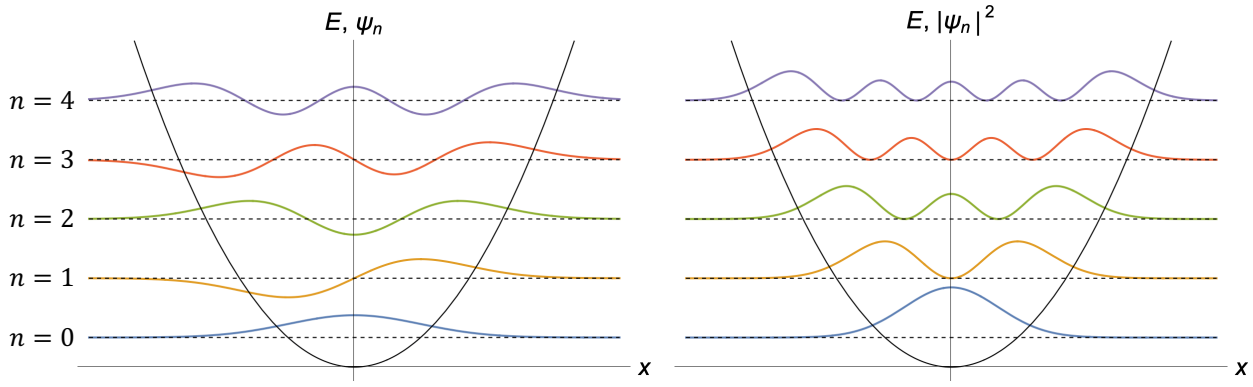
The excited states are then given by

$$\psi_n(x) = \frac{1}{\pi^{1/4}} \frac{1}{\sqrt{2^n n!}} H_n(x) e^{-x^2/2}$$

or in ordinary units,

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) e^{-m\omega x^2/2\hbar}$$

While it is nice to be able to write the eigenfunctions explicitly in terms of Hermite polynomials, for most practical purposes abstract operations involving kets and ladder operators are more useful than explicit representations in terms of wave functions.



7.4 Zero-Point Energy

In classical mechanics, the lowest energy state occurs when $p_x = 0$ and $x = 0$ (the spring is unstretched and the mass is at rest). This configuration is disallowed by the Heisenberg uncertainty relation–after repeated measurements on the system in its ground state, we will find a range of values for both position and momentum, so the expectation value of the energy will be nonzero. To see why, let $\Delta x^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$ and $\Delta \hat{p}_x^2 = \langle \hat{p}_x^2 \rangle - \langle \hat{p}_x \rangle^2$. Then,

$$\langle E \rangle = \frac{1}{2} \left(\langle \hat{p}_x^2 \rangle + \langle \hat{x}^2 \rangle \right) = \frac{1}{2} \left(\Delta p_x^2 + \Delta x^2 - \langle \hat{p}_x \rangle^2 - \langle \hat{x} \rangle^2 \right)$$

The expectation values are easily computing using the ladder operators. Inverting the expressions for a and $a^\dagger$ in terms of $\hat{x}$ and $\hat{p}_x$, we find

$$\hat{x} = \frac{1}{\sqrt{2}} \left(a^\dagger + a \right)$$

$$\hat{p}_x = \frac{i}{\sqrt{2}} \left(a^\dagger - a \right)$$

It follows that for an energy eigenstate $|n\rangle$,

$$\langle n | \hat{x} | n \rangle = \frac{1}{\sqrt{2}} \langle n | \left(a^\dagger + a \right) | n \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{n+1} \langle n | n+1 \rangle + \sqrt{n} \langle n | n-1 \rangle \right) = 0$$

$$\langle n | \hat{p}_x | n \rangle = \frac{i}{\sqrt{2}} \langle n | \left(a^\dagger - a \right) | n \rangle = \frac{i}{\sqrt{2}} \left(\sqrt{n+1} \langle n | n+1 \rangle - \sqrt{n} \langle n | n-1 \rangle \right) = 0$$

Hence,

$$\langle E \rangle = \frac{1}{2} \left(\Delta p_x^2 + \Delta x^2 \right)$$

From the Heisenberg uncertainty relation $\Delta x \Delta p_x \geq 0$, it follows that $\langle E \rangle > 0$. Localizing the particle at the origin will create a spread in momentum, and likewise, trying to put the particle into a state with zero momentum creates a spread in position. Nature chooses a tradeoff in which the particle has both nonzero Δx and Δp_x and, therefore, nonzero energy.

Explicitly, for the ground state,

$$\Delta p_x^2 = \langle \hat{p}_x^2 \rangle = -\frac{1}{2} \langle 0 | \left(a^\dagger - a\right)^2 | 0 \rangle$$

$$= -\frac{1}{2} \langle 0 | \left[a^\dagger a^\dagger - a^\dagger a - a a^\dagger + a a \right] | 0 \rangle$$

$$= \frac{1}{2} \langle 0 | \, a a^\dagger \, | 0 \rangle = \frac{1}{2} \langle 1 | 1 \rangle = \frac{1}{2}$$

and

$$\Delta x^2 = \langle \hat{x}^2 \rangle = \frac{1}{2} \langle 0 | \left(a^\dagger + a\right)^2 | 0 \rangle$$

$$= \frac{1}{2} \langle 0 | \left[a^\dagger a^\dagger + a^\dagger a + a a^\dagger + a a \right] | 0 \rangle$$

$$= \frac{1}{2} \langle 0 | \, a a^\dagger \, | 0 \rangle = \frac{1}{2} \langle 1 | 1 \rangle = \frac{1}{2}$$

Hence, $\Delta x \Delta p_x = 1/2$ or $\hbar/2$ in ordinary units. Evidently, the ground state of the harmonic oscillator is a minimum uncertainty state, which we already knew since the position-space wave function is a Gaussian.

Chapter 8

Wave Mechanics for 3D Motion

8.1 Position

The position eigenstate in three dimensions is given by

$$|\mathbf{r}\rangle = |x, y, z\rangle = |r, \theta, \phi\rangle$$

where any coordinate system that labels each point of space uniquely and smoothly is valid. The position operators for each of the three Cartesian coordinates act exactly as you would expect from the one-dimensional case:

$$\hat{x}\,|\mathbf{r}\rangle = x\,|\mathbf{r}\rangle\,, \quad \hat{y}\,|\mathbf{r}\rangle = y\,|\mathbf{r}\rangle\,, \quad \hat{z}\,|\mathbf{r}\rangle = z\,|\mathbf{r}\rangle$$

Note that the position operators commute with one another, i.e. $[\hat{r}_i, \hat{r}_j] = 0$. The position eigenstates are complete and orthonormal in the sense that

$$\int d^3r \;\; |\mathbf{r}\rangle\,\langle\mathbf{r}| = \iiint dx\; dy\; dz \;\; |x, y, z\rangle\,\langle x, y, z| = \mathbb{1}$$

and

$$\langle \mathbf{r}' | \mathbf{r} \rangle = \delta^3(\mathbf{r}' - \mathbf{r}) = \delta(x' - x)\delta(y' - y)\delta(z' - z)$$

Using the normalization condition, we see that

$$1 = \langle\psi|\psi\rangle = \int d^3r \;|\,\langle\mathbf{r}|\psi\rangle\,|^2 = \int d^3r \;|\psi(\mathbf{r})|^2$$

where $\psi(\mathbf{r}) = \langle\mathbf{r}|\psi\rangle$ is the wave function describing a single particle in 3D Euclidean space, and the quantity $d^3r|\psi(\mathbf{r})|^2$ represents the probability of finding the particle in the state $|\psi\rangle$ in the volume d^3r around $\mathbf{r}$ if a measurement of position is carried out.

8.2 Linear Momentum

To determine the linear momentum operator, we start by introducing a three-dimensional translation operator that satisfies

$$T(\mathbf{a})\,|\mathbf{r}\rangle = |\mathbf{r}+\mathbf{a}\rangle$$

In knee-jerk fashion, consider an infinitesimal displacement $\delta\mathbf{r}$ and expand the translation operator to find order:

$$T(\delta\mathbf{r}) = \mathbb{1} - \frac{i}{\hbar}\delta\mathbf{r}\cdot\mathbf{p}$$

where $\mathbf{p}$, an operator-valued vector (or vector operator), is the generator of translations in 3D Euclidean space and the linear momentum operator in quantum mechanics. Now consider a quantum particle in an arbitrary state $|\psi\rangle$. For an infinitesimal translation,

$$\langle\mathbf{r}|\,T(\delta\mathbf{r})\,|\psi\rangle = \psi(\mathbf{r}-\delta\mathbf{r}) \approx \psi(\mathbf{r}) - \delta\mathbf{r}\cdot\nabla\psi(\mathbf{r})$$

where we've used the Taylor expansion of ψ in the last equality. It follows that

$$\mathbf{p} = -i\hbar\nabla$$

in the position representation. In Cartesian components,

$$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}, \quad \hat{p}_y = -i\hbar\frac{\partial}{\partial y}, \quad \hat{p}_z = -i\hbar\frac{\partial}{\partial z}$$

Using an arbitrary wave function $\psi(\mathbf{r})$, it's easy to verify the following commutation relations:

$$[\hat{r}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

8.3 Orbital Angular Momentum

In chapter #4, we worked out the general theory of rotations and the angular momentum operators $\mathbf{J}$ in quantum mechanics. We will now apply that general framework to the study of a single (spinless) particle moving in three-dimensional space. For the majority of these notes, we'll make no assumptions about the Hamiltonian, but the theory of rotations is especially useful when the particle is under the influence of a central potential $V(r)$, i.e. one that depends only on the distance to some fixed location taken to be the origin.

8.3.1 Particle on a Ring

Before we dive in to the most general problem of a particle moving in three-dimensional space, let's first consider a particle confined to a circular ring of radius a. The circle is centered on the origin and lies in the xy plane. With a reference line along the x axis, let s be the arc length and ϕ be the

angle subtended by a line between the particle and the origin. The generator of translation along the circumference of the circle is

$$\hat{p}_s = i\hbar \left.\frac{\partial T(s)}{\partial s}\right|_{s=0}$$

and since $s = a\phi$, we can associate with the translation operator T a rotation around the z axis. The corresponding generator of rotation is

$$L_z = i\hbar\frac{\partial T}{\partial \phi} = i\hbar\frac{\partial T}{\partial s}\frac{ds}{d\phi} = a\hat{p}_s$$

This expression is analogous to a classical particle executing circular motion with angular momentum $L_z = rp = mvr$. Consider an arbitrary wave function $\psi(\phi)$ which depends on the angular position of the particle. It follows that

$$L_z\psi(\phi) = a\hat{p}\psi(\phi) = -ai\hbar\frac{\partial\psi(\phi)}{\partial\phi}\frac{d\phi}{ds} = -i\hbar\frac{\partial\psi(\phi)}{\partial\phi}$$

We now search for eigenstates of the angular momentum operator such that

$$L_z\psi_m(\phi) = m\hbar\ \psi_m(\phi) = -i\hbar\frac{\partial\psi_m(\phi)}{\partial\phi}$$

where m is simply a unitless constant. This differential equation has the general solution

$$\psi_m(\phi) = Ae^{im\phi}$$

where A is a constant we can determine by enforcing normalization of the wave function:

$$1 = \int_0^{2\pi} d\phi\ |\psi_m(\phi)|^2 = 2\pi|A|^2 \quad \rightarrow \quad A = \frac{1}{\sqrt{2\pi}}$$

Despite there being no physical "boundary" on a ring, there is an important physical constraint that must be satisfied–the wave function must be *single-valued*. Each angular position ϕ is physically equivalent to the point $\phi + 2\pi$ in 3D Euclidean space, so the wave function must satisfy the periodicity condition $\psi(\phi + 2\pi) = \psi(\phi)$. Periodicity is indeed satisfied when

$$e^{i2\pi m} = 1$$

which holds if and only if

$$m = 0, \pm1, \pm2, \pm3, \ldots$$

In other words, m can take only *integer values*. Evidently, the half-integer states are excluded in the case of orbital angular momentum. The eigenstates of L_z are

$$\psi_m(\phi) = \frac{1}{\sqrt{2\pi}}e^{im\phi} \qquad m = 0, \pm1, \pm2, \pm3, \ldots$$

The particle on a ring is a one-dimensional system even though it exists in a two-dimensional space–this is because there is only one degree of freedom ϕ. Let's briefly consider the free-particle Hamiltonian, written in terms of the angular momentum operator:

$$H = \frac{\hat{p}_s^2}{2m} = \frac{L_z^2}{2ma^2} = \frac{L_z^2}{2I}$$

where $I = mr^2$ is the moment of inertia of the particle about the origin. Since $[H, L_z] = 0$, the Hamiltonian and the z component of angular momentum share eigenstates. We can determine the energies of the particle by simply applying the Hamiltonian to an angular momentum eigenstate. For an arbitrary eigenstate $\psi_m(\phi)$,

$$H\psi_m(\phi) = \frac{1}{2I}L_z^2\psi_m(\phi) = \frac{m^2\hbar^2}{2I}\psi_m(\phi)$$

The allowed energies of a particle on a ring are quantized and given by

$$E_m = \frac{m^2\hbar^2}{2I}$$

The eigenstates corresponding to $+|m|$ and $-|m|$ states have the same energy, so there are two energy states at every allowed energy except for the one corresponding to $m = 0$. Thus the particle-on-a-ring system exhibits *degeneracy*, which we first encountered in the free-particle system. For the particle-on-a-ring system, all states are two-fold degenerate except for $m = 0$, which is nondegenerate. The $\pm m$ degeneracy of the energy eigenstates corresponds to the angular momentum states with $L_z = +m\hbar$ and $L_z = -m\hbar$. That is, the two degenerate energy states represent states with opposite components of the angular momentum along the z-axis. The energy is the same regardless of the direction of rotation, which is analogous to the free particle in one dimension where the energy is independent of the direction of travel.

8.3.2 Rotations of a Single-Particle System

In classical mechanics, the angular momentum of a single particle is an ordinary vector defined as $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ where $\mathbf{r}$ is the position vector locating the particle relative to some origin and $\mathbf{p}$ is the linear momentum of the particle. In quantum mechanics, angular momentum is defined as the generator of rotations. We begin with a definition of rotation operators on our single particle system, and then derive their generators. Let R be a proper rotation–an element of SO(3)–and define the unitary operator $U(R)$ by

$$U(R)\,|\mathbf{r}\rangle = |R\mathbf{r}\rangle$$

This equation says "the unitary operator U maps the state vector $|\mathbf{r}\rangle$ corresponding to position vector $\mathbf{r}$ to the state vector $|R\mathbf{r}\rangle$ corresponding to the rotated position vector $R\mathbf{r}$." We can undo

the rotation, so $U(R^{-1})U(R)\,|\mathbf{r}\rangle = |\mathbf{r}\rangle$, and since U is unitary,

$$U^\dagger(R) = U(R^{-1})$$

We can then write the position-space wave function corresponding to a rotation as

$$\langle \mathbf{r}|\, U(R)\,|\psi\rangle = \big\langle R^{-1}\mathbf{r}\big|\psi\big\rangle = \psi(R^{-1}\mathbf{r}) \tag{8.1}$$

This is analogous to $\langle \mathbf{r}|\, T(\mathbf{a})\,|\psi\rangle = \psi(\mathbf{r}-\mathbf{a})$ for the translation operator. Using the axis-angle parameterization, we consider an infinitesimal rotation by $\delta\theta$ about an axis $\hat{\mathbf{n}}$. We have

$$\langle \mathbf{r}|\, U(\hat{\mathbf{n}},\delta\theta)\,|\psi\rangle = \psi\Big(R^{-1}(\hat{\mathbf{n}},\delta\theta)\mathbf{r}\Big) \tag{8.2}$$

where U may be expanded to first order in terms of the generator of rotation as

$$U(\hat{\mathbf{n}},\delta\theta) = \mathbb{1} - \frac{i}{\hbar}\delta\theta\hat{\mathbf{n}}\cdot\mathbf{L} \qquad \text{(operator expression)}$$

and the infinitesimal classical rotation follows from the usual geometric argument

$$R^{-1}(\hat{\mathbf{n}},\delta\theta)\mathbf{r} = \mathbf{r} - \delta\theta\hat{\mathbf{n}}\times\mathbf{r} \qquad \text{(vector expression)}$$

Since we are only considering terms that are first order in $\delta\theta$, we can approximate the wave function by its first order Taylor expansion:

$$\psi\Big(R^{-1}(\hat{\mathbf{n}},\delta\theta)\mathbf{r}\Big) = \psi(\mathbf{r}) - \delta\theta(\hat{\mathbf{n}}\times\mathbf{r})\cdot\nabla\psi$$

At this point, $\mathbf{r}$ may still be considered an ordinary vector in 3D Euclidean space. Substituting the appropriate expansions into Eq. (8.2), we find

$$\psi(\mathbf{r}) - \frac{i}{\hbar}\delta\theta\,(\hat{\mathbf{n}}\cdot\mathbf{L})\,\psi(\mathbf{r}) = \psi(\mathbf{r}) - \delta\theta(\hat{\mathbf{n}}\times\mathbf{r})\cdot\nabla\psi$$

The zeroth order terms cancel, and $\delta\theta$ cancels from the first order terms, so we're left with

$$(\hat{\mathbf{n}}\cdot\mathbf{L})\,\psi(\mathbf{r}) = [(\hat{\mathbf{n}}\times\mathbf{r})\cdot\mathbf{p}]\,\psi(\mathbf{r}) = \hat{\mathbf{n}}\cdot(\mathbf{r}\times\mathbf{p})\psi(\mathbf{r})$$

The position vector $\mathbf{r}$ becomes an operator-valued vector (or vector operator) in the second equality. Let's see how this works by looking at the z component of angular momentum, noting that $\hat{r}_i$ commutes with $\hat{p}_j$ when $i \neq j$:

$$L_z\psi = (x\hat{p}_y - y\hat{p}_x)\psi = (\hat{x}\hat{p}_y - \hat{y}\hat{p}_x)\psi$$

Since ψ is arbitrary, $L_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$, and since $\hat{\mathbf{n}}$ is also arbitrary,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

which is an *operator expression* that mirrors the classical definition of angular momentum! From now on, we refer to generators of rotation satisfying this operator expression as **orbital angular momentum**. Importantly, we derived this expression in a manner consistent with our general treatment of angular momentum in chapter #4, meaning we are free to make use of those results. As we learned in the previous section involving the particle on a ring, only the integer eigenstates are relevant when dealing with orbital angular momentum.

However, the argument that is presented in terms of the single-valuedness of the wave function, while commonly used, is contestable. Single-valuedness can be justified in a *classical field theory*, such as electromagnetism, in which the field is an observable quantity. In quantum theory, the wave function does not have direct physical significance. For the particle on a ring example, all expectation values and probability distributions will remain unchanged if $\psi(\phi + 2\pi) = \pm\psi(\phi)$, and the possibility of a sign change allows for half-integer values of m, i.e., $e^{i2\pi m} = \pm 1$ for $m = 0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \pm 2, \ldots$. Long story short, the conventional boundary conditions applied to classical fields is inadequate for quantum theory.

8.3.3 Orbital Angular Momentum Eigenvalues

In chapter #4, we learned that a component of general angular momentum must have eigenvalues $\hbar m$ where m is either an integer or half-integer. Any further restriction of the orbital angular momentum eigenvalues must come from a special property of the orbital operators that is not possessed by the general angular momentum operators. For example, consider the z component of orbital angular momentum,

$$L_z = xp_y - yp_x, \qquad \text{(operator expression)}$$

From the commutation relations for position and momentum, we can derive a restriction on the eigenvalues of L_z. In a system of units where $\hbar = 1$, we define the following operators:

$$R_1 = \frac{1}{\sqrt{2}}(x + p_y)$$

$$R_2 = \frac{1}{\sqrt{2}}(x - p_y)$$

$$P_1 = \frac{1}{\sqrt{2}}(p_x - y)$$

$$P_2 = \frac{1}{\sqrt{2}}(p_x + y)$$

It can be verified that

$$[R_1, R_2] = [P_1, P_2] = 0, \quad [R_i, P_j] = i\delta_{ij}$$

This is an example of a **canonical transformation** in quantum mechanics. In terms of these new operators, the z component of orbital angular momentum can be written as

$$L_z = \frac{1}{2}(P_1^2 + R_1^2) - \frac{1}{2}(P_2^2 + R_2^2)$$

Comparing with chapter #7, we can see that the angular momentum operator is the difference of two independent harmonic oscillator Hamiltonians, each with mass $m = 1$ and angular frequency $\omega = 1$. Since the two terms commute, the eigenvalues of L_z are just the differences between the eigenvalues of these two terms. Using a key result of chapter #7, the eigenvalues of L_z are

$$\left(n_1 + \frac{1}{2}\right) - \left(n_2 + \frac{1}{2}\right) = n_1 - n_2$$

where n_1 and n_2 are nonnegative integers. Thus we have shown, directly from the properties of the position and momentum operators, that the orbital angular momentum eigenvalues must be integer multiples of $\hbar$, and that half-integer multiples cannot occur!

8.3.4 Eigenstates of Angular Momentum

In accordance with chapter #4, the standard basis of orbital angular momentum is formed by the simultaneous eigenstates of L^2 and L_z, which can be indexed by ℓ and m such that

$$L^2 \left|\ell, m\right\rangle = \ell(\ell+1)\hbar^2 \left|\ell, m\right\rangle$$

$$L_z \left|\ell, m\right\rangle = m\hbar \left|\ell, m\right\rangle$$

The corresponding angular momentum eigenfunctions are $\psi_{\ell m}(\mathbf{r}) = \langle \mathbf{r}|\ell, m\rangle$, and

$$L^2\psi_{\ell m}(\mathbf{r}) = \ell(\ell+1)\hbar^2\psi_{\ell m}(\mathbf{r}) \tag{8.3}$$

$$L_z\psi_{\ell m}(\mathbf{r}) = m\hbar\psi_{\ell m}(\mathbf{r}) \tag{8.4}$$

The key to finding the wave functions is to first consider the "stretched states" with $m = \ell$. States with other values of m can be generated by applying the lowering operator L_-. Since the operator L^2 is pretty complicated, we will work with L_z and the raising operator L_+ for now. For the stretched state $\psi_{\ell\ell}(\mathbf{r})$,

$$L_z\psi_{\ell\ell}(\mathbf{r}) = \ell\hbar\psi_{\ell\ell}(\mathbf{r}) \tag{8.5}$$

$$L_+\psi_{\ell\ell}(\mathbf{r}) = 0 \tag{8.6}$$

While nothing is really convenient about angular momentum, we can improve the situation slightly by using spherical coordinates (r, θ, ϕ) where

$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \end{aligned}$$

The components of angular momentum can be found in spherical coordinates by applying the chain rule and working through a bit of algebra. I'll save you some time:

$$L_x = -i\hbar \left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y} \right) = -i\hbar \left(-\sin\phi \, \frac{\partial}{\partial \theta} - \cot\theta \, \cos\phi \, \frac{\partial}{\partial \phi} \right)$$

$$L_y = -i\hbar \left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} \right) = -i\hbar \left(\cos\phi \, \frac{\partial}{\partial \theta} - \cot\theta \, \sin\phi \, \frac{\partial}{\partial \phi} \right)$$

$$L_z = -i\hbar \left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x} \right) = -i\hbar \, \frac{\partial}{\partial \phi}$$

From L_x and L_y, we can compute the raising and lowering operators, given by

$$L_\pm = L_x \pm iL_y = -i\hbar e^{\pm i\phi} \left(\pm i\frac{\partial}{\partial \theta} - \cot\theta \frac{\partial}{\partial \phi} \right)$$

The component L_z is particularly simple in spherical coordinates—it is the generator of rotations about the z-axis, and ϕ is the azimuthal angle. Another striking feature of the angular momentum operators in spherical coordinates is that they do not depend on the radial coordinate r or the radial differential operator $\partial/\partial r$. The geometrical reason is that rotations in physical space change the direction of vectors, but not their magnitude; the motion of the tip of a given vector takes place on the surface of a sphere. This holds for infinitesimal rotations, which connect nearby points with the same r but different θ and ϕ.

The angular momentum operators are independent of the radial coordinate, so we consider wave functions $f(\theta, \phi)$ defined on the surface of the unit sphere. The wave functions are normalized by integrating over the unit sphere:

$$\langle f|f\rangle = \int d\Omega \; |f(\theta,\phi)|^2 = \int \sin\theta \, d\theta \, d\phi \; |f(\theta,\phi)|^2 = 1$$

where the inner product of two wave functions on the unit sphere is defined as

$$\langle f|g\rangle = \int d\Omega \; f^*(\theta,\phi) g(\theta,\phi)$$

Returning to Eqs. (8.5) and (8.6), we can solve for the stretched eigenfunctions on the unit sphere

$f_{\ell\ell}(\theta,\phi)$. The L_z eigenvalue equation is

$$-i\hbar\frac{\partial}{\partial\phi}f_{\ell\ell}(\theta,\phi)=\ell\hbar f_{\ell\ell}(\theta,\phi)$$

which is almost exactly the same as the particle on a ring! The solution is

$$f_{\ell\ell}(\theta,\phi)=F_{\ell\ell}(\theta)e^{i\ell\phi}$$

where $F_{\ell\ell}(\theta)$ is a function of θ only, and ℓ is an integer. From the L_+ equation, we find

$$0=-i\hbar e^{i\phi}\left(i\frac{\partial}{\partial\theta}-\cot\theta\frac{\partial}{\partial\phi}\right)F_{\ell\ell}e^{i\ell\phi}=-i\hbar e^{i\phi}\left(i\frac{\partial}{\partial\theta}-(i\ell)\cot\theta\frac{\partial}{\partial\phi}\right)F_{\ell\ell}$$

After simplifying the expression, we find

$$\frac{dF_{\ell\ell}}{d\theta}=\ell\cot\theta F_{\ell\ell}$$

which is a separable first-order differential equation. We can therefore apply separation of variables to solve for the function $F_{\ell\ell}(\theta)$. Start by moving all terms related to $F_{\ell\ell}$ to the left-hand side, and all terms involving θ to the right-hand side:

$$\int\frac{dF_{\ell\ell}}{F_{\ell\ell}}=\ell\int\frac{\cos\theta}{\sin\theta}d\theta$$

The integral on the right can be solved by substitution with $u=\sin\theta$. The result is

$$\ln F_{\ell\ell}=\ell\ln\sin\theta+\ln A$$

where $\ln A$ is a constant of integration. It follows that

$$F_{\ell\ell}(\theta)=A\sin^{\ell}\theta$$

and thus,

$$f_{\ell\ell}(\theta,\phi)=A\sin^{\ell}\theta\ e^{i\ell\phi}$$

The constant A is chosen so that $f_{\ell\ell}$ satisfies the normalization condition for wave functions defined on the unit sphere. After a lengthy calculation, we end up with

$$A=\frac{(-1)^{\ell}}{2^{\ell}\ell!}\sqrt{\frac{(2\ell+1)!}{4\pi}}$$

which ensures

$$\langle f_{\ell\ell}|f_{\ell\ell}\rangle=\int d\Omega\ |f_{\ell\ell}|^2=1$$

The resulting wave function $f_{\ell\ell}$ is actually the stretched *spherical harmonic function*

$$Y_{\ell\ell}(\theta,\phi) = \frac{(-1)^\ell}{2^\ell \ell!}\sqrt{\frac{(2\ell+1)!}{4\pi}}\,\sin^\ell\theta\, e^{i\ell\phi}$$

To obtain states corresponding to other values of m, we apply the lowering operator L_-. For the sake of time (and personal sanity), we simply state the result for $m \geq 0$ as

$$Y_\ell^m(\theta,\phi) = \sqrt{\frac{(2\ell+1)}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\,e^{im\phi}P_\ell^m(\cos\theta) \qquad \text{(spherical harmonics)}$$

where $P_\ell^m(\cos\theta)$ are the **associated Legendre functions** defined by

$$P_\ell^m(x) \equiv (-1)^m(1-x^2)^{m/2}\left(\frac{d}{dx}\right)^m P_\ell(x)$$

and $P_\ell(x)$ are the **Legendre polynomials** defined by the *Rodriguez formula*

$$P_\ell(x) \equiv \frac{1}{2^\ell \ell!}\left(\frac{d}{dx}\right)^\ell (x^2-1)^\ell$$

The eigenstates with negative values of m can be obtained from the relation

$$Y_\ell^{-m} = (-1)^m\,(Y_\ell^m)^* \qquad (m>0)$$

where we are using associated Legendre polynomials with the property that for $m > 0$

$$P_\ell^{-m}(x) = (-1)^m\frac{(\ell-m)!}{(\ell+m)!}P_\ell^m(x)$$

In summary, the simultaneous eigenfunctions of L^2 and L_z are the spherical harmonics:

$$L^2Y_\ell^m(\theta,\phi) = \ell(\ell+1)\hbar^2Y_\ell^m(\theta,\phi) \tag{8.7}$$

$$L_zY_\ell^m(\theta,\phi) = m\hbar Y_\ell^m(\theta,\phi) \tag{8.8}$$

A more detailed treatment of spherical harmonics is beyond the scope of this course.

Example 8.1: Orbital Angular Momentum Eigenfunctions

Given the stretched state $Y_\ell^\ell(\theta,\phi)$, compute the position-space eigenfunction $Y_\ell^{\ell-1}(\theta,\phi)$ using the lowering operator L_-.

Solution: In Dirac notation, the lowering operator maps $|\ell\ell\rangle$ to a state proportional to

$|\ell, \ell-1\rangle$. More precisely,

$$L_- |\ell\ell\rangle = \hbar\sqrt{\ell(\ell+1) - \ell(\ell-1)}\,|\ell, \ell-1\rangle = \hbar\sqrt{2\ell}\,|\ell, \ell-1\rangle$$

In terms of the position basis,

$$L_- Y_\ell^\ell = -i\hbar e^{-i\phi}\left(-i\frac{\partial}{\partial\theta} - \cot\theta\frac{\partial}{\partial\phi}\right) Y_\ell^\ell$$

where $Y_\ell^\ell = A_\ell\,(\sin\theta)^\ell\, e^{i\ell\phi}$ and A_ℓ is the normalization constant. After applying the derivatives over θ and ϕ and then simplifying, we find

$$L_- Y_\ell^\ell = -2\ell\hbar A_\ell(\sin\theta)^{\ell-1}\cos\theta\, e^{i(\ell-1)\phi}$$

Hence, the state with $m = \ell - 1$ is given by

$$Y_\ell^{\ell-1}(\theta,\phi) = -\sqrt{2\ell}A_\ell(\sin\theta)^{\ell-1}\cos\theta\, e^{i(\ell-1)\phi}$$

There is no need to verify this is the correct expression; you may simply take for granted that the result is consistent with the spherical harmonics, including the phase. The point is that, even if you weren't given the spherical harmonics, you could still generate all of the eigenfunctions using the lowering operator.

8.3.5 Particle on a Sphere

Consider a free particle confined to the surface of a sphere with radius a and Hamiltonian

$$H = \frac{\mathbf{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2 \qquad \text{(position representation)}$$

where ∇^2 denotes the Laplacian. In spherical coordinates,

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

As it turns out, we can write $\mathbf{p}^2$ in terms of the L^2 operator as

$$\mathbf{p}^2 = -\hbar^2\nabla^2 = -\frac{\hbar^2}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{L^2}{r^2}$$

where

$$L^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

For a particle confined to the sphere with radius a,

$$H = \frac{L^2}{2ma^2} = \frac{L^2}{2I}$$

where $I = ma^2$ is the moment of inertia of the particle about the origin. We can immediately write down the eigenfunctions of the Hamiltonian, since they are the same as the eigenfunctions of L^2, namely spherical harmonics $Y_\ell^m(\theta, \phi)$. Hence,

$$HY_\ell^m(\theta,\phi) = \frac{1}{2I}L^2 Y_\ell^m(\theta,\phi) = \frac{\ell(\ell+1)\hbar^2}{2I}Y_\ell^m(\theta,\phi)$$

and it follows that the energies of a particle on a sphere are

$$E_\ell = \frac{\ell(\ell+1)\hbar^2}{2I}$$

This result may also be applied to the study of diatomic molecules. At low temperatures, diatomic molecules behave like rigid rotators: two atoms separated by a fixed distance (the bond length), rotating about their center of mass. This system has the same rotational degree of freedom as a particle constrained to a sphere. The only thing that changes is the moment of inertia. For a diatomic molecule, $I = \mu r_0^2$ where r_0 is the bond length and μ is the reduced mass given by

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

See Section 5 for an in-depth treatment of the two-body central force problem.

8.4 Central Potential

Consider a particle moving in 3D space under the influence of a conservative central force. The potential energy function $V(r)$ depends only on the particle's distance from the origin; thus we call it a **central potential**. This system exhibits a high degree of rotational symmetry. The Hamiltonian

$$H = \frac{\mathbf{p}^2}{2m} + V(r)$$

commutes with all rotation operators $U(R)$ whose action on wave functions is given by Eq. (8.1). To see why this must be the case, consider the action of $U(R)$ on position and momentum. By definition, $U(R)$ implements a unitary transformation of the state vectors in a way that must correspond to a classical rotation. Hence,

$$U^\dagger(R)\mathbf{r}U(R) = R\mathbf{r} \quad \text{and} \quad U^\dagger(R)\mathbf{p}U(R) = R\mathbf{p}$$

This is the same operator expression we encountered in chapter #4 for the components of angular momentum. The potential energy depends only on the magnitude of $\mathbf{r}$, so

$$U^\dagger(R)V(r)U(R) = V\big(|R\mathbf{r}|\big) = V\,(r) \Rightarrow [V(r), U(R)] = 0$$

The kinetic energy depends on $\mathbf{p}^2 = \mathbf{p}\cdot\mathbf{p}$, so

$$U^\dagger(R)\mathbf{p}^2U(R) = (R\mathbf{p})\cdot(R\mathbf{p}) \Rightarrow [\mathbf{p}^2, U(R)] = 0$$

Since H commutes with $U(R)$ for any rotation R, and $U(R)$ is a function of $\mathbf{L}$, it follows that H commutes with $\mathbf{L}$ (or rather, the components of $\mathbf{L}$) and by extension, H commutes with L^2. In case you don't buy the argument above, I encourage you to compute the relevant commutators. Nonetheless, a particle in a central potential has simultaneous eigenstates for H, L^2, and L_z given by

$$\psi(r,\theta,\phi) = \mathcal{R}(r)Y_\ell^m(\theta,\phi)$$

where $\mathcal{R}(r)$ where is a radial wave function normalized by the integral

$$\int_0^\infty r^2\,dr\,|\mathcal{R}(r)|^2 = 1$$

Substitute ψ into the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

Expand the kinetic energy term in spherical coordinates:

$$\left[-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{L^2}{2mr^2}\right]\psi + V(r)\psi = E\psi$$

Note that L^2 acts non-trivially only on the spherical harmonic terms, bringing out a factor of $\ell(\ell+1)\hbar^2$. The overall factor of Y_ℓ^m cancels and we find an expression only in terms of the radial wave function. The so-called radial equation is

$$\left[-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\ell(\ell+1)\hbar^2}{2mr^2}\right]\mathcal{R} + V(r)\mathcal{R} = E\mathcal{R}$$

This expression can be simplified by defining an effective potential U_{eff} such that

$$-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\mathcal{R}}{\partial r}\right) + V_{\text{eff}}(r)\mathcal{R} = E\mathcal{R}$$

where

$$V_{\text{eff}}(r) = \frac{\ell(\ell+1)\hbar^2}{2mr^2} + V(r)$$

The effective potential contains a *centrifugal term* in addition to the true potential. The radial equation becomes especially familiar by making the substitution

$$u(r) = r\mathcal{R}(r)$$

such that

$$-\frac{\hbar^2}{2m}\frac{du(r)}{dr} + V_{\text{eff}}(r)u(r) = Eu(r)$$

This version is easy to remember because it is almost the same as the one-dimensional Schrödinger equation (usually written in terms of x with $-\infty < x < \infty$). The only differences are the presence of the centrifugal potential in the radial Schrödinger equation and the fact that r lies in the range $0 < r < \infty$. This is completely analogous to the classical particle in a central potential—due to rotational symmetry and conservation of angular momentum, the equation of motion for a single particle is effectively one dimensional despite having three degrees of freedom.

8.5 Two-Body Central Force Motion

The Hamiltonian for two bodies with a potential energy of interaction that depends on the magnitude of the distance separating the two bodies is given by

$$H = \frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} + V\big(|\mathbf{r}_1 - \mathbf{r}_2|\big)$$

where $\mathbf{p}_1$ is the momentum for particle 1 and $\mathbf{p}_2$ is the momentum for particle 2. Technically, a system made of two particles is described by a state living in the tensor product of the individual single-particle vector spaces. A two-particle Hamiltonian is therefore an operator on this tensor product space. We will explore tensor products later in the course (or in Phys 438B) when we study multiparticle quantum systems and/or inclusion of spin states.

The two-body position basis states are described by the state vector $|\mathbf{r}_1, \mathbf{r}_2\rangle$, and the position-space wave function for the system in the state $|\Psi\rangle$ is

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \Psi \rangle = \Psi(\mathbf{r}_1, \mathbf{r}_2)$$

We call $\Psi(\mathbf{r}_1, \mathbf{r}_2)$ the *configuration space wave function* because it depends on the entire configuration of the system. The wave function therefore depends on a $3 + 3 = 6$ dimensional configuration space, while the particles are each moving in a three-dimensional space.

We can define a total translation operator $T(\mathbf{a})$ that translates both particles equally:

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \, T(\mathbf{a}) \, | \Psi \rangle = \Psi(\mathbf{r}_1 - \mathbf{a}, \mathbf{r}_2 - \mathbf{a})$$

The corresponding generator of total translation is the total momentum

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$$

The Hamiltonian commutes with all translations because the distance between the particles is not changed when both particles are displaced by the same amount. It follows that H commutes with the total momentum.

The Hamiltonian is invariant under rotations. The total rotation operator acts on the configuration space wave function according to

$$\langle \mathbf{r}_1, \mathbf{r}_2 | \, U(R) \, | \Psi \rangle = \Psi(R^{-1}\mathbf{r}_1, R^{-1}\mathbf{r}_2)$$

and the corresponding generator of rotations is the total angular momentum

$$\mathbf{L} = \mathbf{L}_1 + \mathbf{L}_2 \quad \text{where } \mathbf{L}_k = \mathbf{r}_k \times \mathbf{p}_k$$

The Hamiltonian commutes with all rotations and hence with the total angular momentum.

To solve the eigenvalue equation $H\Psi = E\Psi$, we first introduce a change of coordinates. The natural coordinates for the two-body central force problem are relative coordinates $\mathbf{r}$ and the center of mass coordinates $\mathbf{R}$. The corresponding position operators are

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$$

$$\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{M}$$

where $M = m_1 + m_2$ is the total mass of the two-particle system. We also define the corresponding generators of translation $\mathbf{p}$ and $\mathbf{P}$, corresponding to translation of $\mathbf{r}$ and $\mathbf{R}$ respectively, which can be written in the position representation as

$$\mathbf{p} = -i\hbar\nabla_{\mathbf{r}} \quad \text{and} \quad \mathbf{P} = -i\hbar\nabla_{\mathbf{R}}$$

By the multidimensional chain rule, we can relate these operators to $\mathbf{p}_1$ and $\mathbf{p}_2$ as

$$\mathbf{p}_1 = -i\hbar\nabla_1 = -i\hbar\left(\nabla_1\mathbf{R}\cdot\nabla_{\mathbf{R}} + \nabla_1\mathbf{r}\cdot\nabla_{\mathbf{r}}\right) = -i\hbar\left(\frac{m_1}{M}\nabla_{\mathbf{R}} + \nabla_{\mathbf{r}}\right)$$

$$\mathbf{p}_2 = -i\hbar\nabla_2 = -i\hbar\left(\nabla_2\mathbf{R}\cdot\nabla_{\mathbf{R}} + \nabla_2\mathbf{r}\cdot\nabla_{\mathbf{r}}\right) = -i\hbar\left(\frac{m_2}{M}\nabla_{\mathbf{R}} - \nabla_{\mathbf{r}}\right)$$

In case this application of the chain rule looks unfamiliar, let's write out the x component of $\mathbf{p}_1$ and use a test wave function $\Psi(\mathbf{R}, \mathbf{r})$:

$$\frac{\partial\Psi}{\partial x_1} = \sum_{i=1}^{3}\left(\frac{\partial R_i}{\partial x_1}\frac{\partial\Psi}{\partial R_i} + \frac{\partial r_i}{\partial x_1}\frac{\partial\Psi}{\partial r_i}\right)$$

Since R_1 and r_1 are the only components that depend on x_1, we find

$$\frac{\partial \Psi}{\partial x_1} = \frac{m_1}{M}\frac{\partial \Psi}{\partial R_1} + \frac{\partial \Psi}{\partial r_1}$$

and hence,

$$\hat{p}_{1x} = -i\hbar\left(\frac{m_1}{M}\frac{\partial}{\partial R_1} + \frac{\partial}{\partial r_1}\right)$$

which is exactly the x component of the expression above. Thus, we've established that

$$\mathbf{p}_1 = \frac{m_1}{M}\mathbf{P} + \mathbf{p}$$

$$\mathbf{p}_2 = \frac{m_2}{M}\mathbf{P} - \mathbf{p}$$

Solving for $\mathbf{P}$ and $\mathbf{p}$, it follows that

$$\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2$$

$$\mathbf{p} = \frac{m_2\mathbf{p}_1 - m_1\mathbf{p}_2}{M}$$

Evidently, the operator $\mathbf{P}$ exactly coincides with the generator of translations defined earlier for the two-particle system. The transformation $(\mathbf{r}_1, \mathbf{r}_2) \mapsto (\mathbf{R}, \mathbf{r})$ is an example of a "canonical transformation" in quantum mechanics. A canonical transformation is one that preserves the canonical commutation relations:

$$[R_i, P_j] = [r_i, p_j] = i\hbar\delta_{ij} \qquad [R_i, r_j] = [R_i, p_j] = [P_i, r_j] = [P_i, p_j] = 0$$

where r_i, R_i, p_i, P_i refer to the components of $\mathbf{r}$, $\mathbf{R}$, $\mathbf{p}$, and $\mathbf{P}$, respectively.

We can now transform the Hamiltonian to the new set of coordinates. The potential energy operator becomes $V(|\mathbf{r}_1 - \mathbf{r}_2|) = V(|\mathbf{r}|) = V(r)$ where r is the distance between the two particles. The kinetic energy operators take a bit more work:

$$\mathbf{p}_1^2 = \frac{m_1^2}{M^2}\mathbf{P}^2 + \mathbf{p}^2 + \frac{m_1}{M}\mathbf{P}\cdot\mathbf{p} + \frac{m_1}{M}\mathbf{p}\cdot\mathbf{P}$$

where $\mathbf{p}$ commutes with $\mathbf{P}$, so

$$\mathbf{p}_1^2 = \frac{m_1^2}{M^2}\mathbf{P}^2 + \mathbf{p}^2 + \frac{2m_1}{M}\mathbf{P}\cdot\mathbf{p}$$

A similar expression holds for $\mathbf{p}_2^2$. The kinetic energy operator is

$$\frac{\mathbf{p}_1^2}{2m_1} + \frac{\mathbf{p}_2^2}{2m_2} = \frac{\mathbf{P}^2}{2M} + \frac{\mathbf{p}^2}{2\mu}$$

where μ denotes the *reduced mass* defined as

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

The Hamiltonian can be written as the sum of a center of mass term and a relative term,

$$H = H_{\rm CM} + H_{\rm rel} = \frac{\mathbf{P}^2}{2M} + \frac{\mathbf{p}^2}{2\mu} + V(r)$$

where

$$H_{\rm CM} = \frac{\mathbf{P}^2}{2M}$$

and

$$H_{\rm rel} = \frac{\mathbf{p}^2}{2\mu} + V(r)$$

The Hamiltonian $H_{\rm CM}$ is a free particle Hamiltonian–the kinetic energy of the center of mass–while $H_{\rm rel}$ is the kinetic energy about the center of mass plus the potential energy. Since $[H_{\rm CM}, H_{\rm rel}] = 0$, the Hamiltonians share an eigenbasis. The eigenstates of $H_{\rm CM}$ are those of the total momentum operator. In position space,

$$\langle \mathbf{R}|\mathbf{P}\rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{P}\cdot\mathbf{R}/\hbar}$$

are the eigenstates of $H_{\rm CM}$. In the exponential, $\mathbf{P}$ is a vector of eigenvalues (P_x, P_y, P_z), not the momentum operator, and the center of mass position vector $\mathbf{R}$ has Cartesian components (R_1, R_2, R_3). It is common to analyze the two body problem in the center of mass frame where $\mathbf{P} = 0$. From now on, we'll concentrate on the relative Hamiltonian

$$H = \frac{\mathbf{p}^2}{2\mu} + V(r)$$

which is nearly identical to the Hamiltonian for a single-particle in a central potential. When the mass of one particle is much greater than the other, say $m_2 \gg m_1$, the reduced mass simplifies to the mass of the lighter particle, the relative position vector simplifies to the position of the lighter particle, and the relative momentum simplifies to the momentum of the lighter particle.

8.6 Normalization of Stretched State

$$1 = \int d\Omega\ |f_{\ell\ell}|^2 = \int_0^\pi \sin\theta\, d\theta \int_0^{2\pi} d\phi\ |A|^2 \sin^{2\ell+1}\theta$$

$$1 = 2\pi |A|^2 \int_0^\pi \sin^{2\ell+1}\theta\ d\theta$$

Consider $I_n = \int_0^\pi \sin^n\theta d\theta$ and do integration by parts with $u = \sin^{n-1}\theta$ and $dv = \sin\theta d\theta$.

$$I_n = \left[-\sin^{n-1}\theta\ \cos\theta\right]_0^\pi + (n-1)\int_0^\pi \sin^{n-2}\theta\ \cos^2\theta\, d\theta$$

First term vanishes at the boundaries and $\cos^2\theta = 1 - \sin^2\theta$, so it follows that

$$I_n = (n-1)I_{n-2} - (n-1)I_n \Rightarrow I_{n-2} = \frac{n}{n-1}I_n$$

Starting with $n = 3$, repeated application up to $n = 2\ell + 1$ gives

$$I_1 = \frac{3}{2}I_3 = \frac{3\cdot 5}{2\cdot 4}I_5 = \frac{3\cdot 5\cdot 7\cdots(2\ell+1)}{2\cdot 4\cdot 6\cdots(2\ell)}I_{2\ell+1}$$

Since $I_1 = \int_0^\pi \sin\theta\, d\theta = 2$,

$$I_{2\ell+1} = 2\prod_{k=1}^{\ell}\frac{2k}{2k+1}$$

$$\prod_{k=1}^{\ell} 2k = 2^\ell \ell!$$

$$\prod_{k=1}^{\ell}(2k+1) = 1\cdot 3\cdot 5\cdots(2\ell+1) = \frac{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6\cdots(2\ell+1)}{2\cdot 4\cdot 6} = \frac{(2\ell+1)!}{2^\ell \ell!}$$

$$I_{2\ell+1} = \frac{2(2^\ell l!)^2}{(2\ell+1)!} = \frac{2^{2\ell+1}(l!)^2}{(2\ell+1)!}$$

$$A = \frac{(-1)^\ell}{2^\ell \ell!}\sqrt{\frac{(2\ell+1)!}{4\pi}}$$

where the phase factor of $(-1)^\ell$ is conventional.

Chapter 9

The Hydrogen Atom

9.1 The Electrostatic Model

In this chapter, we use the *electrostatic model* for the hydrogen atom, in which the electron and proton interact by an electrostatic force determined by the instantaneous positions of the particles. The electrostatic model is an approximation to electrodynamics in which one neglects the delay in the communication between the particles, magnetic effects, and the effects of radiation. As a part of the electrostatic model we shall also assume the electron motion is non-relativistic, something that is justified in the case of hydrogen.

Under these assumptions, the potential energy of interaction between the electron and the proton is

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r} = -\frac{k_e e^2}{r}$$

and the corresponding radial equation is

$$-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2} + \left[\frac{\hbar^2}{2\mu}\frac{\ell(\ell+1)}{r^2} - \frac{k_e e^2}{r}\right]u = Eu$$

where μ is the reduced mass of the electron-proton system

$$\mu = \frac{m_e m_p}{m_e + m_p}$$

To a good approximation, we could take $\mu = m_e$ since the proton is much heavier than the electron. In fact, the ratio of the proton mass to the electron mass is roughly 1836, which is suspiciously close to the pure number $6\pi^5$. The reduced mass differs from the mass of the electron by about 5 parts in 10^4, which is within the limits of experimental precision.

The Coulomb potential admits continuum states (in this case with $E > 0$) and discrete states (with $E < 0$). The Hamiltonian therefore has a mixed spectrum, which implies the bound states are not complete on their own. For now, we are only interested in the bound states since they correspond to a stable hydrogen atom. We'll begin by solving the radial equation, and along the

way, we'll discover the bound state energies match the semi-classical Bohr model you may have learned about in a previous class. This chapter will conclude with a discussion of the hydrogen wave functions.

9.2 The Radial Equation

It is convenient to work with natural units after we identify the relevant physical scales of the hydrogen atom. This may be accomplished either by dimensional analysis or by carefully inspecting the radial equation. Natural units will also make it easier to simulate the hydrogen atom with a computer. The radial equation may be written as

$$\frac{d^2u}{dr^2} + \left[\frac{2\mu E}{\hbar^2} - \frac{\ell(\ell+1)}{r^2} + \frac{2\mu k_e e^2}{\hbar^2}\frac{1}{r}\right] u = 0$$

Each term in the brackets above has units of $1/\text{length}^2$, so let's define a characteristic distance based on the physical quantities that appear in the third term specifically:

$$a \equiv \frac{\hbar^2}{\mu k_e e^2}$$

The distance a is very close to the Bohr radius which is 0.529 Å. Let's use natural units where $\hbar = a = 2\mu = 1$. The energy and momentum scales are then given by

$$[\text{energy}] = \frac{\hbar^2}{2\mu a^2}, \qquad [\text{momentum}] = \frac{\hbar}{a}$$

Lastly, since $E < 0$, we can define a positive constant γ such that

$$\gamma \equiv \sqrt{-E} \quad \Rightarrow \quad E = -\gamma^2$$

In terms of dimensionless parameters, the radial equation becomes

$$\frac{d^2u}{dr^2} + \left[\frac{2}{r} - \frac{\ell(\ell+1)}{r^2} - \gamma^2\right] u = 0$$

9.2.1 Asymptotic Solutions

Figuring out solutions for large and small r is interesting in its own right, and our knowledge of the asymptotic solutions will allow us to simplify the differential equation. We also need to ensure that our solutions are normalizable; an asymptotic solution is the first place to look for any pathological behavior. For large r, the radial equation is approximately

$$\frac{d^2u}{dr^2} = \gamma^2 u$$

The general solution for large r is $u = Ae^{-\gamma r} + Be^{\gamma r}$, but the wave function must vanish at $r \to \infty$, so we must choose $B = 0$. For large r, we find $u = Ae^{-\gamma r}$. Alternatively, when r is small, the centrifugal term dominates, and the radial equation in this limit is

$$\frac{d^2u}{dr^2} = \frac{\ell(\ell+1)}{r^2}u$$

Monomial functions have the property that $(r^s)' = sr^{s-1}$, so with $u = r^s$,

$$s(s-1)u^{s-2} = \ell(\ell+1)u^{s-2} \to s(s-2) = \ell(\ell+1)$$

The general solution for small r is $u = Cr^{\ell+1} + Dr^{-\ell}$. The second term grows without bound for small r (assuming $\ell > 0$), so we must choose $D = 0$. The solution for small r is then $u = Cr^{\ell+1}$. Now that we know the asymptotic solutions, we can find the solution for intermediate r by solving for a function $v(r)$ such that

$$u(r) = r^{\ell+1}e^{-\gamma r}v(r) \quad \text{or} \quad \mathcal{R}(r) = r^{\ell}e^{-\gamma r}v(r)$$

After substituting $u(r)$ as written above into the differential equation (this was quite tedious; you should take moment to make sure it's correct),

$$r\frac{d^2v}{dr^2} + 2(1+\ell-\gamma r)\frac{dv}{dr} + 2(1-\gamma-\gamma\ell)v = 0$$

9.2.2 Series Solution Method

We will use a power series expansion to solve the radial equation for $v(r)$. Assume

$$v(r) = \sum_{k=0}^{\infty} c_k r^k$$

where c_k are coefficients to be determined. While this approach is powerful, it demands caution. One must take care when using the series solution method. The differential equation we aim to solve is *stubborn*. For large r,

$$\frac{d^2v}{dr^2} - 2\gamma\frac{dv}{dr} = 0$$

The non-trivial solution to this asymptotic equation is $e^{2\gamma r}$, and furthermore,

$$u(r) = r^{\ell+1}e^{-\gamma r}e^{2\gamma r} \propto e^{\gamma r}$$

It seems like the solution really wants to grow without bound for large r, and we must eliminate this possibility on physical grounds. If we require the series to terminate for a finite value of the index, we can remove the exponential solution.

After substituting the series solution,

$$\sum_k k(k+1)c_{k+1}r^k + 2(\ell+1)\sum_k (k+1)c_{k+1}r^k$$
$$-2\gamma\sum_k kc_k r^k + 2(1-\gamma-\gamma\ell)\sum_k c_k r^k = 0$$

In order for all terms of the series to sum to zero for any and all values of r, the coefficient of each power of r must vanish. Hence,

$$k(k+1)c_{k+1} + 2(\ell+1)(k+1)c_{k+1} - 2\gamma k c_k + 2(1-\gamma-\gamma\ell)c_k = 0$$

$$c_{k+1} = \frac{2\gamma(k+\ell+1)-2}{(k+1)(k+2\ell+2)}c_k$$

This is a recurrence relation for the coefficients; the starting coefficient c_0 determines all of the remaining coefficients in the function $v(r)$. Ultimately, c_0 is determined by normalization.

9.2.3 Termination Condition

Notice that for large values of k,

$$c_{k+1} \approx \frac{2\gamma}{k+1}c_k \Rightarrow c_k = \frac{(2\gamma)^k}{k!}$$

which is the same recurrence relation for $e^{2\gamma r}$. There must exist $k_{\max}$ such that $c_{k_{\max}+1} = 0$. From the recurrence relation, we can see that the coefficients will vanish if

$$\gamma(k_{\max}+\ell+1) - 1 = 0$$

Since $k_{\max}$ and ℓ are integers, it follows that

$$n = k_{\max} + \ell + 1 = 1, 2, 3, \ldots$$

is also an integer, and by the termination condition,

$$\gamma = \frac{1}{n}$$

Enforcing the boundary conditions on the radial wave function has lead to quantization of energy! More precisely, we find that

$$E = -\gamma^2 = -\frac{1}{n^2} \quad \text{(natural units)}$$

$$= -\frac{\hbar^2}{2\mu a^2}\frac{1}{n^2} \quad \text{(ordinary units)}$$

The boundary conditions also constrain the values of ℓ. For a given n, $\ell = n - k_{\max} - 1$ has a maximum value when $k_{\max} = 0$, so $\ell_{\max} = n - 1$. We now have all the "quantum numbers" of the hydrogen atom! Let's summarize what we know so far:

1. **Principal quantum number:** $n = 1, 2, 3, \ldots, \infty$ (also called the shell number)
2. **Angular momentum quantum number:** $\ell = 0, 1, 2, \ldots n - 1$
3. **Magnetic quantum number:** $m = -\ell, -\ell + 1, \ldots, 0, \ldots, \ell - 1, \ell$

The magnetic quantum number is named after its relevance to the splitting of spectral lines when the atom interacts with a magnetic field.

9.3 Hydrogen Energies and Spectra

Approximating the reduced mass as the mass of the electron alone, we arrive at the following set of discrete energies labeled by the principle quantum number:

$$E_n = -\frac{\hbar^2}{2\mu a^2}\frac{1}{n^2} = -\frac{m_e c^2}{2n^2}\left(\frac{k_e e^2}{\hbar c}\right)^2 = -\frac{\alpha^2}{2n^2} m_e c^2$$

where $\alpha = 1/137$ is the **fine structure constant**. In this form, it is easy to see that the result has units of energy, roughly on the order of 10 eV. Somewhat more precisely,

$$\boxed{E_n = -\frac{13.6 \text{ eV}}{n^2} = -\frac{1}{n^2}\text{Ryd}}$$

where Ryd stands for "Rydberg." Here are some important features of the hydrogen energies:

1. There are an infinite number of bound states.
2. The energy levels are degenerate. For each energy E_n, there are n possible ℓ states and for each ℓ, there are $2\ell + 1$ possible m states. The total number of states at each energy level E_n is the sum of these possibilities:

$$\sum_{\ell=0}^{n-1}(2\ell + 1) = n^2$$

   If we include the spin of the electron, the total number is $2n^2$. The m-degeneracy is related to spherical symmetry while ℓ-degeneracy is related to a special dynamical symmetry of the $1/r$ potential (Runge-Lenz symmetry). Roughly speaking, m-degeneracy reflects no preferred direction in space, while the ℓ-degeneracy reflects the fact that in a $1/r$ potential, different orbital shapes can have the same energy.

Hydrogen atoms absorb or emit light when electrons transition between energy levels. The photon energy always matches the energy difference:

$$E_{\text{photon}} = \Delta E = |E_f - E_i|$$

The energy of a photon is related to its wavelength:

$$E_{\text{photon}} = \hbar\omega = hf = \frac{hc}{\lambda}$$

Hence, the wavelengths of light that can be emitted or absorbed by hydrogen are given by

$$\boxed{\frac{1}{\lambda} = R_\infty \left| \frac{1}{n_i^2} - \frac{1}{n_f^2} \right|}$$

where

$$R_\infty = \frac{\alpha^2 m_e c^2}{4\pi\hbar c} = \frac{m_e e^4}{8\varepsilon_0 h^2 c} \approx 10.97 \times 10^6 \text{ m}^{-1}$$

Not all transitions are allowed. Conservation of angular momentum leads to the following **selection rules for electric-dipole transitions:**

$$\Delta\ell = \ell_f - \ell_i = \pm 1$$

$$\Delta m = m_f - m_i = 0, \pm 1$$

The photon itself has spin angular momentum (helicity) of 1, so the atom's angular momentum must change as a result of absorption or emission of a single photon.

9.4 Radial Wave Function

The radial wave function with dimensions back in place is

$$\mathcal{R}_{n\ell}(r) = \left(\frac{r}{a}\right)^\ell e^{-r/na} v(r/a)$$

where

$$v(r/a) = \sum_{k=0}^{n-\ell-1} c_k (r/a)^k$$

and

$$c_{k+1} = \frac{2\gamma(k+\ell+1) - 2}{(k+1)(k+2\ell+2)} c_k$$

9.4.1 Example: Ground State

The state with lowest energy, the ground state of the hydrogen atom, is labeled by $n = 1$ and $\ell = 0$. The ground state radial wave function is

$$\mathcal{R}_{10}(r) = c_0 e^{-r/a}$$

where c_0 is determined by normalization of the radial wave function:

$$1 = \int_0^\infty dr\, r^2 |\mathcal{R}_{10}(r)|^2 = \int_0^\infty dr\, r^2 |c_0|^2 e^{-2r/a} = |c_0|^2 \cdot \frac{a^3}{4}$$

The full ground state radial wave function can be written as

$$\mathcal{R}_{10}(r) = \frac{2}{a^{3/2}} e^{-r/a}$$

9.4.2 Example: Excited States

The following states all have the same energy:

$$|n, \ell, m\rangle = \Big\{ |2,0,0\rangle\,, |2,1,-1\rangle\,, |2,1,0\rangle\, |2,1,1\rangle \Big\}$$

For the state $|2,0,0\rangle$, $k_{\max} = 1$, and $v(r/a) = c_0 + c_1(r/a)$. By the recursion relation,

$$c_1 = -\frac{1}{2} c_0$$

so the radial wave function is

$$\mathcal{R}_{20}(r) = c_0 e^{-r/2a}(1 - r/2a)$$

For states with $n = 2$ and $\ell = 1$, $k_{\max} = 0$, so

$$\mathcal{R}_{21}(r) = c_0 r e^{-r/2a}$$

Note that in each of the expressions above, the term c_0 is particular to that solution–they are not the same, but they are all determined by normalization.

The radial wave functions can be written in terms of special functions called associated Laguerre polynomials $L_q^p(x)$, where q indicates the degree of the polynomial. In terms of associated Laguerre polynomials, the radial wave functions are

$$\boxed{\mathcal{R}_{n\ell}(r) = \sqrt{\left(\frac{2}{na}\right)^3 \frac{(n-\ell-1)!}{2n(n+\ell)!}} \left(\frac{2r}{na}\right)^\ell \left[L_{n-\ell-1}^{2\ell+1}(2r/na)\right] e^{-r/na}}$$

where

$$L_q^p(x) = \frac{x^{-p} e^x}{q!} \frac{d^q}{dx^q} \left(e^{-x} x^{q+p}\right) = (-1)^p \frac{d^p}{dx^p} L_{p+q}(x)$$

and $L_q(x)$ are the ordinary Laguerre polynomials defined as

$$L_q(x) = \frac{e^x}{q!} \frac{d^q}{dx^q} \left(e^{-x} x^q\right)$$

The radial wave function $R_{n\ell}$ has $n - \ell - 1$ nodes and $n - \ell$ antinodes. For a given n, $R_{n\ell}$ has fewer

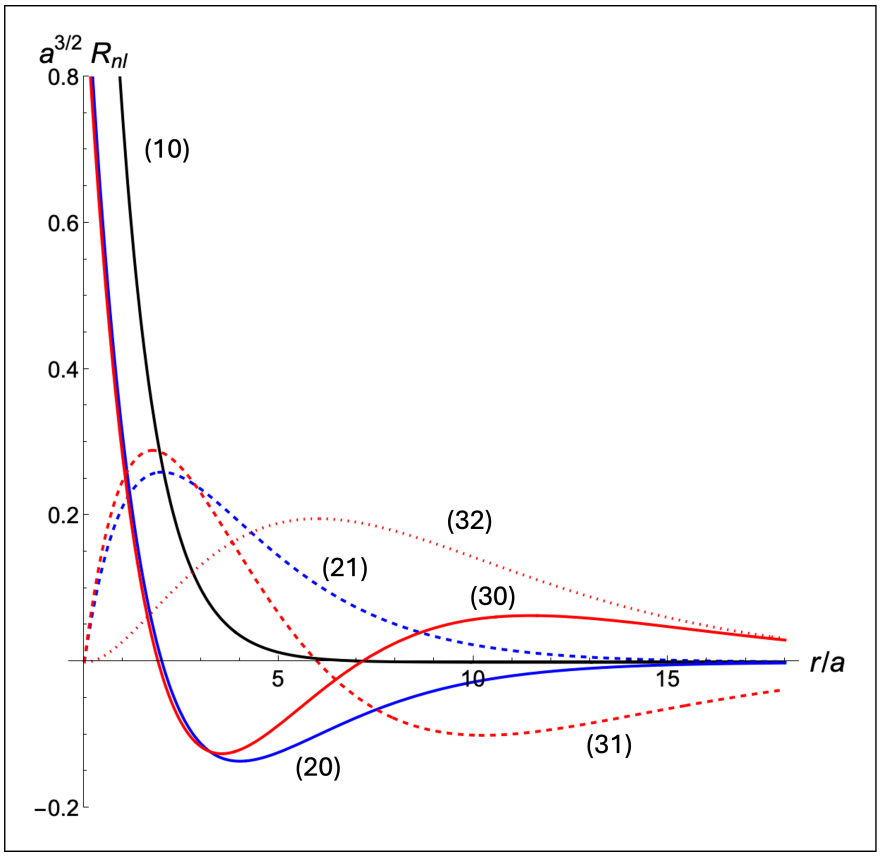


Figure 9.1: Radial wave functions for the ground state (10) and the first few excited states.

nodes for higher ℓ states, and the angular wave functions "compensate" by having more nodes. See Figure 9.1 for a visualization of the radial wave functions.

9.5 Full Hydrogen Wave Functions

We can recombine the three separated parts of the wave function to form the full three- dimensional energy eigenstate wave functions of the hydrogen atom:

$$\boxed{\psi_{n\ell m}(r,\theta,\phi) = \mathcal{R}_{n\ell}(r) Y_\ell^m(\theta,\phi)}$$

These states are also eigenstates of the angular momentum operators L^2 and L_z; they are simultaneous eigenstates of H, L^2, and L_z altogether. The three eigenvalue equations are

$$H\psi_{n\ell m}(r,\theta,\phi) = -\frac{13.6 \text{ eV}}{n^2}\,\psi_{n\ell m}(r,\theta,\phi)$$

$$L^2\psi_{n\ell m}(r,\theta,\phi) = \ell(\ell+1)\hbar^2\,\psi_{n\ell m}(r,\theta,\phi)$$

$$L_z\psi_{n\ell m}(r,\theta,\phi) = m\hbar\,\psi_{n\ell m}(r,\theta,\phi)$$

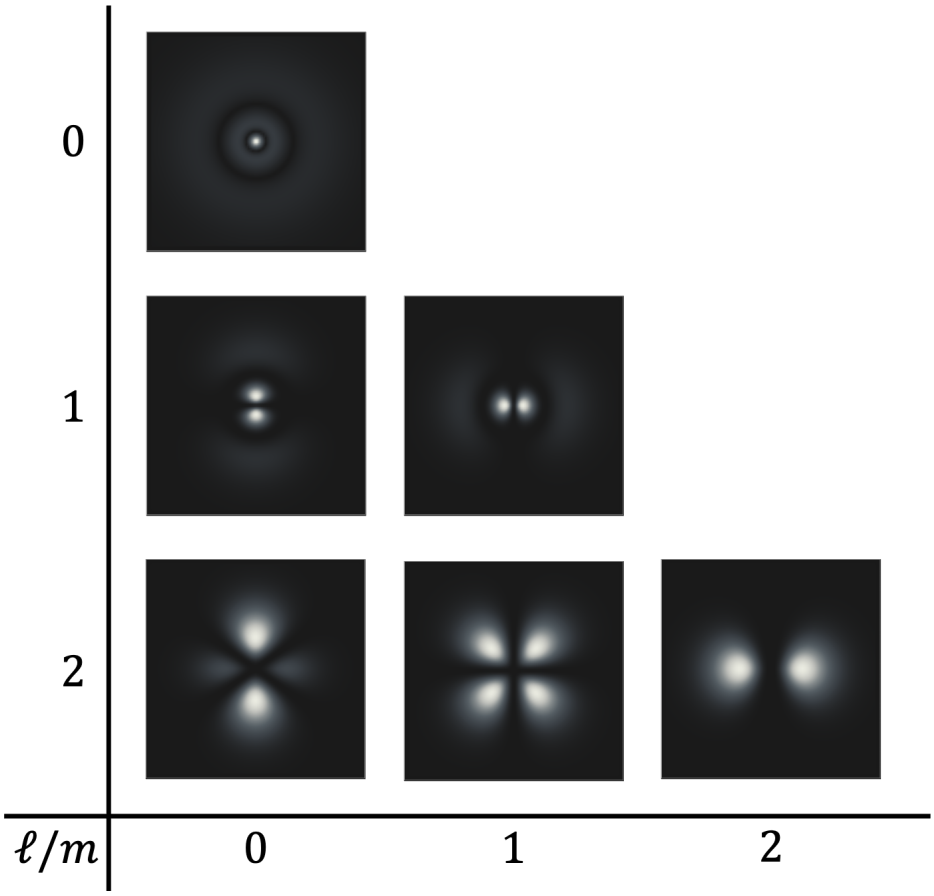


Figure 9.2: Hydrogen atom probability densities $|\psi_{n\ell m}|^2$ for $n = 3$, shown in the xz-plane ($y = 0$) passing through the nucleus at the origin. Rows correspond to $\ell = 0, 1, 2$ (top to bottom) and rows to $m = 0, 1, 2$ (left to right where allowed). Bright regions indicate high probability density. Radial nodes (spherical shells) are given by $n - \ell - 1$: the $3s$ state ($\ell = 0$) has 2 radial nodes, $3p$ ($\ell = 1$) has 1 radial node, and $3d$ ($\ell = 2$) has no radial nodes. Angular nodes are determined by ℓ: 0 for s, 1 for p (a nodal plane), and 2 for d states (combinations of nodal planes/cones), visible as directions where the density vanishes.

The normalization condition for the full wave function is the three-dimensional integral

$$\begin{aligned} 1 &= \int d^3r \, |\psi_{n\ell m}(r, \theta, \phi|^2 \\ &= \left\{ \int_0^\infty dr \, r^2 |\mathcal{R}_{n\ell}(r)|^2 \right\} \left\{ \int_0^\pi \int_0^{\pi/2} d\theta d\phi \, \sin\theta |Y_\ell^m(\theta, \phi)|^2 \right\} \end{aligned}$$

See Figure 9.2 for a static visualization of the hydrogen wave functions.

9.6 Summary

For the bound states of hydrogen, the spatial wavefunctions have these properties:

1. They separate into radial and angular parts:

$$\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_{\ell m}(\theta,\phi).$$

2. They are simultaneous eigenfunctions of the hydrogen Hamiltonian, L^2, and L_z, with quantum numbers

$$n = 1, 2, \ldots, \qquad \ell = 0, 1, \ldots, n-1, \qquad m = -\ell, \ldots, \ell.$$

3. They are normalized and orthogonal:

$$\int \psi^*_{n\ell m}\psi_{n'\ell'm'}\, d^3r = \delta_{nn'}\delta_{\ell\ell'}\delta_{mm'}.$$

4. Their energies depend only on n:

$$E_n = -\frac{13.6 \text{ eV}}{n^2},$$

so each energy level has n^2 spatial states, ignoring spin.

5. Their parity under spatial inversion $\mathbf{r} \to -\mathbf{r}$ is $(-1)^\ell$:

$$\psi_{n\ell m}(-\mathbf{r}) = (-1)^\ell \psi_{n\ell m}(\mathbf{r})$$

6. A state has $n - \ell - 1$ radial nodes and ℓ angular nodes; the total number of nodes is $n - 1$.

7. Near the origin, the radial part behaves as $R_{n\ell}(r) \propto r^\ell$; bound-state wavefunctions decay exponentially at large r.

8. A stationary state's full time-dependent wavefunction is $\psi_{n\ell m}(\mathbf{r})e^{-iE_nt/\hbar}$, so its probability density is time-independent.

Chapter 10

Quantum Foundations

10.1 Introduction

Quantum foundations is the part of physics that asks not just *how to calculate* with quantum mechanics, but *what the theory is really saying about the world*—about reality, causality, probability, space, and time. It probes whether quantum mechanics offers a complete description of physical systems, or whether deeper principles or structures might lie beneath its remarkably successful formalism. Quantum foundations sits at the intersection of physics, mathematics, and philosophy, and over the last few decades it has become an increasingly sharp, experimentally-driven science.

Furthermore, it has repeatedly generated *new science and technology* rather than remaining a purely interpretive enterprise. John Bell's analysis of locality, originally aimed at clarifying the implications of entanglement, led to device-independent quantum cryptography and certified randomness. The 2022 Nobel prize in physics was awarded jointly to Alain Aspect, John F. Clauser, and Anton Zeilinger "for experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science." Efforts to understand measurement and classical emergence through *decoherence*, a concept invented by David Bohm of Bohmian mechanics or pilot-wave theory, underpin modern noise models and error correction. David Deutsch's many-worlds–inspired question about computation in a quantum universe launched quantum computing itself. The history of quantum foundations shows that probing the *conceptual structure* of a theory can reshape both our picture of the world and our technological reach.

We begin in Sec. 2 by introducing the Einstein–Podolsky–Rosen (EPR) argument in Bohm's spin formulation, reviewing the addition of two spin-1/2 systems and deriving the singlet state. This leads naturally to the notion of *entanglement*. In Sec. 3 we turn to Bell's theorem: we exploit the rotational invariance of the singlet state to compute the quantum spin correlation function, and then contrast this prediction with what would follow from local hidden-variable models using the "instruction set" picture. The resulting contradiction highlights the tension between locality and hidden variables and sets the stage for a discussion of modern Bell tests and the no-communication theorem.

10.2 Einstein-Padolsky-Rosen (EPR)

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen published a paper [1] presenting a thought experiment designed to expose the shortcomings of quantum mechanics. They considered two particles with *perfectly correlated positions and momenta* and claimed that, if measurements on one cannot instantaneously affect the other, then the distant particle must possess definite values of both quantities, implying that quantum mechanics is incomplete. By incomplete, they mean to reject the idea that the state vector (or wave function) is a complete description of the system. There should exist additional variables (later on, these were called hidden variables) that specify the state of the system.

David Bohm reformulated EPR using two spin-1/2 particles, making the argument cleaner and experimentally accessible[1]. In the EPRB thought experiment, an unstable particle with spin 0 decays into two spin-1/2 particles, which by conservation of angular momentum must have opposite spin components and by conservation of linear momentum must travel in opposite directions. For example, a neutral pi meson decays into an electron and a positron: $\pi^0 \to e^- + e^+$. Observers A and B are on opposite sides of the decaying particle and each has a Stern-Gerlach apparatus to measure the spin component of the particle headed in its direction. Whenever one observer measures spin up along a given direction, then the other observer measures spin down along that same direction.

10.2.1 Adding Two Spin-1/2 Particles

Before it decays, the neutral pion is in the spin angular momentum state $|s = 0, m = 0\rangle$. Since angular momentum is conserved in all fundamental interactions, this must be the spin angular momentum state of the composite system comprised of two spin-1/2 particles. The first postulate of quantum mechanics states that all systems have an associated state-vector space, and the goal of this section is to explore the combined spin state of two spin-1/2 particles. The particles have spin operators $\mathbf{S}_1$ and $\mathbf{S}_2$. We start with the *uncoupled basis*, which simultaneously diagonalizes the commuting operators S_{1z} and S_{2z}. Conceptually, it is possible for observer A or B to measure either spin up or spin down in their apparatus, and the possible measurement outcomes can be enumerated as

$$\{(\uparrow,\uparrow),(\uparrow,\downarrow),(\downarrow,\uparrow),(\downarrow,\downarrow)\}$$

where the first item in each tuple indicates the spin measured by A and the second by B.

The composite system must allow you to describe all joint measurement outcomes, allow independent operations on each part, and reduce to the usual description when the systems don't interact. The mathematical structure that does exactly this is the *tensor product*.

[1] Technically, in order to use an S-G apparatus, the particles must be neutral. Otherwise the Lorentz force screws up the measurement. This is why modern Bell tests use entangled photons.

The **tensor product** of two vector spaces is itself a vector space whose elements are linear combinations of product states denoted $|\psi\rangle \otimes |\phi\rangle$ satisfying bilinearity:

$$\Big(a\,|\psi_1\rangle + b\,|\psi_2\rangle\Big) \otimes |\phi\rangle = a\,|\psi_1\rangle \otimes |\phi\rangle + b\,|\psi_2\rangle \otimes |\phi\rangle$$
$$|\psi\rangle \otimes \Big(c\,|\phi_1\rangle + d\,|\phi_2\rangle\Big) = c\,|\psi\rangle \otimes |\phi_1\rangle + d\,|\psi\rangle \otimes |\phi_2\rangle$$

The inner product in a tensor product space is defined multiplicatively as

$$\langle \psi_1 \otimes \phi_1 | \psi_2 \otimes \phi_2 \rangle = \langle \psi_1 | \psi_2 \rangle \, \langle \phi_1 | \phi_2 \rangle$$

Lastly, for linear operators A on the first space and B on the second, the tensor product $A \otimes B$ is the unique linear operator satisfying

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = (A\,|\psi\rangle) \otimes (B\,|\phi\rangle)$$

Note that there exist vectors in the tensor product space that cannot be written as product states, and likewise, there are linear operators which may not be expressed as a tensor product of two linear operators acting on each space as above.

In the case of two spin-1/2 particles, the product states of the uncoupled basis are

$$|{+}{+}\rangle = |{+}z\rangle_1 \otimes |{+}z\rangle_2$$
$$|{+}{-}\rangle = |{+}z\rangle_1 \otimes |{-}z\rangle_2$$
$$|{-}{+}\rangle = |{-}z\rangle_1 \otimes |{+}z\rangle_2$$
$$|{-}{-}\rangle = |{-}z\rangle_1 \otimes |{-}z\rangle_2$$

and the operators S_{1z} and S_{2z} can be written as

$$S_{1z} = S_z \otimes \mathbb{1} \quad \text{and} \quad S_{2z} = \mathbb{1} \otimes S_z$$

Let's verify the product states of the uncoupled basis really are eigenstates of the spin operators above. From the properties of a tensor product space, we find

$$S_{1z}\,|{+}{+}\rangle = (S_z\,|{+}z\rangle) \otimes |{+}z\rangle = \frac{\hbar}{2}\,|{+}z\rangle \otimes |{+}z\rangle = \frac{\hbar}{2}\,|{+}{+}\rangle$$

By similar calculations,

$$S_{1z}\,|{+}{\pm}\rangle = \frac{\hbar}{2}\,|{+}{\pm}\rangle\,,\; S_{1z}\,|{-}{\pm}\rangle = -\frac{\hbar}{2}\,|{-}{\pm}\rangle\,,\; S_{2z}\,|{\pm}{+}\rangle = \frac{\hbar}{2}\,|{+}{\pm}\rangle\,,\; S_{2z}\,|{\pm}{-}\rangle = -\frac{\hbar}{2}\,|{\pm}{-}\rangle\,,$$

Hence, the uncoupled basis simultaneously diagonalizes the spin operators, and it's not too sur-

prising such a basis exists given $[S_{1z}, S_{2z}] = 0$, as mentioned earlier.

Whether our system consists of a single particle or many, the theory of rotations we developed in chapter #4 can be applied to any system residing in 3D Euclidean space. On the tensor product space, we postulate a unitary transformation corresponding to local rotations of each system about the same axis by the same angle:

$$U_{\text{tot}}(\boldsymbol{\theta}) = U_1(\boldsymbol{\theta}) \otimes U_2(\boldsymbol{\theta})$$

Expanding for small $\boldsymbol{\theta}$ (you know the drill),

$$U_{\text{tot}}(\boldsymbol{\theta}) = \Big(\mathbb{1} - \frac{i}{\hbar}\boldsymbol{\theta}\cdot\mathbf{S}_1 + \cdots\Big) \otimes \Big(\mathbb{1} - \frac{i}{\hbar}\boldsymbol{\theta}\cdot\mathbf{S}_2 + \cdots\Big)$$

$$= \mathbb{1}\otimes\mathbb{1} - \frac{i}{\hbar}\boldsymbol{\theta}\cdot\big(\mathbf{S}_1\otimes\mathbb{1} + \mathbb{1}\otimes\mathbf{S}_2\big) + O(\theta^2)$$

The first term is the identity on the tensor product space, and the second term contains the generator of rotations in the tensor product space. We identify the **total spin angular momentum** as the tensor product operator

$$\mathbf{S} = \mathbf{S}_1\otimes\mathbb{1} + \mathbb{1}\otimes\mathbf{S}_2$$

often abbreviated as $\mathbf{S}_1 + \mathbf{S}_2$. It's exactly what you would have guessed, perhaps naively.

We now seek simultaneous eigenstates, the *coupled basis*, of $\mathbf{S}^2$ and S_z. Since $\mathbf{S}$ is an angular momentum, its components will satisfy the usual commutation relations, and we can simply make use of our results from chapter #4. Notice the largest value of m, related to the eigenvalue of S_z ($\hbar m$), is given by $1/2 + 1/2 = 1$. There is only one product state with this value of m, and that's $|{+}{+}\rangle$. Hence,

$$|s = 1, m = 1\rangle = |1,1\rangle = |{+}{+}\rangle$$

We can generate the remaining states with $s = 1$ by applying the lowering operator. Starting with the state $|1,1\rangle$,

$$S_-\,|1,1\rangle = (S_{1-} + S_{2-})\,|{+}{+}\rangle = \hbar\,|{-}{+}\rangle + \hbar\,|{+}{-}\rangle$$

and on the other hand, with $s = 1$ and $m = 1$,

$$S_-\,|1,1\rangle = \hbar\sqrt{2}\,|1,0\rangle$$

Hence,

$$|s = 1, m = 0\rangle = |1,0\rangle = \frac{1}{\sqrt{2}}\Big(\,|{+}{-}\rangle + |{-}{+}\rangle\,\Big)$$

Applying S_- once more, we find

$$|s = 1, m = -1\rangle = |1,-1\rangle = |{-}{-}\rangle$$

So far, we've found a "triplet" of states with $s = 1$. If we look at the possible values of m in the uncoupled basis, we only have three of the four values ($m = +1, 0, -1$, just once each), so there must be a single state with $m = 0$ remaining. This state necessarily has a total spin value of 0. Physically, there can be different spin configurations that give the same total z-projection. Since the S_z operator is two-fold degenerate in the uncoupled basis, it must be two-fold degenerate in the coupled basis. A change of basis will only map the vectors $|+-\rangle$ and $|-+\rangle$ into linear combinations of themselves. The remaining spin state of the coupled basis must have $m = 0$ and be orthogonal to $|0, 0\rangle$. The so-called "singlet" state is

$$|0, 0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle - |-+\rangle \Big)$$

In summary, we've derived the simultaneous eigenstates of $\mathbf{S}^2$ and S_z:

$$|1, 1\rangle = |++\rangle$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle + |-+\rangle \Big)$$

$$|1, -1\rangle = |--\rangle$$

$$|0, 0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle - |-+\rangle \Big)$$

We'll complete the topic of angular momentum addition in Physics 438B. In this chapter, we really only care about the singlet state. This is the combined spin state of the electron and the position after the neutral pion decays.

10.2.2 Elements of Reality

In the original EPR paper, the authors conjectured: "If, without in any way disturbing a system, we can predict with certainty the value of a physical quantity, then there exists an element of physical reality corresponding to this quantity." Suppose observer A measures S_{1z} of the electron with a suitable S-G apparatus and finds $+\hbar/2$. The measurement collapses the state to $|+-\rangle$, allowing observer A to predict with certainty that observer B will measure $-\hbar/2$ for the positron's spin along z. This perfect anti-correlation, occurring even when the measurements are spacelike separated, appears to involve "spooky action at a distance." How could the positron "know" the electron's outcome? EPR assumed that such spooky action is impossible, so they argued that the distant particle must already know what measurement result it would reveal, implying there is an element of reality not encoded in the quantum state. Therefore, they concluded that quantum mechanics is incomplete[2].

One important consequence of the EPR argument, and the work it inspired, was to force the

[2]There are interesting things to say about incompatible observables, but this is the crux of the argument.

physics community to take the concept of *entanglement* seriously. Shortly after the EPR paper was published, Schrödinger coined the term entanglement (Verschränkung) which he claimed was *the* characteristic feature of quantum mechanics (whatever that means). "The best possible knowledge of a whole does not include the best possible knowledge of its parts." Notice that the singlet state *cannot* be written as a product state. In other words, there exist no vectors $|\psi\rangle$ or $|\phi\rangle$ such that

$$\frac{1}{\sqrt{2}}\Big(|+-\rangle - |-+\rangle \Big) = |\psi\rangle \otimes |\phi\rangle \qquad \text{(impossible)}$$

According to quantum mechanics, the electron and positron are therefore *entangled*. The composite system is in a perfectly well-defined state, but neither particle can be assigned its own independent pure state. In Physics 438B, we'll learn how to assign "mixed states" to the electron and positron in situations like this.

10.3 Bell's Theorem

An important result of special relativity is the concept of *causal locality*: causal influences should not be able to travel faster than light. In 1964, John Bell wrote a paper "On the Einstein-Padolsky-Rosen Paper" [2] where he confronts us with a logical fork . We must either accept causal nonlocality or assert both the incompleteness of quantum theory and the existence of a causally local way for measurement results to be "predetermined." The conclusion of Bell's paper was that all roads lead to nonlocality:

> "In a theory in which parameters are added to quantum mechanics to determine the results of individual measurements, without changing the statistical predictions, there must be a mechanism whereby the setting of one measuring device can influence the reading of another instrument, however remote. Moreover, the signal involved must propagate instantaneously, so that such a theory could not be Lorentz invariant."

For now we'll put aside the question of whether Bell's treatment is sufficiently general or necessarily implies that quantum theory is causally nonlocal. Bell himself realized that the 1964 argument left room for confusion and had technical limitations. Over the next two decades he wrote several more papers to clarify, generalize, and strengthen the result. A more conservative conclusion of his 1964 paper is to rule out measurement deterministic hidden-variables theories obeying causally local dynamics.

10.3.1 Rotational Invariance

The singlet state, denoted henceforth as $|\Psi_-\rangle$, is *rotationally invariant* in the sense that

$$\Big(U(R) \otimes U(R) \Big) |\Psi_-\rangle = |\Psi_-\rangle$$

for proper rotation R by any angle θ about any axis. We can therefore choose a rotation R that maps $\hat{\mathbf{z}} \to \hat{\mathbf{n}}$ such that

$$U(R)\,|+z\rangle = |+n\rangle \quad \text{and} \quad U(R)\,|-z\rangle = |-n\rangle$$

The singlet state is now expressible in terms of the S_n eigenstates as

$$|\Psi_-\rangle = \frac{1}{\sqrt{2}}\Big(\,|+-\rangle_n - |-+\rangle_n\,\Big)$$

where, for example, $|+-\rangle_n = |+n\rangle \otimes |-n\rangle$. Since the singlet has the same form in every spin basis, the measurement results for spin along the same direction are perfectly anti-correlated for all directions of physical space.

It is easy to show that $|\Psi_-\rangle$ is rotationally invariant. Consider a finite rotation by $\boldsymbol{\theta}$,

$$\Big(U(R)\otimes U(R)\Big)\,|\Psi_-\rangle = \exp\left(-\frac{i}{\hbar}\boldsymbol{\theta}\cdot(\mathbf{S}_1+\mathbf{S}_2)\right)|\Psi_-\rangle$$

Since $|\Psi_-\rangle$ is a spin state with $s=0$, $\mathbf{S}^2\,|\Psi_-\rangle = 0$, and it follows that $\mathbf{S}\,|\Psi_-\rangle = 0$. The singlet state $|\Psi_-\rangle$ is also an eigenstate of $\mathbf{S}$ with eigenvalue 0. Hence,

$$\exp\left(-\frac{i}{\hbar}\boldsymbol{\theta}\cdot(\mathbf{S}_1+\mathbf{S}_2)\right)|\Psi_-\rangle = \exp(0)\,|\Psi_-\rangle = |\Psi_-\rangle$$

Observers A and B can measure spin along the directions $\mathbf{a}$ and $\mathbf{b}$ respectively, where both $\mathbf{a}$ and $\mathbf{b}$ are unit vectors. When $\mathbf{a}=\mathbf{b}$, the spins will be anti-correlated no matter which direction the vectors point in physical space. In general, for some nonzero angle between $\mathbf{a}$ and $\mathbf{b}$, the spin measurements will not be anti-correlated, but they will still display correlations that can be inferred from a statistical distribution of measurement outcomes. Intuitively, we expect the correlations to only depend on the angle between $\mathbf{a}$ and $\mathbf{b}$ due to the rotational invariance of the singlet state.

10.3.2 Spin Correlation Function

Observers A and B measure a spin component along directions $\mathbf{a}$ and $\mathbf{b}$, getting outcomes $\pm\hbar/2$, or ± 1 after dividing by $\hbar/2$. This is equivalent to A and B measuring the observables $\boldsymbol{\sigma}\cdot\mathbf{a}$ and $\boldsymbol{\sigma}\cdot\mathbf{b}$ respectively, where $\boldsymbol{\sigma}$ is the vector of Pauli matrices $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Since the measurement outcomes for each observer are random,

$$\langle\boldsymbol{\sigma}\cdot\mathbf{a}\rangle = 0, \quad \text{and} \quad \langle\boldsymbol{\sigma}\cdot\mathbf{b}\rangle = 0,$$

for the singlet state. The *local statistics* don't tell us anything interesting. We quantify the correlations between the subsystems using the **spin correlation function** defined as

$$E(\mathbf{a},\mathbf{b}) \equiv \langle(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{b})\rangle$$

When $\mathbf{b} = \mathbf{a}$, we can use rotational invariance of the singlet state to find

$$\begin{aligned} E(\mathbf{a},\mathbf{a}) &= \langle\Psi_-|\,(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{a})\,|\Psi_-\rangle \\ &= \frac{1}{\sqrt{2}}\Big(\,\langle+-|_a - \langle-+|_a\,\Big)\Big((\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{a})\Big)\frac{1}{\sqrt{2}}\Big(\,|+-\rangle_a - |-+\rangle_a\,\Big) \end{aligned}$$

where

$$(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{a})\,|+-\rangle_a = -\,|+-\rangle_a$$

$$(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{a})\,|-+\rangle_a = -\,|-+\rangle_a$$

so

$$(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{a})\Big(\,|+-\rangle_a - |-+\rangle_a\,\Big) = -\,|+-\rangle_a + |-+\rangle_a$$

The product states are orthonormal, so it follows that

$$E(\mathbf{a},\mathbf{a}) = \frac{1}{2}(-1-1) = -1$$

We could have guessed this result. The expectation value corresponds to the average of many measurement results, and since the results are anti-correlated, when observer A measures $+1$, B measures -1. Similarly, when A measures -1, B measures $+1$. The product of their measurement outcomes is always -1. Nonetheless, it's good to see how the tensor product works even when the answer is obvious.

There's a very nice way to calculate the correlation function for arbitrary $\mathbf{a}$ and $\mathbf{b}$ that relies on rotational invariance of $|\Psi_-\rangle$. Begin by writing

$$(\boldsymbol{\sigma}\cdot\mathbf{a})\otimes(\boldsymbol{\sigma}\cdot\mathbf{b}) = \sum_{i,j} a_i b_j\ \sigma_i\otimes\sigma_j$$

where i and $j = 1,2,3$ corresponds to x, y, z respectively. The correlation function is

$$E(\mathbf{a},\mathbf{b}) = \sum_{i,j} a_i b_j C_{ij}$$

where $C_{ij} = \langle\Psi_-|\,\sigma_i\otimes\sigma_j\,|\Psi_-\rangle$ are the matrix elements to be determined. Earlier, we found that $\mathbf{S}\,|\Psi_-\rangle = 0$, and it follows that $\boldsymbol{\sigma}_1\,|\Psi_-\rangle = -\boldsymbol{\sigma}_2\,|\Psi_-\rangle$. Furthermore,

$$\langle\sigma_i\otimes\sigma_j\rangle = -\langle\mathbb{1}\otimes\sigma_i\sigma_j\rangle = -\langle\mathbb{1}\otimes(\delta_{ij}\,\mathbb{1} + i\epsilon_{ijk}\sigma_k)\rangle$$

where the last equality follows from properties of the Pauli matrices. When $i = j$, the term in parentheses becomes the identity, so $C_{ij} = -1$. When $i \neq j$, only a term proportional to σ_k is left (where $k \neq i$ and $k \neq j$), and we already know the expectation value of σ_k is zero in the singlet

state, so $C_{ij} = 0$. Combining these results, we find that $C_{ij} = -\delta_{ij}$ and thus,

$$E(\mathbf{a}, \mathbf{b}) = -\sum_{i,j} a_i b_j \delta_{ij} = -\mathbf{a} \cdot \mathbf{b}$$

If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then we have

$$\boxed{E(\mathbf{a}, \mathbf{b}) = -\cos\theta}$$

Clearly, when $\theta = 0$, the "measurement settings" are aligned and $E(\mathbf{a}, \mathbf{b}) = -1$. When $\theta = \pi/2$, the two axes are orthogonal to each other, and we don't predict any correlations in the data. For example, if A measures spin up along the x direction, they will have nothing to say about what B will measure along y.

Having obtained the spin correlation function, it is worth pausing to ask what this quantity means operationally. In an actual experiment, A and B do not directly measure a product of outcomes; they record long lists of individual outcomes, and from these lists they count how often their results agree and how often they disagree. Let the outcomes henceforth be denoted $A = \pm 1$ for observer A and $B = \pm 1$ for observer B. Let the disagreement rate r be the probability that $A \neq B$. The agreement rate, given by $1 - r$, is the probability that $A = B$. Because the product AB equals -1 when the outcomes disagree and $+1$ when they agree, the correlation function can be written as

$$E(\mathbf{a}, \mathbf{b}) = (-1)(r) + (+1)(1 - r) = 1 - 2r$$

and the disagreement rate is given by

$$r = \frac{1 - E(\mathbf{a}, \mathbf{b})}{2}$$

Substituting the prediction of quantum mechanics, the disagreement rate becomes

$$\boxed{r(\theta) = \frac{1 + \cos\theta}{2} = \cos^2\frac{\theta}{2}}$$

This reformulation makes two important points clear. First, it connects the abstract correlator to directly observable frequencies in an experiment. "Bell tests" are ultimately about how often two distant observers agree or disagree. Second, it highlights the striking content of the singlet correlations. When the analyzers are aligned, the outcomes always disagree; when they are orthogonal, agreement and disagreement are equally likely; and when they are opposite, the outcomes always agree.

10.3.3 Instruction Sets

In this section, we present an argument that arrives at the same conclusion as Bell's 1964 paper, but from a different perspective. This argument is similar to the one presented in your textbook;

it relies on the "instruction set" conception of hidden variables. In this approach, a local hidden variable theory assigns to each particle a set of instructions specifying in advance the outcomes of measurements along any two directions **a** and **b**. This is arguably a strong assumption for a hidden-variables theory, but it is likely close to what EPR had in mind. If the particles do carry a local set of instructions, then no influence traveling faster than light between the two observers would be needed to explain the correlations.

The argument is one by contradiction. Suppose the particles carry a set of instructions denoted λ such that outcome A depends only on λ and **a** (not on **b**); likewise, B depends only on λ and **b** (not on **a**). Furthermore, assume the outcome statistics will reproduce the predictions of quantum mechanics; the disagreement rate must be given by $\cos^2(\theta/2)$ where θ is the angle between **a** and **b**. Since **a** and **b** are co-planar unit vectors, we can specify the measurement settings by angles θ_a and θ_b above some reference line, e.g., a common x-axis. We denote the outcomes by $A = A(\theta_a)$ and $B = B(\theta_b)$.

Whenever $\theta_a = \theta_b$, the measurement settings are aligned, and $r(0) = 1$ means that

$$A(\theta_a) = -B(\theta_b) \text{ always when } \theta_a = \theta_b \tag{10.1}$$

Suppose $\theta_a = +60^\circ$ and $\theta_b = 0^\circ$. The disagreement rate is $r(60^\circ) = 0.75$, and since we know $B(0^\circ) = -A(0^\circ)$ always,

$$A(+60^\circ) = A(0^\circ) \text{ for } 75\% \text{ of runs} \tag{10.2}$$

By similar reasoning used above, when $\theta_a = 0^\circ$ and $\theta_b = -60^\circ$, we find that

$$A(-60^\circ) = A(0^\circ) \text{ for } 75\% \text{ of runs} \tag{10.3}$$

Imagine that we have two sets of data; one which satisfies (10.2) and another that satisfies (10.3). Two sets of size 75% must overlap by **at least** 50% (see Figure 10.1), meaning

$$A(+60^\circ) = A(-60^\circ) \text{ for at least } 50\% \text{ of runs} \tag{10.4}$$

Since $A(-60^\circ) = -B(-60^\circ)$ always, this means $r(120^\circ) \geq 0.50$. Our assumptions have led to a constraint on the disagreement rate, which is clearly violated by the quantum prediction:

$$r(120^\circ) = \cos^2(120^\circ/2) = \cos^2(60^\circ) = 0.25$$

The argument shows that three assumptions cannot all hold simultaneously. The first is that quantum mechanics makes correct predictions, namely $r(\theta) = \cos^2(\theta/2)$. The second is that each particle carries predetermined outcomes—instruction sets $A(\theta_a)$ and $B(\theta_b)$ defined for all possible settings simultaneously, before any measurement is made. The third is locality: that each outcome depends only on the local detector setting, so A depends only on θ_a and B depends only on θ_b, with neither influenced by the distant observer's choice. The overlap argument leading to (10.4) relies

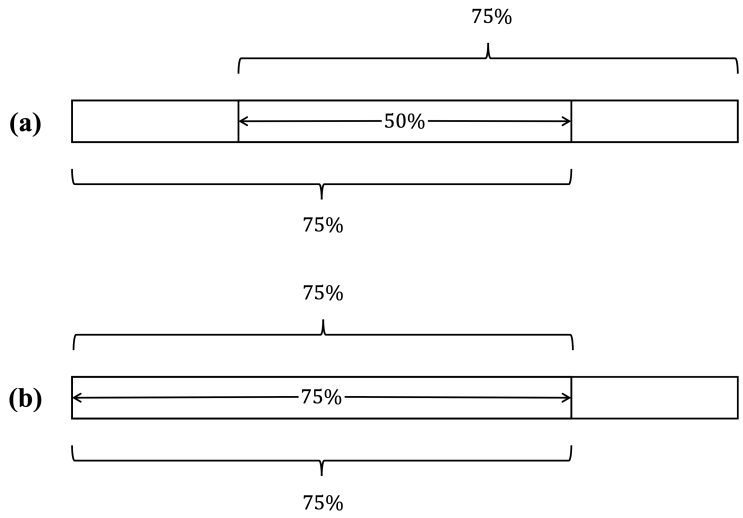
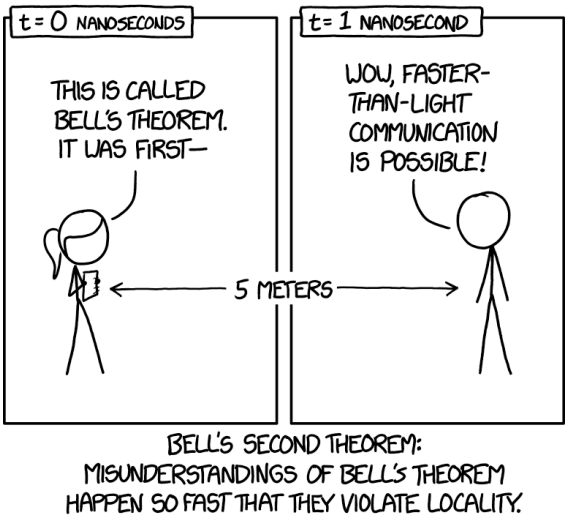


Figure 10.1: Illustrations of two overlapping sets. (a) Minimum overlap of two sets of size 75% is 50%. (b) Maximum overlap between the two sets is 75%.

on all three at once—it requires the outcomes at $+60°$, $0°$, and $-60°$ to be simultaneously well-defined for the same particle on the same run, and it requires that $B(\theta) = -A(\theta)$ holds regardless of which settings are actually chosen that run. Since experiments confirm the quantum predictions, we are forced to conclude that no theory built on local instruction sets can reproduce the observed correlations.

This does not necessarily imply the existence of faster-than-light signals or causal influences. Rather, it shows that quantum phenomena exhibit nonlocal correlations, or statistical relationships between distant measurements that cannot be explained by any local mechanism. Whether or not the correlations can be explained by some "spooky action at a distance," the **no-communication theorem** (or no-signaling theorem) states that entanglement cannot be used to transmit information faster than light. Even though entangled particles exhibit strong correlations, no local operation on one particle can influence the measurement-outcome statistics of the other in a way that could be used for communication. [By local operations, we mean local unitary operations or local measurements.]



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Chapter 11

The Quantum Formalism

11.1 Introduction

Welcome to Phys 438B! This chapter consolidates the mathematical structure of quantum mechanics in a form suitable for the topics that follow. Rather than revisiting the full axiomatic development from 438A, the goal here is to sharpen and extend the formalism in directions that will be used repeatedly. A key extension introduced is the density operator, which allows a unified treatment of pure states and nonpure states (also called "mixed states"). The density operator is essential when dealing with composite systems, where even a perfectly well-defined global state may induce mixed states locally. The density operator formalism serves as the natural bridge between single-particle quantum mechanics and many-body and even open systems.

We also develop the structure of composite systems using tensor products. This provides the setting for entanglement, correlations, and the addition of angular momentum. The latter will be treated both as a physical example and as a representation-theoretic construction, emphasizing how total angular momentum emerges from combining subsystems. Spin systems, in particular, offer a clean and finite-dimensional context in which these ideas can be worked out explicitly.

Another organizing principle introduced here is the role of generators of transformations. Continuous symmetries, such as translations and rotations, are associated with Hermitian operators that generate unitary transformations. In this framework, operators are not assigned by analogy with classical expressions, but are defined by how they act on states under symmetry transformations. This approach has several advantages: it avoids ambiguities associated with promoting classical expressions to operators; it applies equally well in systems without a clear classical starting point (e.g., spin); it makes symmetry principles primary, clarifying the connection between conservation laws and operators; lastly, it generalizes naturally to more abstract settings, including quantum information and field theory.

The intent of this chapter is not only to review familiar concepts, but to reframe them in a way that supports a more systematic and abstract use of quantum mechanics. The topics that follow rely heavily on this formulation, and benefit from treating it as the primary language rather than as background material.

11.2 Postulates of Quantum Mechanics

The connection between quantum theory and linear algebra was first exposed by Heisenberg's matrix mechanics and later extended by Schrödinger's wave mechanics. A unified theory based on a mathematical structure consisting of *linear operators acting on a vector space* was first formulated by Paul Dirac and later strengthened by John von Neumann, who developed a large portion of functional analysis. Informally, functional analysis is an extension of linear algebraic concepts to infinite-dimensional vector spaces, and while it underlies the formalism of quantum mechanics, we will emphasize only the most essential structures and defer a full treatment to advanced courses.

The postulates of quantum mechanics dictate how to treat a quantum mechanical system mathematically and how to interpret the mathematics to learn about the physical system in question. These postulates cannot be proven, but they have been successfully tested by many experiments, so we accept them as an accurate way to describe quantum mechanical systems. Quantum mechanics has five basic rules, and I've come up with a silly mnemonic to help you remember them:

Small **O**bjects **B**reak **C**lassical **T**heories

11.2.1 The Density Operator

A *quantum experiment* consists of two basic stages: *preparation* and *measurement*. The preparation procedure specifies how the system is produced, while the measurement procedure determines which observable is probed and how outcomes are recorded. In Chapter #1, we considered a Stern–Gerlach experiment in which a beam of thermal silver atoms passes through an analyzer. By blocking the atoms emerging from one output port and allowing only those from the other port to continue, we effectively prepare the system in a definite spin state. Repeating this procedure produces an ensemble of atoms all in the same state, corresponding to a pure spin state consistent with the selected measurement outcome.

Now consider instead the atoms emerging directly from the oven, before any such filtering is applied. The beam contains atoms in a variety of spin states, with no preparation that singles out a definite orientation. Measurements performed on this ensemble yield probabilistic outcomes that cannot be attributed to a single state vector. Rather, the beam must be described statistically, as a collection of systems prepared under identical macroscopic conditions but not in identical quantum pure states.

This distinction motivates a more general description of quantum states. While a state vector suffices for ensembles prepared in a definite pure state, it is not adequate for situations involving incomplete preparation or statistical mixtures. To describe such ensembles, we introduce the **density operator**, which encodes all measurable statistical properties of the system and provides a unified framework for both pure states and mixtures.

As a concrete motivating example, suppose that 50% of the atoms emerging from the oven are found to have $S_z = \hbar/2$ and 50% have $S_z = -\hbar/2$. From this information alone, *we cannot determine the quantum state of the atoms.* In particular, these measurement outcomes are consistent

with very different physical situations. One possibility is that each atom is prepared in the same pure state

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + e^{i\phi}\,|-z\rangle \Big),$$

for some phase ϕ. In this case, the probabilistic outcomes arise from quantum superposition. Alternatively, the beam may consist of a statistical mixture: 50% of the atoms are in the pure state $|+z\rangle$, while the remaining 50% are in the pure state $|-z\rangle$. Informally, the probabilities reflect "classical uncertainty" about which state a given atom occupies. These two situations are fundamentally different, even though they produce identical statistics for measurements of S_z. The difference becomes apparent when we measure a different observable, such as S_x. The coherent superposition exhibits interference effects that depend on the phase ϕ, while the statistical mixture does not.

This example illustrates that specifying probabilities for a single observable is not sufficient to characterize a quantum state. More generally, a complete description must encode the outcomes of all possible measurements. The appropriate object that captures this information, and distinguishes between coherent superpositions and statistical mixtures, is the density operator.

Postulate 1

For every quantum system, there is a complex Hilbert space, and the **states** of the system are all *Hermitian*, *positive*, *trace-class linear maps* ρ for which $\operatorname{Tr}\rho = 1$.

Here are several important properties of the density operator:

(i) Hermitian: $\rho^\dagger = \rho$

(ii) Positive semi-definite: $\langle\psi|\,\rho\,|\psi\rangle \geq 0$ for all $|\psi\rangle$ in the Hilbert space.

(iii) An operator A is **trace-class** if, for any orthonormal basis $\{e_n\}$,

$$\sum_n \langle e_n|\,A\,|e_n\rangle < \infty$$

(iv) Let A be a trace-class operator. Then the **trace** of A is defined as

$$\operatorname{Tr} A = \sum_n \langle e_n|\,A\,|e_n\rangle$$

One can show that the trace does not depend on the choice of orthonormal basis. In a finite-dimensional vector space, the trace of a matrix is the sum of its diagonal entries.

(v) A state is called **pure** if there exists a ket $|\psi\rangle$ such that for all $|\phi\rangle$

$$\rho\,|\phi\rangle = \langle\psi|\phi\rangle\,|\psi\rangle$$

Since a pure state density operator acts as a projection onto a one-dimensional subspace, it can be written in the form

$$\rho = |\psi\rangle \langle\psi| .$$

In this sense, the operator $|\psi\rangle \langle\psi|$ represents the same physical state as the ket $|\psi\rangle$.

This representation is not unique. If $|\psi'\rangle = e^{i\alpha} |\psi\rangle$, then

$$|\psi'\rangle \langle\psi'| = |\psi\rangle \langle\psi| ,$$

so both kets correspond to the same density operator and therefore the same physical state. Thus, pure states are associated not with individual kets, but with one-dimensional subspaces (or equivalently, rays) in Hilbert space.

Example 11.1: Mixed State

Consider a beam of silver atoms emerging directly from an oven and entering a Stern–Gerlach apparatus. Suppose that measurements of S_z yield the outcomes $+\hbar/2$ and $-\hbar/2$ with equal probability. The atoms are produced in a thermal environment and emerge from the oven with *random phases* and no well-defined relative coherence between spin components. Each atom may be viewed as being prepared independently, with no phase relation maintained across the ensemble. This lack of coherence implies that the ensemble cannot be described by a single state vector. Instead, the beam is appropriately described as a statistical mixture, with density operator

$$\rho = \tfrac{1}{2} |+z\rangle \langle+z| + \tfrac{1}{2} |-z\rangle \langle-z| = \tfrac{1}{2} \mathbb{1}.$$

We now phrase the remaining postulates in terms of the density operator ρ. The reader is encouraged to verify that, for a pure state ρ, these reduce to the postulates introduced in Chapter #2. While we will primarily consider pure states, the more general formulation of the quantum state is both more precise and will play an essential role in later chapters.

Postulate 2

Every measurable physical quantity, called an **observable**, is described by a self-adjoint linear operator. The result of measuring an observable is always an eigenvalue of the corresponding operator, and the operator's eigenvectors form a basis for the Hilbert space.

Postulate 3

The probability of obtaining an eigenvalue a_n in the measurement of an observable A on a quantum system in the state ρ is given by the **Born rule**:

$$p(a_n) = \mathrm{Tr}(\rho P_n) \tag{11.1}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n. The expectation value of A may be written as

$$\langle A \rangle = \mathrm{Tr}(\rho A) \tag{11.2}$$

Postulate 4

If a measurement of an observable A on a quantum system in the state ρ yields the result a_n, the quantum system irreversibly **collapses** to a new state

$$\rho' = \frac{P_n \rho P_n}{\mathrm{Tr}(\rho P_n)} \tag{11.3}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n.

Postulate 5

The **time evolution** of a closed quantum system is described by a *unitary transformation*. That is, the state of the system at time t_0 is related to the state of the system at time t by a unitary operator U which depends on t and t_0 such that

$$\rho(t) = U(t, t_0)\rho(t_0)U^\dagger(t, t_0) \tag{11.4}$$

For continuous evolution, the state obeys the *von Neumann equation* given by

$$i\hbar \frac{d\rho(t)}{dt} = [H(t), \rho(t)] \tag{11.5}$$

where the Hamiltonian $H(t)$ is the observable associated with the total energy of the system and $\hbar$ is the reduced Planck's constant (a universal constant).

11.3 Generators of the Galilei Group

Having established the formal structure of quantum theory, we now apply it to the study of a single *nonrelativistic particle*. By *particle*, we mean a system for which position is, in principle, an observable that can be measured with arbitrary precision. In practice, such localization is limited, and ultimately bounded by relativistic and quantum field–theoretic considerations, which we will not pursue here. By *nonrelativistic*, we mean that the relevant symmetries are those of Galilean transformations relating inertial observers. These symmetries will play a central role, as their generators define the fundamental observables of the theory. Although these symmetries are evidently not fundamental to nature, they provide an accurate and sufficient framework for the systems considered in this course.

If two observers are in inertial frames of reference, their spacetime coordinates are related by the transformation

$$(\mathbf{x}, t) \mapsto (R\mathbf{x} + \mathbf{u}t + \mathbf{a}, t + s)$$

where R denotes a rotation matrix, $\mathbf{u}$ is a velocity vector corresponding to a boost, $\mathbf{a}$ is a displacement vector corresponding to a translation in space, and s is a real number corresponding to a translation in time. These transformations form a continuous group: the Galilei group. Since the laws of nonrelativistic physics are invariant under these transformations, any quantum theory describing such physics must realize this symmetry. This is the core principle underlying our correspondence rules, whereby we identify particular dynamical variables with particular operators.

By Wigner's theorem, any *continuous symmetry transformation* connected to the identity must be induced by a unitary operator acting on the underlying Hilbert space. Galilean transformations are continuous and connected to the identity, so they are represented by unitary operators. Thus, for each Galilean transformation g there is a unitary operator $U(g)$. By Stone's theorem, if $U(g(\epsilon)) = U(\epsilon)$ is an element of a strongly continuous one-parameter unitary subgroup, then there exists a unique self-adjoint operator K such that

$$U(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon K\right), \tag{11.6}$$

where K is called the generator of the transformation, and ϵ is a continuous parameter. For the Galilei group, the generators are:

A **symmetry transformation** is essentially a mapping that preserves all physical predictions. Let U be a unitary operator implementing a continuous symmetry transformation. The predictions of quantum theory are invariant under the following unitary transformation:

$$|\psi\rangle \mapsto U|\psi\rangle$$

$$\rho \mapsto U\rho U^\dagger$$

$$A \mapsto UAU^\dagger$$

Transformation	**Unitary Operator**	**Generator**
Spatial translation ($\mathbf{a}$)	$U(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{a}\cdot\mathbf{p}\right)$	p_i (momentum)
Time translation (s)	$U(s) = \exp\left(\frac{i}{\hbar}\,sH\right)$	$-H$ ($-$Hamiltonian*)
Rotation ($R(\boldsymbol{\theta})$)	$U(R) = \exp\left(-\frac{i}{\hbar}\,\boldsymbol{\theta}\cdot\mathbf{J}\right)$	J_i (angular momentum)
Galilean boost ($\mathbf{u}$)	$U(\mathbf{u}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{u}\cdot\mathbf{G}\right)$	G_i (boost generator)

These transformations preserve all probabilities and expectation values:

$$\langle A'\rangle = \langle A\rangle, \quad \mathrm{Tr}(\rho A) = \mathrm{Tr}\big(\rho' A'\big)$$

Our goal is to capture the group structure in the operator representations. Any sequence of spacetime transformations is equivalent to another transformation in the group, so the composition of unitary transformations must respect this property. In fact, the unitary transformation operator U is defined only up to an overall phase, since U and $e^{i\gamma}U$ induce the same transformation on ρ and all observables. Hence, the product of two unitary operators implementing group elements may differ from the operator for the composed group element only by a phase. Generally speaking, this means

$$U(g_1)U(g_2) = e^{i\gamma(g_1,g_2)}U(g_1 \circ g_2) \tag{11.7}$$

for some real-valued function $\gamma(g_1, g_2)$. In many cases, we can determine the phase by additional considerations.

The discussion presented so far is only the starting point of a more detailed analysis, in which the *commutation relations* among the generators are fixed by consistency conditions and transformation properties. In a full treatment[1] , one systematically examines how successive infinitesimal transformations compose, and uses these relations, together with the requirement that the generators transform appropriately under rotations and other symmetries, to determine the "algebra" of the generators. This procedure ultimately leads to the commutation relations of the Galilei group, including the appearance of a central term proportional to the identity. Rather than carry out this derivation in detail, we will simply state the resulting commutation relations, which encode the essential structure:

$$[r_i, p_j] = [G_i, v_j] = i\hbar\delta_{ij}, \quad \text{and} \quad [G_i, p_j] = i\hbar m\delta_{ij}$$

where m is a real-valued constant and v_i is the ith Cartesian component of the **velocity operator**

[1] See Ballentine, 1998.

defined as

$$\mathbf{v} = \frac{i}{\hbar}[H, \mathbf{r}] \tag{11.8}$$

For a free particle with no internal degrees of freedom (i.e., spin), one can derive the following relations between the operators:

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{J} = \mathbf{L} = \mathbf{r} \times \mathbf{p}$$

$$H = \tfrac{1}{2}m\mathbf{v}^2 + E_0$$

where $\mathbf{L}$ denotes the orbital angular momentum (a special case of generalized angular momentum), and E_0 is a multiple of the identity. If m denotes the *mass of the particle*, then these expressions are the familiar forms of momentum, angular momentum, and energy from classical mechanics. In the words of Julain Schwinger, we borrow only names from classical mechanics—we've derived these operators from quantum-mechanical principles[2].

For a particle with no internal degrees of freedom interacting with external fields, the most general Hamiltonian consistent with the symmetry requirements is

$$H = \frac{\big[\mathbf{p} - \mathbf{A}(\mathbf{r})\big]^2}{2m} + V(\mathbf{r}) \tag{11.9}$$

where $\mathbf{A}(\mathbf{r})$ is a *vector potential* and $V(\mathbf{r})$ is a *scalar potential*. Vector potentials, as we'll see in the next chapter, are relevant whenever interactions are velocity dependent yet energy conserving. The relation between the momentum and velocity operators is altered by the presence of a vector potential:

$$\mathbf{v} = \frac{\mathbf{p} - \mathbf{A}(\mathbf{r})}{m} \tag{11.10}$$

Observe that adding a constant multiple of the identity to the Hamiltonian does not affect any commutators with observables, and therefore leaves the dynamics unchanged. Hence, the Hamiltonian is defined only up to an additive constant, which we absorb into the definition of $V(\mathbf{r})$. Both $\mathbf{A}(\mathbf{r})$ and $V(\mathbf{r})$ may be time-dependent but must be independent of $\mathbf{p}$.

11.3.1 Spin Degrees of Freedom

In proving the relations summarized thus far in this chapter, we made use of two important mathematical theorems, which we state here without proof:

Theorem. Any operator that commutes with (r_1, r_2, r_3) must be a function of the operators in that set, and likewise for (p_1, p_2, p_3).

[2]Another popular strategy to obtain operators for dynamical variables is called "canonical quantization."

Theorem. For a particle with no internal degrees of freedom, any operator that commutes with $\mathbf{r}$ and $\mathbf{p}$ must be a multiple of the identity.

We now use these results to motivate the introduction of *spin degrees of freedom*. With $\mathbf{J}$ as the generator of rotations (generalized angular momentum) and $\mathbf{L} \equiv \mathbf{r} \times \mathbf{p}$ as the orbital angular momentum, we define the operator

$$\mathbf{S} \equiv \mathbf{J} - \mathbf{L}$$

Using the transformation properties of $\mathbf{r}$ and $\mathbf{p}$ under rotations:

$$[J_i, r_j] = i\hbar\epsilon_{ijk} r_k, \quad [J_i, p_j] = i\hbar\epsilon_{ijk} p_k$$

together with the identities

$$[L_i, r_j] = i\hbar\epsilon_{ijk} r_k, \quad [L_i, p_j] = i\hbar\epsilon_{ijk} p_k$$

it follows immediately that

$$[S_i, r_j] = 0, \quad \text{and} \quad [S_i, p_j] = 0$$

Thus, $\mathbf{S}$ commutes with both $\mathbf{r}$ and $\mathbf{p}$. By Theorem 2, $\mathbf{S}$ must act trivially (i.e., as a multiple of the identity) on a particle with no internal degrees of freedom. However, this is incompatible with with the angular momentum algebra

$$[S_i, S_j] = i\hbar\epsilon_{ijk} S_k$$

unless $\mathbf{S} = 0$. Therefore, for a particle with no internal degrees of freedom, $\mathbf{J} = \mathbf{L}$.

If however, the particle does have internal degrees of freedom, $\mathbf{S}$ may not be proportional to the identity, but it cannot act on the same Hilbert space as $\mathbf{r}$ and $\mathbf{p}$. We introduce an *internal Hilbert space* on which $\mathbf{S}$ acts nontrivially. These internal degrees of freedom are identified with the particle's *spin*. This analysis does not *require* spin; it merely shows that the formalism admits such additional structure. Whether $\mathbf{S} \neq 0$ is ultimately determined by experiment, and empirical observations show that particles carry intrinsic angular momentum independent of their spatial motion.

11.4 Composite Systems

So far we have worked with systems whose degrees of freedom are *kinematically independent*. This includes two important cases: a single particle with multiple types of degrees of freedom (e.g., orbital position/momentum and spin) and multiple commuting observables (e.g., r_1, r_2, and r_3). The defining feature in these examples is that the corresponding operators commute, and the system admits a description in which each degree of freedom can be specified independently. The transition to **composite systems** is conceptually straightforward. We extend this structure to

systems composed of multiple subsystems (e.g., multiple particles). The key requirement is, if subsystems can be prepared and manipulated independently, then the formalism must allow a joint description compatible with their separate descriptions.

Let the operator $A^{(1)}$ represent an observable of component 1, and let $B^{(2)}$ represent an observable of component 2. If $A^{(1)} |a_m\rangle^{(1)} = a_m |a_m\rangle^{(1)}$ and $B^{(2)} |b_n\rangle^{(2)} = b_n |b_n\rangle^{(2)}$, then for every m and n, there must be a joint eigenvector such that

$$A^{(1)} |a_m, b_n\rangle = a_m |a_m, b_n\rangle$$

$$B^{(2)} |a_m, b_n\rangle = b_n |a_m, b_n\rangle$$

These equations are satisfied if the joint eigenvectors are of the product form

$$|a_m, b_n\rangle = |a_n\rangle^{(1)} |b_n\rangle^{(2)} = |a_m\rangle \otimes |b_n\rangle$$

where $\otimes$ denotes the tensor product, and the vectors $\{|a_m\rangle \otimes |b_n\rangle\}$ span a tensor product space. The tensor product between a vector in an M-dimensional space and an N-dimensional space is a vector in an MN-dimensional space.

The **tensor product** of two vector spaces is itself a vector space whose elements are linear combinations of product states denoted $|\psi\rangle \otimes |\phi\rangle$ satisfying bilinearity:

$$\Big(a |\psi_1\rangle + b |\psi_2\rangle\Big) \otimes |\phi\rangle = a |\psi_1\rangle \otimes |\phi\rangle + b |\psi_2\rangle \otimes |\phi\rangle$$
$$|\psi\rangle \otimes \Big(c |\phi_1\rangle + d |\phi_2\rangle\Big) = c |\psi\rangle \otimes |\phi_1\rangle + d |\psi\rangle \otimes |\phi_2\rangle$$

The inner product in a tensor product space is defined multiplicatively as

$$\langle \psi_1 \otimes \phi_1 | \psi_2 \otimes \phi_2 \rangle = \langle \psi_1 | \psi_2 \rangle \langle \phi_1 | \phi_2 \rangle$$

Lastly, for linear operators A on the first space and B on the second, the tensor product $A \otimes B$ is the unique linear operator satisfying

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = (A |\psi\rangle) \otimes (B |\phi\rangle)$$

Note that there exist vectors in the tensor product space that cannot be written as product states, and likewise, there are linear operators which may not be expressed as a tensor product of two linear operators acting on each space as above.

11.4.1 Statistical Independence

For a two-component system, there exist joint probability distributions that generally do not factorize. This is the basis for entanglement. If, however, the preparations of components 1 and 2 are independent and do not influence each other, then the joint probability distribution for observables $A^{(1)}$ and $B^{(2)}$ should obey the condition of *statistical independence*. If the state of the system is represented by a product state $|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle$, then

$$\text{Prob}(A = a_m \;\&\; B = b_n|\Psi) = |\langle a_m, b_n|\Psi\rangle|^2 = |\langle a_m|\psi\rangle|^2 \, |\langle b_n|\phi\rangle|^2$$

This factorization holds more generally if the state is represented by an operator of the factored form $\rho = \rho^{(1)} \otimes \rho^{(2)}$. It should be emphasized that this factored, or uncorrelated, state is very special and corresponds to independent preparations of the separate components.

11.5 Addition of Angular Momenta

In this section, we present the general formalism corresponding to the problem of adding two commuting angular momenta. Review Chapter #10 for the special case of adding two spin-1/2 particles. At the end of this section, we'll consider some essential examples that will be important for later chapters.

Consider a two-component system, each of which has angular momentum degrees of freedom. For convenience, we'll refer to component 1 as "particle 1" and component 2 as "particle 2," and we denote the corresponding angular momentum operators as $\mathbf{J}_1$ and $\mathbf{J}_2$, although each component may correspond to the orbital and spin degrees of freedom for a single particle (e.g., spin-orbit coupling). Consider the following commuting set of operators:

$$J_1^2 = \mathbf{J}_1 \cdot \mathbf{J}_1 = J_{1x}^2 + J_{1y}^2 + J_{1z}^2$$

$$J_2^2 = \mathbf{J}_2 \cdot \mathbf{J}_2 = J_{2x}^2 + J_{2y}^2 + J_{2z}^2$$

and the corresponding z-components, J_{1z} and J_{2z}. Since these operators commute, they admit a common set of eigenstates. The joint eigenstates are given by tensor products of the individual angular momentum eigenstates:

$$|j_1, m_1\rangle^{(1)} \, |j_2, m_2\rangle^{(2)} = |m_1; m_2\rangle \,,$$

where we drop the labels j_1 and j_2 since they are usually fixed. These eigenstates form the "uncoupled basis," where each factor in the tensor product is an eigenstate of J_i^2 and J_{iz} for $i = 1, 2$. The product space has dimension $(2j_1+1) \times (2j_2+1)$. It is straightforward to verify that the uncoupled basis is complete and orthonormal.

The **total angular momentum operator** is the generator of rotations in the product space,

and it is defined as

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2 = \mathbf{J}_1 \otimes \mathbb{1} + \mathbb{1} \otimes \mathbf{J}_2 \tag{11.11}$$

It follows that J_1^2, J_2^2, J^2, and J_z mutually commute and therefore admit a common set of eigenstates. In contrast, J_{1z} and J_{2z} cannot be included in this set, since they do not commute with J^2. This non-commutativity arises from the cross term $\mathbf{J}_1 \cdot \mathbf{J}_2$ in J^2, whose components (e.g., $J_{1x}J_{2x}$, $J_{1y}J_{2y}$) do not commute with J_{1z} or J_{2z}.

Designate the joint eigenstates of J_1^2, J_2^2, J^2, and J_z as $|j_1, j_2, j, m\rangle$ such that

$$J_1^2 |j_1, j_2, j, m\rangle = j_1(j_1+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J_2^2 |j_1, j_2, j, m\rangle = j_2(j_2+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J^2 |j_1, j_2, j, m\rangle = j(j+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J_z |j_1, j_2, j, m\rangle = m\hbar |j_1, j_2, j, m\rangle$$

Since j_1 and j_2 are usually fixed, we'll use shorthand notation where $|j_1, j_2, j, m\rangle \to |j, m\rangle$. The vectors $|j, m\rangle$ form a complete and orthonormal basis called the "coupled basis," and they span the same product space as the uncoupled basis. The problem of "angular momentum addition" now reduces to finding a transformation between the uncoupled and coupled bases of the product space, i.e., we seek a change of basis from

$$\{|m_1; m_2\rangle\} \to \{|j, m\rangle\}$$

11.5.1 Clebsch-Gordan Coefficients

The uncoupled basis is a *complete basis*, meaning for fixed j_1 and j_2, the uncoupled basis admits the following resolution of the identity on the tensor product space:

$$\sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} |m_1; m_2\rangle \langle m_1; m_2| = \mathbb{1}$$

The coupled basis may be obtained from the uncoupled basis by means of the following transformation, which is obtained by inserting the resolution of the identity above:

$$|j, m\rangle = \Big(\sum_{m_1, m_2} |m_1; m_2\rangle \langle m_1; m_2| \Big) |j, m\rangle$$

$$= \sum_{m_1, m_2} \langle m_1; m_2 | j, m\rangle |m_1; m_2\rangle$$

$$= \sum_{m_1,m_2} C^{j_1 j_2 j}_{m_1 m_2 m} \, |m_1; m_2\rangle$$

The coefficients $C^{j_1 j_2 j}_{m_1 m_2 m} = \langle m_1; m_2 | j, m\rangle$ are called the **Clebsch-Gordan coefficients**, and although they depend on j_1, j_2, j, m_1, m_2, and m, they also obey *selection rules* whereby the coefficients vanish unless certain conditions are satisfied. For example, $C^{j_1 j_2 j}_{m_1 m_2 m}$ vanishes unless $m = m_1 + m_2$. Since $J_z = J_{1z} + J_{2z}$,

$$0 = \langle m_1; m_2 | \left(J_z - J_z^{(1)} - J_z^{(2)} \right) |j, m\rangle$$

$$= \hbar \, (m - m_1 - m_2) \, \langle m_1; m_2 | j, m\rangle$$

which shows that $C^{j_1 j_2 j}_{m_1 m_2 m}$ is nonzero only when $m - m_1 - m_2 = 0$. Evidently, the uncoupled basis states are also eigenvectors of the operator J_z with eigenvalue $(m_1 + m_2)\hbar$.

For a fixed value of m, the eigenvectors of J_z with eigenvalue $m\hbar$ form a subspace (or eigenspace). Both the coupled states $|j, m\rangle$ and uncoupled states $|m_1; m_2\rangle$ with $m_1 + m_2 = m$ are eigenstates of J_z with eigenvalue $m\hbar$. Since the dimension of a subspace (like any vector space) is invariant under a change of basis, the number of coupled states with a given m must equal the number of uncoupled states with $m_1 + m_2 = m$. The eigenspace having the maximum value of $m = j_1 + j_2$ is one dimensional, i.e., there is only one state with $m = j_1 + j_2$. In the uncoupled basis, that state is $|m_1 = j_1; m_2 = j_2\rangle$, where both m_1 and m_2 take their maximum values. In the coupled basis, the state with $m = j_1 + j_2$ must belong to the set of states (called a multiplet) with $j = j_{\max} \equiv j_1 + j_2$. Hence, the uncoupled state $|m_1 = j_1; m_2 = j_2\rangle$ differs only by an overall phase from the coupled state $|j_{\max}, j_{\max}\rangle$.

In the set of states with $j = j_{\max}$, there is a single state with $m = j_{\max} - 1$. However, in the uncoupled basis, the value of $m = m_1 + m_2 = j_{\max} - 1$ occurs *twice*: the corresponding states are $|j_1 - 1; j_2\rangle$ and $|j_1; j_2 - 1\rangle$. Since the $m = j_{\max} - 1$ eigenspace is two dimensional, there must be another coupled state with this value of m. That state must have $j = j_{\max} - 1$, since the $j = j_{\max}$ state has already been accounted for. We can continue this process as long as the allowed ranges of m_1 and m_2 have not been exhausted. At $m = j_{\max} - 2$, the uncoupled basis contains three states, while the $j = j_{\max}$ and $j = j_{\max} - 1$ multiplets each account for one coupled state with this value of m. The remaining state must therefore have $j = j_{\max} - 2$. Continuing downward in this way, each successive value $j = j_1 + j_2, j_1 + j_2 - 1, \ldots$, appears exactly once. The sequence terminates when the total number of states in the coupled basis equals the dimension of the product space.

Since there are $2j+1$ states in the coupled basis for each j, and the product space has dimension $(2j_1 + 1) \times (2j_2 + 1)$, there must exist a unique $j_{\min}$ such that

$$\sum_{j=j_{\min}}^{j_{\max}} (2j + 1) = (2j_1 + 1)(2j_2 + 1).$$

On the left-hand side, we can evaluate the summation explicitly. Let $N = j_{\max} - j_{\min}$ (a non-

negative integer) and re-index the sum by setting $k = j_{\max} - j$ so that k runs from 0 to N as j runs from $j_{\max}$ down to $j_{\min}$. Consequently,

$$\begin{aligned}\sum_{j=j_{\min}}^{j_{\max}} (2j+1) &= \sum_{k=0}^{N} [2j_{\max} + 1 - 2k] \\ &= (2j_{\max}+1)\sum_{k=0}^{N} 1 - 2\sum_{k=0}^{N} k \\ &= (2j_{\max}+1)(N+1) - N(N+1) \\ &= (N+1)(2j_{\max}+1-N)\end{aligned}$$

Equating to the right-hand side and expanding in terms of j_1 and j_2, we find

$$(j_1 + j_2 - j_{\min} + 1)(j_1 + j_2 + j_{\min} + 1) = (2j_1+1)(2j_2+1)$$

or

$$j_{\min}^2 = (j_1 - j_2)^2 \quad \Rightarrow \quad j_{\min} = |j_1 - j_2|$$

The Clebsch-Gordan coefficients vanish unless m and j satisfy the following **selection rules**:

$$\boxed{\begin{aligned} &m = m_1 + m_2, \\ &j = |j_1 - j_2|,\ |j_1 - j_2| + 1,\ \ldots,\ j_1 + j_2. \end{aligned}} \tag{11.12}$$

The Clebsch-Gordan coefficients are taken to be real by convention, and furthermore, for each allowed value of j, the coupled basis states $|j, m\rangle$ are defined such that the Clebsch–Gordan coefficient for the "highest-weight" state satisfies

$$\langle m_1 = j_1\,;\, m_2 = j - j_1 | j, j\rangle > 0, \tag{11.13}$$

i.e., it is real and strictly positive. All other Clebsch–Gordan coefficients are then determined by the action of the ladder operators and by orthonormality[3]. While the Clebsch-Gordan coefficients are tabulated in many books, we'll briefly summarize a procedure to generate them before looking at a few important examples in more detail.

The Clebsch–Gordan coefficients for the highest- and lowest-weight states of the $j = j_1 + j_2$ multiplet are equal to one:

$$\langle m_1 = j_1; m_2 = j_2 | j_1 + j_2, j_1 + j_2\rangle = \langle m_1 = -j_1; m_2 = -j_2 | j_1 + j_2, -(j_1 + j_2)\rangle = 1. \tag{11.14}$$

[3]The phase convention mentioned above is chosen to be consistent with the phase convention used in defining the action of the ladder operators in Chapter #4.

The other coefficients can be related to each other by recursion relations, which can be derived by evaluating the matrix elements of $J_\pm = J_{1\pm} + J_{2\pm}$. From,

$$J_\pm |j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\,|j,m\pm 1\rangle$$

$$J_{1\pm} |m_1;m_2\rangle = \hbar\sqrt{j_1(j_1+1) - m_1(m_1\pm 1)}\,|m_1\pm 1;m_2\rangle$$

$$J_{2\pm} |m_1;m_2\rangle = \hbar\sqrt{j_2(j_2+1) - m_2(m_2\pm 1)}\,|m_1;m_2\pm 1\rangle$$

we get the following recursion relation:

$$\begin{aligned}\sqrt{j(j+1) - m(m\pm 1)}\, C^{j_1 j_2 j}_{m_1 m_2,\, m\pm 1} = {} & \sqrt{j_1(j_1+1) - m_1(m_1\pm 1)}\, C^{j_1 j_2 j}_{m_1\pm 1,\, m_2,\, m} \\ & + \sqrt{j_2(j_2+1) - m_2(m_2\pm 1)}\, C^{j_1 j_2 j}_{m_1,\, m_2\pm 1,\, m}.\end{aligned} \tag{11.15}$$

By orthonormality of the coupled and uncoupled bases, we can derive the following properties of the Clebsch-Gordan coefficients:

$$\sum_{m_1,m_2} C^{j_1 j_2 j}_{m_1 m_2 m}\, C^{j_1 j_2 j'}_{m_1 m_2 m'} = \delta_{jj'}\,\delta_{mm'} \tag{11.16}$$

$$\sum_{j,m} C^{j_1 j_2 j}_{m_1 m_2 m}\, C^{j_1 j_2 j}_{m_1' m_2' m} = \delta_{m_1 m_1'}\,\delta_{m_2 m_2'} \tag{11.17}$$

$$\sum_{m_1,m_2} \left(C^{j_1 j_2 j}_{m_1 m_2 m}\right)^2 = 1 \tag{11.18}$$

11.5.2 Two Spin-1/2 Particles

Let's revisit a familiar problem to learn how the coefficients work. Let $j_1 = \frac{1}{2}$, $j_2 = \frac{1}{2}$. Then $m_1, m_2 = \pm\frac{1}{2}$ and the uncoupled basis states are

$$\left|\tfrac{1}{2};\tfrac{1}{2}\right\rangle,\ \left|\tfrac{1}{2};-\tfrac{1}{2}\right\rangle,\ \left|-\tfrac{1}{2};\tfrac{1}{2}\right\rangle,\ \left|-\tfrac{1}{2};-\tfrac{1}{2}\right\rangle,$$

which we relabel using $\pm$ to indicate *spin up* and *spin down*:

$$|{+}{+}\rangle,\ |{+}{-}\rangle,\ |{-}{+}\rangle,\ |{-}{-}\rangle.$$

By the selection rules, the allowed values of j are 0 and 1. Starting with $j = 1$, the state with $m = 1$ is given by

$$|1,1\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},\frac{1}{2},1}\,|{+}{+}\rangle \quad\Rightarrow\quad C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},\frac{1}{2},1} = 1$$

and the state with $m = -1$ is

$$|1,-1\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},-\frac{1}{2},-1}\,|--\rangle \quad \Rightarrow \quad C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},-\frac{1}{2},-1} = 1$$

The state with $m = 0$ takes a bit more work:

$$|1,0\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},-\frac{1}{2},0}\,|+-\rangle + C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},\frac{1}{2},0}\,|-+\rangle$$

Using the recursion relation with $m_1 = 1/2$ and $m_2 = -1/2$,

$$\sqrt{2}C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},-\frac{1}{2},0} = 1$$

and with $m_1 = -1/2$ and $m_2 = 1/2$,

$$\sqrt{2}C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},\frac{1}{2},0} = 1$$

Hence,

$$|1,0\rangle = \tfrac{1}{\sqrt{2}}\Big(\,|+-\rangle + |-+\rangle\,\Big)$$

Having generated the triplet for $j = 1$, consider

$$|0,0\rangle = C^{\frac{1}{2}\frac{1}{2}0}_{\frac{1}{2},-\frac{1}{2},0}\,|+-\rangle + C^{\frac{1}{2}\frac{1}{2}0}_{-\frac{1}{2},\frac{1}{2},0}\,|-+\rangle$$

Since the first coefficient is positive, and the basis states are orthogonal, we have

$$|0,0\rangle = \tfrac{1}{\sqrt{2}}\Big(\,|+-\rangle - |-+\rangle\,\Big)$$

Example 11.2: Hyperfine Structure of the Hydrogen Ground State

In Chapter 9, we found that the energy eigenvalues of the Hamiltonian for the hydrogen atom are given by $E_n = -13.6\ \text{eV}/n^2$, where n is a positive integer. As it turns out, there are a number of corrections that must be made to our original model, and one of those corrections involves the intrinsic spins of both the electron and the proton. Since the proton has a magnetic moment (proportional to its spin), the proton is a source of a magnetic field. The electron interacts with this magnetic field, producing an interaction energy proportional to the intrinsic spins of both particles.

If we restrict our analysis to the ground state of hydrogen, a state with zero orbital angular momentum, the spin-spin Hamiltonian can be expressed in the simple form

$$H = \frac{2A}{\hbar^2}\mathbf{S}_1 \cdot \mathbf{S}_2$$

where $\mathbf{S}_1$ and $\mathbf{S}_2$ denote the spin angular momenta of the electron and proton respectively, and $A > 0$ is a constant. As it turns out, the eigenstates of this Hamiltonian are precisely the coupled eigenstates of two spin-1/2 particles. From the identity,

$$\mathbf{S}^2 = (\mathbf{S}_1 + \mathbf{S}_2)^2 = \mathbf{S}_1^1 + \mathbf{S}_2^2 + 2\mathbf{S}_1 \cdot \mathbf{S}_2,$$

we find

$$2\mathbf{S}_1 \cdot \mathbf{S}_2 = \mathbf{S}^2 - \mathbf{S}_1^1 - \mathbf{S}_2^2,$$

and since the coupled basis is a simultaneous eigenstate of $\mathbf{S}_1^2$, $\mathbf{S}_2^2$, and $\mathbf{S}^2$ (also S_z^2), it follows that a coupled basis vector is also an eigenvector of $\mathbf{S}_1 \cdot \mathbf{S}_2$. The corresponding eigenvalue equations of the Hamiltonian are

$$H\,|1, \pm 1\rangle = H\,|1, 0\rangle = \frac{A}{2}\,|1, 1\rangle\,, \quad H\,|0, 0\rangle = -\frac{3A}{2}\,|0, 0\rangle$$

The triplet states are therefore degenerate with energy $A/2$ and the singlet state has energy $-3A/2$. The energy difference is given by

$$\Delta E = \left| \frac{A}{2} - \left(-\frac{3A}{2} \right) \right| = 2A$$

A photon emitted or absorbed in making a transition between these two energy levels must therefore have a wavelength λ determined by $hc/\lambda = 2A$. For hydrogen, this wavelength can be determined by spectroscopic methods and happens to be 21 cm.

11.5.3 Addition of Orbital and Spin Angular Momentum

An important application of angular momentum addition in atomic physics is the coupling of the orbital angular momentum $\mathbf{L}$ and spin angular momentum $\mathbf{S}$ of electrons. The total angular momentum is defined as

$$\mathbf{J} = \mathbf{L} + \mathbf{S}.$$

For a single electron (e.g., hydrogen), the spin is $s = \frac{1}{2}$. The allowed values of the total angular momentum j are

$$j = \ell + s,\, \ell + s - 1,\, \ldots,\, |\ell - s|.$$

Thus,

$$j = \begin{cases} \ell + \frac{1}{2},\ \ell - \frac{1}{2}, & \ell \geq 1, \\ \frac{1}{2}, & \ell = 0. \end{cases}$$

It is conventional to use lower-case letters (j, ℓ, s) for a single electron and upper-case letters (J, L, S) for the total angular momenta of the atom.

For atoms with multiple electrons, one typically first adds all orbital angular momenta to obtain

L, then all spins to obtain **S**, and finally couples **L** and **S** to obtain **J** (LS coupling). Atomic states are labeled using spectroscopic notation

$$^{2S+1}L_J$$

where S and J are quantum numbers and L is denoted by a letter:

L	0	1	2	3	4	$\cdots$
	S	P	D	F	G	$\cdots$

Chapter 12

Charged Particle in a Magnetic Field

12.1 Classical Electromagnetism

We begin with a brief review of the classical description of a charged particle in an electromagnetic field. While these ideas are developed more fully in courses such as Physics 408, only a limited subset will be needed here. In particular, we introduce the scalar and vector potentials, which provide the most general description of particle-field interactions consistent with Galilean relativity (we encountered this idea in Chapter #11).

Some of the formalism—especially in its Lagrangian and Hamiltonian form—may be unfamiliar. However, the essential results can be followed without a comprehensive background in electromagnetism. The goal of this review is to highlight that several features which may appear distinctly quantum, such as the significance of potentials, already arise at the classical level. A working knowledge of classical mechanics, including Lagrangian and Hamiltonian formulations, will be assumed.

The electric and magnetic fields, $\mathbf{E}$ and $\mathbf{B}$, enter the Lagrangian formulation of classical mechanics through the scalar and vector potentials, $\mathbf{A}$ and ϕ, defined such that

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t} \tag{12.1}$$

$$\mathbf{B} = \nabla \times \mathbf{A} \tag{12.2}$$

In classical E&M, the fields are considered real physical fields defined at every point in space and time. Operationally, they are defined by the forces they produce on charges, and dynamically, they are independent degrees of freedom governed by Maxwell's equations. They can exist and propagate even in the absence of sources, carry energy and momentum, and interact with charges and currents. The potentials that are used to describe the fields are not considered physical; they are viewed as auxiliary quantities that are useful within a mathematical framework, but do not themselves represent any observable features of nature.

Within this classical framework, the potentials are not unique. The fields $\mathbf{E}$ and $\mathbf{B}$ are unaffected

by the following transformation:

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \phi \mapsto \phi' = \phi - \frac{\partial\Lambda}{\partial t} \tag{12.3}$$

where Λ is an arbitrary real scalar function of position and time. Eq. (12.3) is called a **gauge transformation** and the theory is called **gauge invariant**. In particular, the fields are called gauge invariant quantities under this transformation. Informally, gauge invariance reflects a redundancy in how we describe the system, not a symmetry that changes its physical state. The "extra freedom" only concerns the mathematical representation and indicates that our variables contain more information than the physics actually requires.

The Lagrangian[1] for a charged particle of mass m and charge q in an arbitrary electromagnetic field is given by

$$L(\mathbf{r}, \mathbf{v}, t) = \tfrac{1}{2}m\mathbf{v}^2 + q\mathbf{v} \cdot \mathbf{A}(\mathbf{r}, t) - q\phi(\mathbf{r}, t) \tag{12.4}$$

where $\mathbf{r}$ and $\mathbf{v}$ are the position and velocity of the particle. The Euler-Lagrange equations,

$$\frac{d}{dt}\left[\frac{\partial L}{\partial v_\alpha}\right] - \frac{\partial L}{\partial r_\alpha} = 0, \quad (\alpha = 1, 2, 3), \tag{12.5}$$

lead to the correct Newtonian equation of motion in terms of the Lorentz force law:

$$m\frac{d\mathbf{v}}{dt} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right) \tag{12.6}$$

From the Lagrangian, we can calculate the *canonical momentum* for a particle in an electromagnetic field:

$$p_\alpha = \frac{\partial L}{\partial v_\alpha} \;\Rightarrow\; \mathbf{p} = m\mathbf{v} + q\mathbf{A} \tag{12.7}$$

Notice that $\mathbf{p}$ changes under the gauge transformation given by Eq. (12.3), and hence lacks direct physical significance. Its importance is primarily mathematical, arising from its conservation when the system is invariant under spatial translations. More precisely, when the Lagrangian is invariant under spatial translation, the canonical momentum is conserved, and not the more intuitive quantity $m\mathbf{v}$.

The classical Hamiltonian for a particle in an electromagnetic field is given by

$$H = \mathbf{v} \cdot \mathbf{p} - L = \frac{1}{2m}\left(\mathbf{p} - q\mathbf{A}\right)^2 + q\phi \tag{12.8}$$

Since the first term is really just $\frac{1}{2}mv^2$, the Hamiltonian corresponds to the total energy: the sum of the kinetic energy and the electric potential energy. Hamilton's equations yield the familiar Newtonian equation of motion.

[1] See Taylor, *Classical Mechanics*, 2005.

12.2 Quantum Theory

The most general Hamiltonian for a spinless particle of mass m that respects the Galilean symmetries of spacetime is given by

$$H = \frac{1}{2m}\left(\mathbf{p} - \alpha\mathbf{A}\right)^2 + \beta\phi$$

where $\mathbf{A}$ is a vector potential, ϕ is a scalar potential, and the scalar constants α and β are included to account for the observation that particles with the same mass may move differently in a region of space containing the same $\mathbf{A}$ and ϕ, possibly as a result of some other physical characteristic (e.g., the charge). In a nonrelativistic theory, there doesn't seem to be any clear arguments that $\alpha = \beta = q$, the charge of the particle. For simplicity, we'll take for granted that the coupling of the particle to the electromagnetic field is proportional to the particle's charge q and rewrite the Hamiltonian in the more familiar form

$$H = \frac{1}{2m}\left(\mathbf{p} - q\mathbf{A}\right)^2 + q\phi \tag{12.9}$$

12.2.1 Heisenberg Picture & Gauge Transformation

For simplicity of notation we shall not distinguish between the Schrödinger and Heisenberg operators (see Chapter #3 for a discussion of the Heisenberg picture). In the Heisenberg picture, the rate of change of the position operator is

$$\frac{d\mathbf{r}}{dt} = \frac{i}{\hbar}[H, \mathbf{r}] = \frac{\mathbf{p} - q\mathbf{A}}{m} \tag{12.10}$$

which is just the velocity operator $\mathbf{v}$. Since expectation values of $\mathbf{v}$ are tied to measurable changes in the expectation values of $\mathbf{r}$, we regard the velocity operator as more *physically relevant* than either $\mathbf{p}$ or $\mathbf{A}$. The Heisenberg equation of motion for $m\mathbf{v}$ is

$$m\frac{d\mathbf{v}}{dt} = \frac{im}{\hbar}[H, \mathbf{v}] + m\frac{\partial\mathbf{v}}{\partial t} \tag{12.11}$$

and after some tedious calculations, we ultimately find

$$m\frac{d\mathbf{v}}{dt} = -q\nabla\phi - q\,\frac{\partial\mathbf{A}}{\partial t} + \frac{q}{2}\Big[\mathbf{v}\times(\nabla\times\mathbf{A}) - (\nabla\times\mathbf{A})\times\mathbf{v}\Big] \tag{12.12}$$

The operators $\mathbf{E} = -\nabla\phi - \partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$ are tied to measurable changes in $\mathbf{v}$, so we regard them as physically relevant. The field operators are invariant under

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \phi \mapsto \phi' = \phi - \frac{\partial\Lambda}{\partial t} \tag{12.13}$$

and furthermore, the Schrödinger equation in the coordinate representation is invariant under the transformation above as long as we include

$$\psi \mapsto \psi' = e^{i(q/\hbar)\Lambda}\psi \tag{12.14}$$

which, as we know, leaves the state of the system unchanged, e.g., $|\psi|^2$ is invariant.

Although it appears the velocity operator is gauge dependent, *its expectation value remains unchanged.* The expectation value of momentum, however, is not gauge invariant. For this reason, the physical significance of a result will usually be more apparent if it is expressed in terms of the velocity, rather than in terms of the momentum. It is therefore worth examining the mathematical properties of the velocity operator. For instance, consider the commutator of the velocity components with each other:

$$[v_x, v_y] = -\frac{q}{m^2}\Big([A_x, p_y] + [p_x, A_y]\Big)$$

Consider an arbitrary position-space wave function $\psi(\mathbf{r})$:

$$\begin{aligned}
[v_x, v_y]\psi &= \frac{i\hbar q}{m^2}\left(A_x\frac{\partial}{\partial y} - \frac{\partial}{\partial y}A_x + \frac{\partial}{\partial x}A_y - A_y\frac{\partial}{\partial x}\right)\psi \\
&= \frac{i\hbar q}{m^2}\left(-\frac{\partial A_x}{\partial y} + \frac{\partial A_y}{\partial x}\right)\psi \\
&= \frac{i\hbar q}{m^2}B_z\psi
\end{aligned}$$

Since this result holds for an arbitrary wave function, we can write $[v_x, v_y] = (i\hbar q/m^2)B_z$, which may be generalized to

$$[v_i, v_j] = \frac{i\hbar q}{m^2}\epsilon_{ijk}B_k \tag{12.15}$$

In words, the commutator of two components of velocity is proportional to the magnetic field in the remaining direction. If the magnetic field is zero in any particular direction, the velocity components perpendicular to that direction will commute with one another, and thus will share a common set of eigenstates.

12.3 Motion in a Uniform Magnetic Field

Consider a charged particle moving in a region of space where all field components are zero except B_z, which is a multiple of the identity. Although many choices for the potentials are consistent with this arrangement, we will assume the vector potential is static and the scalar potential vanishes.

Since B_x and B_y are both zero, it follows that v_z commutes with v_x and v_y. The Hamiltonian is

$$H = \frac{1}{2}m(v_x^2 + v_y^2) + \frac{1}{2}mv_z^2 = H_{xy} + H_z \tag{12.16}$$

where the operators $H_{xy} = \frac{1}{2}m(v_x^2 + v_y^2)$ and $H_z = \frac{1}{2}mv_z^2$ commute with each other, and thus have common eigenstates. Every eigenvalue of H is just the sum of an eigenvalue of H_{xy} and an eigenvalue of H_z.

We can derive the energy levels by first making the follow substitutions:

$$v_x = \frac{\sqrt{\hbar|q|B_z}}{m}\,Q, \quad \text{and} \quad v_y = \frac{\sqrt{\hbar|q|B_z}}{m}\frac{q}{|q|}\,P \tag{12.17}$$

From the commutation relation for the velocity components, it follows that $[Q, P] = i$ and the transverse Hamiltonian becomes

$$H_{xy} = \frac{\hbar|q|B_z}{m}\frac{1}{2}\Big(Q^2 + P^2\Big) \tag{12.18}$$

which is structurally identical to the Hamiltonian for a harmonic oscillator! The eigenvalues of the transverse Hamiltonian are therefore given by

$$E_{xy} = \Big(n + \tfrac{1}{2}\Big)\hbar\omega_c \tag{12.19}$$

where n is a nonnegative integer and $\omega_c = |q|B_z/m$ is the **cyclotron frequency**. The eigenvalue spectrum of H_z is continuous from $-\infty$ to ∞, so the energy eigenvalues for a charged particle in a uniform static magnetic field are

$$E_n(v_z) = \Big(n + \tfrac{1}{2}\Big)\hbar\omega_c + \tfrac{1}{2}mv_z^2 \tag{12.20}$$

The motion parallel to the magnetic field is not coupled to the transverse motion, and is unaffected by the field. The classical motion in the plane perpendicular to the field is in a circular orbit with angular frequency ω_c. Periodic motions correspond to discrete energy levels whose separation is $\hbar\omega_c$.

12.3.1 Energy Eigenfunctions

Although we were primarily interested in the energy eigenvalues, we can also determine the energy eigenfunctions. Let's ignore the motion in the z direction and confine our attention to the transverse motion. There are a few choices for the vector potential; let's go with $A_y = xB_z$. This gives the correct magnetic field with only a z component B_z. The Hamiltonian,

$$H = \frac{p_x^2}{2m} + \frac{1}{2m}\Big(p_y - xqB_z\Big)^2 \tag{12.21}$$

clearly commutes with p_y, but not with p_x due to the presence of x. We can therefore search for common eigenstates of H and p_y, with the latter being plane waves proportional to $e^{ik_y y}$ where $\hbar k_y$ is the momentum eigenvalue. In the coordinate representation, the energy eigenvalue equation $H\psi = E\psi$ is

$$\left[-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) + \frac{q^2 B_z^2}{2m}\, x^2 + i\hbar \frac{qB_z}{m}\, x\, \frac{\partial}{\partial y} \right] \psi = E\psi \tag{12.22}$$

Since ψ is a common eigenstate of H and p_y, we will substitute

$$\psi(x, y) = e^{ik_y y}\phi(x) \tag{12.23}$$

and reduce the partial differential equation to an ordinary differential equation

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{\hbar^2 k_y^2}{2m} + \frac{q^2 B_z^2}{2m} x^2 - \hbar k_y \frac{qB_z}{m}\, x \right] \phi = E\phi \tag{12.24}$$

The second term in the square brackets above can be written as a complete square by defining $x_0 = \hbar k_y / qB_z$. The differential equation then takes the form

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2} m\omega_c^2 (x - x_0)^2 \right] \phi = E\phi \tag{12.25}$$

which is, unsurprisingly, exactly the differential equation for a simple harmonic oscillator. The function $\phi(x)$ is a harmonic oscillator eigenfunction, and the complete energy eigenfunction for the transverse Hamiltonian is

$$\psi(x, y) = N H_n\left(\sqrt{\frac{m\omega_c}{\hbar}}\,(x - x_0)\right) \exp\left(-\frac{m\omega_c}{2\hbar}(x - x_0)^2 + ik_y y\right) \tag{12.26}$$

where N is a normalization constant.

12.3.2 Degeneracy

Since the energy depends only on n, there is degeneracy in the energy eigenstates corresponding to possible values of k_y. For each fixed n, there is a family of states labeled by k_y with the same energy. Different values of k_y correspond to different values of x_0, the center of the effective harmonic oscillator potential. Therefore, each eigenstate corresponds to the same cyclotron motion, but with a different guiding center position. The magnetic field fixes the energy of the orbit, but not where the orbit sits in space.

If the space available to the particle is infinite, then k_y is continuous and there are infinitely many possible values of x_0. Let's put the particle in a box of size $L_x \times L_y$. With the gauge potential $A_y = xB_z$, the Hamiltonian is translationally invariant in the y direction, so it is natural to impose

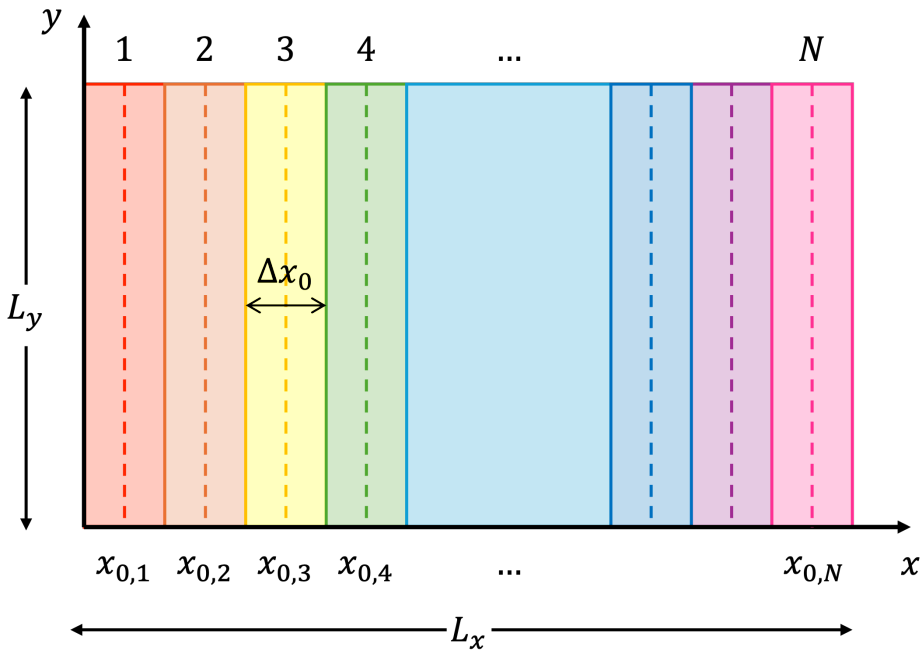


Figure 12.1: Representation of *Landau levels* in the x–y plane for a region of size $L_x \times L_y$. Each vertical band (parallel to the y-axis) corresponds to a quantum state labeled by its guiding center x_0, with states extended along y and localized in x about x_0, and adjacent centers separated by Δx_0. The states are indexed by $k_y = 2\pi j/L_y$ with $j \in \mathbb{Z}$, where $\pm j$ correspond to plane waves propagating in the $\pm y$ directions and $j = 0$ to a non-propagating state, so the total number of states is approximately twice the number of positive j values (plus one for $j = 0$). Each band occupies an effective area $\Delta A = L_y \Delta x_0 = \Phi_0/B_z$, corresponding to one flux quantum, and the total number of such bands within $0 \le x \le L_x$ is proportional to the Landau level degeneracy $N \propto \Phi/\Phi_0$.

periodic boundary conditions there, i.e., $\psi(x, y + L_y) = \psi(x, y)$. This condition leads to

$$k_y = \frac{2\pi j}{L_y}, \quad j \in \mathbb{Z}$$

Since the wave function is localized around the point x_0, it is natural to confine the center to the region $0 \le x_0 \le L_x$. We'll assume the orbits are small compared to the overall size of the box so we can ignore the small fraction of those states lying close to the boundaries. In this limit, the number of degenerate states is approximately

$$2j_{\max} = 2\,\frac{|q|B_z}{2\pi\hbar} L_x L_y = 2\,\frac{\Phi}{\Phi_0} \tag{12.27}$$

where $\Phi = B_z L_x L_y$ is the *total magnetic flux* passing through the region $L_x \times L_y$ and $\Phi_0 = 2\pi\hbar/|q|$ is the *magnetic flux quantum*. There is a wonderful geometrical interpretation: each state is associated with an area of magnitude Φ_0/B_z in the plane perpendicular to the magnetic field. The quantity Φ_0 is a natural unit of magnetic flux. Thus the degeneracy factor of an energy level, called a Landau level in this context, is simply equal to the number of units of magnetic flux passing through the

system.

12.4 Aharonov–Bohm Effect

Suppose a particle is moving in a region of space with a time-independent vector potential $\mathbf{A}$ *and* $\mathbf{B} = 0$. There are situations where $\mathbf{A}$ may be nonzero but we can effectively remove it from the quantum description with a gauge transformation. In the coordinate representation, the Schrödinger equation is

$$i\hbar\frac{\partial\psi}{\partial t} = \left[\frac{1}{2m}\left(-i\hbar\nabla - q\mathbf{A}\right)^2\right]\psi$$

We can simplify the Schrödinger equation with the following gauge transformation:

$$\mathbf{A} \mapsto \mathbf{A}' = \mathbf{A} + \nabla\Lambda, \quad \psi \mapsto \psi' = e^{i(q/\hbar)\Lambda}\psi,$$

where

$$\Lambda(\mathbf{r}) \equiv -\int_{\mathbf{r_0}}^{\mathbf{r}} \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}'$$

such that $\nabla\Lambda = -\mathbf{A}$ and $\mathbf{A}' = 0$. Note that the "gauge potential" Λ is independent of the path taken from some reference point $\mathbf{r}_0$ to the point $\mathbf{r}$ since $\nabla\times\mathbf{A} = 0$, *assuming the space is simply connected*. Since we've effectively removed the vector potential, the Schrödinger equation becomes

$$i\hbar\frac{\partial\psi'}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi'$$

which says that ψ' satisfies the Schrödinger equation without the vector potential $\mathbf{A}$. As it turns out, there are circumstances in which the gauge potential cannot be globally defined, and $\mathbf{A}$ cannot simply be removed.

In order for Λ to be a well-defined scalar function, we need the line integral of $\mathbf{A}$ to be zero for every closed loop. In a simply connected region, the condition $\nabla\times\mathbf{A} = 0$ implies exactly that according to Stokes' theorem. To see where the argument above fails, consider two different paths, C_1 and C_2, from $\mathbf{r}_0$ to $\mathbf{r}$ and define $\Lambda_1(\mathbf{r})$ along C_1 and $\Lambda_2(\mathbf{r})$ along C_2. Their difference is given by

$$\Lambda_1(\mathbf{r}) - \Lambda_2(\mathbf{r}) = -\oint_C \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}'$$

where C is a closed loop formed by going out along C_1 and back along C_2. Applying Stokes' theorem, we find that the difference is related to the magnetic flux:

$$\oint_C \mathbf{A}(\mathbf{r}')\cdot d\mathbf{r}' = \int_S (\nabla\times\mathbf{A})\cdot d\mathbf{a} = \int_S \mathbf{B}\cdot d\mathbf{a} = \Phi_B$$

where S is a surface bounded by the closed curve C. Whenever the magnetic flux is nonzero, the difference between Λ_1 and Λ_2 will be nonzero, and the function cannot be well-defined.

For example, consider a particle that is confined to move on a ring of radius a surrounding a

solenoid of radius $b < a$. If the solenoid is extremely long, the magnetic field inside it is uniform, and the field outside is zero. Let $\Lambda_1(\theta)$ correspond to a path taken from a reference line where $\theta = 0$ to the angle θ around the ring and $\Lambda_2(\theta)$ correspond to a path taken from $\theta = 0$ all the way around the ring, and then to θ. Both paths encircle the solenoid in the same direction as the current. Geometrically, we have

$$\int_{C_2} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}' = \oint_C \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}' + \int_{C_1} \mathbf{A}(\mathbf{r}') \cdot d\mathbf{r}',$$

or

$$\Lambda_2(\mathbf{r}) = \Lambda_1(\mathbf{r}) - \Phi_B$$

Using single-valuedness of the wave function,

$$\psi_2'(\theta + 2\pi) = e^{i(q/\hbar)\Lambda_2}\psi(\theta + 2\pi) = e^{i(q/\hbar)(\Lambda_1 - \Phi_B)}\psi(\theta) = e^{-i(q/\hbar)\Phi_B}\psi_1'(\theta)$$

which can be thought of as a "twisted boundary condition" in the sense that ψ_1' and ψ_2' are both solutions to the same free-particle Schrödinger equation, but the nonzero magnetic flux enclosed by the loop leads to an overall phase difference as the ring is encircled. We can easily determine the energy eigenvalues using the free-particle solution[2] $\psi' = e^{ik\theta}$. Imposing the twisted boundary condition,

$$e^{ik(\theta+2\pi)} = e^{-i(q/\hbar)\Phi_B}e^{ik\theta} \Rightarrow k = n - \frac{q\Phi_B}{2\pi\hbar},\ n \in \mathbb{Z}$$

Therefore, the energy eigenvalues are

$$E_n = \frac{\hbar^2}{2I}\left(n - \frac{q\Phi_B}{2\pi\hbar}\right)^2$$

so the effect of the flux is simply to shift the integer quantum number $n \to n - \Phi_B/\Phi_0$. The solenoid lifts the two-fold degeneracy of the particle-on-a-ring: positive n, representing a particle traveling in the *same direction* as the current in the solenoid, has a somewhat lower energy (assuming q is positive) than negative n, describing a particle traveling in the *opposite direction*. More importantly, the allowed energies clearly depend on the field inside the solenoid, even though the field at the location of the particle is zero!

Aharonov and Bohm proposed an experiment in which a beam of electrons is split in two, and they pass either side of a long solenoid before recombining. The beams are kept well away from the solenoid itself, so they encounter only regions where $\mathbf{B} = 0$. The magnetic flux contained within a closed path around the solenoid is not zero, and the two beams arrive with different phases:

$$\text{phase difference } \propto \frac{\Phi_B}{\Phi_0}$$

This phase shift leads to measurable interference, which has been confirmed experimentally. An-

[2]See Chapter #8, *Particle on a Ring*

other consequence of the Aharonov–Bohm (AB) effect was the discovery that magnetic flux is likely not quantized; if the total magnetic flux were quantized in integer multiples of Φ_0, no interference pattern would be observed.

The AB effect is called a *topological effect*, in that the effect depends on the flux encircled by the paths available to the particle, even though the paths may never approach the region of the flux. Since the magnetic force on the charge vanishes on all possible paths of the particle, one might wonder whether the charge is necessary for the effect. According to the theory, the effect depends on the dimensionless ratio $q\Phi_B/h$, which is proportional to the charge, and it has been experimentally confirmed that no AB effect occurs if neutrons are used instead of electrons.

Chapter 13

Time-Independent Perturbation Theory

13.1 Introduction

There are relatively few Hamiltonians for which the energy eigenvalue problem

$$H\left|\psi\right\rangle = E\left|\psi\right\rangle$$

can be solved exactly. In many physically important situations, however, the Hamiltonian is close to another Hamiltonian whose eigenvalue problem we already know how to solve. The Hamiltonian we can solve exactly is called the unperturbed Hamiltonian, and the left-over portion is the so-called perturbation. We assume the perturbation is small so that we can use approximations to find corrections to the exact solutions. Our goal in this section is to discover how energies and energy eigenstates of the unperturbed Hamiltonian are affected by the perturbation. Before we develop the full machinery of time-independent perturbation theory, we consider a simple exactly solvable problem. We'll solve it and then expand the solutions in small parameters. This will give us some intuition for perturbation theory and provide a useful point of comparison for the general formalism.

13.2 A Spin-1/2 Example

Ignoring orbital degrees of freedom (measurements of $\mathbf{r}, \mathbf{p}$, etc.), a particle with spin will precess in a magnetic field. The Hamiltonian for a spin-1/2 system with an applied magnetic field in the z-direction, $\mathbf{B} = B_0\hat{\mathbf{z}}$, can be written as

$$H_0 = -\boldsymbol{\mu}\cdot\mathbf{B} = \omega_0 S_z = \frac{\hbar}{2}\begin{pmatrix}\omega_0 & 0\\ 0 & -\omega_0\end{pmatrix}$$

where ω_0 is the Larmor frequency, proportional to the strength of the applied magnetic field B_0, and the matrix representation of H_0 is written in the ordered basis $\{|+z\rangle, |-z\rangle\}$. The energy eigenstates are the spin-up and spin-down states $|\pm z\rangle$ and the corresponding energy eigenvalues are

$$E_{\pm}^{(0)} = \pm\frac{\hbar\omega_0}{2},$$

where the superscript zero on the energy indicates the order of the solution—these are the unperturbed energies.

We perturb the system by changing the magnetic field. Since any change can be decomposed into a component along the original field direction and a component perpendicular, we write the new total field as $\mathbf{B} = B_0\hat{\mathbf{z}} + B_1\hat{\mathbf{z}} + B_2\hat{\mathbf{x}}$, and the Hamiltonian becomes

$$H = \omega_0 S_z + \omega_1 S_z + \omega_2 S_x = \frac{\hbar}{2}\begin{pmatrix} \omega_0+\omega_1 & \omega_2 \\ \omega_2 & -\omega_0-\omega_1 \end{pmatrix},$$

where ω_1 is proportional to B_1 and ω_2 is proportional to B_2. We have already solved the eigenvalue problem $H|\psi\rangle = E|\psi\rangle$ for this Hamiltonian in Chapter 3, so I will simply write the results below:

$$H|\pm n\rangle = E_{\pm}|\pm n\rangle,$$

where $|\pm n\rangle$ are the eigenstates of $S_n = \hat{\mathbf{n}}\cdot\mathbf{S}$ and $E_{\pm}$ are the corresponding eigenvalues. In this case, the direction $\hat{\mathbf{n}}$ of the magnetic field is given by the polar angle $\theta = \tan^{-1}\frac{\omega_2}{(\omega_0+\omega_1)}$ and the azimuthal angle $\phi = 0$ on the x-z plane. Explicitly, we have the energy eigenstates

$$|+n\rangle = \cos\frac{\theta}{2}|+z\rangle + \sin\frac{\theta}{2}|-z\rangle$$

$$|-n\rangle = -\sin\frac{\theta}{2}|+z\rangle + \cos\frac{\theta}{2}|-z\rangle$$

and the energies are

$$E_{\pm} = \pm\frac{\hbar\omega_0}{2}\sqrt{\left(1+\frac{\omega_1}{\omega_0}\right)^2 + \left(\frac{\omega_2}{\omega_0}\right)^2}.$$

Given the complete solution, we can now consider the perturbing field to be weak in the sense that $\omega_1 \ll \omega_0$ and $\omega_2 \ll \omega_0$. Notice that when $\omega_1 = \omega_2 = 0$, the problem reduces to that of spin precession about the z direction, i.e.,

$$|\pm n\rangle \to |\pm z\rangle, \quad E_{\pm} \to \pm\frac{\hbar\omega_0}{2}.$$

Let's start by expanding the energies to second order in ω_1/ω_0 and ω_2/ω_0:

$$E_{\pm} = \pm\frac{\hbar\omega_0}{2}\sqrt{1 + \frac{2\omega_1}{\omega_0} + \frac{\omega_1^2}{\omega_0^2} + \frac{\omega_2^2}{\omega_0^2}}$$

$$\approx \pm\frac{\hbar\omega_0}{2}\left[1+\frac{1}{2}\left(\frac{2\omega_1}{\omega_0}+\frac{\omega_1^2}{\omega_0^2}+\frac{\omega_2^2}{\omega_0^2}\right)-\frac{1}{8}\left(\frac{2\omega_1}{\omega_0}+\frac{\omega_1^2}{\omega_0^2}+\frac{\omega_2^2}{\omega_0^2}\right)^2\right]$$

$$=\pm\frac{\hbar\omega_0}{2}\left[1+\frac{\omega_1}{\omega_0}+\frac{\omega_2^2}{2\omega_0^2}\right]=\pm\frac{\hbar\omega_0}{2}\pm\frac{\hbar\omega_1}{2}\pm\frac{\hbar\omega_2^2}{4\omega_0}$$

Observe that the first "correction" to a given energy is linear or first order in the perturbation and equal to the *diagonal term* of the perturbation Hamiltonian, and the second correction is quadratic in the perturbation and proportional to the square of the *off-diagonal term* in the perturbation Hamiltonian. This general pattern of corrections is characteristic of perturbation theory.

As for the energy eigenstates, note that to first order,

$$\sin\frac{\theta}{2}\approx\frac{\omega_2}{2\omega_0},\quad\text{and}\quad\cos\frac{\theta}{2}\approx 1.$$

Hence,

$$|+n\rangle\approx|+z\rangle+\frac{\omega_2}{2\omega_0}|-z\rangle$$

$$|-n\rangle=|-z\rangle-\frac{\omega_2}{2\omega_0}|+z\rangle$$

We conclude that the first-order eigenstate correction depends only on the off-diagonal matrix element, not on the diagonal elements. We can easily verify that the eigenstates are normalized to first order in the perturbation, e.g.,

$$\langle+n|+n\rangle=1+\left(\frac{\omega_2}{2\omega_0}\right)^2\approx 1.$$

13.3 Nondegenerate Perturbation Theory

Suppose we know how to solve the eigenvalue problem for a Hamiltonian H_0, written as

$$H_0\left|n^{(0)}\right\rangle=E_n^{(0)}\left|n^{(0)}\right\rangle,\tag{13.1}$$

where the eigenstates form a complete orthonormal basis in the sense that

$$\left\langle m^{(0)}\middle|n^{(0)}\right\rangle=\delta_{nm},\quad\text{and}\quad\sum_m\left|m^{(0)}\right\rangle\left\langle m^{(0)}\right|=\mathbb{1}.$$

Technically, the expressions above are only relevant to the discrete portion of the Hamiltonian's spectrum. For the continuous portion, when relevant, it's understood that the Kronecker delta is replaced by a Dirac delta, and the sum over states becomes an integral. We will stick to the discrete notation in these notes.

In perturbation theory, we consider adding a perturbation Hamiltonian V such that

$$H = H_0 + V, \tag{13.2}$$

and the goal is to approximate the eigenvalues and eigenstates of H using the known eigenvalues and eigenstates of H_0. It's useful to introduce a parameter λ and consider the family

$$H = H_0 + \lambda V. \tag{13.3}$$

The parameter λ allows us to interpolate between the unperturbed Hamiltonian ($\lambda = 0$) and the complete Hamiltonian containing the perturbation ($\lambda = 1$).

Assume that $E_n^{(0)}$ is a nondegenerate eigenvalue of H_0. As we increase λ, the unperturbed eigenvalue will change smoothly to the value E, and the corresponding eigenstate becomes $|\psi\rangle$. When $\lambda = 0$, $E = E_n^{(0)}$ and $|\psi\rangle = \left|n^{(0)}\right\rangle$. For $\lambda \neq 0$, the exact eigenvalue equation is

$$(H_0 + \lambda V)\,|\psi\rangle = E\,|\psi\rangle\,. \tag{13.4}$$

Let P denote the projector onto the unperturbed state $\left|n^{(0)}\right\rangle$, i.e., $P = \left|n^{(0)}\right\rangle\left\langle n^{(0)}\right|$, and let Q be its orthogonal complement, which simply means $Q = \mathbb{1} - P$. Recall that any vector in $\mathbb{R}^3$ may be decomposed into one component that is parallel to some unit vector $\hat{\mathbf{n}}$ and another component which is perpendicular. This decomposition works even in abstract vector spaces. For example, since $P + Q = \mathbb{1}$, we may write

$$|\psi\rangle = P\,|\psi\rangle + Q\,|\psi\rangle\,. \tag{13.5}$$

The state $|\psi\rangle$ is only defined up to an overall normalization factor, and it will be convenient to normalize according to the condition

$$\left\langle n^{(0)}\middle|\psi\right\rangle = 1 \tag{13.6}$$

This is called "intermediate normalization," and it has the following wonderful consequence:

$$P\,|\psi\rangle = \left|n^{(0)}\right\rangle\left\langle n^{(0)}\middle|\psi\right\rangle = \left|n^{(0)}\right\rangle \tag{13.7}$$

and therefore,

$$|\psi\rangle = \left|n^{(0)}\right\rangle + Q\,|\psi\rangle\,. \tag{13.8}$$

Thus, at least formally, $Q\,|\psi\rangle$ constitutes a "correction" to the original state. Because

$$Q = \sum_{m\neq n} \left|m^{(0)}\right\rangle\left\langle m^{(0)}\right|,$$

the correction consists entirely of states other than $\left|n^{(0)}\right\rangle$.

Since we already know the component of $|\psi\rangle$ along $\left|n^{(0)}\right\rangle$, we only need to solve for the remaining

component $Q|\psi\rangle$. From the eigenvalue equation, it follows that

$$Q(E - H_0)|\psi\rangle = \lambda QV|\psi\rangle \tag{13.9}$$

Since Q is constructed from eigenvectors of H_0, it commutes with H_0, and we find

$$(E - H_0)Q|\psi\rangle = \lambda QV|\psi\rangle \tag{13.10}$$

Note that the operator $E - H_0$ is invertible as long as E does not lie within the spectrum of H_0 (it's not one of the eigenvalues of H_0). This is obviously a problem when $\lambda = 0$, since $E = E_n^{(0)}$. We can get around this by simply restricting the inverse to the subspace spanned by all eigenstates except for $\left|n^{(0)}\right\rangle$. In other words, define the reduced inverse operator

$$R = \sum_{m \neq n} \frac{\left|m^{(0)}\right\rangle \left\langle m^{(0)}\right|}{E - E_m^{(0)}}. \tag{13.11}$$

The inverse operator is explained a bit further in Appendix A. As long as $E \neq E_m^{(0)}$ for $m \neq n$, the operator R exists. From orthonormality of the eigenstates, it follows that

$$R(E - H_0)Q = Q, \quad \text{and} \quad RQ = R. \tag{13.12}$$

The expressions above may or may not be immediately obvious to you. When in doubt, write them out! Start by expanding R and Q in terms of the (unperturbed) energy eigenstates. If apply R from the left to both sides of Eq. (13.10), we find

$$Q|\psi\rangle = \lambda RV|\psi\rangle\,, \tag{13.13}$$

and as a consequence,

$$\boxed{|\psi\rangle = \left|n^{(0)}\right\rangle + \lambda RV|\psi\rangle\,.} \tag{13.14}$$

This expression allows us to assess the correction to the unperturbed state in more detail. The size of the correction is ultimately controlled by the strength with which V couples different unperturbed states compared with their energy separation. For a weak perturbation, where $E \approx E_n^{(0)}$ and $|\psi\rangle \approx \left|n^{(0)}\right\rangle$, the coefficients are approximately

$$\frac{\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle}{E_n^{(0)} - E_m^{(0)}}$$

when $\lambda = 1$. Perturbation theory is therefore expected to work well when these dimensionless ratios are small.

We'll return to our implicit expression for $|\psi\rangle$ once we've made progress on determining the

energy E. Taking the inner product of the eigenvalue equation with the state $\left|n^{(0)}\right\rangle$, we find

$$\left\langle n^{(0)}\right| (H_0 + \lambda V) \left|\psi\right\rangle = E \left\langle n^{(0)}\middle|\psi\right\rangle .$$

Since $H \left|n^{(0)}\right\rangle = E_n^{(0)} \left|n^{(0)}\right\rangle$ and $\left\langle n^{(0)}|\psi\right\rangle = 1$, it follows that

$$\boxed{E = E_n^{(0)} + \lambda \left\langle n^{(0)}\right| V \left|\psi\right\rangle} \tag{13.15}$$

It should be emphasized at this stage that the expressions for E and $|\psi\rangle$ are exact (and implicit).

13.3.1 First-Order Approximation

Eq. (13.14) is an implicit expression for $|\psi\rangle$ since it appears on both the left- and right-hand sides. We can substitute the expression for $|\psi\rangle$ into the right-hand side to find

$$|\psi\rangle = \left|n^{(0)}\right\rangle + \lambda RV\Big(\left|n^{(0)}\right\rangle + \lambda RV \left|\psi\right\rangle \Big) = \left|n^{(0)}\right\rangle + \lambda RV \left|n^{(0)}\right\rangle + \lambda^2 RVRV \left|\psi\right\rangle . \tag{13.16}$$

If we neglect the final term, which is second order in the perturbation, we obtain an expression for $|\psi\rangle$ in terms of $\left|n^{(0)}\right\rangle$ that is valid to first order. If we substitute the first-order approximation for $|\psi\rangle$ into the expression for E and neglect terms that are second order and higher, we arrive at a first-order approximation for the energy:

$$\boxed{E^{[1]} = E_n^{(0)} + \lambda \left\langle n^{(0)}\right| V \left|n^{(0)}\right\rangle} \tag{13.17}$$

The first-order approximation to the energy eigenstate is then given by

$$\boxed{\left|\psi^{[1]}\right\rangle = \left|n^{(0)}\right\rangle + \lambda \sum_{m\neq n} \frac{\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle}{E_n^{(0)} - E_m^{(0)}} \left|m^{(0)}\right\rangle} \tag{13.18}$$

since

$$R(E^{[1]}) = \sum_{m\neq n} \frac{\left|m^{(0)}\right\rangle \left\langle m^{(0)}\right|}{E_n^{(0)} + \lambda \left\langle n^{(0)}\right| V \left|n^{(0)}\right\rangle - E_m^{(0)}} = \sum_{m\neq n} \frac{\left|m^{(0)}\right\rangle \left\langle m^{(0)}\right|}{E_n^{(0)} - E_m^{(0)}} + O(\lambda). \tag{13.19}$$

It is common to refer to the correction terms themselves, so we denote the first-order correction to the energy $\left|E_n^{(1)}\right\rangle$ and the first-order correction to the energy eigenstate $\left|n^{(1)}\right\rangle$. Thus,

$$E^{[1]} = E_n^{(0)} + \lambda E_n^{(1)}, \quad \text{and} \quad \left|\psi^{[1]}\right\rangle = \left|n^{(0)}\right\rangle + \lambda \left|n^{(1)}\right\rangle \tag{13.20}$$

where

$$E_n^{(1)} = \left\langle n^{(0)}\right| V \left|n^{(0)}\right\rangle , \quad \text{and} \quad \left|n^{(1)}\right\rangle = \sum_{m\neq n} \frac{\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle}{E_n^{(0)} - E_m^{(0)}} \left|m^{(0)}\right\rangle \tag{13.21}$$

As you can see, the first-order correction to the energy is a diagonal matrix element of the perturbation Hamiltonian and the first-order correction to the state is a linear combination of the off-diagonal matrix elements of V and all states orthogonal to the unperturbed state.

13.3.2 Second-Order Approximation

Beginning with the exact expression for the energy, Eq. (13.15), and successively substituting the implicit expression for $|\psi\rangle$, we find

$$\begin{aligned} E &= E_n^{(0)} + \lambda \left\langle n^{(0)} \right| V \Big(\left| n^{(0)} \right\rangle + \lambda RV \left|\psi\right\rangle \Big) \\ &= E_n^{(0)} + \lambda \left\langle n^{(0)} \right| V \left| n^{(0)} \right\rangle + \lambda^2 \left\langle n^{(0)} \right| VRV \Big(\left| n^{(0)} \right\rangle + \lambda RV \left|\psi\right\rangle \Big) \\ &= E_n^{(0)} + \lambda \left\langle n^{(0)} \right| V \left| n^{(0)} \right\rangle + \lambda^2 \left\langle n^{(0)} \right| VRV \left| n^{(0)} \right\rangle + O(\lambda^3) \end{aligned}$$

Thus, to second order,

$$E^{[2]} = E_n^{(0)} + \lambda \left\langle n^{(0)} \right| V \left| n^{(0)} \right\rangle + \lambda^2 \left\langle n^{(0)} \right| VRV \left| n^{(0)} \right\rangle \tag{13.22}$$

Since the final term is already proportional to λ^2, $R(E)$ need only be evaluated using the zeroth-order energy $E_n^{(0)}$ in this term. Using the definition of R then gives

$$\boxed{E_n^{(2)} = \left\langle n^{(0)} \right| VRV \left| n^{(0)} \right\rangle = \sum_{m \neq n} \frac{|\left\langle m^{(0)} \right| V \left| n^{(0)} \right\rangle|^2}{E_n^{(0)} - E_m^{(0)}}} \tag{13.23}$$

To obtain the state through second order, we similarly continue the successive substitution in the exact equation for $|\psi\rangle$:

$$\begin{aligned} |\psi\rangle &= \left| n^{(0)} \right\rangle + \lambda RV \left| n^{(0)} \right\rangle + \lambda^2 RVRV \Big(\left| n^{(0)} \right\rangle + \lambda RV \left|\psi\right\rangle \Big) \\ &= \left| n^{(0)} \right\rangle + \lambda RV \left| n^{(0)} \right\rangle + \lambda^2 RVRV \left| n^{(0)} \right\rangle + O(\lambda^3) \end{aligned}$$

There is an additional subtlety in obtaining the second-order correction to the state since R depends on the energy E. Since the first term containing R is multiplied by λ, we must know $R(E)$ through first order in λ in order to determine $|\psi\rangle$ through second order. Using the second-order approximation to the energy, we have

$$R(E^{[2]}) = \sum_{m \neq n} \frac{\left| m^{(0)} \right\rangle \left\langle m^{(0)} \right|}{E_n^{(0)} - E_m^{(0)}} - \lambda \left\langle n^{(0)} \right| V \left| n^{(0)} \right\rangle \sum_{m \neq n} \frac{\left| m^{(0)} \right\rangle \left\langle m^{(0)} \right|}{\left(E_n^{(0)} - E_m^{(0)} \right)^2} + O(\lambda^2) \tag{13.24}$$

Although the explicit calculation is not shown, this follows from the series expansion

$$\frac{1}{a+\lambda b+\lambda^2 c} = \frac{1}{a} - \lambda\frac{b}{a^2} + O(\lambda^2)$$

Notice that the λ^2 correction to the energy does not contribute to R at the order required. Substituting this expansion for R and collecting all terms at each order of λ gives the second-order correction to the energy eigenstate:

$$\boxed{\begin{aligned}\left|n^{(2)}\right\rangle = &\sum_{m\neq n}\sum_{k\neq n}\frac{\left\langle m^{(0)}\right|V\left|k^{(0)}\right\rangle\left\langle k^{(0)}\right|V\left|n^{(0)}\right\rangle}{\left(E_n^{(0)}-E_m^{(0)}\right)\left(E_n^{(0)}-E_k^{(0)}\right)}\left|m^{(0)}\right\rangle \\ &-\sum_{m\neq n}\frac{\left\langle n^{(0)}\right|V\left|n^{(0)}\right\rangle\left\langle m^{(0)}\right|V\left|n^{(0)}\right\rangle}{\left(E_n^{(0)}-E_m^{(0)}\right)^2}\left|m^{(0)}\right\rangle\end{aligned}} \tag{13.25}$$

13.3.3 Summary of Key Results

Define

$$V_{mn} \equiv \left\langle m^{(0)}\right|V\left|n^{(0)}\right\rangle, \quad \text{and} \quad \Delta_{nm} \equiv E_n^{(0)} - E_m^{(0)}.$$

The second-order approximation to the energy is

$$E^{[2]} = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)}$$

and the second-order approximation to the state is

$$\left|\psi^{[2]}\right\rangle = \left|n^{(0)}\right\rangle + \lambda\left|n^{(1)}\right\rangle + \lambda^2\left|n^{(2)}\right\rangle$$

The first- and second-order corrections to the energy are

$$E_n^{(1)} = V_{nn}$$

and

$$E_n^{(2)} = \sum_{m\neq n}\frac{|V_{mn}|^2}{\Delta_{nm}}$$

The corresponding corrections to the state are

$$\left|n^{(1)}\right\rangle = \sum_{m\neq n}\frac{V_{mn}}{\Delta_{nm}}\left|m^{(0)}\right\rangle$$

and

$$\left|n^{(2)}\right\rangle = \sum_{m\neq n}\sum_{k\neq n}\frac{V_{mk}V_{kn}}{\Delta_{nm}\Delta_{nk}}\left|m^{(0)}\right\rangle - \sum_{m\neq n}\frac{V_{nn}V_{mn}}{\Delta_{nm}^2}\left|m^{(0)}\right\rangle$$

These results have been obtained using intermediate normalization, $\langle n^{(0)}|\psi\rangle = 1$. Consequently, all corrections to the state are orthogonal to $|n^{(0)}\rangle$. Normalization of the complete approximate state may be imposed afterward if needed.

13.3.4 Setting $\lambda = 1$

We introduced $H = H_0 + \lambda V$ to construct a perturbative solution. The physical Hamiltonian is $H = H_0 + V$, which corresponds to $\lambda = 1$. Therefore, after constructing the approximation to the desired order, we set $\lambda = 1$. For example, the energy through second order is

$$E_n \approx E_n^{(0)} + V_{nn} + \sum_{m\neq n} \frac{|V_{mn}|^2}{\Delta_{nm}}$$

Setting $\lambda = 1$ does not mean we are assuming 1 is a small number. The actual measure of the strength of the perturbation appears in dimensionless quantities such as $|V_{mn}/\Delta_{nm}|$ Thus, λ serves primarily as a bookkeeping parameter that organizes the successive corrections. The physical smallness of the perturbation is determined by the matrix elements of V relative to the relevant unperturbed energy gaps.

Example 13.1: Harmonic Oscillator with a Linear Perturbation

Consider a one-dimensional harmonic oscillator with mass m and angular frequency ω. The unperturbed Hamiltonian for this system is

$$H_0 = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

A weak linear potential is added as a perturbation, namely

$$V(x) = f_0 x,$$

where f_0 is a small positive constant. The full Hamiltonian is $H = H_0 + V$.

(a) Solve the full Hamiltonian by completing the square. In other words, determine the exact solutions to the energy eigenvalue equation.

(b) Expand the exact energies and eigenstates for small f_0.

(c) Use first-order perturbation theory to find the energy corrections,

(d) and then find the corrections to the energy eigenstates.

Solution. **(a)** The Hamiltonian is quadratic in x even with a linear term. Completing the

square in x, we find

$$\frac{1}{2}m\omega^2 x^2 + \lambda f_0 x = \frac{m\omega^2}{2}\left(x^2 + \frac{2\lambda f_0}{m\omega^2}x\right)$$
$$= \frac{m\omega^2}{2}\left(x + \frac{\lambda f_0}{m\omega^2}\right)^2 - \frac{\lambda^2 f_0^2}{2m\omega^2}$$

This potential describes a harmonic oscillator with a shifted equilibrium position

$$x_0 = -\frac{\lambda f_0}{m\omega^2}$$

and a constant energy shift. Thus, the exact energy eigenvalues are

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{\lambda^2 f_0^2}{2m\omega^2}$$

The exact eigenstates, in position space, are the harmonic oscillator eigenstates centered at the displaced equilibrium position:

$$\psi_n(x) = \psi_n^{(0)}\left(x + \frac{\lambda f_0}{m\omega^2}\right)$$

(b) Evidently, the leading order correction to the energy is second order:

$$E_n^{(1)} = 0$$
$$E_n^{(2)} = -\frac{f_0^2}{2m\omega^2}$$

From the exact eigenfunctions, we can Taylor expand about $f_0 = 0$ to find the first-order correction to the state:

$$\psi_n(x) = \psi_n^{(0)}\left(x + \frac{\lambda f_0}{m\omega^2}\right) \approx \psi_n^{(0)} + \frac{\lambda f_0}{m\omega^2}\frac{d\psi_n^{(0)}}{dx}$$

Therefore, the first-order correction is

$$\psi_n^{(1)}(x) = \frac{f_0}{m\omega^2}\frac{d\psi_n^{(0)}}{dx}$$

It will be convenient to work with the energy eigenkets rather than the position-space wave functions. Using

$$p = -i\sqrt{\frac{m\hbar\omega}{2}}\left(a - a^\dagger\right)$$

and $d/dx = ip/\hbar$, we have

$$\frac{d}{dx}|n\rangle = \sqrt{\frac{m\omega}{2\hbar}}\left(\sqrt{n}\,|n-1\rangle - \sqrt{n+1}\,|n+1\rangle\right)$$

Hence,

$$\left|n^{(1)}\right\rangle = \frac{f_0}{\sqrt{2m\hbar\omega^3}}\left(\sqrt{n}\,|n-1\rangle - \sqrt{n+1}\,|n+1\rangle\right)$$

(c) According to first-order perturbation theory,

$$E_n^{(1)} = \left\langle n^{(0)}\right| V \left|n^{(0)}\right\rangle = f_0 \left\langle n^{(0)}\right| x \left|n^{(0)}\right\rangle$$

The diagonal matrix elements are the expectation values of x in the energy eigenstates. Since the energy eigenstates are even or odd about the center of the potential ($x = 0$ for the unperturbed system), the diagonal elements vanish. Thus, $E_n^{(1)} = 0$.

(d) The first-order correction to the eigenstate is

$$\left|n^{(1)}\right\rangle = \sum_{m\neq n} \frac{\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle}{E_n^{(0)} - E_m^{(0)}} \left|m^{(0)}\right\rangle.$$

Using

$$x = \sqrt{\frac{\hbar}{2m\omega}}\left(a + a^\dagger\right),$$

we find

$$x\left|n^{(0)}\right\rangle = \sqrt{\frac{\hbar}{2m\omega}}\left(\sqrt{n}\,|n-1\rangle + \sqrt{n+1}\,|n+1\rangle\right).$$

The only nonzero matrix elements are

$$\langle n-1|\, x\,|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}$$

and

$$\langle n+1|\, x\,|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n+1}$$

and the corresponding energy gaps are

$$E_n^{(0)} - E_{n-1}^{(0)} = \hbar\omega, \quad \text{and} \quad E_n^{(0)} - E_{n+1}^{(0)} = -\hbar\omega$$

Therefore, first-order correction is

$$\left|n^{(1)}\right\rangle = \frac{f_0\sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}}{\hbar\omega}\,|n-1\rangle - \frac{f_0\sqrt{\frac{\hbar}{2m\omega}}\sqrt{n+1}}{\hbar\omega}\,|n+1\rangle,$$

or

$$\left|n^{(1)}\right\rangle = \frac{f_0}{\sqrt{2m\hbar\omega^3}}\left(\sqrt{n}\,|n-1\rangle - \sqrt{n+1}\,|n+1\rangle\right).$$

Example 13.2: Infinite Square Well with a Linear Perturbation

Consider a particle of mass m confined to an infinite square well extending from $x = 0$ to $x = L$. The unperturbed Hamiltonian is denoted H_0. A weak linear potential is added as a perturbation, namely

$$V(x) = f_0 x,$$

where f_0 is a small positive constant. The full Hamiltonian is $H = H_0 + V$.

(a) Find the first-order corrections to the energy eigenvalues $E_n^{(1)}$.

(b) Find the off-diagonal matrix elements $\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle$, $m \neq n$.

(c) Determine the first-order corrections to the energy eigenstates $\left|n^{(1)}\right\rangle$.

(d) Identify a dimensionless measure of the strength of the perturbation and state a condition under which first-order perturbation theory should be reliable.

Solution. The unperturbed eigenstates and energies are

$$\psi_n^{(0)} = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right), \quad E_n^{(0)} = \frac{n^2\pi^2\hbar^2}{2mL^2}.$$

(a) Formally, first-order correction to the energy eigenvalue is

$$E_n^{(1)} = \left\langle n^{(0)}\right| V \left|n^{(0)}\right\rangle = f_0 \left\langle n^{(0)}\right| x \left|n^{(0)}\right\rangle$$

In position space,

$$E_n^{(1)} = \frac{2f_0}{L}\int_0^L x\sin^2\left(\frac{n\pi x}{L}\right)dx$$

Using $2\sin^2(\theta) = 1 - \cos 2\theta$,

$$E_n^{(1)} = \frac{f_0}{L}\int_0^L x\left[1 - \cos\left(\frac{2n\pi x}{L}\right)\right]dx = \boxed{\frac{f_0 L}{2}}$$

(b) We solved for the off-diagonal matrix elements of the position operator in Group Activity #1, but we'll re-derive the result here anyways. Letting $V_{mn} = \left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle$,

$$V_{mn} = \frac{2f_0}{L}\int_0^L x\sin\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)dx.$$

Using $2\sin A\sin B = \cos(A-B) - \cos(A+B)$,

$$V_{mn} = \frac{f_0}{L}\int_0^L x\left[\cos\left(\frac{(m-n)\pi x}{L}\right) - \cos\left(\frac{(m+n)\pi x}{L}\right)\right]dx.$$

Since

$$\frac{\partial}{\partial a}\sin(ax) = x\cos(ax),$$

it follows that

$$\int x\cos(ax)dx = \frac{d}{da}\int \sin(ax)dx = \frac{d}{da}\left(-\frac{\cos(ax)}{a}\right) = \frac{\cos(ax)}{a^2} + \frac{x\sin(ax)}{a}$$

In our case, $a = (m\pm n)\pi/L$, so the product ax evaluates to $(m\pm n)\pi$ or 0 under the definite integral, and in both cases $\sin(ax) = 0$. Hence,

$$V_{mn} = \frac{f_0}{L}\frac{L^2}{\pi^2}\left[\left(\frac{\cos((m-n)\pi)-1}{(m-n)^2}\right) - \left(\frac{\cos((m+n)\pi)-1}{(m+n)^2}\right)\right]$$

If $m+n$ is even, so is $m-n$, and $\cos((m\pm n)\pi) = 1$. We can see above that both terms vanish, so $V_{mn} = 0$ when $m+n$ is even. On the other hand, if $m+n$ is odd, so is $m-n$, and $\cos((m\pm n)\pi) = -1$. For $m+n$ odd, we find

$$V_{mn} = -\frac{2f_0L}{\pi^2}\left[\frac{1}{(m-n)^2} - \frac{1}{(m+n)^2}\right] = -\frac{8f_0L}{\pi^2}\frac{mn}{(m^2-n^2)^2}$$

Therefore,

$$\boxed{V_{mn} = \begin{cases} -\dfrac{8f_0L}{\pi^2}\dfrac{mn}{(m^2-n^2)^2}, & m+n \text{ odd}, \\ 0, & m+n \text{ even}. \end{cases}}$$

It is interesting that the perturbation mixes only states with opposite parity.

(c) Formally, the first-order correction to the state is

$$\left|n^{(1)}\right\rangle = \sum_{m\neq n}\frac{\left\langle m^{(0)}\right| V \left|n^{(0)}\right\rangle}{E_n^{(0)} - E_m^{(0)}}\left|m^{(0)}\right\rangle.$$

We've already found the off-diagonal matrix elements of the perturbation, so all we need are the energy gaps:

$$E_n^{(0)} - E_m^{(0)} = \frac{\pi^2\hbar^2}{2mL^2}(n^2-m^2).$$

Thus, for $m+n$ odd,

$$\frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}} = \frac{16mf_0L^3}{\pi^4\hbar^2}\frac{mn}{(m^2-n^2)^3}$$

and the first-order correction to the energy eigenstate is

$$\left|n^{(1)}\right\rangle = \frac{16mf_0L^3}{\pi^4\hbar^2} \sum_{\substack{m\neq n \\ m+n \text{ odd}}} \frac{mn}{(m^2-n^2)^3} \left|m^{(0)}\right\rangle.$$

In position space,

$$\boxed{\psi_n^{(1)}(x) = \frac{16mf_0L^3}{\pi^4\hbar^2} \sum_{\substack{m\neq n \\ m+n \text{ odd}}} \frac{mn}{(m^2-n^2)^3} \psi_m^{(0)}(x).}$$

(d) The coefficients in the first-order correction to the eigenstate contain the dimensionless quantity

$$\frac{mf_0L^3}{\hbar^2}$$

Therefore, first-order perturbation theory should be reliable when $f_0 \ll \hbar^2/mL^3$.

13.4 Degenerate Perturbation Theory

Now suppose the unperturbed energy $E_n^{(0)}$ is d_n-fold degenerate, i.e.,

$$H_0 \left|n,a\right\rangle = E_n^{(0)} \left|n,a\right\rangle, \quad a = 1,\ldots,d_n. \tag{13.26}$$

Here n labels a distinct unperturbed energy eigenspace while a labels the different states within that space. The states satisfy

$$\langle n,a|m,b\rangle = \delta_{mn}\delta_{ab}$$

The projector onto the n-eigenspace associated with $E_n^{(0)}$ is

$$P_n = \sum_{a=1}^{d_n} \left|n,a\right\rangle \left\langle n,a\right| \tag{13.27}$$

while its orthogonal complement is

$$Q_n = \mathbb{1} - P_n = \sum_{m\neq n} \sum_{a=1}^{d_m} \left|m,a\right\rangle \left\langle m,a\right| \tag{13.28}$$

where d_m denotes the degeneracy of the mth eigenspace.

As before, we consider an exact eigenstate $|\psi\rangle$ satisfying $(H_0+\lambda V)\left|\psi\right\rangle = E\left|\psi\right\rangle$, and decompose the state as

$$\left|\psi\right\rangle = P_n\left|\psi\right\rangle + Q_n\left|\psi\right\rangle \tag{13.29}$$

There are many states with the same unperturbed energy, so we do not yet know which linear combinations of them will correspond to the perturbed eigenstates. In the end, we may find multiple solutions with different perturbed energies, in which case we say that the degeneracy has been "lifted" by the perturbation.

The determination of $Q_n |\psi\rangle$ is mostly the same as the nondegenerate case, with the key difference being a new definition for the reduced inverse:

$$R_n = \sum_{m \neq n} \sum_{a=1}^{d_m} \frac{|m,a\rangle \langle m,a|}{E - E_m^{(0)}} \tag{13.30}$$

The new implicit equation for $|\psi\rangle$ is

$$|\psi\rangle = P_n |\psi\rangle + \lambda R_n V |\psi\rangle \tag{13.31}$$

This is almost identical to the expression obtained in the nondegenerate case. The difference is that $P_n |\psi\rangle$ is now an unknown vector within a multidimensional subspace. We therefore need another equation to determine $P_n |\psi\rangle$. Apply P_n to the eigenvalue equation:

$$P_n(H_0 + \lambda V) |\psi\rangle = E P_n |\psi\rangle \tag{13.32}$$

Since $P_n H_0 = E_n^{(0)} P_n$, we find

$$(E - E_n^{(0)}) P_n |\psi\rangle = \lambda P_n V |\psi\rangle \tag{13.33}$$

Substituting Eq. (13.31) into the right-hand side of Eq. (13.33), we find

$$(E - E_n^{(0)}) P_n |\psi\rangle = \lambda P_n V P_n |\psi\rangle + O(\lambda^2) \tag{13.34}$$

Since $P_n |\psi\rangle$ is a linear combination of the eigenstates within the degenerate subspace,

$$P_n |\psi\rangle = \sum_{a=1}^{d_n} c_a |n,a\rangle \tag{13.35}$$

where c_a are expansion coefficients. Taking the inner product with $|n,a\rangle$,

$$\langle n,a| P_n |\psi\rangle = \langle n,a|\psi\rangle = c_a. \tag{13.36}$$

We then take the inner product of Eq. (13.34) with $|n,a\rangle$ and ignore terms that are second order and higher in the perturbation. The first-order approximation to the energy is given implicitly by

$$\left(E^{[1]} - E_n^{(0)}\right) c_a = \lambda \sum_b \langle n,a| V |n,b\rangle c_b \tag{13.37}$$

Set $\lambda = 1$ to obtain the physical Hamiltonian and define

$$E_n^{(1)} = E^{[1]} - E_n^{(0)}, \quad \text{and} \quad V_{ab}^{(n)} = \langle n, a | V | n, b \rangle \tag{13.38}$$

Eq. (13.37) then takes the form of an eigenvalue equation:

$$\sum_b V_{ab}^{(n)} c_b = E_n^{(1)} c_a \tag{13.39}$$

Thus, finding the first-order corrections reduces to an ordinary $d_n \times d_n$ matrix eigenvalue problem. The eigenvalues of the perturbation matrix V_{ab} give the possible first-order corrections $E_n^{(1)}$ of the originally degenerate energy $E_n^{(0)}$, while the corresponding eigenvectors determine the particular linear combinations of the degenerate unperturbed states that become the perturbed eigenstates. If the eigenvalues of V_{ab} are distinct, the degeneracy is lifted at first order; if some remain equal, the degeneracy is only partially lifted (or not at all).

Once these linear combinations have been determined, the first-order correction to each state is found in essentially the same way as in the nondegenerate case. The only difference is that the sum over intermediate states must exclude the entire degenerate eigenspace associated with $E_n^{(0)}$, rather than a single state.

Example 13.3: A 3-Level System with Degeneracy

Consider a three-level quantum system with the following unperturbed Hamiltonian

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \Delta \end{pmatrix}, \quad \Delta > 0,$$

and the perturbation

$$V = \begin{pmatrix} 0 & g & a \\ g & 0 & b \\ a & b & 0 \end{pmatrix}$$

Assume that g, a, b are small compared to Δ. Observe that the first two unperturbed states are degenerate. First, determine the first-order energy corrections and the corresponding zeroth-order states selected by the perturbation within the degenerate subspace. Then determine the first-order corrections to these states. Lastly, compute the first-order corrections to the energy and state of the single nondegenerate eigenstate.

Solution. The unperturbed Hamiltonian is already diagonal in the standard orthonormal basis $\{|i\rangle\}$, i.e.,

$$H_0 |1\rangle = 0, \quad H_0 |2\rangle = 0, \quad H_0 |3\rangle = \Delta |3\rangle$$

The zero-energy eigenspace is two-dimensional with basis $|1,1\rangle = |1\rangle$ and $|1,2\rangle = |2\rangle$ with $E_1^{(0)} = 0$. The remaining state belongs to a different eigenspace with $E_2^{(0)} = \Delta$ and $|2,1\rangle = |3\rangle$. The projector onto the degenerate eigenspace is

$$P_1 = |1\rangle\langle 1| + |2\rangle\langle 2| = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and the perturbation restricted to this subspace is

$$V_1 = \begin{pmatrix} \langle 1|V|1\rangle & \langle 1|V|2\rangle \\ \langle 2|V|1\rangle & \langle 2|V|2\rangle \end{pmatrix} = \begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix}$$

We must now diagonalize this matrix to find the unperturbed states and first-order energy corrections. The characteristic equation is

$$\det \begin{pmatrix} -E^{(1)} & g \\ g & -E^{(1)} \end{pmatrix} = 0, \quad \text{or} \quad \left(E^{(1)}\right)^2 - g^2 = 0$$

The corresponding energy shifts are

$$E_+^{(1)} = g, \quad \text{and} \quad E_-^{(1)} = -g.$$

For $E_+^{(1)} = g$,

$$\begin{pmatrix} 0 & g \\ g & 0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = g \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

which gives $c_1 = c_2$. Choosing normalized coefficients, let

$$\left|\psi_+^{(0)}\right\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right)$$

be the first eigenstate to zeroth order. Similarly, for $E_-^{(1)} = -g$, we find $c_1 = -c_2$, and therefore

$$\left|\psi_-^{(0)}\right\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle - |2\rangle\right)$$

is the second eigenstate to zeroth order. Thus, the originally degenerate energy levels split at first order:

$$\boxed{E_+^{[1]} = g, \quad E_-^{[1]} = -g}$$

and the correct zeroth states are given by

$$\boxed{\left|\psi_+^{(0)}\right\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right), \quad \left|\psi_-^{(0)}\right\rangle = \frac{1}{\sqrt{2}}\left(|1\rangle - |2\rangle\right)}$$

First-order corrections to the degenerate states. At this point, the degeneracy problem has been resolved within the original degenerate subspace. We can now use the same formula as in nondegenerate perturbation theory:

$$\left|\psi_\pm^{(1)}\right\rangle = \sum_{m\neq 1}\sum_b \frac{\langle m,b|\, V \left|\psi_\pm^{(0)}\right\rangle}{E_1^{(0)} - E_m^{(0)}} |m,b\rangle$$

where $|m,b\rangle$ denotes the zeroth-order eigenstates outside the degenerate subspace. In this case, there is only one such state, namely $|2,1\rangle = |3\rangle$. Thus,

$$\left|\psi_\pm^{(1)}\right\rangle = \frac{\langle 3|\, V \left|\psi_\pm^{(0)}\right\rangle}{0-\Delta} |3\rangle$$

For the state $\left|\psi_+^{(1)}\right\rangle$,

$$\langle 3|\, V \left|\psi_+^{(0)}\right\rangle = \frac{1}{\sqrt{2}}\left(\langle 3|\, V\, |1\rangle + \langle 3|\, V\, |2\rangle\right) = \frac{a+b}{\sqrt{2}}$$

Therefore,

$$\boxed{\left|\psi_+^{(1)}\right\rangle = -\frac{a+b}{\sqrt{2}\,\Delta}|3\rangle}$$

Similarly,

$$\langle 3|\, V \left|\psi_-^{(0)}\right\rangle = \frac{a-b}{\sqrt{2}}$$

and therefore,

$$\boxed{\left|\psi_-^{(1)}\right\rangle = -\frac{a-b}{\sqrt{2}\,\Delta}|3\rangle}$$

First-order corrections for the nondegenerate state. The first-order correction to the energy E_2 (the energy of the *second* eigenspace) is simply given by

$$\boxed{E_2^{(1)} = \langle 3|\, V\, |3\rangle = 0}$$

In other words, the energy does not change to first order in the perturbation. The first-order

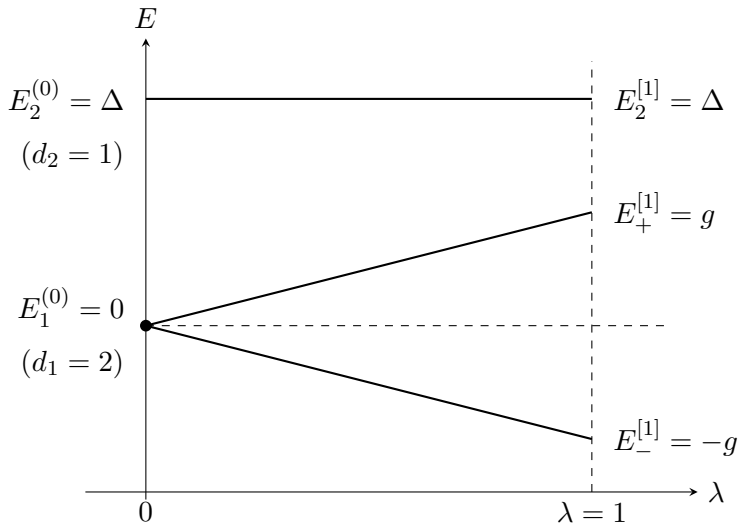
correction to $|2,1\rangle = |3\rangle$ is

$$\left|3^{(1)}\right\rangle = \frac{\left\langle \psi_+^{(0)} \right| V \left|3\right\rangle}{\Delta} \left|\psi_+^{(0)}\right\rangle + \frac{\left\langle \psi_-^{(0)} \right| V \left|3\right\rangle}{\Delta} \left|1_-^{(0)}\right\rangle$$

After working through some calculations, we find a rather simple result:

$$\boxed{\left|3^{(1)}\right\rangle = \frac{a}{\Delta}\left|1\right\rangle + \frac{b}{\Delta}\left|2\right\rangle}$$

See the figure below for a schematic representation of the energy levels as a function of λ, illustrating how the perturbation lifts the degeneracy at first order.



13.4.1 Finding Good States in Degenerate Perturbation Theory

Theorem: Let H_0 have a degenerate eigenspace with energy $E_n^{(0)}$, and let V be the perturbation. Suppose A is a Hermitian operator satisfying

$$[A, H_0] = 0, \qquad [A, V] = 0.$$

The states within the degenerate eigenspace of H_0 can be chosen to be eigenstates of A, and furthermore, if

$$A\left|n,\alpha\right\rangle = a_\alpha \left|n,\alpha\right\rangle,$$

then states with different A-eigenvalues cannot be coupled by the perturbation:

$$\text{If } \; a_\alpha \neq a_\beta \quad \Longrightarrow \quad \left\langle n,\alpha\right| V \left|n,\beta\right\rangle = 0.$$

Consequently, if the eigenvalues a_α are all distinct within the degenerate eigenspace, the basis $\{|n,\alpha\rangle\}$ already diagonalizes V within that subspace. These are therefore the appropriate states to use in degenerate perturbation theory.

Proof. Because

$$[A, H_0] = 0,$$

the operator A preserves each eigenspace of H_0. Indeed, if

$$H_0 |n,\alpha\rangle = E_n^{(0)} |n,\alpha\rangle ,$$

then

$$H_0 A |n,\alpha\rangle = A H_0 |n,\alpha\rangle = E_n^{(0)} A |n,\alpha\rangle .$$

Thus $A|n,\alpha\rangle$ is also contained in the same degenerate eigenspace. Since A is Hermitian, we can diagonalize it within that eigenspace and choose the states $|n,\alpha\rangle$ to satisfy

$$A |n,\alpha\rangle = a_\alpha |n,\alpha\rangle .$$

Now use $[A, V] = 0$ and consider the matrix element

$$\langle n,\alpha| AV |n,\beta\rangle = \langle n,\alpha| VA |n,\beta\rangle .$$

Since $|n,\alpha\rangle$ and $|n,\beta\rangle$ are eigenstates of A,

$$\langle n,\alpha| AV |n,\beta\rangle = a_\alpha \langle n,\alpha| V |n,\beta\rangle .$$

On the other hand,

$$\langle n,\alpha| VA |n,\beta\rangle = a_\beta \langle n,\alpha| V |n,\beta\rangle .$$

Since $AV = VA$, these must be equal:

$$a_\alpha \langle n,\alpha| V |n,\beta\rangle = a_\beta \langle n,\alpha| V |n,\beta\rangle .$$

Therefore,

$$(a_\alpha - a_\beta) \langle n,\alpha| V |n,\beta\rangle = 0.$$

Hence, whenever

$$a_\alpha \neq a_\beta,$$

we must have

$$\langle n,\alpha| V |n,\beta\rangle = 0.$$

Thus, V does not mix states having different eigenvalues of A.

Connection to Degenerate Perturbation Theory

Normally, for a d_n-fold degenerate level, we construct

$$V_{\alpha\beta} = \langle n, \alpha | \, V \, | n, \beta \rangle$$

and diagonalize this $d_n \times d_n$ matrix. The theorem in the previous section gives a shortcut. If a Hermitian operator A commutes with both H_0 and V, and A has distinct eigenvalues within the degenerate subspace, then

$$V_{\alpha\beta} = 0 \qquad (\alpha \neq \beta),$$

so the perturbation matrix is already diagonal in the A-eigenbasis. The corresponding first-order energy shifts are simply

$$\boxed{E^{(1)}_{n,\alpha} = \langle n, \alpha | \, V \, | n, \alpha \rangle \, .}$$

An important qualification is that A must distinguish the states. If two states within the degenerate subspace have the same A-eigenvalue, the theorem does not tell us that V cannot mix them. In that remaining subspace, we may still need to diagonalize V.

We can make a slight modification to Example 3.1 to illustrate the usefulness of this theorem. Let $a = b$ in the perturbation V and consider the Hermitian matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

It is easy to verify that $[H_0, A] = [V, A] = 0$. The eigenvectors of A are

$$|+\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle), \qquad |-\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle), \qquad |3\rangle \, ,$$

with eigenvalues $+1, -1, +1$, respectively. It's okay that A has a degenerate eigenvalue of $+1$ because A is nondegenerate in the degenerate eigenspace of H_0. Thus, the two states within the degenerate $E = 0$ subspace of H_0 have distinct eigenvalues of A, and are precisely the linear combinations we found previously using degenerate perturbation theory. The theorem therefore provides a shortcut for identifying the "good" states without explicitly diagonalizing the perturbation within the degenerate subspace.

The operator A has a simple interpretation: it exchanges the states $|1\rangle$ and $|2\rangle$ while leaving $|3\rangle$ unchanged,

$$A \, |1\rangle = |2\rangle \, , \qquad A \, |2\rangle = |1\rangle \, , \qquad A \, |3\rangle = |3\rangle \, .$$

When $a = b$, both H_0 and V are invariant under this exchange—in either matrix, if you swap rows 1 and 2, and then swap columns 1 and 2, you'll find that you're left with the same matrix you started with. In this sense, the exchange of $|1\rangle$ and $|2\rangle$ is a *symmetry* of both the unperturbed

system and the perturbation. The relations

$$[A, H_0] = 0, \qquad [A, V] = 0$$

can be viewed as a consequence of this symmetry.

The significance of this observation extends beyond the present example. When a Hamiltonian possesses a symmetry, its eigenstates can often be classified according to how they transform under the corresponding symmetry operator. Here, the states

$$\frac{1}{\sqrt{2}}\left(|1\rangle + |2\rangle\right), \qquad \frac{1}{\sqrt{2}}\left(|1\rangle - |2\rangle\right)$$

are eigenstates of the exchange symmetry with eigenvalues $+1$ and -1, respectively. Since the perturbation also respects this symmetry, it cannot mix states belonging to these different symmetry classes. This is why the symmetry eigenstates are precisely the "good" states for degenerate perturbation theory. We will encounter similar ideas later in the course.

13.5 Inverse Operators

Consider the operator $B = c\mathbb{1} - A$ where c is a scalar and the operator A is Hermitian. We are interested in finding the inverse operator, B^{-1}, such that $B^{-1}B = BB^{-1} = \mathbb{1}$, and possibly the conditions for its existence. On the eigenspace of A,

$$B\,|a_n\rangle = (c - a_n)\,|a_n\rangle\,.$$

Provided $c \neq a_n$, we can divide both sides by $c - a_n$ and write

$$B\left(\frac{1}{c - a_n}\right)|a_n\rangle = |a_n\rangle\,.$$

Furthermore, if c cannot be found anywhere within the spectrum of A, then the equation above holds for all n. Since A is Hermitian, it has a complete, orthonormal set of eigenvectors. Thus,

$$\mathbb{1} = \sum_n |a_n\rangle\langle a_n| = B\sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n} = \sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n} B,$$

and it follows that

$$B^{-1} = (c\mathbb{1} - A)^{-1} = \sum_n \frac{|a_n\rangle\langle a_n|}{c - a_n}$$

Chapter 14

Perturbations of Hydrogen

14.1 Introduction

The nonrelativistic hydrogen Hamiltonian provides one of the central exactly solvable models of quantum mechanics, but it does not describe every feature of the observed hydrogen spectrum. External electric and magnetic fields shift and split the energy levels, while relativistic effects produce smaller corrections even for an isolated atom. These effects are weak (or can be made weak) compared with the Coulomb interaction and therefore provide natural applications of time-independent perturbation theory.

We begin with the Stark effect, which results from placing the atom in an external electric field. We then consider two contributions to the fine structure of hydrogen: the relativistic correction to the electron's kinetic energy and the coupling between its orbital and spin angular momenta. Finally, we will consider the Zeeman effect due to an external magnetic field, distinguishing the regimes in which the magnetic interaction is weak or strong compared with spin-orbit coupling. These perturbations may break symmetries of H_0, lift degeneracies, and select new linear combinations of the unperturbed hydrogen states as the relevant energy eigenstates.

14.2 The Stark Effect

The *Stark effect* refers to the shift and splitting of atomic energy levels caused by an external electric field. We will consider a hydrogen atom placed in a uniform, time-independent electric field

$$\mathbf{E} = \mathcal{E}_0 \hat{\mathbf{z}} \tag{14.1}$$

We take the proton to be fixed at the origin and neglect spin throughout this section. The unperturbed Hamiltonian of the hydrogen atom in the electrostatic model is taken as

$$H_0 = \frac{\mathbf{p}^2}{2m_e} - \frac{k_e e^2}{r} \tag{14.2}$$

with energy eigenstates $|n, \ell, m\rangle$ and corresponding energies

$$E_n^{(0)} = -\frac{13.6 \text{ eV}}{n^2} \tag{14.3}$$

Since the energy depends only on the principal quantum number n, states with different values of ℓ and m may be degenerate.

The uniform electric field in Eq. (14.1) may be derived from the scalar potential

$$\phi(\mathbf{r}) = -\mathbf{E} \cdot \mathbf{r} = -\mathcal{E}_0 z \tag{14.4}$$

and since the electron has charge $-e$, the potential energy is given by

$$V = -e\phi = e\mathcal{E}_0 z \tag{14.5}$$

We will assume the electric field is weak so that V may be treated as a perturbation.

14.2.1 Selection Rules

Before calculating the energy shifts, it's useful to determine which unperturbed states can be connected by the perturbation. We'll find that many of the matrix elements vanish as a result of symmetry. In spherical coordinates,

$$z = r\cos\theta$$

and we notice that the angular dependence of z is proportional to the spherical harmonic $Y_1^0(\theta, \phi)$. In particular,

$$z = r\sqrt{\frac{4\pi}{3}} Y_1^0(\theta, \phi).$$

Consider the matrix element $\langle n', \ell', m'|z|n, l, m\rangle$. Since the hydrogen wave functions separate into radial and angular parts, we can restrict our attention to the angular integral

$$I = \int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m)$$

The ϕ-dependence of Y_ℓ^m is given by $e^{im\phi}$, so the ϕ-dependent part of the angular integral is

$$\int_0^{2\pi} d\phi\, e^{-im'\phi} e^{im\phi} = \int_0^{2\pi} d\phi\, e^{i(m-m')\phi} = 2\pi\delta_{mm'}.$$

Therefore, the matrix element vanishes unless $m' = m$ or $\Delta m = 0$.

To determine the allowed values of ℓ, we begin by proving the following spherical-harmonic identity:

$$\cos\theta Y_\ell^m = C_{\ell m} Y_{\ell+1}^m + D_{\ell m} Y_{\ell-1}^m$$

where $C_{\ell m}$ and $D_{\ell m}$ are constants that depend on ℓ and m. The θ-dependence of Y_ℓ^m is given by

the associated Legendre polynomial $P_\ell^m(\cos\theta)$, which satisfies the recurrence relation:

$$(\ell - m + 1)\, P_{\ell+1}^m(x) = (2\ell + 1)\, x\, P_\ell^m(x) - (\ell + m)\, P_{\ell-1}^m(x)$$

where $x = \cos\theta$. Rearrange the expression and define constants $A_{\ell m}$ and $B_{\ell m}$ such that

$$x\, P_\ell^m(x) = A_{\ell m} P_{\ell+1}^m(x) + B_{\ell m} P_{\ell-1}^m(x)$$

The spherical-harmonic identity follows directly from the expression above with the appropriate definitions for $C_{\ell m}$ and $D_{\ell m}$. Substituting the identity into the angular integral,

$$\int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m) = C_{\ell m} \int d\Omega\, (Y_{\ell'}^{m'})^* (Y_{\ell+1}^m) + D_{\ell m} \int d\Omega\, (Y_{\ell'}^{m'})^* (Y_{\ell-1}^m),$$

and using orthonormality,

$$\int d\Omega\, (Y_{\ell'}^{m'})^* \cos\theta (Y_\ell^m) = C_{\ell m} \delta_{\ell',\ell+1} \delta_{m'm} + D_{\ell m} \delta_{\ell',\ell-1} \delta_{m'm}.$$

Consequently, the angular integral can only be nonzero if $m' = m$, which we already knew, and $\ell' = \ell + 1$ or $\ell' = \ell - 1$. Putting the two results together,

$$\big\langle n', \ell', m' \big| z \big| n, l, m \big\rangle \neq 0 \quad \text{iff} \quad \Delta\ell = \pm 1,\ \Delta m = 0 \tag{14.6}$$

It follows that every diagonal matrix element of the perturbation vanishes, but this does not imply that every first-order Stark shift vanishes. If an energy level is degenerate, the off-diagonal matrix elements of V within the degenerate eigenspace must also be considered.

14.2.2 The Ground State

The ground state of hydrogen $|1,0,0\rangle$ is nondegenerate. Its first-order energy correction is

$$E_1^{(1)} = \langle 1,0,0|V|1,0,0\rangle = e\mathcal{E}_0 \langle 1,0,0|z|1,0,0\rangle = 0 \tag{14.7}$$

The second-order energy correction is formally given by

$$E_1^{(2)} = e^2 \mathcal{E}_0^2 \sum_{n' \neq 1} \sum_{\ell', m'} \frac{|\langle n', \ell', m'|z|1,0,0\rangle|^2}{E_1^{(0)} - E_{n'}^{(0)}} \tag{14.8}$$

where the selection rules restrict the contributing intermediate states to those with $\ell' = 1$ and $m' = 0$. Since every excited state has energy greater than the ground state, the second-order correction is negative. Strictly speaking, the complete spectral summation also includes the continuum states, but there's a convenient way to get around that complication. The second-order correction to the energy can be directly calculated from the first-order correction in the state. In Chapter 13, we

used intermediate normalization to show that

$$E = E_n^{(0)} + \lambda \left\langle n^{(0)} \middle| V \middle| \psi \right\rangle, \tag{14.9}$$

in the nondegenerate case. If we substitute the first-order approximation for $|\psi\rangle$ into Eq. (14.9), we obtain an approximation to E that is valid to second order. The second-order correction to the energy can also be written as

$$E_n^{(2)} = \left\langle n^{(0)} \middle| V \middle| n^{(1)} \right\rangle \tag{14.10}$$

where $\left|n^{(1)}\right\rangle$ is the first-order correction to the state. To first-order, we have

$$|\psi\rangle = \left|n^{(0)}\right\rangle + \lambda \left|n^{(1)}\right\rangle + O(\lambda^2), \quad E = E_n^{(0)} + \lambda E_n^{(1)} + O(\lambda^2).$$

Substituting these expressions into the energy eigenvalue equation,

$$(H_0 + \lambda V)\,|\psi\rangle = E\,|\psi\rangle\,,$$

and neglecting terms that are second order and higher, we find

$$H_0 \left|n^{(0)}\right\rangle + \lambda \left(H_0 \left|n^{(1)}\right\rangle + V \left|n^{(0)}\right\rangle\right) = E_n^{(0)} \left|n^{(0)}\right\rangle + \lambda \left(E_n^{(0)} \left|n^{(1)}\right\rangle + E_n(1) \left|n^{(0)}\right\rangle\right).$$

Note that $H_0 \left|n^{(0)}\right\rangle = E_n^{(0)} \left|n^{(0)}\right\rangle$ and $E_n^{(1)} = 0$ for the hydrogen ground state in this perturbation. Therefore, the first-order correction to the energy eigenstate satisfies

$$(H - E_n^{(0)}) \left|n^{(1)}\right\rangle = -V \left|n^{(0)}\right\rangle \tag{14.11}$$

If we can solve Eq. (14.11) to determine the first-order correction, then we can use it to calculate the second-order correction to the energy using Eq. (14.10).

Define natural units (also called atomic units) such that $\hbar = m_e = e = a_0 = 1$, where a_0 is the Bohr radius (in terms of the electron's mass). In these units, the relevant energy and momentum scales are

$$[\text{energy}] = \frac{\hbar^2}{m_e a_0^2}, \quad \text{and} \quad [\text{momentum}] = \frac{\hbar}{a_0},$$

and from the definition of the Bohr radius,

$$a_0 = \frac{\hbar^2}{m_e k_e e^2}$$

it follows that $k_e = 1$. In natural units, the unperturbed Hamiltonian is

$$H_0 = \frac{1}{2}\mathbf{p}^2 - \frac{1}{r}$$

and the perturbation may be written as

$$V = \mathcal{E}_0\, r\cos\theta.$$

where the electric field is measured in units of

$$[\text{electric field}] = \frac{\hbar^2}{m_e a^3}\frac{1}{e} = \frac{k_e e}{a_0^2}.$$

The unperturbed ground-state energy is $E_1^{(0)} = -1/2$. In position-space, Eq. (14.11) becomes

$$\left(-\frac{1}{2}\nabla^2 - \frac{1}{r} + \frac{1}{2}\right)\psi_1^{(1)}(\mathbf{r}) = -\mathcal{E}_0\, r\cos\theta\,\psi_{100}(\mathbf{r}), \tag{14.12}$$

where in natural units, the (unperturbed) ground-state wave function is

$$\psi_{100}(\mathbf{r}) = \frac{1}{\sqrt{\pi}}e^{-r}.$$

Since the right-hand side is proportional to $\cos\theta$ and ψ_{100}, we introduce the ansatz

$$\psi_1^{(1)}(\mathbf{r}) = -\mathcal{E}_0 f(r)\cos\theta\,\psi_{100}(r) \tag{14.13}$$

for an undetermined function $f(r)$. Substituting the ansatz into Eq. (14.12), and working through some tedious calculations, we arrive at the following differential equation for $f(r)$:

$$-\frac{1}{2}f''(r) + \left(1 - \frac{1}{r}\right)f'(r) + \frac{f(r)}{r^2} = r \tag{14.14}$$

At first it might seem like we've traded one hard problem for another since we have to solve a differential equation, but as it turns out, the solution satisfying the appropriate boundary conditions is given by

$$f(r) = r + \frac{1}{2}r^2, \tag{14.15}$$

as you can easily verify. The radial part of the first-order correction is $R_1^{(1)}(r) \propto f(r)e^{-r}$. Given the solution for $f(r)$ above, $R_1^{(1)} < \infty$ for $r \to 0$ and $R_1^{(1)} \to 0$ for $r \to \infty$. Hence, it is a good solution (see Chapter 9 if you need a refresher on boundary conditions).

Given $\psi_1^{(1)}(\mathbf{r})$, the second-order correction to the ground-state energy can be written as

$$E_1^{(2)} = -\mathcal{E}_0^2\left\langle\left(r^2 + \frac{1}{2}r^3\right)\cos^2\theta\right\rangle_{100}, \tag{14.16}$$

where the angled brackets indicate an expectation value in the unperturbed ground state. The

ground state is spherically symmetric, so the radial and angular parts factorize as

$$E_1^{(2)} = -\mathcal{E}_0^2 \left(\langle r^2 \rangle_{100} + \left\langle \frac{1}{2} r^3 \right\rangle_{100} \right) \langle \cos^2 \theta \rangle_{100} \,, \tag{14.17}$$

For the ground state of hydrogen, if k is a nonnegative integer, then

$$\langle r^k \rangle_{100} = \frac{(k+2)!}{2^{k+1}}, \tag{14.18}$$

and it follows that $\langle r^2 \rangle = 3$ and $\langle r^3 \rangle = 15/2$. Finally, the angular expectation value gives a factor of $1/3$. Altogether, the second-order correction to the energy is

$$E_1^{(2)} = -\frac{9}{4} \mathcal{E}_0^2 \tag{14.19}$$

We convert to ordinary units by replacing

$$E_1^{(2)} \to \frac{E_1^{(2)}}{\hbar^2 / m_e a_0^2}, \quad \text{and} \quad \mathcal{E}_0 \to \frac{\mathcal{E}_0}{k_e e / a_0^2}.$$

Since $k_e = 1/4\pi\varepsilon_0$, we can write the final form of the correction as

$$\boxed{E_1^{(2)} = -9\pi\varepsilon_0 a_0^3 \mathcal{E}_0^2} \tag{14.20}$$

The ground-state energy shift is quadratic in the applied electric field and is therefore called the *quadratic Stark effect*. This energy shift is related to the atomic polarizability of ground-state hydrogen, and it has been measured experimentally. Early experiments used the deflection of atomic hydrogen in an inhomogeneous static electric field, while later measurements used interferometry to determine the polarizability with greater precision.

14.2.3 The First Excited Energy Level

The $n = 2$ energy eigenspace is four-dimensional, with basis

$$|2,0,0\rangle \,, \qquad |2,1,0\rangle \,, \qquad |2,1,1\rangle \,, \qquad |2,1,-1\rangle \,. \tag{14.21}$$

These states all have the same unperturbed energy

$$E_2^{(0)} = -\frac{13.6 \text{ eV}}{4}.$$

Because this energy is degenerate, we must diagonalize the perturbation within the $n = 2$ eigenspace. Let

$$V_{\alpha\beta}^{(2)} = \langle 2, \alpha | V | 2, \beta \rangle \,,$$

where α and β label the four states within the $n = 2$ eigenspace.

The selection rule $\Delta m = 0$ prevents the perturbation from coupling states with different values of m. The selection rule $\Delta\ell = \pm 1$ also prevents V from coupling two $2p$ states to one another. The only nonzero matrix elements within the $n = 2$ eigenspace are therefore

$$\langle 2,0,0|V|2,1,0\rangle \qquad \text{and} \qquad \langle 2,1,0|V|2,0,0\rangle\,.$$

Using the standard hydrogen wave functions,

$$\langle 2,0,0|z|2,1,0\rangle = -3a_0, \tag{14.22}$$

and hence

$$\langle 2,0,0|V|2,1,0\rangle = -3ea_0\mathcal{E}_0. \tag{14.23}$$

In the ordered basis

$$\{|2,0,0\rangle\,, |2,1,0\rangle\,, |2,1,1\rangle\,, |2,1,-1\rangle\}\,,$$

the perturbation matrix is

$$V^{(2)} = 3ea_0\mathcal{E}_0 \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{14.24}$$

The first-order energy corrections are the eigenvalues of this matrix:

$$\boxed{E^{(1)}_{2,+} = +3ea_0\mathcal{E}_0, \qquad E^{(1)}_{2,-} = -3ea_0\mathcal{E}_0, \qquad E^{(1)}_{2,0} = 0,} \tag{14.25}$$

where the zero-order correction remains two-fold degenerate. The corresponding zeroth-order states that evolve smoothly into eigenstates of the perturbed Hamiltonian are

$$|2,+\rangle = \frac{1}{\sqrt{2}}\left(|2,0,0\rangle - |2,1,0\rangle\right), \tag{14.26}$$

$$|2,-\rangle = \frac{1}{\sqrt{2}}\left(|2,0,0\rangle + |2,1,0\rangle\right), \tag{14.27}$$

together with the two unshifted states

$$|2,1,1\rangle\,, \qquad |2,1,-1\rangle\,. \tag{14.28}$$

The states $|2,0,0\rangle$ and $|2,1,0\rangle$ individually have definite parity and therefore have no permanent electric dipole moment along the z-axis. The linear combinations selected by the perturbation do not have definite parity. Indeed,

$$\langle 2,\pm|z|2,\pm\rangle = \pm 3a_0, \tag{14.29}$$

so these states possess equal and opposite electric dipole moments along the direction of the applied

field. The applied electric field lifts the degeneracy of the first excited energy level by selecting states with charge distributions that are polarized either parallel or antiparallel to the field. Because the resulting energy shifts are proportional to the field strength $\mathcal{E}_0$, the splitting is known as the *linear Stark effect*.

The qualitative difference between the $n = 1$ and $n = 2$ Stark effects arises from degeneracy. The nondegenerate ground state has no first-order energy shift and exhibits a quadratic Stark effect. The degenerate $n = 2$ level contains opposite-parity states that can be coupled directly by the perturbation, producing a linear Stark effect.

14.3 Relativistic Corrections

According to the special theory of relativity, the relativistic energy of a particle is given by

$$E = \sqrt{m^2c^4 + p^2c^2}. \tag{14.30}$$

where p is the relativistic momentum. Using a binomial expansion in powers of p/mc,

$$E = mc^2 \left[1 + \frac{1}{2}\left(\frac{p}{mc}\right)^2 - \frac{1}{8}\left(\frac{p}{mc}\right)^4 + \cdots\right]. \tag{14.31}$$

When the relativistic momentum p is sufficiently small compared to mc, we can use the first three terms to approximate the relativistic energy as

$$E \approx mc^2 + \frac{p^2}{2m} - \frac{p^4}{8m^3c^2}. \tag{14.32}$$

The first term is the rest energy of the particle and the remaining terms constitute an approximation to the kinetic energy. In particular, we can treat the third term as a perturbation to the nonrelativistic kinetic energy $p^2/2m$. Let the perturbation Hamiltonian be

$$V = -\frac{\mathbf{p}^4}{8m_e^3c^2} \tag{14.33}$$

where $\mathbf{p}$ is now the momentum operator and m_e is the mass of the electron.

Although V has arisen as the next term in a low-momentum expansion, we should still verify that it is small compared with the unperturbed hydrogen Hamiltonian. The characteristic momentum scale of an electron in hydrogen is $\hbar/a_0$ and the rest energy of the electron is m_ec^2. We can therefore write the perturbation Hamiltonian as

$$V = -\frac{1}{8}\left(\frac{\hbar/a_0}{m_ec}\right)^4 m_ec^2 \left(\frac{\mathbf{p}}{\hbar/a_0}\right)^4 = -\frac{1}{8}\alpha^4 m_ec^2 \left(\frac{\mathbf{p}}{\hbar/a_0}\right)^4,$$

where $\alpha \approx 1/137$ is the *fine-structure constant*. The operator in parentheses is dimensionless, and its expectation value in a hydrogen energy eigenstate is generally of order one. The characteristic

size of the relativistic correction is therefore $\alpha^4 m_e c^2$. By comparison, the characteristic scale of the nonrelativistic hydrogen binding energy is $\alpha^2 m_e c^2$. Since the relativistic correction is smaller than the nonrelativistic energy by a factor of order α^2, V may be treated as a perturbation.

Now that we've identified the perturbation Hamiltonian, we can apply perturbation theory to determine the resulting energy corrections. The unperturbed hydrogen energies are degenerate with respect to ℓ and m, so we should use degenerate perturbation theory. However, $\mathbf{p}^4$ commutes with both L^2 and L_z, so the perturbation is already diagonal in the basis $|n, \ell, m\rangle$ within each degenerate energy eigenspace. The first-order corrections are simply the diagonal matrix elements of the perturbation. In atomic units, $c = 1/\alpha$, so

$$V = -\frac{\alpha^2}{8}\,\mathbf{p}^4 \tag{14.34}$$

and the first-order relativistic energy correction is

$$E^{(1)}_{\text{rel}} = \langle n, \ell, m|V|n, \ell, m\rangle = -\frac{\alpha^2}{8}\,\langle n, \ell, m|\mathbf{p}^4|n, \ell, m\rangle \tag{14.35}$$

Using

$$\mathbf{p}^2 = 2\left(H_0 + \frac{1}{r}\right),$$

it follows that

$$\begin{aligned}
E^{(1)}_{\text{rel}} &= -\frac{\alpha^2}{8}\,\langle n, \ell, m|\mathbf{p}^2\mathbf{p}^2|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\,\langle n, \ell, m|\left(H_0 + \frac{1}{r}\right)\left(H_0 + \frac{1}{r}\right)|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\,\langle n, \ell, m|\left(E^{(0)}_n + \frac{1}{r}\right)\left(E^{(0)}_n + \frac{1}{r}\right)|n, \ell, m\rangle \\
&= -\frac{\alpha^2}{2}\left[\left(E^{(0)}_n\right)^2 + 2E^{(0)}_n\left\langle\frac{1}{r}\right\rangle_{n\ell m} + \left\langle\frac{1}{r^2}\right\rangle_{n\ell m}\right]
\end{aligned}$$

The two spatial integrals implied in the expression above are much simpler than the ones we would have obtained from the matrix elements of $\mathbf{p}^4$ directly. The radial operator is independent of the angular variables, so the angular parts of the integrals are unity and the matrix elements reduce to radial integrals:

$$\left\langle\frac{1}{r}\right\rangle_{n\ell m} = \int_0^\infty \frac{1}{r}\mathcal{R}^2_{n\ell}(r)r^2 dr = \frac{1}{n^2} \tag{14.36}$$

$$\left\langle\frac{1}{r^2}\right\rangle_{n\ell m} = \int_0^\infty \frac{1}{r^2}\mathcal{R}^2_{n\ell}(r)r^2 dr = \frac{1}{n^3\left(\ell + \frac{1}{2}\right)} \tag{14.37}$$

Using $E_n^{(0)} = -1/2n^2$, the first-order relativistic correction in atomic units is

$$E_{\text{rel}}^{(1)} = -\frac{1}{2}\alpha^2 \left[\frac{1}{n^3\left(\ell + \frac{1}{2}\right)} - \frac{3}{4n^4} \right] \tag{14.38}$$

To convert the expression to ordinary units, multiply by the energy scale

$$\frac{\hbar^2}{m_e a_0^2} = \alpha^2 m_e c^2.$$

Hence,

$$\boxed{E_{\text{rel}}^{(1)} = -\frac{1}{2}\alpha^4 m_e c^2 \left[\frac{1}{n^3\left(\ell + \frac{1}{2}\right)} - \frac{3}{4n^4} \right]} \tag{14.39}$$

As we expected, the relativistic correction is α^2 times smaller than the Bohr energy.

State	$E_{\text{rel}}^{(1)}$ $(10^{-4}\,\text{eV})$
$1s$	-9.06
$2s$	-1.47
$2p$	-0.264

Table 14.1: First-order relativistic corrections.

14.4 Spin-Orbit Coupling

The electron's orbital motion and intrinsic spin are accompanied by magnetic moments. In the electron's rest frame, the electric field of the proton is partly experienced as a magnetic field, which interacts with the electron's spin magnetic moment. After accounting for the relativistic Thomas-precession correction, this interaction is described by the spin-orbit Hamiltonian

$$V_{\text{SO}} = \frac{k_e e^2}{2m_e c^2 r^3} \mathbf{L} \cdot \mathbf{S} \tag{14.40}$$

In atomic units, where $\hbar = m_e = a_0 = e = 1$ and $\alpha = 1/c$,

$$V_{\text{SO}} = \frac{\alpha^2}{2r^3} \mathbf{L} \cdot \mathbf{S} \tag{14.41}$$

For low-energy states with small angular-momentum quantum numbers,

$$r \approx a_0, \quad L \approx S \approx \hbar,$$

so the characteristic size of the spin-orbit interaction is $V_{\text{SO}} \approx \alpha^2$ in atomic units, or $\alpha^4 m_e c^2$ in ordinary units (recall the energy unit may be written as $\alpha^2 m_e c^2$). Evidently, the spin-orbit interaction is on the same order of magnitude as the relativistic correction we studied in the

previous section.

Including the electron's spin degrees of freedom, the eigenstates of the unperturbed Hamiltonian in the uncoupled basis are

$$|n,\ell,m\rangle\,|s,m_s\rangle = |n,\ell,m;m_s\rangle$$

where $|n,\ell,m\rangle$ denotes the electron's orbital state and $|s,m_s\rangle$ describes its spin state. The operator $\mathbf{L}\cdot\mathbf{S}$ is not diagonal in the uncoupled basis, but it is diagonal in the *coupled basis*, defined as the simultaneous eigenstates of L^2, S^2, J^2 and J_z, where

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \tag{14.42}$$

is the total angular momentum. Denote the elements of the coupled basis as $|n,\ell,j,m_j\rangle$. Since the electron has $s = 1/2$, the possible values of j are

$$j = \begin{cases} \ell \pm \frac{1}{2}, & \ell > 0, \\ \\ \frac{1}{2}, & \ell = 0. \end{cases} \tag{14.43}$$

Using

$$J^2 = (\mathbf{L}+\mathbf{S})\cdot(\mathbf{J}+\mathbf{S}) = L^2 + S^2 + 2\mathbf{L}\cdot\mathbf{S},$$

we obtain

$$\mathbf{L}\cdot\mathbf{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right),$$

and furthermore, since $s = 1/2$,

$$\mathbf{L}\cdot\mathbf{S}\,|n,\ell,j,m_j\rangle = \frac{1}{2}\left[j(j+1) - \ell(\ell+1) - \frac{3}{4}\right]|n,\ell,j,m_j\rangle\,. \tag{14.44}$$

Using time-independent perturbation theory, the first-order correction to the energy is

$$E^{(1)}_{\rm SO} = \langle n,\ell,j,m_j|V_{\rm SO}|n,\ell,j,m_j\rangle \tag{14.45}$$

The radial and angular factors separate, and thus

$$E^{(1)}_{\rm SO} = \frac{\alpha^2}{2}\left\langle\frac{1}{r^3}\right\rangle_{n\ell}\langle\mathbf{L}\cdot\mathbf{S}\rangle_{\ell j} \tag{14.46}$$

For $\ell \neq 0$,

$$\left\langle\frac{1}{r^3}\right\rangle_{n\ell} = \frac{1}{n^3\ell(\ell+1/2)(\ell+1)}, \tag{14.47}$$

and we conclude that

$$E^{(1)}_{\rm SO} = \frac{\alpha^2}{4n^3}\,\frac{j(j+1)-\ell(\ell+1)-3/4}{\ell(\ell+1/2)(\ell+1)}. \tag{14.48}$$

For $j = \ell + 1/2$, the orbital and spin angular momenta are more nearly aligned, so $\langle\mathbf{L}\cdot\mathbf{S}\rangle > 0$,

and the spin-orbit interaction raises the energy. For $j = \ell - 1/2$, the angular momenta are more nearly oppositely aligned, so $\langle \mathbf{L} \cdot \mathbf{S} \rangle < 0$, and the spin-orbit interaction lowers the energy. The correction does not depend on m_j, so the degeneracy among states with different values of m_j remains. Spin-orbit coupling instead lifts the degeneracy between states with the same n and ℓ but different values of j. For $\ell = 0$, $\mathbf{L} \cdot \mathbf{S} = 0$, so there simply is no spin-orbit correction. A complete relativistic treatment of the s-states also contains the *Darwin term*.

Restoring ordinary units, the first-order energy correction is

$$E_{\text{SO}}^{(1)} = \frac{\alpha^4 m_e c^2}{4n^3} \, \frac{j(j+1) - \ell(\ell+1) - 3/4}{\ell(\ell+1/2)(\ell+1)}, \tag{14.49}$$

which is on the order of $\alpha^4 m_e c^2$ for states with low angular-momentum quantum numbers. Adding the spin-orbit correction to the relativistic correction gives the complete fine-structure formula

$$E_{\text{fs}}^{(1)} = -\frac{\alpha^4 m_e c^2}{2n^4} \left[\frac{n}{j+1/2} - \frac{3}{4} \right]. \tag{14.50}$$

Since the unperturbed hydrogen energies are

$$E_n^{(0)} = -\frac{\alpha^2 m_e c^2}{2n^2} = -\frac{13.6 \text{ eV}}{n^2}, \tag{14.51}$$

the energy levels, including the leading fine-structure correction, are

$$\boxed{E_{nj} = -\frac{13.6 \text{ eV}}{n^2} \left[1 + \frac{\alpha^2}{n^2} \left(\frac{n}{j+1/2} - \frac{3}{4} \right) \right].} \tag{14.52}$$

The correction depends on n and j, but not separately on ℓ or m_j. Thus, it follows from our corrected model that states with the same n and j remain degenerate.

14.5 Zeeman Effect

When an atom is placed in an external magnetic field, its energy levels generally split into several closely spaced levels. This phenomenon is called the Zeeman effect. The splitting arises because the magnetic moments associated with the atom's orbital and spin angular momenta interact with the applied magnetic field. The Zeeman effect applies to all atoms, but we will focus primarily on hydrogen.

14.5.1 Zeeman Effect without Spin

We begin by ignoring the electron's intrinsic spin. Even though the electron's orbital and spin magnetic moments contribute equally to the effect, we will use this unrealistic model because it is an easy calculation and it introduces us to the essential features of the Zeeman effect. Because we are ignoring spin, the eigenstates of the hydrogen Hamiltonian are $|n\ell m_\ell\rangle$ and the energies are

$E_n^{(0)} = -13.6 \text{ eV}/n^2$. We perturb the system by applying a magnetic field $\mathbf{B}$ along the z axis. The resultant Zeeman perturbation Hamiltonian is

$$V_{\text{Z}} = g_\ell \mu_B \frac{1}{\hbar} \mathbf{L} \cdot \mathbf{B} = g_\ell \mu_B \frac{B}{\hbar} L_z, \tag{14.53}$$

where μ_B is the Bohr magneton and $g_\ell = 1$ is the electron's orbital g-factor. The magnetic moment of the proton is about 1000 times smaller than that of the electron, so we will neglect the proton contribution to the Zeeman effect.

Since the zeroth-order energy eigenstates are eigenstates of the perturbation, the first-order energy corrections are the diagonal elements of the perturbation Hamiltonian:

$$\boxed{E_n^{(1)} = \langle n\ell m_\ell | V_{\text{Z}} | n\ell m_\ell \rangle = g_\ell \mu_B B m_\ell.} \tag{14.54}$$

For a given value of ℓ, the magnetic quantum number takes the values $m_\ell = -\ell, -\ell+1, \ldots, \ell$, so an orbital level splits into $2\ell + 1$ equally spaced levels. The separation between adjacent levels is $\Delta E = \mu_B B$. This result is called the normal Zeeman effect. It correctly describes systems in which spin plays no role, but it is not sufficient for a single electron such as the electron in hydrogen. The normal Zeeman effect is proportional to the orbital g-factor, the Bohr magneton μ_B, the magnetic field strength B, and the magnetic quantum number m_ℓ. This general form is repeated when we study the Zeeman effect in more realistic models. The g-factor and the particular magnetic quantum number change as we include other magnetic moments in the model.

14.5.2 Including Electron Spin

Including spin, the full Zeeman interaction is given by the Hamiltonian

$$V_{\text{Z}} = \frac{\mu_B B}{\hbar} \Big(g_\ell L_z + g_e S_z \Big) \tag{14.55}$$

Taking $g_\ell = 1$ and $g_e = 2$, one might be tempted to conclude that the Zeeman shift is

$$\Delta E_{\text{Z}} = \mu_B B (m_\ell + 2 m_s).$$

but this is only correct when the energy eigenstates can be labeled by the quantum numbers m_ℓ and m_s. Recall that the spin-orbit interaction is diagonal in the coupled basis while the Zeeman interaction is diagonal in the uncoupled basis. Which basis we choose for the zeroth-order energy eigenstates depends on the relative strength of the interactions.

14.5.3 Strong Magnetic Field

In the strong-field regime, the Zeeman interaction is strong enough to disrupt the coupling between $\mathbf{L}$ and $\mathbf{S}$. We include the Zeeeman Hamiltonian in zeroth order and treat the fine structure as a perturbation. The uncoupled basis, denoted $|n\ell m_\ell m_s\rangle$, is the preferred basis because the Zeeman

Hamiltonian is diagonal in that basis. The zeroth-order energies are

$$\begin{aligned} E_n^{(0)} &= -\frac{13.6\text{ eV}}{n^2} + \langle n\ell m_\ell m_s | V_{\text{Z}} | n\ell m_\ell m_s \rangle \\ &= -\frac{13.6\text{ eV}}{n^2} + \mu_B B(m_\ell + 2m_s) \end{aligned} \tag{14.56}$$

Now we treat the fine structure as a perturbation to the zeroth-order states that include the Zeeman interaction. The relativistic Hamiltonian is diagonal in the both the coupled and uncoupled bases, but the spin-orbit Hamiltonian is diagonal only in the coupled basis. Fortunately, the off-diagonal matrix elements of the spin-orbit Hamiltonian do not couple any of the states that are degenerate with respect to the Zeeman Hamiltonian. Hence we can use nondegenerate perturbation theory to find the first-order fine-structure corrections. In the uncoupled basis, the spin-orbit energy corrections are the expectation values

$$\begin{aligned} E_{\text{SO}}^{(1)} &= \frac{k_e e^2}{2m_e^2 c^2} \langle m\ell m_\ell m_s | \left(\frac{1}{r^3} \mathbf{L} \cdot \mathbf{S} \right) | m\ell m_\ell m_s \rangle \\ &= \frac{k_e e^2}{2m_e^2 c^2} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} \langle m_\ell m_s | \tfrac{1}{2} \left(L_+ S_- + L_- S_+ \right) + L_z S_z | m_\ell m_s \rangle \end{aligned} \tag{14.57}$$

Because the ladder operators connect adjacent states, they produce no diagonal matrix elements, so the first-order correction reduces to

$$E_{\text{SO}}^{(1)} = \frac{k_e e^2}{2m_e^2 c^2} \left\langle \frac{1}{r^3} \right\rangle_{n\ell} m_\ell m_s \hbar^2 \tag{14.58}$$

Using Eq. (14.47) for the radial matrix elements (in ordinary units), we obtain

$$E_{\text{SO}}^{(1)} = \frac{1}{2} \alpha^4 m_e c^2 \frac{m_\ell m_s}{n^3 \ell(\ell + 1/2)(\ell + 1)}. \tag{14.59}$$

Adding together the relativistic and spin-orbit corrections yields the fine-structure shifts

$$\boxed{E_{\text{fs}}^{(1)} = \frac{1}{2} \alpha^4 m_e c^2 \left[\frac{3}{4n^3} - \frac{\ell(\ell + 1) - m_\ell m_s}{n^3 \ell(\ell + 1/2)(\ell + 1)}. \right]} \tag{14.60}$$

14.5.4 Weak Magnetic Field

In the weak-field regime, the zeroth-order states are the coupled basis states $|n\ell j m_j\rangle$, in which the fine-structure Hamiltonian is diagonal. Our task is to determine how the Zeeman perturbation modifies the corresponding zeroth-order energies. The fine-structure energy depends on j, but not on m_j. Thus, for fixed n, ℓ, and j, the $2j + 1$ states with different values of m_j all have the same unperturbed energy. Each j-level therefore contains a $(2j + 1)$-dimensional degenerate subspace.

We must consequently use degenerate perturbation theory, which ordinarily requires us to di-

agonalize the perturbation within each degenerate subspace. In this case, however, the coupled states $|n\ell j m_j\rangle$ already form a basis in which $V_{\rm Z}$ is diagonal within each fixed-j subspace. Although the Zeeman perturbation is not diagonal in the entire coupled basis (it can couple states with the same m_j but different j), perturbation theory requires only that it be diagonalized within each degenerate subspace. The different j-levels are separated by the fine-structure splitting and are therefore nondegenerate in the weak-field treatment.

The simplest way to understand this result is to consider the commutator of $V_{\rm Z}$ with J_z. Since J_z commutes with L_z and S_z, it follows that $[V_{\rm Z}, J_z] = 0$, and furthermore, the states $|n\ell j m_j\rangle$ must be the "good states" to use for perturbation theory (see Chapter #13). Hence,

$$E_{\rm Z}^{(1)} = \langle n\ell j m_j | V_{\rm Z} | n\ell j m_j \rangle \,, \tag{14.61}$$

and all that remains is to compute the expectation value. There is an elegant way to calculate the first-order correction that comes from the Wigner-Eckhart theorem. The only formula we need from the Wigner-Eckhart theorem relates the matrix element of any general vector component A_z to the matrix element of the total angular momentum component J_z:

$$\big\langle j m_j \big| A_z \big| j m_j' \big\rangle = \frac{\langle j m_j | \mathbf{A}\cdot\mathbf{J} | j m_j \rangle}{\hbar^2 j(j+1)} \big\langle j m_j \big| J_z \big| j m_j' \big\rangle \tag{14.62}$$

This is known as the projection theorem, and there is a heuristic way of understanding it using a classical model. The classical picture is that, within a state of definite total angular momentum j, the only distinguished direction is the direction of $\mathbf{J}$. Therefore, the part of any vector $\mathbf{A}$ that can survive averaging is its projection along $\mathbf{J}$. Classically, decompose

$$\mathbf{A} = \mathbf{A}_\parallel + \mathbf{A}_\perp, \quad \text{where} \quad \mathbf{A}_\parallel = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2}\,\mathbf{J}.$$

If $\mathbf{A}$ precesses about $\mathbf{J}$, the perpendicular component continually changes direction and averages to zero, i.e., $\overline{\mathbf{A}}_\perp = 0$. Thus,

$$\overline{\mathbf{A}} = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2}\mathbf{J}, \quad \text{and} \quad \overline{A}_z = \frac{\mathbf{A}\cdot\mathbf{J}}{J^2} J_z.$$

Because J_z is diagonal in the coupled basis, the only nonzero matrix elements have $m_j = m_j'$, and the projection theorem reduces to

$$\langle j m_j | A_z | j m_j \rangle = \frac{m_j}{\hbar j(j+1)} \langle j m_j | \mathbf{A}\cdot\mathbf{J} | j m_j \rangle \tag{14.63}$$

Using

$$\mathbf{L}\cdot\mathbf{J} = \tfrac{1}{2}\left(\mathbf{J}^2 + \mathbf{L}^2 - \mathbf{S}^2\right), \quad \text{and} \quad \mathbf{S}\cdot\mathbf{J} = \tfrac{1}{2}\left(\mathbf{J}^2 + \mathbf{S}^2 - \mathbf{L}^2\right),$$

the diagonal matrix elements required in the projection theorem are

$$\langle \mathbf{L} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} \left[j(j+1) + \ell(\ell+1) - s(s+1) \right] \tag{14.64}$$

$$\langle \mathbf{S} \cdot \mathbf{J} \rangle = \frac{\hbar^2}{2} \left[j(j+1) + s(s+1) - \ell(\ell+1) \right] \tag{14.65}$$

Using these coefficients in the projection theorem yields the Lz and Sz matrix elements in the coupled basis within a given j subspace:

$$\langle L_z \rangle = \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} \, m_j \tag{14.66}$$

$$\langle S_z \rangle = \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \, m_j \tag{14.67}$$

The resulting first-order Zeeman correction to the energy can be written in the standard Zeeman form as

$$E_{\mathrm{Z}}^{(1)} = g_j \mu_B B m_j, \tag{14.68}$$

where g_j is the Landé g-factor given by

$$g_j = g_\ell \frac{j(j+1) + \ell(\ell+1) - s(s+1)}{2j(j+1)} + g_e \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)} \tag{14.69}$$

The Landé g-factor accounts for the relative contributions of the magnetic moments due to the spin and orbital angular momenta caused by their differing magnitudes and differing alignments. For hydrogen, with only one electron, $s = 1/2$ and $j = \ell \pm 1/2$. Using $g_\ell = 1$ and $g_e = 2$, the Landé g-factor for hydrogen is

$$g_j = 1 \pm \frac{1}{2\ell + 1}. \tag{14.70}$$

Consequently, the Zeeman energy correction is dependent on j and ℓ:

$$\boxed{E_{\mathrm{Z}}^{(1)} = \mu_B B m_j \left(1 \pm \frac{1}{2\ell + 1} \right).} \tag{14.71}$$

14.6 Summary: Choosing the Right States

For a nondegenerate unperturbed state $|n\rangle$, the first-order energy correction is simply

$$E_n^{(1)} = \langle n|V|n\rangle \, .$$

For a degenerate energy level, however, an arbitrary unperturbed basis state does not necessarily evolve into an energy eigenstate when the perturbation is turned on. One must first diagonalize the perturbation within the degenerate eigenspace. Once these "good states" have been identified,

their first-order energy corrections are again given by expectation values:

$$E_n^{(1)} = \langle n|V|n\rangle\,.$$

Thus, throughout this chapter, the important question was whether the states being used already diagonalized the perturbation *within each degenerate subspace*.

14.6.1 Stark Effect

The hydrogen ground state $|1,0,0\rangle$ is nondegenerate, so ordinary nondegenerate perturbation theory applies:

$$E_1^{(1)} = \langle 1,0,0|eE_0 z|1,0,0\rangle = 0.$$

The expectation value vanishes by parity, so the leading ground-state Stark shift is second order. The $n=2$ level is degenerate, and the states $|2,0,0\rangle$ and $|2,1,0\rangle$ are coupled by the perturbation. Their individual expectation values vanish, but they are not the appropriate states for first-order perturbation theory. Diagonalizing V within the $n=2$ eigenspace gives

$$\frac{1}{\sqrt{2}}\left(|2,0,0\rangle \pm |2,1,0\rangle\right),$$

together with the uncoupled states $|2,1,\pm1\rangle$. These states diagonalize the perturbation within the degenerate subspace, so their first-order shifts may be calculated as expectation values:

$$E_{2,\pm}^{(1)} = \langle 2,\pm|V|2,\pm\rangle = \pm 3ea_0E_0, \qquad E_{2,1,\pm1}^{(1)} = 0.$$

14.6.2 Relativistic Correction

The Coulomb energies are degenerate with respect to ℓ and m, so degenerate perturbation theory is formally required. Nevertheless, the relativistic perturbation

$$V_{\text{rel}} = -\frac{\mathbf{p}^4}{8m_e^3c^2}$$

is rotationally invariant and commutes with L^2 and L_z. It therefore does not mix states with different ℓ or m within a fixed-n eigenspace. The states $|n,\ell,m\rangle$ already diagonalize the perturbation within every degenerate energy eigenspace, and hence

$$E_{\text{rel}}^{(1)} = \langle n,\ell,m|V_{\text{rel}}|n,\ell,m\rangle\,.$$

14.6.3 Spin-Orbit Coupling

The spin-orbit perturbation is proportional to

$$V_{\text{SO}} \propto \frac{1}{r^3}\mathbf{L}\cdot\mathbf{S}.$$

It is not diagonal in the uncoupled basis $|n, \ell, m_\ell, m_s\rangle$, because the ladder-operator terms in $\mathbf{L} \cdot \mathbf{S}$ mix different values of m_ℓ and m_s. It is diagonal in the coupled basis $|n, \ell, j, m_j\rangle$, since

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}\left(J^2 - L^2 - S^2\right).$$

The coupled states are therefore the appropriate zeroth-order states, and

$$E_{\mathrm{SO}}^{(1)} = \langle n, \ell, j, m_j | V_{\mathrm{SO}} | n, \ell, j, m_j \rangle .$$

The result depends on n, ℓ, and j, but not on m_j, so the degeneracy among the different m_j states is not lifted.

14.6.4 Strong-Field Zeeman Effect

In the strong-field regime, the Zeeman interaction is included in the zeroth-order Hamiltonian because it is stronger than spin-orbit coupling. The Zeeman Hamiltonian is diagonal in the uncoupled basis $|n, \ell, m_\ell, m_s\rangle$, so these are the appropriate zeroth-order states. The fine-structure Hamiltonian is then treated as the perturbation. The spin-orbit operator is not completely diagonal in the uncoupled basis, but its off-diagonal ladder-operator terms do not connect states that have the same zeroth-order Zeeman energy. Therefore, within every degenerate zeroth-order subspace, the fine-structure perturbation is already diagonal in the relevant uncoupled states. See the soultions to Homework #4, Problem #3. Its first-order correction can then be calculated using

$$E_{\mathrm{fs}}^{(1)} = \langle n, \ell, m_\ell, m_s | V_{\mathrm{fs}} | n, \ell, m_\ell, m_s \rangle .$$

In particular, only the diagonal term $L_z S_z$ contributes to the first-order expectation value of the spin-orbit interaction.

14.6.5 Weak-Field Zeeman Effect

In the weak-field regime, the fine structure is included in the zeroth-order Hamiltonian, so the appropriate states are the coupled states $|n, \ell, j, m_j\rangle$. States with different m_j remain degenerate before the magnetic field is applied. We must therefore determine whether the Zeeman perturbation is diagonal within each degenerate subspace. Because

$$[V_{\mathrm{Z}}, J_z] = 0,$$

the Zeeman perturbation cannot connect states with different m_j. Within a fixed n, ℓ, and j subspace, each value of m_j labels a distinct state, so the coupled basis diagonalizes V_{Z} within that subspace. Moreover, V_{Z} does not mix different values of ℓ, while states with different j are separated

by the fine-structure splitting and are not degenerate in the weak-field treatment. Therefore,

$$E_Z^{(1)} = \langle n, \ell, j, m_j | V_Z | n, \ell, j, m_j \rangle = g_j \mu_B B m_j.$$

The strong- and weak-field limits require different states because different interactions define the zeroth-order Hamiltonian:

Regime	Dominant interaction	Appropriate states
Strong magnetic field	V_{Z}	$\lvert n, \ell, m_\ell, m_s \rangle$
Weak magnetic field	V_{fs}	$\lvert n, \ell, j, m_j \rangle$

In each case, the appropriate states are those that diagonalize the zeroth-order Hamiltonian and the perturbation within any remaining degenerate subspaces.

Chapter 15

Identical Particles & Electronic Structure

15.1 Introduction

Thus far, we have considered either single-particle systems or systems in which the constituent particles can be distinguished from one another. Nature, however, contains many systems composed of *identical particles*. Every electron has exactly the same mass, charge, and spin as every other electron; the same is true of protons, photons, and many other particles. Unlike classical objects, identical quantum particles possess no hidden serial numbers that allow us to track their individual identities. This fundamental indistinguishability requires us to reconsider how multiparticle states are represented and leads to some of the most important consequences of quantum mechanics, including the distinction between bosons and fermions and the Pauli exclusion principle.

We will apply these concepts to the *electronic structure of solids*. The Hamiltonian determines which single-electron states and energies are available, while Fermi statistics determine how those states may be occupied. We will first consider the free-electron gas, in which conduction electrons move through a uniform background, and then the tight-binding model, in which electrons begin in localized atomic orbitals and acquire an amplitude to move between neighboring atoms. These models illustrate how the behavior of identical fermions, together with the atomic structure of a material, gives rise to Fermi energies, degeneracy pressure, and electronic energy bands.

15.2 The Principle of Indistinguishability

Particles are said to be identical when they belong to the same species and possess no observable intrinsic property by which one can be distinguished from another. It follows that exchanging the labels assigned to identical particles cannot produce an observable change. This is the principle of indistinguishability:

> Dynamical states that differ only by a permutation of identical particles cannot be

distinguished by any observation whatsoever.

This principle does not say that two identical particles must occupy the same state, nor does it say that they cannot be distinguished approximately by their locations. If one electron is localized near the Earth and another near the Moon, it is useful to speak of "the electron near the Earth" and "the electron near the Moon." These descriptions identify spatial states, however, not intrinsic particle identities. If the two electrons exchanged places, with all observable properties exchanged along with them, there would be no experiment that could reveal that an exchange had occurred.

Consider a system of just two identical particles. Any arbitrary vector in the two-particle state space can be expressed as a linear combination of tensor product states as

$$|\Psi\rangle = \sum_{\alpha,\beta} c_{\alpha\beta} \, |\alpha\rangle \, |\beta\rangle$$

where $|\alpha\rangle \, |\beta\rangle = |\alpha\rangle \otimes |\beta\rangle$. In this notation, recall that $|\alpha\rangle$ refers to the first particle and $|\beta\rangle$ refers to the second. A wave function in the two-particle space has the form

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Big(\langle \mathbf{r}_1| \otimes \langle \mathbf{r}_2| \Big) \, |\Psi\rangle = \sum_{\alpha,\beta} c_{\alpha\beta} \, \langle \mathbf{r}_1|\alpha\rangle \, \langle \mathbf{r}_2|\beta\rangle$$

Let P_{12} denote a transposition operator; a transposition is a permutation that interchanges exactly two elements of a set while leaving all other elements fixed. Define P_{12} by the relation

$$P_{12} \, |\alpha\rangle \, |\beta\rangle = |\beta\rangle \, |\alpha\rangle \, .$$

The operator P_{12} is (i) its own inverse, (ii) Hermitian, and (iii) unitary. Its effect on a two-particle wave function is

$$P_{12}\Psi(\mathbf{r}_1, \mathbf{r}_2) = \Psi(\mathbf{r}_2, \mathbf{r}_1)$$

As you can see, the transposition operator simply exchanges the labels of the particles. Consider the effect of the exchange operator on the state $|\mathbf{r}_1; +\rangle \, |\mathbf{r}_2; -\rangle$, which represents a spin-up particle at location $\mathbf{r}_1$ and a spin-down particle at location $\mathbf{r}_2$. The exchange operator returns the state $|\mathbf{r}_2; -\rangle \, |\mathbf{r}_1; +\rangle$. In this example, there is still a particle near $\mathbf{r}_1$ that is spin-up and another near $\mathbf{r}_2$ that is spin-down. This example was chosen carefully to make the following point: *if the two particles are truly identical, the labels 1 and 2 have no direct physical meaning.* Interchanging them cannot produce an observable change.

Let A be an operator that represents a dynamical observable and let $|\Psi\rangle$ represent a state of a system of identical particles. The state obtained by interchanging particles i and j is described by the state $P_{ij} \, |\Psi\rangle$, and according to the principle of indistinguishability,

$$\langle \Psi | P_{ij}^{\dagger} \, A \, P_{ij} | \Psi \rangle = \langle \Psi | A | \Psi \rangle \, . \tag{15.1}$$

This must hold for any $|\Psi\rangle$, and since both A and $P_{ij}^\dagger\, A\, P_{ij}$ are Hermitian, it follows that

$$P_{ij}^\dagger A P_{ij} = A, \quad \text{or} \quad [P_{ij}, A] = 0. \tag{15.2}$$

Since A represents an arbitrary observable, we have shown that all physical observables must be permutation-invariant. In a system of two identical particles, P_{12} is the only nontrivial permutation operator, and since $[A, P_{12}] = 0$, the operators A and P_{12} possess a complete set of common eigenvectors. Let $|\Phi\rangle$ denote an eigenvector of the permutation P_{12}, i.e., $P_{12}\,|\Phi\rangle = \lambda\,|\Phi\rangle$, where λ is a real number since P_{12} is Hermitian. Applying P_{12} to the eigenvalue equation, we find

$$P_{12}^2\,|\Phi\rangle = \lambda P_{12}\,|\Phi\rangle = \lambda^2\,|\Phi\rangle \tag{15.3}$$

Since $P_{12}^2 = \mathbb{1}$, it follows that $\lambda = \pm 1$, and we call the eigenstates *symmetric* or *antisymmetric* under the interchange of two identical particles:

$$P_{12}\,|\psi_S\rangle = |\psi_S\rangle\,, \quad \text{and} \quad P_{12}\,|\psi_A\rangle = -\,|\psi_A\rangle \tag{15.4}$$

where $|\psi_S\rangle$ is a symmetric state and $|\psi_A\rangle$ is antisymmetric. A physical observable cannot map a symmetric state into an antisymmetric state or vice versa. Using $[P_{12}, A] = 0$,

$$P_{12}A\,|\psi_S\rangle = AP_{12}\,|\psi_S\rangle = A\,|\psi_S\rangle\,, \quad \text{and} \quad P_{12}A\,|\psi_A\rangle = AP_{12}\,|\psi_A\rangle = -A\,|\psi_A\rangle\,,$$

so $A\,|\psi_S\rangle$ remains symmetric and $A\,|\psi_A\rangle$ remains antisymmetric.

The Hamiltonian is a physical observable, so it must also be invariant under permutations. Consequently, exchange symmetry is conserved by the time evolution. A state that is initially symmetric remains symmetric, and a state that is initially antisymmetric remains antisymmetric. A typical two-particle Hamiltonian has the form

$$H = \frac{\mathbf{p}_1^2}{2m} + \frac{\mathbf{p}_2^2}{2m} + V_0(\mathbf{r}_1) + V_0(\mathbf{r}_2) + V(\mathbf{r}_1, \mathbf{r}_2)$$

Under the exchange $\mathbf{r}_1 \leftrightarrow \mathbf{r}_2$, $\mathbf{p}_1 \leftrightarrow \mathbf{p}_2$, H remains unchanged provided $V(\mathbf{r}_1, \mathbf{r}_2) = V(\mathbf{r}_2, \mathbf{r}_1)$, which is automatically satisfied by any central potential $V(\mathbf{r}_1, \mathbf{r}_2) = V\left(|\mathbf{r}_2 - \mathbf{r}_1|\right)$.

15.3 The Symmetrization Postulate

The principle of indistinguishability requires observable physics to be invariant under permutations, but it does not, by itself, require every allowed state vector to be symmetric or antisymmetric. The stronger restriction on the allowed states is supplied by the symmetrization postulate:

> The states of a system of identical particles are either completely symmetric or completely antisymmetric under the interchange of any pair of particles.

In nature, the exchange symmetry of a system of identical particles is determined by the spin of the particles. Identical particles with integer spin occupy symmetric states and are called **bosons**, whereas identical particles with half-integer spin occupy antisymmetric states and are called **fermions**. This relationship is known as the **spin-statistics connection**. In relativistic quantum field theory, it follows from the spin-statistics theorem; in nonrelativistic quantum mechanics, it must be introduced as an empirical fact.

Example 15.1: Two Identical Particles in a Box

The symmetrization postulate applies to the complete state, including both spatial and spin degrees of freedom. If the state factorizes as

$$|\Psi\rangle = |\psi\rangle_{\text{spatial}} \, |\chi\rangle_{\text{spin}} \, ,$$

then the exchange symmetries of the two factors must combine to give the symmetry of the complete state. In this example, suppose the spin state is symmetric, i.e.,

$$P_{12} \, |\chi\rangle_{\text{spin}} = + \, |\chi\rangle_{\text{spin}} \, .$$

The spatial state must then be symmetric for bosons and antisymmetric for fermions:

$$P_{12} \, |\psi\rangle_{\text{spatial}} = \begin{cases} + \, |\psi\rangle_{\text{spatial}} \, , & \text{bosons,} \\ - \, |\psi\rangle_{\text{spatial}} \, , & \text{fermions.} \end{cases}$$

We will suppress the unchanged spin factor and work only with the spatial wave function.

Example: Suppose we have two noninteracting particles, both of mass m, in an infinite square well. The one-particle states are

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad E_n = n^2 K,$$

where $K = \pi^2\hbar^2/2ma^2$. If the particles are distinguishable, the composite wave function is

$$\psi_{n_1,n_2}(x_1, x_2) = \psi_{n_1}(x_1)\psi_{n_2}(x_2), \quad E_{n_1,n_2} = (n_1^2 + n_2^2)K.$$

For example, the ground state is

$$\psi_{11} = \frac{2}{a} \sin\left(\frac{\pi x_1}{a}\right) \sin\left(\frac{\pi x_2}{a}\right), \quad E_{11} = 2K,$$

and the first excited state is two-fold degenerate:

$$\psi_{12} = \frac{2}{a}\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right), \quad E_{12} = 5K,$$

$$\psi_{21} = \frac{2}{a}\sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right), \quad E_{21} = 5K,$$

If, however, the two particles are identical bosons, the ground state is unchanged, but the first excited state is nondegenerate:

$$\psi_S = \frac{\sqrt{2}}{a}\left[\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) + \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right], \quad E_S = 5K.$$

If the particles are identical fermions, there is no state with energy $2K$; the ground state is

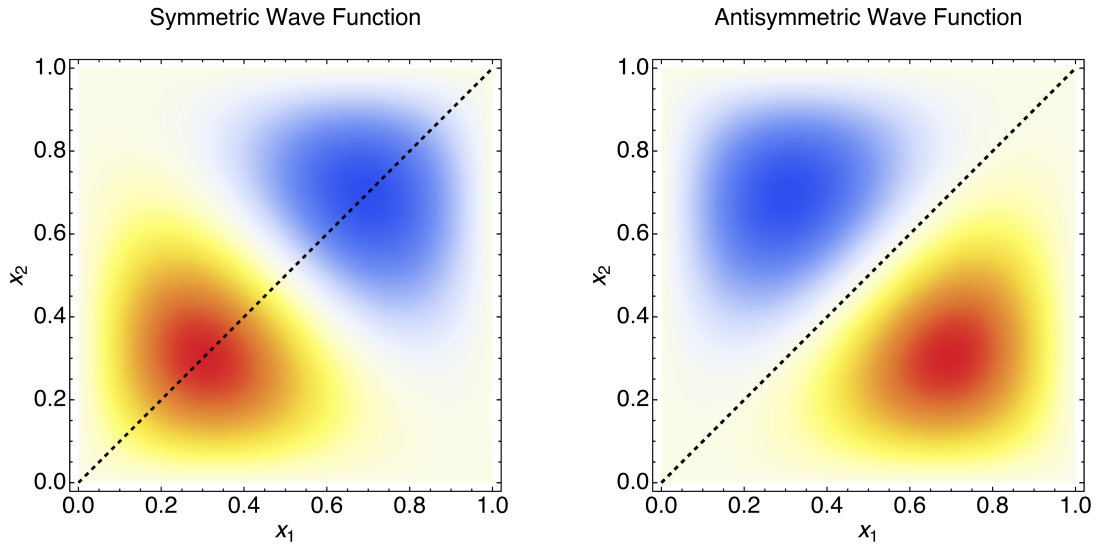
$$\psi_A = \frac{\sqrt{2}}{a}\left[\sin\left(\frac{\pi x_1}{a}\right)\sin\left(\frac{2\pi x_2}{a}\right) - \sin\left(\frac{2\pi x_1}{a}\right)\sin\left(\frac{\pi x_2}{a}\right)\right], \quad E_A = 5K.$$

There is no state with energy $2K$ because such a state would require both particles to occupy the one-particle ground state, $n_1 = n_2 = 1$. Antisymmetrizing the product state gives

$$\psi_{11}^{(A)}(x_1, x_2) = \frac{1}{\sqrt{2}}\left[\psi_1(x_1)\psi_1(x_2) - \psi_1(x_2)\psi_1(x_1)\right] = 0$$

This is an example of the **Pauli exclusion principle**: two identical fermions cannot occupy the same complete one-particle state. In the present example, their spin state is symmetric, so placing both particles in the same spatial state would make the complete state vanish. If the two fermions instead had an antisymmetric spin state, they could occupy the same spatial state because the complete state would still be antisymmetric.

The figure below shows the symmetric and antisymmetric wave functions plotted against x_1 and x_2. The exchange $x_1 \leftrightarrow x_2$ corresponds to reflection across the dashed line $x_1 = x_2$. The symmetric wave function is unchanged by this reflection; points mirrored across the diagonal have the same value and therefore the same color. The antisymmetric wave function changes sign; regions of positive amplitude on one side correspond to regions of negative amplitude on the other. Consequently, the antisymmetric wave function vanishes along $x_1 = x_2$, producing the nodal line visible in the plot. Thus, for this state, the two fermions have zero probability of being found at the same position.



Mean-square separation

We might try to characterize the locations of the particles by calculating $\langle x_1 \rangle$ and $\langle x_2 \rangle$. For both the symmetric and antisymmetric states, we find $\langle x_1 \rangle = \langle x_2 \rangle = a/2$. Thus, if we temporarily retain the artificial particle labels, the average position associated with each label is the center of the well. Strictly speaking, however, x_1 and x_2 are not individually physical observables for a system of identical particles. Under exchange,

$$P_{12}^\dagger x_1 P_{12} = x_2, \qquad P_{12}^\dagger x_2 P_{12} = x_1,$$

so neither operator commutes with P_{12}. The sum of the positions is permutation-invariant:

$$P_{12}^\dagger (x_1 + x_2) P_{12} = x_1 + x_2.$$

It is therefore a physical observable, and its expectation value is $\langle x_1 + x_2 \rangle = a$. Equivalently, the center-of-mass coordinate $X = (x_1 + x_2)/2$ is physical and satisfies $\langle X \rangle = a/2$. The signed difference $x_1 - x_2$ is not permutation-invariant:

$$P_{12}^\dagger (x_1 - x_2) P_{12} = -(x_1 - x_2).$$

Its sign depends on which identical particle was arbitrarily assigned the label 1. Nevertheless, its expectation value vanishes in both states:

$$\langle x_1 - x_2 \rangle_S = \langle x_1 - x_2 \rangle_A = 0.$$

This simply reflects the absence of any physical distinction between the two particle labels.

The square of the separation is permutation-invariant:

$$P_{12}^{\dagger}(x_1 - x_2)^2 P_{12} = (x_1 - x_2)^2.$$

It is therefore a physical observable. Using

$$(x_1 - x_2)^2 = x_1^2 + x_2^2 - 2x_1x_2,$$

we find

$$\langle x_1^2 + x_2^2 \rangle = a^2 \left(\frac{2}{3} - \frac{5}{8\pi^2} \right)$$

for both exchange symmetries. The correlation term, however, depends on the relative sign in the wave function:

$$\langle x_1 x_2 \rangle_{S/A} = \frac{a^2}{4} \pm \frac{256a^2}{81\pi^4},$$

where the upper sign applies to the symmetric state and the lower sign to the antisymmetric state. Consequently,

$$\big\langle (x_1 - x_2)^2 \big\rangle_S = a^2 \left(\frac{1}{6} - \frac{5}{8\pi^2} - \frac{512}{81\pi^4} \right)$$

and

$$\big\langle (x_1 - x_2)^2 \big\rangle_A = a^2 \left(\frac{1}{6} - \frac{5}{8\pi^2} + \frac{512}{81\pi^4} \right).$$

Numerically,

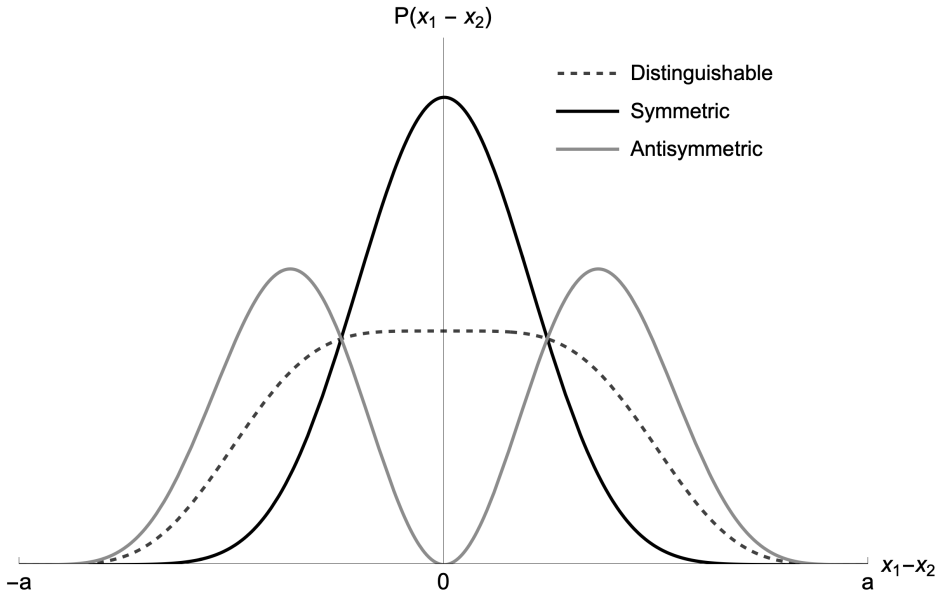
$$\big\langle (x_1 - x_2)^2 \big\rangle_S \approx 0.0384a^2,$$

whereas

$$\big\langle (x_1 - x_2)^2 \big\rangle_A \approx 0.1682a^2.$$

Although the particles do not interact, the antisymmetric state has a much larger mean-square separation than the symmetric state. The two states have the same average center-of-mass position and the same individual one-particle position distribution, but they possess different spatial correlations. Symmetric exchange favors configurations in which the particles are closer together, while antisymmetric exchange suppresses such configurations and forces the wave function to vanish when $x_1 = x_2$.

The system behaves as if there were an attractive force between identical bosons pulling them closer together and a repulsive force between identical fermions pushing them apart. We call it an **exchange force**, but it's not really a force at all. Rather, it is a consequence of the symmetrization requirement, which is purely quantum-mechanical with no classical analog. The exchange force is best understood as a statistical correlation, not a dynamical interaction.



Probability density for the signed relative coordinate $x_1 - x_2$, for two particles in the first excited configuration of an infinite square well of width a, with one particle in each of the $n = 1$ and $n = 2$ states. The dashed curve is for distinguishable particles; the black and gray curves show, respectively, symmetric and antisymmetric spatial states.

Including spin states

Thus far, we assumed the two particles had a symmetric spin state. Let's remove that restriction and determine all allowed states for two identical spin-1/2 particles. The particles are fermions, so the complete state must be antisymmetric. Because the Hamiltonian in this example is spin-independent, we can choose a basis of energy eigenstates that factor into spatial and spin parts.

The four spin states of two spin-1/2 particles consist of one antisymmetric singlet and three symmetric triplet states. The antisymmetric singlet is

$$|0,0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle - |-+\rangle \Big),$$

and the three symmetric states comprising the triplet are

$$|1,1\rangle = |++\rangle\,,$$

$$|1,0\rangle = \frac{1}{\sqrt{2}}\Big(|+-\rangle + |-+\rangle \Big),$$

$$|1,-1\rangle = |--\rangle\,.$$

It is interesting that the coupled spin states emerge naturally from the symmetrization requirement. Using ket notation for the spin state, the nondegenerate ground state is

$$|\Psi_0(x_1, x_2)\rangle = \psi_1(x_1)\psi_1(x_2)\,|0,0\rangle\,, \quad E_{11} = 2K$$

The antisymmetric spin singlet allows the two fermions to share the same lowest-energy spatial orbital without violating the Pauli exclusion principle.

At the first-excited energy level, $n_1 = 1$ and $n_2 = 2$, so $E_{12} = 5K$. The symmetric spatial state must be paired with the antisymmetric singlet state:

$$\frac{1}{\sqrt{2}}\Big[\psi_1(x_1)\psi_2(x_2) + \psi_2(x_1)\psi_1(x_2)\Big]\;|0,0\rangle\,.$$

The antisymmetric spatial state must be paired with one of the three symmetric triplet states:

$$\frac{1}{\sqrt{2}}\Big[\psi_1(x_1)\psi_2(x_2) - \psi_2(x_1)\psi_1(x_2)\Big]\;|1,m\rangle\,, \quad m = -1, 0, 1$$

15.4 The Free Electron Gas

Consider N noninteracting electrons confined to a cubic box of side length L and volume $V = L^3$. The walls are impenetrable, so the single-electron spatial wave functions vanish at the boundaries. The normalized single-particle wave functions are

$$\psi_{n_x,n_y,n_z} = \sqrt{\frac{8}{V}}\,\sin\left(\frac{n_x\pi x}{L}\right)\sin\left(\frac{n_y\pi y}{L}\right)\sin\left(\frac{n_z\pi z}{L}\right) \tag{15.5}$$

where $n_x, n_y, n_z = 1, 2, 3, \ldots$ The corresponding energies are

$$E_{n_x,n_y,n_z} = \frac{\pi^2\hbar^2}{2mL^2}\left(n_x^2 + n_y^2 + n_z^2\right) = \frac{\hbar^2 k^2}{2m} \tag{15.6}$$

where k is the magnitude of the wave vector

$$\mathbf{k} = \frac{n_x\pi}{L}\hat{\mathbf{x}} + \frac{n_y\pi}{L}\hat{\mathbf{y}} + \frac{n_z\pi}{L}\hat{\mathbf{z}} \tag{15.7}$$

The spacing between adjacent values of k_i is $\Delta k_i = \pi/L$, so each spatial state occupies the k-space volume $\Delta^3 k = \pi^3/V$. Because n_i are positive, the allowed points lie in the positive octant of k-space.

At zero temperature, the electrons fill the lowest energy one-particle states. Since electrons are spin-1/2, they cannot all occupy the same lowest energy state; each spatial state can accommodate two electrons, one spin-up and the other spin-down. The occupied states fill the positive-octant portion of a sphere in k-space of radius k_F, called the Fermi wave vector. The total number of

electrons that occupy states inside this region is approximately

$$N = 2 \times \frac{\frac{1}{8}\left(\frac{4}{3}\pi k_F^3\right)}{\pi^3/V} = \frac{V k_F^3}{3\pi^2} \tag{15.8}$$

The occupied states do not form an exact spherical octant because the allowed points are discrete. The sphere-counting argument becomes accurate in the thermodynamic limit, where boundary corrections are negligible. Therefore,

$$\boxed{k_F = \left(\frac{3\pi^2 N}{V}\right)^{1/3} = \left(3\pi^2 n\right)^{1/3}} \tag{15.9}$$

where $n = N/V$ is the free electron number density (the number of free electrons per unit volume). The Fermi energy is the energy of the highest occupied one-electron state at zero temperature:

$$\boxed{E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}\left(3\pi^2 n\right)^{2/3}} \tag{15.10}$$

Both k_F and E_F are determined by the electron number density $n = N/V$. For a particular metal under ordinary conditions, the conduction-electron density is approximately fixed by its atomic density and the number of conduction electrons contributed by each atom. For example, the density of copper is 8.96 g/cm^3, and its atomic weight is 63.5 g/mol. If we assume that each atom contributes one conduction electron, then copper has an electron number density of 8.49×10^{28} m^{-3} and a corresponding Fermi energy of about 7.0 eV.

The total ground-state energy of the electron gas may be calculated using the one-electron density of states $g(E)$, defined so that

$$g(E)\, dE$$

is the number of available one-electron states with energies between E and $E + dE$. At zero temperature, every one-electron state below the Fermi energy is occupied, so

$$N = \int_0^{E_F} g(E) dE \tag{15.11}$$

and the total ground-state energy is

$$U = \int_0^{E_F} E\, g(E) dE \tag{15.12}$$

The number of electrons occupying single-particle states within a shell of thickness dk is

$$dN = \frac{2 \times \frac{1}{8}\left(4\pi k^2 dk\right)}{\pi^3/V} = \frac{V}{\pi^2} k^2 dk$$

We can find the density of states $g(E) = dN/dE$ for the hard-wall electron gas using the chain rule:

$$dN = g(E)dE = g(E)\frac{dE}{dk}dk,$$

where

$$E = \frac{\hbar^2 k^2}{2m} \Rightarrow \frac{dE}{dk} = \frac{\hbar^2 k}{m}.$$

Comparing our two expressions for dN, we find

$$g(E)\frac{\hbar^2 k}{m} = \frac{V}{\pi^2}k^2$$

and using $k = \sqrt{2mE}/\hbar$, we obtain

$$\boxed{g(E) = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}} \tag{15.13}$$

The total ground-state energy is therefore

$$U = \frac{V}{2\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^{E_F} E^{3/2}\,dE$$

$$= \frac{V}{5\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E_F^{5/2}$$

Using the expression for N in terms of the Fermi energy,

$$\boxed{U = \frac{3}{5}NE_F,} \tag{15.14}$$

which means that the average energy per electron is $\frac{3}{5}E_F$. It makes sense that the average energy would be less than the Fermi energy since it represents the highest available single-electron energy.

If we hold the number of electrons fixed, the ground-state energy of the gas decreases as the box expands. The electron gas exerts an outward pressure on the walls of the container, much like an ordinary gas. In this case, however, the temperature is zero, so the pressure is not caused by thermal agitation. The Pauli exclusion principle forces electrons to occupy high-energy states, producing a nonzero kinetic energy and pressure even at zero temperature. This purely quantum-mechanical pressure is called **degeneracy pressure**. To calculate it, recall that when a gas expands quasistatically by an amount dV, the work it performs is

$$dW = P\,dV.$$

This work comes at the expense of the gas's internal energy, so

$$dU = -P\,dV.$$

Therefore, at fixed particle number,

$$P = -\left(\frac{\partial U}{\partial V}\right)_N,$$

and the degeneracy pressure of the free-electron gas can be written as

$$\boxed{P = \frac{2}{3}\frac{U}{V} = \frac{2}{5}nE_F} \tag{15.15}$$

Interestingly, the expression $P = 2U/(3V)$ is the same as that for an ideal gas, where U is the internal energy of the gas at temperature T. This relation holds for both a classical ideal gas and a noninteracting quantum gas because the one-particle energy has the quadratic form $E \propto p^2$. At $T = 0$, a classical ideal gas would have $U = P = 0$, whereas the "Fermi gas" retains kinetic energy as a result of the Pauli exclusion principle. Using copper as an example, the degeneracy pressure comes out to 3.8×10^{10} Pa or 38 GPa. Degeneracy pressure makes a large contribution to copper's resistance to compression, but it's not the whole explanation. Core-electron Pauli repulsion, electrostatic interactions, and the band structure supply most of the remaining stiffness.

15.5 The Tight-Binding Model

The free-electron gas provides a useful first approximation to the electrons in a metal. In that model, the positively charged ions are replaced by a uniform background, and the conduction electrons move almost freely throughout the solid. The Pauli exclusion principle determines how the available one-electron states are filled, leading to the Fermi energy and degeneracy pressure. What the model leaves out, however, is the discrete atomic structure of the solid. A real crystal consists of atoms arranged in a *periodic lattice*, and the electrons experience the potential of each atomic nucleus together with the fields produced by the other electrons.

We will now approach the problem of electronic structure from a complementary limit using the *tight-binding model*. Imagine beginning with a collection of widely separated atoms. Each atom possesses localized electronic states with discrete energies. If the atoms are brought together to form a crystal, neighboring atomic wave functions begin to overlap. An electron initially associated with one atom then acquires some amplitude to move, or "hop," to a neighboring atom. The resulting energy eigenstates are no longer localized on individual atoms but extend throughout the lattice.

For simplicity, consider a one-dimensional lattice of N identical atoms separated by a distance a, with the nth atom located at $x_n = na$. When the electron sits on the nth atom in isolation, it occupies the state $|n\rangle$ with energy E_0 where $n = 0, 1, \ldots, N-1$. We take these states to be orthonormal; more refined treatments include orbital overlap. In addition to the on-site energy, we

also include nearest-neighbor hopping terms that describe an electron hopping from one atom to neighboring atoms. The tight-binding Hamiltonian is

$$H = E_0 \sum_n |n\rangle \langle n| - t \sum_n \Big(|n\rangle \langle n+1| + |n+1\rangle \langle n| \Big) \tag{15.16}$$

where $t > 0$ is the hopping parameter. We will also impose periodic boundary conditions such that $|N\rangle = |0\rangle$. This might seem arbitrary, but it's merely a mathematical convenience: it eliminates special boundary sites (edge effects) and makes every lattice site equivalent. Different boundary conditions lead to slightly different states for finite N, but their predictions for bulk properties approach the same result as N becomes large. Periodic boundary conditions should not be confused with the physical periodicity of the lattice.

15.5.1 Discrete Translation Operator

Our lattice has discrete translational symmetry; it's invariant under spatial translation in either direction by the lattice spacing a. Define the discrete translation operator by its action on the localized basis states:

$$T_a |n\rangle = |n+1\rangle \tag{15.17}$$

From Eq. (15.17), it follows that T_a is unitary. Furthermore, the tight-binding Hamiltonian is invariant under discrete translation:

$$T_a H T_a^\dagger = E_0 \sum_n |n+1\rangle \langle n+1| - t \sum_n \Big(|n+1\rangle \langle n+2| + |n+2\rangle \langle n+1| \Big).$$

Relabeling the summation index, we find $T_a H T_a^\dagger = H$ or $[H, T_a] = 0$. Thus, the energy eigenstates can be chosen to be eigenstates of discrete translation. Because T_a is unitary, its eigenvalues must satisfy $|\lambda_k| = 1$. Thus, $\lambda_k = e^{-i\theta_k}$. Furthermore, for an eigenstate $|k\rangle$ of the translation operator,

$$T_a^n |k\rangle = e^{-in\theta_k} |k\rangle$$

and thus $|k\rangle$ is an eigenstate with eigenvalue $e^{-in\theta_k}$. Since the operator T_a^n corresponds to translation by na, we define the "wave number" $k = \theta_k/a$ such that

$$T_a^n |k\rangle = e^{-ikna} |k\rangle \tag{15.18}$$

We'll call $|k\rangle$ a momentum eigenstate; more precisely, k is called the crystal wave number or Bloch wave number, while $\hbar k$ is the crystal momentum. The momentum eigenstate can be expanded in terms of localized states as

$$|k\rangle = \sum_m c_m |m\rangle \tag{15.19}$$

Substituting Eq. (15.19) into Eq. (15.18) with $n = 1$ gives

$$\sum_m c_m \left|m+1\right\rangle = \sum_m c_m e^{-ika} \left|m\right\rangle$$

Relabel the summation index on the left-hand side $m + 1 \to m$ such that

$$\sum_m c_{m-1} \left|m\right\rangle = \sum_m c_m e^{-ika} \left|m\right\rangle$$

Comparing coefficients, $c_{m-1} = c_m e^{-ika}$ or $c_m = e^{ika} c_{m-1}$, which is a recursion relation. Repeated application of the recursion gives

$$c_m = c_0 e^{ikma}$$

The momentum eigenstate is

$$\left|k\right\rangle = c_0 \sum_n e^{ikna} \left|n\right\rangle$$

and c_0 can be fixed by normalization:

$$\langle k|k\rangle = |c_0|^2 \sum_{n,m} e^{ik(n-m)a} \delta_{nm} = |c_0|^2 \sum_{n=0}^{N-1} 1 = |c_0|^2 N = 1$$

Hence, the momentum eigenstates are

$$\left|k\right\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} \left|n\right\rangle \tag{15.20}$$

Each state $|n\rangle$ is localized near the nth lattice site. By contrast, each momentum eigenstate $|k\rangle$ is a coherent superposition extending over the entire lattice, in which the amplitude on each site differs from that on the preceding site by the same phase factor e^{ika}. The state is therefore delocalized but possesses a definite spatial phase pattern across the lattice.

15.5.2 Allowed Wave Numbers & Energies

The periodic boundary conditions restrict the possible values of k. Translating through all N sites returns the state to its original site, so it must be the case that

$$T_a^N = \mathbb{1} \qquad \text{(periodic boundary conditions)}$$

Thus,

$$T_a^N \left|k\right\rangle = e^{-ikNa} \left|k\right\rangle = \left|k\right\rangle$$

which requires

$$e^{-ikNa} = 1.$$

This is a transcendental equation with the solution

$$kNa = 2\pi j, \quad j = 0, 1, 2, \ldots, N-1.$$

The allowed wave numbers are

$$k_j = \frac{2\pi j}{Na}, \quad j = 0, 1, 2, \ldots, N-1. \tag{15.21}$$

and the momentum eigenstates are

$$\boxed{|k_j\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{i2\pi jn/N} |n\rangle} \tag{15.22}$$

They form the discrete Fourier transform of the localized basis. Their orthogonality follows from

$$\langle k_{j'} | k_j \rangle = \frac{1}{N} \sum_{n=0}^{N-1} e^{i(k_j - k_{j'})na} = \delta_{jj'}.$$

In this model, each allowed translation eigenvalue corresponds to a one-dimensional eigenspace. Since $[H, T_a] = 0$,

$$T_a H |k\rangle = H T_a |k\rangle = e^{-ika} H |k\rangle .$$

Thus, $H|k\rangle$ belongs to the same one-dimensional translation eigenspace as $|k\rangle$, so it must be proportional to $|k\rangle$. Therefore, $|k\rangle$ is also an energy eigenstate. We can use the energy eigenvalue equation to determine the allowed energies of the system:

$$H |k\rangle = c_0 \sum_n e^{ikna} \Big(E_0 |n\rangle - t |n-1\rangle - t |n+1\rangle \Big)$$

Relabeling the sums in the hopping terms gives

$$\begin{aligned} H |k\rangle &= c_0 \sum_n e^{ikna} \Big(E_0 |n\rangle - te^{ika} |n\rangle - te^{-ika} |n\rangle \Big) \\ &= \Big[E_0 - t(e^{ika} + e^{-ika}) \Big] |k\rangle \\ &= [E_0 - 2t\cos(ka)] |k\rangle \end{aligned}$$

Hence, the allowed energies are

$$\boxed{E_j = E_0 - 2t\cos\left(\frac{2\pi j}{N}\right), \quad j = 0, 1, 2, \ldots, N-1.} \tag{15.23}$$

We may refer to the expression $E(k) = E_0 - 2t\cos(ka)$ as well.

15.5.3 The Brillouin Zone

The wave number k is not unique; any k that differs by an integer multiple of $2\pi/a$ is associated with the same momentum eigenstate. Indeed, for any integer ℓ,

$$|k + 2\pi\ell/a\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{ikna} e^{i2\pi\ell n} |n\rangle = |k\rangle$$

Thus, the wave numbers k and $k + 2\pi\ell/a$ belong to the same equivalence class. We may therefore choose one interval of width $2\pi/a$ containing one representative of each equivalence class. The conventional choice is the **first Brillouin zone**:

$$\boxed{-\frac{\pi}{a} \leq k < \frac{\pi}{a}}$$

The Brillouin zone does not impose an additional physical restriction on the electron. It simply avoids labeling the same crystal state more than once. For a finite chain, the first Brillouin zone contains N allowed values of k, separated by

$$\Delta k = \frac{2\pi}{Na}.$$

As N becomes macroscopically large, these values become extremely closely spaced and k may be treated as continuous.

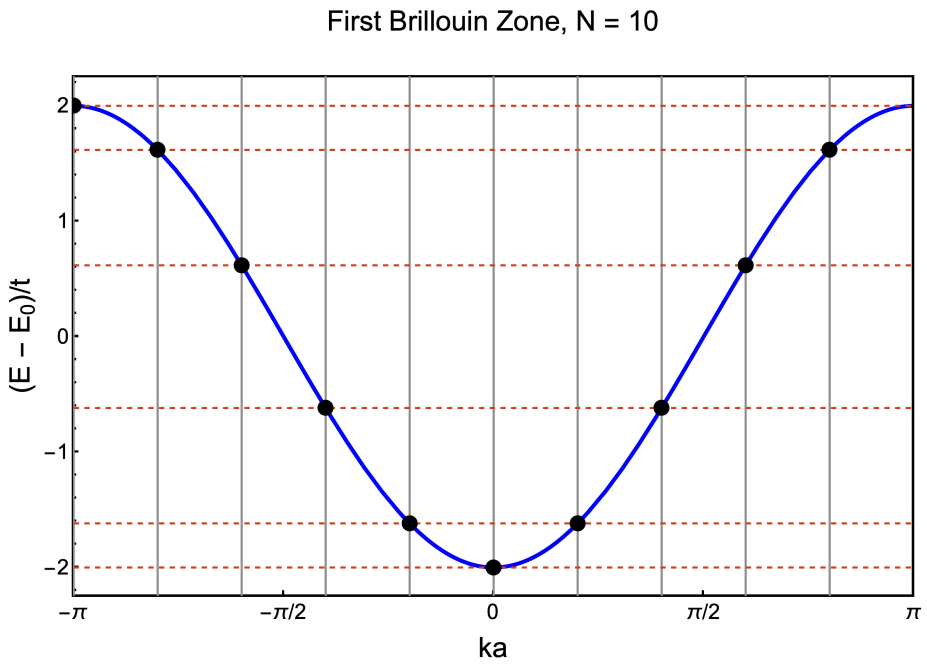
When the hopping is neglected, every localized state $|n\rangle$ has the same energy E_0. The energy E_0 is therefore N-fold degenerate. Turning on the hopping term couples the localized states and distributes the N crystal states over the energies

$$E(k) = E_0 - 2t\cos(ka)$$

Since $-1 \leq \cos(ka) \leq 1$, the allowed energies satisfy

$$E_0 - 2t \leq E(k) \leq E_0 + 2t.$$

In other words, the original atomic energy level has broadened into an *energy band* of width $4t$. Strictly speaking, the finite crystal still has N discrete spatial states. For macroscopic N, however, the energy levels are so closely spaced that they appear to form a continuous band. The spectrum is shown in the figure below for $N = 10$ and $t > 0$.



In summary, when $t = 0$, the localized states $|n\rangle$ are degenerate energy eigenstates with energy E_0. Adding any nonzero hopping term, no matter how small, couples neighboring sites, and the energy eigenstates become delocalized Bloch states $|k\rangle$. In each Bloch state, the electron has the same probability of being found at any atom and the probability amplitude acquires a consistent phase difference ka from one site to the next. At the same time, hopping lifts the original N-fold degeneracy and spreads the single atomic energy E_0 into a band of width $4t$. As $t \to 0$, the band becomes arbitrarily narrow, but the Bloch states remain delocalized for every $t \neq 0$; only at exactly $t = 0$ can the localized states themselves be chosen as energy eigenstates.

15.5.4 Effective Mass

For small crystal momentum, $ka \ll 1$ and $\cos(ka) \approx 1 - (ka)^2/2$. The energy is therefore

$$E(k) \approx E_0 - 2t + ta^2k^2 = E_{\min} + \frac{\hbar^2 k^2}{2m^*}$$

where

$$m^* = \frac{\hbar^2}{2ta^2}$$

is the *effective mass* near the bottom of the band. In terms of the crystal momentum $p_{\text{cr}} = \hbar k$,

$$E(k) \approx E_{\min} + \frac{p_{\text{cr}}^2}{2m^*}$$

Thus, an electron near the bottom of the energy band responds approximately like a free particle with energy $p^2/2m$, but with the bare electron mass replaced by an effective mass determined by

the curvature of the band. More generally,

$$\frac{1}{m^*(k)} = \frac{1}{\hbar^2}\frac{d^2E}{dk^2} = \frac{2ta^2}{\hbar^2}\cos(ka).$$

15.5.5 Filling the Band with Electrons

We now place N_e noninteracting electrons in the lattice at zero temperature. Since the Hamiltonian is spin independent, each spatial state $|k\rangle$ has two associated spin states; one spin up and the other spin down. The band contains N spatial one-electron states and can therefore accommodate $2N$ electrons. The many-electron ground state is the antisymmetrized state obtained by filling the lowest-energy spin orbitals, with no more than one electron in each spin orbital.

At zero temperature, the electrons occupy the lowest available energies up to the Fermi energy. If a band is partially filled, the Fermi energy lies within it, so occupied states are separated from unoccupied states by arbitrarily small energies. An applied electric field can then shift the distribution of occupied crystal momenta, producing an imbalance between electrons moving in opposite directions and hence an electrical current. A partially filled band therefore describes a conductor in the noninteracting band picture.

A completely filled band behaves differently. Every state in the band is occupied, so the Pauli exclusion principle prevents electrons from rearranging themselves within the band in response to a weak electric field. Furthermore, the currents associated with states of opposite crystal momentum cancel. If the filled band is separated from the next unoccupied band by an energy gap, a weak field cannot promote electrons into that band either. The material is then a **band insulator**. In our tight-binding model, there is no energy gap because there's only one energy band.

A real crystal generally possesses many energy bands. The highest completely filled band is called the **valence band**, while the next higher band is called the **conduction band**. In an insulator or semiconductor, these bands are separated by an energy gap. A semiconductor has a relatively small gap, so thermal energy or absorbed light can excite electrons from the valence band into the conduction band. The excited electrons can then move through the conduction band and carry current, while the vacancies they leave behind, called **holes**, can also move through the valence band and contribute to the current. An ideal intrinsic semiconductor is therefore insulating at zero temperature but becomes conducting as electrons are excited across the gap.

15.5.6 Bonding & Antibonding

The tight-binding energy is minimized when $\cos(ka) = 1$, which occurs at $k = 0$. The corresponding state is

$$|k_{\min}\rangle = \frac{1}{\sqrt{N}}\sum_n |n\rangle$$

and its energy is $E_{\min} = E_0 - 2t$. All amplitudes have the same sign. In other words, the amplitudes from the two neighboring sites reinforce the amplitude at site n, and the minus sign in the hopping Hamiltonian lowers the energy. This uniform phase pattern is called the bonding state.

The energy is maximized when $\cos(ka) = -1$, which occurs at the edge of the Brillouin zone where $k = \pm\pi/a$. The maximum-energy state is

$$|k_{\max}\rangle = \frac{1}{\sqrt{N}} \sum_n e^{i\pi n} |n\rangle = \frac{1}{\sqrt{N}} \sum_n (-1)^n |n\rangle$$

and its energy is $E_{\max} = E_0 + 2t$. The amplitudes from the neighboring sites are opposite in phase to the amplitude at site n, so the hopping term raises the energy. This alternating phase pattern is called the antibonding state.

The terms "bonding" and "antibonding" apply most naturally to a real-space interpretation if the states $|n\rangle$ represent localized atomic orbitals $\phi_n(x)$. The corresponding wave function is

$$\psi_k(x) = \frac{1}{\sqrt{N}} \sum_n e^{ikna} \phi_n(x).$$

For $k = 0$, neighboring orbitals enter with the same sign and reinforce one another where they overlap. The resulting bonding wave function varies relatively smoothly through the lattice and has the lowest energy. For $k = \pm\pi/a$, neighboring orbitals enter with opposite signs and tend to cancel in their overlap regions, typically producing additional nodes. The resulting antibonding wave function varies more rapidly in space and has the highest energy.

Chapter 16

The Quantized Electromagnetic Field

16.1 Classical Source-Free Electromagnetism

Throughout these notes, we take $\epsilon_0 = \mu_0 = c = 1$. Consequently, electric and magnetic fields have the same units, the *vacuum dispersion relation* is $\omega_{\mathbf{k}} = |\mathbf{k}|$, and the classical energy is

$$H_{\rm cl} = \frac{1}{2}\int_{\mathbb{R}^3} d^3r \, \left(\mathbf{E}^2 + \mathbf{B}^2\right). \tag{16.1}$$

We will retain $\hbar$, since it identifies the genuinely quantum statements. In vacuum, and in the absence of charges and currents,

$$\nabla\cdot\mathbf{E} = 0, \qquad \nabla\cdot\mathbf{B} = 0, \tag{16.2}$$

$$\nabla\times\mathbf{E} = -\partial_t\mathbf{B}, \qquad \nabla\times\mathbf{B} = \partial_t\mathbf{E}. \tag{16.3}$$

These are the source-free Maxwell equations. With suitable initial data and appropriate boundary conditions, they determine the subsequent evolution of the electromagnetic field.

16.1.1 The Wave Equation

Take the curl of Faraday's law:

$$\nabla\times(\nabla\times\mathbf{E}) = -\partial_t(\nabla\times\mathbf{B}) = -\partial_t^2\mathbf{E}.$$

Using

$$\nabla\times(\nabla\times\mathbf{E}) = \nabla(\nabla\cdot\mathbf{E}) - \nabla^2\mathbf{E} = -\nabla^2\mathbf{E},$$

where the last equality uses $\nabla\cdot\mathbf{E} = 0$, we find

$$(\partial_t^2 - \nabla^2)\mathbf{E} = 0. \tag{16.4}$$

The same calculation with $\mathbf{B}$ gives

$$(\partial_t^2 - \nabla^2)\mathbf{B} = 0. \tag{16.5}$$

The *wave equations* are necessary consequences of Maxwell's equations. They are not by themselves sufficient: a solution of the vector wave equation must also satisfy the divergence constraints and the first-order curl equations.

The general solutions to Eqs. (16.4) & (16.5) are well known and correspond to *linear combinations of traveling plane waves*. We can write a particular solution to Eq. (16.4) as

$$\mathbf{E}(\mathbf{r},t) = \mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \mathbf{C}_\mathbf{k}^*\, e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} = 2\,\mathrm{Re}\left[\mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)}\right] \tag{16.6}$$

where $\mathbf{C}_\mathbf{k}$ is a complex amplitude vector, $\mathbf{k}$ is the wavevector, and $\omega > 0$ is the angular frequency of the wave. The complex amplitude vector encodes the real amplitude, direction of the electric field (the polarization), and the phase constant of the wave, while the wavevector describes the direction of wave propagation. In order for Eq. (16.6) to be a valid solution to the wave equation, we must have $\omega_\mathbf{k} = |\mathbf{k}|$. To construct the most general solution to the wave equation, we must combine all the different plane-wave solutions. In general, this superposition takes the form of an integral over all possible values of $\mathbf{k}$. We can simplify the situation by restricting the field to a finite cubic region and imposing periodic boundary conditions. This is merely a mathematical convenience. Once the construction is complete, we can take the limit $L \to \infty$ and recover the continuum description. Under periodic boundary conditions, we require

$$\mathbf{E}(\mathbf{r} + L\,\hat{\mathbf{r}}_j, t) = \mathbf{E}(\mathbf{r}, t) \tag{16.7}$$

where r_j is a Cartesian coordinate. A plane wave must therefore satisfy

$$e^{i\mathbf{k}\cdot(\mathbf{r}+L\hat{\mathbf{r}}_j)} = e^{i\mathbf{k}\cdot\mathbf{r}}, \quad \text{or} \quad e^{ik_jL} = 1, \tag{16.8}$$

so the allowed components of the wavevector are given by $k_jL = 2\pi m_j$, where $m_j \in \mathbb{Z}$. The allowed wavevectors form a discrete lattice in wavevector space.

The most general solution to the wave equation in a finite periodic box can be written as

$$\mathbf{E}(\mathbf{r},t) = \sum_\mathbf{k} \mathbf{C}_\mathbf{k}\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{c.c.}, \tag{16.9}$$

where c.c. denotes the complex conjugate of the first term in the equation. Applying Gauss's law for the electric field,

$$0 = \nabla\cdot\mathbf{E} = i\sum_\mathbf{k} (\mathbf{k}\cdot\mathbf{C}_\mathbf{k})\, e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{c.c.} \tag{16.10}$$

and it follows that $\mathbf{k}\cdot\mathbf{C}_\mathbf{k} = 0$. Hence, $\mathbf{C}_\mathbf{k}$ lies in a two dimensional plane orthogonal to $\mathbf{k}$. For each

$\mathbf{k} \neq 0$, we choose two real, orthonormal polarization vectors $\{\hat{\mathbf{e}}_{\mathbf{k}\mu}\}$, $\mu = 1, 2$, such that

$$\mathbf{C}_{\mathbf{k}} = \sum_{\mu=1}^{2} C_{\mathbf{k}\mu}\, \hat{\mathbf{e}}_{\mathbf{k}\mu} \tag{16.11}$$

In many applications, it is convenient to choose the set $\left\{\hat{\mathbf{e}}_{\mathbf{k}1}, \hat{\mathbf{e}}_{\mathbf{k}2}, \hat{\mathbf{k}}\right\}$ as a right-handed set of orthogonal unit vectors. Thus, for each nonzero wavevector $\mathbf{k}$, the electric field has two independent transverse polarization components. The complex scalar amplitudes $C_{\mathbf{k}\mu}$ completely specify the amplitude, phase, and polarization of the corresponding plane-wave mode. We may therefore label the classical modes of the free electromagnetic field by the pair $(\mathbf{k}, \mu)$, $\mu = 1, 2$ with angular frequency $\omega_{\mathbf{k}} = |\mathbf{k}|$. It will be convenient to label the modes with one index, i.e., $\lambda = (\mathbf{k}, \mu)$. The electric field can now be written as

$$\mathbf{E}(\mathbf{r}, t) = \sum_{\lambda} C_{\lambda}\, \hat{\mathbf{e}}_{\lambda}\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.12}$$

Starting with Eq. (16.5), we can apply a similar line of reasoning to derive an expression for the magnetic field as a superposition of plane-wave modes:

$$\mathbf{B}(\mathbf{r}, t) = \sum_{\lambda} D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.13}$$

From $\nabla \cdot \mathbf{B} = 0$, it follows that $\mathbf{k} \cdot \hat{\mathbf{b}}_{\lambda} = 0$. Applying Faraday's law,

$$\nabla \times \mathbf{E} = -\partial_t \mathbf{B}, \tag{16.14}$$

we find that

$$i \sum_{\lambda} C_{\lambda}\, (\mathbf{k} \times \hat{\mathbf{e}}_{\lambda})\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} = i \sum_{\lambda} D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, \omega_{\mathbf{k}} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.15}$$

Equating coefficients,

$$C_{\lambda}\, (\mathbf{k} \times \hat{\mathbf{e}}_{\lambda}) = D_{\lambda}\, \hat{\mathbf{b}}_{\lambda}\, \omega_{\mathbf{k}} \tag{16.16}$$

Taking the magnitude of both sides, noting that $\hat{\mathbf{e}}_{\lambda}$ is perpendicular to $\mathbf{k}$ and $\omega_{\mathbf{k}} = |\mathbf{k}|$, we find $C_{\lambda} = D_{\lambda}$, and furthermore,

$$\hat{\mathbf{b}}_{\lambda} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\lambda} \tag{16.17}$$

Hence, the magnetic field is

$$\mathbf{B}(\mathbf{r}, t) = \sum_{\lambda} C_{\lambda}\, \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\lambda}\right)\, e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_{\mathbf{k}} t)} + \text{c.c.} \tag{16.18}$$

It happens that Eqs. (16.12) & (16.18) are the most general solutions to Maxwell's equations in a finite periodic box. Applying the Ampère-Maxwell equation only tells us that $\omega_{\mathbf{k}} = |\mathbf{k}|$, which we

found by solving the wave equation.

16.1.2 Field Energy

We now wish to compute the field energy in terms of the solutions for $\mathbf{E}$ and $\mathbf{B}$. For each mode, define

$$\phi_{\mathbf{k}}(\mathbf{r}, t) = \mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t,$$

and write the fields as

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_+^*, \qquad \mathbf{E}_+ = \sum_\lambda C_\lambda \hat{\mathbf{e}}_\lambda e^{i\phi_{\mathbf{k}}},$$

$$\mathbf{B} = \mathbf{B}_+ + \mathbf{B}_+^*, \qquad \mathbf{B}_+ = \sum_\lambda C_\lambda \hat{\mathbf{b}}_\lambda e^{i\phi_{\mathbf{k}}}.$$

According to Eq. (16.1), for the total energy, we must integrate $\mathbf{E}^2$ and $\mathbf{B}^2$ over the cube. First consider the electric-field contribution to the energy. Expanding,

$$\mathbf{E}^2 = \mathbf{E}_+ \cdot \mathbf{E}_+ + \mathbf{E}_+^* \cdot \mathbf{E}_+ + \mathbf{E}_+ \cdot \mathbf{E}_+^* + \mathbf{E}_+^* \cdot \mathbf{E}_+^*$$

Isolating one of the mixed terms, we find

$$\int_V \mathbf{E}_+ \cdot \mathbf{E}_+^* \, d^3r = \sum_{\mathbf{k},\mu} \sum_{\mathbf{k}',\nu} C_{\mathbf{k}\mu} C_{\mathbf{k}'\nu}^* \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}'\nu}\right) e^{-i(\omega_k - \omega_{k'})t} \int_V e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \, d^3r$$

For a periodic box, we have the following orthogonality condition:

$$\int_V e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \, d^3r = V \delta_{\mathbf{k},\mathbf{k}'}.$$

Hence,

$$\int_V \mathbf{E}_+ \cdot \mathbf{E}_+^* \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2$$

The other mixed term is just the complex conjugate of the expression above and gives the same result. Together they contribute $2V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2$. As for the unmixed terms, define

$$F = \sum_{\mathbf{k},\mu} \sum_{\mathbf{k}',\nu} C_{\mathbf{k}\mu} C_{\mathbf{k}'\nu} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{\mathbf{k}'\nu}\right) e^{-i(\omega_k + \omega_{k'})t} \int_V e^{i(\mathbf{k}+\mathbf{k}')\cdot\mathbf{r}} \, d^3r$$

such that

$$\int_V \mathbf{E}^2 \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2 + F + F^*.$$

The orthogonality relation selects $\mathbf{k}' = -\mathbf{k}$, and since $\omega_{-\mathbf{k}} = \omega_{\mathbf{k}}$, we find

$$F = V \sum_{\mathbf{k},\mu,\nu} C_{\mathbf{k}\mu} C_{-\mathbf{k}\nu} \left(\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\nu}\right) e^{-2i\omega_k t}$$

By a similar line of reasoning, we arrive at an equivalent expression for the magnetic-field contribution to the field energy:

$$\int_V \mathbf{B}^2 \, d^3r = V \sum_{\mathbf{k},\mu} |C_{\mathbf{k}\mu}|^2 + G + G^*,$$

where

$$G = V \sum_{\mathbf{k},\mu,\nu} C_{\mathbf{k}\mu} C_{-\mathbf{k}\nu} \left(\hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\nu} \right) e^{-i2\omega_k t}$$

Since $\hat{\mathbf{b}}_{\mathbf{k}\mu} = \hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu}$, using the vector identity $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$,

$$\hat{\mathbf{b}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{b}}_{-\mathbf{k}\nu} = \left(\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{\mathbf{k}\mu} \right) \cdot \left(-\hat{\mathbf{k}} \times \hat{\mathbf{e}}_{-\mathbf{k}\nu} \right) = -\hat{\mathbf{e}}_{\mathbf{k}\mu} \cdot \hat{\mathbf{e}}_{-\mathbf{k}\nu}$$

Therefore, $G = -F$, and the unmixed terms cancel in the total energy:

$$H_{\text{cl}} = 2V \sum_{\lambda} |C_\lambda|^2 \tag{16.19}$$

Thus, although the fields are constructed by superposing many plane waves, the energy contains no coupling between distinct modes. It is a sum of nonnegative contributions, one from each mode $(\mathbf{k}, \mu)$. In this sense, the normal modes of the free electromagnetic field are *energetically independent*.

16.1.3 Oscillator Form

Since the modes are energetically independent, let

$$H_{\text{cl}} = \sum_{\lambda} H_\lambda, \quad H_\lambda = 2V|C_\lambda|^2 \tag{16.20}$$

Define the time-dependent mode amplitude

$$C_\lambda(t) = C_\lambda e^{-i\omega_{\mathbf{k}} t} \tag{16.21}$$

It is complex, so it contains two real quantities. Define the following change of variables:

$$\begin{aligned} Q_\lambda(t) &= \frac{\sqrt{V}}{i\omega_{\mathbf{k}}} \left[C_\lambda(t) - C_\lambda^*(t) \right] \\ P_\lambda(t) &= -\sqrt{V} \left[C_\lambda(t) + C_\lambda^*(t) \right] \end{aligned} \tag{16.22}$$

Inverting the expressions above, we find

$$C_\lambda(t) + C_\lambda^*(t) = -\frac{P_\lambda(t)}{\sqrt{V}}, \qquad C_\lambda(t) - C_\lambda^*(t) = \frac{i\omega_{\mathbf{k}} Q_\lambda(t)}{\sqrt{V}},$$

and it follows that

$$C_\lambda(t) = \frac{-P_\lambda(t) + i\omega_{\mathbf{k}} Q_\lambda(t)}{2\sqrt{V}}. \tag{16.23}$$

Therefore,

$$|C_\lambda|^2 = \frac{P_\lambda(t)^2 + \omega_{\mathbf{k}}^2 Q_\lambda(t)^2}{4V} \tag{16.24}$$

Finally, since $|C_\lambda(t)| = |C_\lambda|$,

$$H_\lambda = 2V|C_\lambda|^2 = \frac{1}{2}\left(P_\lambda^2 + \omega_{\mathbf{k}}^2 Q_\lambda^2\right). \tag{16.25}$$

While the expression above may have the *mathematical form* of a simple harmonic oscillator, it remains to be shown that the variables Q_λ and P_λ behave dynamically as expected. From Eqs. (16.21) & (16.22), we find

$$\dot{Q}_\lambda = P_\lambda, \quad \text{and} \quad \dot{P}_\lambda = -\omega_{\mathbf{k}}^2 Q_\lambda, \tag{16.26}$$

which are precisely the equations of motion for a simple harmonic oscillator of mass $m = 1$. At this stage, we have shown that the source-free electromagnetic field can be decomposed into a collection of independent simple harmonic oscillators, one for each mode.

16.2 Quantizing the Field

To construct a quantum description that preserves the Maxwell dynamics, we retain the mode expansion but "promote" the real mode variables Q_λ and P_λ to Hermitian operators. We require their Heisenberg equations, generated by the oscillator Hamiltonian, to reproduce the classical equations of motion. Because the electric and magnetic fields are linear combinations of these mode variables, this correspondence ensures that the resulting operator fields obey the source-free Maxwell equations.

Since each oscillator is independent, we consider the Heisenberg evolution of just one mode. The Heisenberg equation for Q is

$$\dot{Q} = \frac{1}{i\hbar}[Q, H], \tag{16.27}$$

where we will drop the λ subscript for now. Since $H = \frac{1}{2}(P^2 + \omega^2 Q^2)$,

$$\begin{aligned}\dot{Q} &= \frac{1}{2i\hbar}[Q, P^2] \\ &= \frac{1}{2i\hbar}\big([Q, P]P + P[Q, P]\big)\end{aligned}$$

Letting $K = [Q, P]$, in order to reproduce Eq. (16.26), we require

$$\{K, P\} = 2i\hbar P, \tag{16.28}$$

where $\{K, P\} = KP + PK$ denotes the *anticommutator* of K and P. Similarly, the Heisenberg equation for P is

$$\dot{P} = \frac{1}{i\hbar}[P, H] = -\frac{\omega^2}{2i\hbar}\{K, Q\}, \tag{16.29}$$

so we require

$$\{K, Q\} = 2i\hbar Q \tag{16.30}$$

The simplest way to satisfy Eqs. (16.28) & (16.30) is to set $K = i\hbar$, where we omit the identity. While this is precisely the choice that leads to a correct description of the quantized electromagnetic field, it is worth mentioning that it is not the most general solution if our only constraint is to satisfy the Maxwell equations.

Consider $K = i\hbar(1+G)$, where G is any linear operator satisfying $\{G, P\} = \{G, Q\} = 0$. Because Q and P are Hermitian, their commutator is anti-Hermitian. So $K^\dagger = -K$, and consequently, $G^\dagger = G$. Furthermore, since G anticommutes with Q and P, it commutes with their squares:

$$[G, Q^2] = [G, P^2] = 0$$

This implies that G commutes with the Hamiltonian and is therefore conserved. Hence,

$$[Q, P] = i\hbar + i\hbar G, \quad \text{where} \quad \{G, P\} = \{G, Q\} = 0, \tag{16.31}$$

and G is a conserved quantity. At this stage, the general solution includes $G \neq 0$.

16.2.1 Creation and Annihilation Operators

When we studied the quantum harmonic oscillator in Chapter 7, we introduced *ladder operators* and used them to determine the oscillator energies and energy eigenstates. Define

$$a = \sqrt{\frac{\hbar\omega}{2\hbar}}Q + \frac{i}{\sqrt{2m\omega}}P, \qquad a^\dagger = \sqrt{\frac{\hbar\omega}{2\hbar}}Q - \frac{i}{\sqrt{2m\omega}}P, \tag{16.32}$$

A direct calculation gives

$$[a, a^\dagger] = \frac{1}{i\hbar}[Q, P] = 1 + G, \tag{16.33}$$

and the anticommutation conditions imply $\{G, a\} = \{G, a^\dagger\} = 0$. Furthermore, a and $a^\dagger$ satisfy

$$\dot{a} = -i\omega a, \quad \dot{a}^\dagger = +i\omega a^\dagger \tag{16.34}$$

as Heisenberg operators. In terms of the ladder operators, the Hamiltonian becomes

$$H = \frac{\hbar\omega}{2}\left(aa^\dagger + a^\dagger a\right) = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right) + \frac{\hbar\omega}{2}G \tag{16.35}$$

We are now in a good place to further constrain the quantized field. Photon-counting experi-

ments show that excitations of a mode are detected as discrete quanta, each with energy $\hbar\omega$, and that a mode can contain any nonnegative integer number of photons, consistent with their bosonic statistics. We therefore choose the standard bosonic representation, in which $a^\dagger a$ is the photon-number operator and $a^\dagger$ raises that number by one while a lowers it by one. As we will show, this representation has $[a, a^\dagger] = 1$; comparing with the generalized relation gives $G = 0$. The Maxwell equations supply the oscillator dynamics while identifying the mode operators with the observed photon-counting states supplies the additional physical input that is needed. In other words, we choose $G = 0$ because photons are observable and obey bosonic statistics.

In summary, each mode of the quantized electromagnetic field behaves like a simple harmonic oscillator with quantized excitations that we call *photons*. The commutation relation $[a, a^\dagger] = 1$ encodes the bosonic nature of photons in a particular mode, and the energy eigenstates indicate the number of photons occupying the mode. Since the modes are independent, the full set of bosonic commutation relations, with $\lambda = (\mathbf{k}, \mu)$, is

$$\boxed{\left[a_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = \delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'},} \qquad \boxed{\left[a_{\mathbf{k}\mu}, a_{\mathbf{k}'\mu'}\right] = \left[a^\dagger_{\mathbf{k}\mu}, a^\dagger_{\mathbf{k}'\mu'}\right] = 0.} \tag{16.36}$$

The total field Hamiltonian is

$$\boxed{H = \sum_\lambda \hbar\omega_{\mathbf{k}} \left(N_\lambda + \frac{1}{2}\right),} \tag{16.37}$$

where $N_\lambda = a^\dagger_\lambda a_\lambda$ is the number operator for each mode. The lowest energy state of the Hamiltonian is called the **vacuum state** and is denoted $|0\rangle$. The vacuum state satisfies $a_\lambda |0\rangle = 0$ for all modes. The vacuum state is a tensor product of lowest energy states for each mode, i.e.,

$$|0\rangle = |0_{\mathbf{k}_1, s_1}\rangle \otimes |0_{\mathbf{k}_2, s_2}\rangle \otimes \cdots, \tag{16.38}$$

and the vacuum energy, also called zero-point energy, is

$$E_{\text{vac}} = \sum_\lambda \frac{\hbar\omega_{\mathbf{k}}}{2} \tag{16.39}$$

Unless there is some cutoff that limits the number of oscillators with arbitrarily high frequencies, this sum diverges because there are an infinite number of modes. Nonetheless, it is convenient to treat E_{vac} as if it were finite. This constant has no effect on the free-field dynamics because it commutes with every operator.

If we apply the creation operator for a particular mode to the vacuum state, we obtain

$$\begin{aligned} a^\dagger_{\mathbf{k}\mu}|0\rangle &= |0_{\mathbf{k}_1,\mu_1}\rangle \otimes |0_{\mathbf{k}_2,\mu_2}\rangle \otimes \cdots a^\dagger_{\mathbf{k}\mu}|0_{\mathbf{k}\mu}\rangle \otimes \cdots \\ &= |0_{\mathbf{k}_1,\mu_1}\rangle \otimes |0_{\mathbf{k}_2,\mu_2}\rangle \otimes \cdots |1_{\mathbf{k}\mu}\rangle \otimes \cdots \end{aligned} \tag{16.40}$$

For simplicity, we denote the single-photon state by $|1_{\mathbf{k}\mu}\rangle$ or $|1_\lambda\rangle$. We denote the state containing n photons in the mode $(\mathbf{k},\mu)$ by

$$|n_{\mathbf{k}\mu}\rangle, \qquad n_{\mathbf{k}\mu} = 0, 1, 2, \ldots. \tag{16.41}$$

This notation describes the occupation of a single mode. A general photon-number state of the entire field must specify the occupation number of every mode. We write such a state as

$$\big|\{n_{\mathbf{k}\mu}\}\big\rangle, \tag{16.42}$$

where the braces denote the complete collection of occupation numbers.

When the ladder operators act on a state of the entire field, they change only the occupation number of the specified mode:

$$a_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu}}\,\big|\ldots, n_{\mathbf{k}\mu}-1, \ldots\big\rangle, \tag{16.43}$$

$$a^\dagger_{\mathbf{k}\mu}\big|\{n_{\mathbf{q}\sigma}\}\big\rangle = \sqrt{n_{\mathbf{k}\mu}+1}\,\big|\ldots, n_{\mathbf{k}\mu}+1, \ldots\big\rangle. \tag{16.44}$$

All other occupation numbers remain unchanged. For a single mode with n photons,

$$|n_\lambda\rangle = \frac{\left(a^\dagger_\lambda\right)^n}{\sqrt{n!}}|0\rangle \tag{16.45}$$

and for an arbitrary photon-number state,

$$|\{n_\sigma\}\rangle = \prod_\lambda \frac{\left(a^\dagger_\lambda\right)^{n_\lambda}}{\sqrt{n_\lambda!}}|0\rangle \tag{16.46}$$

Since creation operators for distinct modes commute, the order of the product does not matter. The photon-number states are orthonormal, so an arbitrary field state can be written as

$$|\Psi\rangle = \sum_{\{n_\lambda\}} \psi_{\{n_\lambda\}}\,|\{n_\lambda\}\rangle \tag{16.47}$$

16.2.2 Field Operators

We now turn our attention to the field operators. Begin by writing the fields in terms of (Q_λ, P_λ). Then use Eq. (16.32) to determine the field expansions in terms of a_λ and $a^\dagger_\lambda$. This results in the

same field operators as the direction substitution

$$C_\lambda(t) \longrightarrow i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, a_\lambda(t), \qquad C^*_\lambda(t) \longrightarrow -i\sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, a^\dagger_\lambda(t), \tag{16.48}$$

in the classical expressions. Thus, the electric-field operator becomes

$$\boxed{\mathbf{E}(\mathbf{r},t) = i\sum_\lambda \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, \hat{\mathbf{e}}_\lambda \left[a_\lambda e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)} - a^\dagger_\lambda e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_{\mathbf{k}}t)}\right],} \tag{16.49}$$

and the magnetic-field operator is

$$\boxed{\mathbf{B}(\mathbf{r},t) = i\sum_\lambda \sqrt{\frac{\hbar\omega_{\mathbf{k}}}{2V}}\, \hat{\mathbf{b}}_\lambda \left[a_\lambda e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)} - a^\dagger_\lambda e^{-i(\mathbf{k}\cdot\mathbf{r}-\omega_k t)}\right].} \tag{16.50}$$

We see that in a quantum-mechanical treatment of the electromagnetic field, the field itself becomes an operator that annihilates and creates photons. Assuming that our procedure has given us the correct expressions for the electric- and magnetic-field operators, we now turn our attention to the genuinely quantum-mechanical features of the quantized electromagnetic field. In a photon-number state, the expectation values of the electric and magnetic fields vanish. This is because each field operator contains creation and annihilation operators, which map a photon-number state to orthogonal states. The expectation values of the squared fields, however, do not vanish. Thus, a photon-number state can possess nonzero electromagnetic energy and *nonzero field fluctuations* even though its mean electric and magnetic fields are zero. The vacuum provides the simplest example: its mean fields vanish, but its mean-square fields are nonzero.

A nonzero mean electric or magnetic field requires a coherent superposition of states with different photon numbers. More precisely, because the field operators change photon number by one, the state must contain coherence between photon-number states whose occupation numbers differ by one. An incoherent mixture of different photon numbers still has a vanishing mean field. **Coherent states** are the most important example of states with nonzero mean fields. Their field expectation values have the same plane-wave form as classical electromagnetic fields, while their photon numbers are not definite. When the mean photon number is large, the relative photon-number fluctuations become small, and the coherent state increasingly resembles a classical wave.

Example 16.1: Photon-Number States

Let $|\{n_{\mathbf{k}\mu}\}\rangle$ denote a photon-number state in a finite periodic box. Calculate the expectation values of $\mathbf{E}$, $\mathbf{B}$, $\mathbf{E}^2$, and $\mathbf{B}^2$. Use the last two expressions to calculate the expectation value of H_{field} and confirm that it gives the correct energy.

Solution. For every mode,

$$\langle\{n\}|a_{\mathbf{k}\mu}|\{n\}\rangle = 0, \qquad \langle\{n\}|a^\dagger_{\mathbf{k}\mu}|\{n\}\rangle = 0\rangle\,,$$

because the ladder operators change the photon occupation number. Therefore,

$$\langle\mathbf{E}(\mathbf{r},t)\rangle = 0, \qquad \langle\mathbf{B}(\mathbf{r},t)\rangle = 0.$$

For products of ladder operators,

$$\left\langle a_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}\right\rangle = \left\langle a^\dagger_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}'\mu'}\right\rangle = 0,$$

while

$$\left\langle a^\dagger_{\mathbf{k}\mu}a_{\mathbf{k}'\mu'}\right\rangle = n_{\mathbf{k}\mu}\delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'}$$

and

$$\left\langle a_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}'\mu'}\right\rangle = (n_{\mathbf{k}\mu}+1)\,\delta_{\mathbf{k}\mathbf{k}'}\delta_{\mu\mu'}.$$

Using $\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu} = 1$, we obtain

$$\begin{aligned}\langle\mathbf{E}^2(\mathbf{r},t)\rangle &= \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left[\left\langle a_{\mathbf{k}\mu}a^\dagger_{\mathbf{k}\mu}\right\rangle + \left\langle a^\dagger_{\mathbf{k}\mu}a_{\mathbf{k}\mu}\right\rangle\right]\\ &= \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left[(n_{\mathbf{k}\mu}+1)+n_{\mathbf{k}\mu}\right].\end{aligned}$$

Hence,

$$\boxed{\langle\mathbf{E}^2(\mathbf{r},t)\rangle = \sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{2V}\left(2n_{\mathbf{k}\mu}+1\right).}$$

The magnetic field calculation gives the same result, i.e., $\left\langle\mathbf{E}^2\right\rangle = \left\langle\mathbf{B}^2\right\rangle$. The expectation value of the field Hamiltonian is

$$\begin{aligned}\langle H\rangle &= \frac{1}{2}\int_V d^3r\,\left(\langle\mathbf{E}^2\rangle + \langle\mathbf{B}^2\rangle\right)\\ &= \frac{1}{2}V\sum_{\mathbf{k},\mu}\frac{\hbar\omega_{\mathbf{k}}}{V}\left(2n_{\mathbf{k}\mu}+1\right)\\ &= \sum_{\mathbf{k},\mu}\hbar\omega_{\mathbf{k}}\left(n_{\mathbf{k}\mu}+\frac{1}{2}\right).\end{aligned}$$

This agrees with the result obtained from computing the expectation value directly using $\langle\{n\}|H|\{n\}\rangle$; it's just the energy eigenvalue of the photon-number state.

Hence, for an electromagnetic field prepared in a photon-number state,

$$\langle \mathbf{E} \rangle = 0, \qquad \langle \mathbf{E}^2 \rangle > 0.$$

The first equation says that the probability distribution for a suitable electric-field measurement is centered at zero. The second says that the distribution has nonzero width. Therefore, individual measurements need not return zero; repeated measurements on identically prepared states produce different outcomes whose average approaches zero.

16.2.3 Field Momentum

Classically, we know that the electromagnetic field carries momentum as well as energy. In units with $c = 1$, the momentum density of the electromagnetic field is

$$\mathbf{g}(\mathbf{r}, t) = \mathbf{E} \times \mathbf{B} \tag{16.51}$$

so the total field momentum is

$$\mathbf{P} = \int d^3r \; \mathbf{E} \times \mathbf{B} \tag{16.52}$$

For a plane wave, the fields are transverse and $\mathbf{B} = \hat{\mathbf{k}} \times \mathbf{E}$, so its momentum points in the direction of propagation $\hat{\mathbf{k}}$. For the quantized field, despite the ordering ambiguity, Eq. (16.52) is the correct expression for the total field momentum operator. As an operator, it represents the *generator of translations of the field itself*. Substituting the mode expansions for $\mathbf{E}$ and $\mathbf{B}$ into Eq. (16.52),

$$\mathbf{P} = \sum_{\mathbf{k},\mu} \hbar \mathbf{k} a^\dagger_{\mathbf{k}\mu} a_{\mathbf{k}\mu} = \sum_{\mathbf{k},\mu} \hbar \mathbf{k} N_{\mathbf{k},\mu}, \tag{16.53}$$

where we've used the fact that $\sum_{\mathbf{k},\mu} \hbar \mathbf{k} = 0$ since for every value of $\mathbf{k}$, there is a $-\mathbf{k}$ to cancel it.

As expected, the vacuum state has no momentum. Applying the momentum operator to a single-photon state,

$$\mathbf{P} \, |1_{\mathbf{k}\mu}\rangle = \sum_{\mathbf{k}',\mu'} \hbar \mathbf{k}' N_{\mathbf{k}'\mu} \, |1_{\mathbf{k}\mu}\rangle = \hbar \mathbf{k} \, |1_{\mathbf{k}\mu}\rangle$$

Evidently, a single photon carries momentum $\hbar \mathbf{k}$ in addition to its energy $\hbar\omega$. Since $\omega = |\mathbf{k}|$, we find that the energy of a photon is related to its momentum: $E = P$. If we interpret E and P as the relativistic energy and momentum, this would imply that photons are massless according to the energy-momentum relation from special relativity.

16.2.4 Angular Momentum

The classical field also carries angular momentum. For a localized field configuration, its total angular momentum about the origin is

$$\mathbf{J} = \int d^3r\, \mathbf{r} \times \big(\mathbf{E} \times \mathbf{B}\big) \tag{16.54}$$

Once again, despite the ordering ambiguity, Eq. (16.54) is the correct expression for the total angular momentum operator. As an operator it represents the *generator of rotations of the field.* A rotation of the electromagnetic field changes both where the field is evaluated and its vector components.

For a single photon, the most direct intrinsic angular-momentum property is its **helicity**, defined as the component of its spin along its direction of motion:

$$\Lambda_{\mathbf{k}} = \hat{\mathbf{k}} \cdot \mathbf{J}. \tag{16.55}$$

For a plane wave, a rotation about $\hat{\mathbf{k}}$ leaves the spatial dependence of the wave unchanged because the field is uniform on the surface of each plane perpendicular to $\hat{\mathbf{k}}$. Its angular momentum about that axis therefore acts on the transverse polarization components. Let $|1_{\mathbf{k}1}\rangle$ and $|1_{\mathbf{k}2}\rangle$ denote one-photon states with the two linear polarizations and define the **circularly polarized states**

$$|1_{\mathbf{k},\sigma}\rangle = \frac{1}{\sqrt{2}}\big(|1_{\mathbf{k}1}\rangle + i\sigma|1_{\mathbf{k}2}\rangle\big), \qquad \sigma = \pm 1. \tag{16.56}$$

$$\Lambda_{\mathbf{k}}|1_{\mathbf{k},\sigma}\rangle = \sigma\hbar|1_{\mathbf{k},\sigma}\rangle. \tag{16.57}$$

Thus, the two allowed helicity eigenvalues are $+\hbar$ and $-\hbar$. The photon has an intrinsic spin of 1 because the electromagnetic field transforms as a vector under rotations. But of course, we would not be speaking of photons if it weren't for the bosonic commutation relations we used to arrive at the photon picture. This is our first glimpse into the spin-statistics theorem!

16.3 Jaynes-Cummings Model

Having constructed a quantum description of the electromagnetic field, we can now ask how the field interacts with matter. In general, this is a complicated problem: an atom has many energy levels, and the field contains many modes, all of which may participate in the interaction. We will develop a broader treatment of light–matter interactions in the next chapter. For now, we will focus on a simple model that isolates some of the essential physics.

The Jaynes–Cummings model describes a two-level atom interacting with a single mode of the quantized electromagnetic field. The two-level approximation retains just two atomic states, while the single-mode approximation keeps just one field oscillator. With the dipole and rotating-wave

approximations, the Hamiltonian can be written as

$$H = \hbar\omega\, a^\dagger a + \frac{\hbar\omega_0}{2}\sigma_z + \hbar g\left(\sigma_+ a + \sigma_- a^\dagger\right), \tag{16.58}$$

where the operators σ_z, σ_+, and σ_- act on the atomic states while the ladder operators $a, a^\dagger$ act on the photon-number states. The parameter ω refers to the frequency of the field, ω_0 is related to the energy separation between the two atomic states, and g describes the strength of the interaction between the atom and the field. Despite its simplicity, this model captures a central feature of light–matter interaction: the atom and field can exchange discrete excitations. It will let us examine that exchange and the resulting atom–field dynamics in a setting simple enough to solve exactly.

16.3.1 Derivation of the Hamiltonian

Consider a two-level atom with ground state $|g\rangle$, excited state $|e\rangle$, and energy separation $\hbar\omega_0$. The free atomic Hamiltonian may be written as

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\sigma_z, \tag{16.59}$$

where σ_z is a Pauli operator. We may formulate the atomic Hamiltonian in matrix representation by choosing the standard orthonormal basis for $\{|g\rangle, |e\rangle\}$. More precisely, we set

$$|e\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |g\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \tag{16.60}$$

so that

$$\sigma_z = |e\rangle\langle e| - |g\rangle\langle g| = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}, \tag{16.61}$$

and

$$H_{\text{atom}} = \frac{\hbar\omega_0}{2}\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}. \tag{16.62}$$

We also define the atomic ladder operators

$$\sigma_+ = |e\rangle\langle g| = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}, \quad \sigma_- = |g\rangle\langle e| = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix}. \tag{16.63}$$

The two-level system is interacting with a single mode of the electromagnetic field characterized by angular frequency ω, wavevector $\mathbf{k}$, and polarization μ. Neglecting the zero-point energy, the free-field Hamiltonian for this mode is

$$H_{\text{field}} = \hbar\omega\, a^\dagger a \tag{16.64}$$

The free Hamiltonian of the combined system, neglecting interactions, is therefore

$$H_0 = \frac{\hbar\omega_0}{2}\,\sigma_z + \hbar\omega\, a^\dagger a \tag{16.65}$$

Assuming the wavelength of the radiation is much larger than the characteristic size of the two-level system, we use the **electric-dipole approximation** to model the interaction Hamiltonian as

$$V = -\mathbf{d}\cdot\mathbf{E}(\mathbf{r}_0) \tag{16.66}$$

here $\mathbf{d}$ is the dipole operator of the two-level system and $\mathbf{r}_0$ is its location. Within the two-level subspace, the dipole operator may be written in matrix representation as

$$\mathbf{d} = \begin{pmatrix} \mathbf{d}_{ee} & \mathbf{d}_{eg} \\ \mathbf{d}_{ge} & \mathbf{d}_{gg} \end{pmatrix} \tag{16.67}$$

where $\mathbf{d}_{ij} = \langle i|\mathbf{d}|j\rangle$ denotes a matrix element of the dipole operator in the atomic states. Many two-atoms lack a permanent dipole moment in their energy eigenstates, so we will set the diagonal terms to zero. Since $\mathbf{d}$ must be Hermitian, $\mathbf{d}_{eg} = \mathbf{d}_{ge}^*$. In operator form,

$$\mathbf{d} = \mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-. \tag{16.68}$$

For the selected mode, the Schrödinger-picture electric-field operator at the position of the two-level atom is

$$\mathbf{E}(\mathbf{r}_0) = i\mathcal{E}\,\hat{\mathbf{e}}\left(a e^{i\mathbf{k}\cdot\mathbf{r}_0} - a^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_0}\right) = \mathcal{E}\,\hat{\mathbf{e}}\left(a + a^\dagger\right), \tag{16.69}$$

where $\mathcal{E} = \sqrt{\hbar\omega/2V}$, and in the second equality, we've absorbed the constant phases coming from i and $e^{i\mathbf{k}\cdot\mathbf{r}_0}$ into the definitions of the ladder operators. In other words,

$$ia e^{i\mathbf{k}\cdot\mathbf{r}_0} \to a, \qquad -ia^\dagger e^{-i\mathbf{k}\cdot\mathbf{r}_0} \to a^\dagger$$

This redefinition does not affect the bosonic commutation relations, and therefore has no theoretical or empirical consequence.

The interaction Hamiltonian is given by

$$\begin{aligned} V &= -\left(\mathbf{d}_{eg}\sigma_+ + \mathbf{d}_{ge}\sigma_-\right)\cdot\left[\mathcal{E}\hat{\mathbf{e}}(a + a^\dagger)\right] \\ &= -\mathcal{E}\left[\left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_-\right](a + a^\dagger). \end{aligned}$$

The atomic and field operators act on independent vector spaces, so they commute. We may expand the product:

$$V = -\mathcal{E}\Big[\left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ a + \left(\mathbf{d}_{eg}\cdot\hat{\mathbf{e}}\right)\sigma_+ a^\dagger + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_- a + \left(\mathbf{d}_{ge}\cdot\hat{\mathbf{e}}\right)\sigma_- a^\dagger\Big].$$

Define the complex coupling constant g via

$$\hbar g = -\mathcal{E}\left(\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right).$$

For a real polarization vector, $\mathbf{d}_{ge} \cdot \widehat{\mathbf{e}} = \left(\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right)^*$, and futhermore, $\hbar g^* = -\mathcal{E}\left(\mathbf{d}_{ge} \cdot \widehat{\mathbf{e}}\right)$. The interaction Hamiltonian becomes

$$V = \left(\hbar g \sigma_+ + \hbar g^* \sigma_-\right)\left(a + a^\dagger\right)$$

The phase of g can be absorbed into the phases of $|e\rangle$ and $|g\rangle$. More precisely, let $g = |g|e^{i\delta}$ in polar form, where $\delta \in \mathbb{R}$. Making the substitutions $|\tilde{e}\rangle = |e\rangle\, e^{i\delta/2}$, $|\tilde{g}\rangle = |g\rangle\, e^{-i\delta/2}$, we find

$$V = \hbar|g|\left(\,|\tilde{e}\rangle\langle\tilde{g}| + |\tilde{g}\rangle\langle\tilde{e}|\,\right)\left(a + a^\dagger\right)$$

Since the atomic states are only well-defined up to an overall phase, we can simply use a new basis in terms of $|\tilde{e}\rangle$ and $|\tilde{g}\rangle$, and we will subsequently relabel these states $|e\rangle$ and $|g\rangle$. We may therefore choose g to be real and positive. Its magnitude is

$$g = \frac{\mathcal{E}}{\hbar}\left|\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right| = \sqrt{\frac{\omega}{2\hbar V}}\;\left|\mathbf{d}_{eg} \cdot \widehat{\mathbf{e}}\right|.$$

In the end, our interaction Hamiltonian may be written in the form

$$V = \hbar g\left(\sigma_+ a + \sigma_+ a^\dagger + \sigma_- a + \sigma_- a^\dagger\right).$$

Let's study the action of this Hamiltonian on the arbitrary product states $|g\rangle|n\rangle$ and $|e\rangle|n\rangle$, where the first term in the product refers to the atomic state and the second term refers to the photon-number state with $n > 0$. The term $\sigma_+ a$ excites the two-level system while annihilating a photon. It therefore describes *photon absorption*. The term $\sigma_- a^\dagger$ lowers the two-level system while creating a photon, thus describing *photon emission*. The free-energy difference between states involved in photon absorption or emission is $\hbar|\omega - \omega_0|$. which is small when the field is nearly resonant with the transition.

The remaining terms simultaneously excite both systems or simultaneously lower both systems. These terms couple states whose free energies differ by $\hbar(\omega + \omega_0)$. Consequently, the relative phases of the coupled states vary rapidly at frequency $\omega + \omega_0$, causing the effects of these terms to largely average out over the longer time scale of the near-resonant dynamics. When the coupling is weak compared with $\omega + \omega_0$, we therefore neglect these two terms. This is the **rotating-wave approximation**, and we write

$$V \approx V_{\text{RWA}} = \hbar g\left(\sigma_+ a + \sigma_- a^\dagger\right)$$

The rotating wave approximation is valid in the near-resonant, weak-coupling regime. Remind me to provide a better explanation when we cover time-dependent perturbation theory!

Adding everything together, we find Eq. (16.58) in its conventional form. The Jaynes–Cummmings model applies approximately to systems such as atoms in optical or microwave cavities, Rydberg atoms, quantum dots, and superconducting qubits coupled to resonators. Its validity relies on the single-mode, two-level, electric-dipole, and rotating-wave approximations, together with the neglect of dissipation. It captures many of the central features of quantum light–matter interaction, including photon absorption and emission, vacuum Rabi oscillations, and atom–field entanglement.

16.3.2 Excitation Conservation

Let n, the number of photons, be the excitation number of the field. Likewise, the atom has an excitation number of 1 in its excited state and 0 in its ground state. Define the **total excitation number** N as the sum of excitations and the **excitation number operator** as

$$N_{\text{tot}} = a^\dagger a + |e\rangle\langle e| \tag{16.70}$$

As it turns out, the Jaynes-Cummings Hamiltonian commutes with N_{tot}. This means the total excitation number is conserved, and we can search for common eigenstates of N_{tot} and H. Consider the action of the number operator on the state $|g\rangle|n\rangle$ and $|e\rangle|n\rangle$:

$$N_{\text{tot}}|g\rangle|n\rangle = n|g\rangle|n\rangle, \quad N_{\text{tot}}|e\rangle|n\rangle = (n+1)|g\rangle|n\rangle \tag{16.71}$$

Thus, the eigenvalue $N = 0$ has the single state $|g\rangle|0\rangle$. For each $N \geq 1$, the eigenvalue N is twofold degenerate, with an eigenspace spanned by $|g\rangle|N\rangle$ and $|e\rangle|N-1\rangle$. Commutation guarantees that H and N_{tot} share eigenspaces; within each degenerate eigenspace of N_{tot}, we must diagonalize H to find the shared eigenvectors.

Let's construct the Hamiltonian within the eigenspace $\{|e\rangle|n\rangle, |g\rangle|n+1\rangle\}$. The total excitation number in this eigenspace is $n+1$. The free Hamiltonian is already diagonal in this eigenspace; we need to compute the action of the interaction term:

$$V|e\rangle|n\rangle = \hbar g\sqrt{n+1}\,|g\rangle|n+1\rangle \tag{16.72}$$

and

$$V|g\rangle|n+1\rangle = \hbar g\sqrt{n+1}\,|e\rangle|n\rangle \tag{16.73}$$

Evidently, the free Hamiltonian accounts for the diagonal terms while the interaction Hamiltonian accounts for off-diagonal terms. This is easier to see in matrix notation. Let

$$|e\rangle|n\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \qquad |g\rangle|n+1\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \tag{16.74}$$

In matrix representation, the Hamiltonian in a fixed-excitation-number eigenspace is

$$H_n = \begin{pmatrix} n\hbar\omega + \hbar\omega_0/2 & \hbar g\sqrt{n+1} \\ \hbar g\sqrt{n+1} & (n+1)\hbar\omega - \hbar\omega_0/2 \end{pmatrix} \tag{16.75}$$

The eigenvalues of H_n are found to be

$$E_\pm = \hbar\omega\left(n + \frac{1}{2}\right) \pm \frac{\hbar}{2}\sqrt{(\omega_0 - \omega)^2 + 4g^2(n+1)}, \tag{16.76}$$

and an orthonormal set of corresponding eigenvectors may be written as

$$\begin{aligned} |n+\rangle &= \cos\frac{\theta_n}{2}\,|e\rangle|n\rangle + \sin\frac{\theta_n}{2}\,|g\rangle|n+1\rangle, \\ |n-\rangle &= \sin\frac{\theta_n}{2}\,|e\rangle|n\rangle - \cos\frac{\theta_n}{2}\,|g\rangle|n+1\rangle. \end{aligned} \tag{16.77}$$

where the angle θ_n is defined through

$$\tan\theta_n = \frac{2g\sqrt{n+1}}{\omega_0 - \omega}. \tag{16.78}$$

16.3.3 Rabi Oscillations

If the field initially contains n photons and the atom is excited, the initial state is

$$|\Psi(0)\rangle = |e\rangle|n\rangle.$$

Since the Jaynes–Cummings Hamiltonian preserves excitation number, the state at any later time is spanned by $|e\rangle\,|n\rangle$ and $|g\rangle\,n+1$ as we found in the last section. We write

$$|\Psi(t)\rangle = c_e(t)|e\rangle|n\rangle + c_g(t)|g\rangle|n+1\rangle.$$

Define the **detuning** $\Delta = \omega_0 - \omega$ and the **generalized Rabi frequency** $\Omega_n = \sqrt{\Delta^2 + 4g^2(n+1)}$. Evolving the initial state with the fixed-excitation Hamiltonian gives

$$|c_g(t)|^2 = \frac{4g^2(n+1)}{\Omega_n^2}\sin^2\left(\frac{\Omega_n t}{2}\right). \tag{16.79}$$

This is the probability that the atom has made a transition to its ground state while the field has gained one photon. The probability oscillates with angular frequency Ω_n, since

$$\sin^2\left(\frac{\Omega_n t}{2}\right) = \frac{1 - \cos(\Omega_n t)}{2}.$$

Its period is $2\pi/\Omega_n$. On resonance, $\omega = \omega_0$, it reduces to

$$|c_g(t)|^2 = \sin^2\!\big(gt\sqrt{n+1}\big). \tag{16.80}$$

The atom is fully in its ground state at $t = \pi/(2g\sqrt{n+1})$, and the excitation flows back into the atom. Increasing g increases the oscillation frequency, so the atom and field exchange energy more rapidly. Increasing the initial photon number also speeds up the exchange, through the factor $\sqrt{n+1}$. The n-dependent contribution reflects **stimulated emission**—photons already in the mode enhance the emission process. Stimulated emission is the basic amplification process used in lasers. The formula also applies when $n = 0$ and therefore describes **spontaneous emission**.

Example 16.2: Rabi Oscillations

Let's work through a simpler example and show more details of the calculation. Consider a two-level system resonantly coupled to a single mode of the electromagnetic field, so that $\omega = \omega_0$. Initially, the two-level system is in its excited state and the cavity mode is in the vacuum state:

$$|\psi(0)\rangle = |e,0\rangle\,.$$

Determine the state $|\psi(t)\rangle$ at time t. Then find the probability that the system has emitted a photon into the cavity and the earliest time at which this probability reaches its maximum value.

Solution. The Jaynes–Cummings Hamiltonian conserves the total number of excitations. The initial state $|e,0\rangle$ contains one excitation, so the evolution remains in the one-excitation subspace. Hence,

$$|\psi(t)\rangle = c_e(t)\,|e,0\rangle + c_g(t)\,|g,1\rangle\,.$$

The coefficient $c_e(t)$ is the amplitude for the two-level system to remain excited with no photon in the cavity. The coefficient $c_g(t)$ is the amplitude for the two-level system to be in its ground state with one photon in the cavity.

In the ordered basis $\{|e,0\rangle\,, |g,1\rangle\}$, the Hamiltonian is

$$H = \begin{pmatrix} \langle e,0|H|e,0\rangle & \langle e,0|H|g,1\rangle \\ \langle g,1|H|e,0\rangle & \langle g,1|H|g,1\rangle \end{pmatrix} = \hbar \begin{pmatrix} \omega & g \\ g & \omega \end{pmatrix}.$$

The eigenvectors are easy to find; denote them by

$$|+\rangle = \frac{1}{\sqrt{2}}\left(|e,0\rangle + |g,1\rangle\right)$$

and

$$|-\rangle = \frac{1}{\sqrt{2}}\left(|e,0\rangle - |g,1\rangle\right)$$

The corresponding energies are $E_\pm = \hbar(\omega \pm g)$. The time-evolution operator is

$$U(t) = e^{-iHt/\hbar} = \frac{e^{-i(\omega+g)t}}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} + \frac{e^{-i(\omega-g)t}}{2}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}.$$

We separate the common factor of $e^{-i\omega t}$ and simplify using Euler's identity:

$$U(t) = e^{-i\omega t}\begin{pmatrix}\cos(gt) & -i\sin(gt)\\ -i\sin(gt) & \cos(gt)\end{pmatrix}.$$

The complete time-evolved state is

$$|\psi(t)\rangle = e^{-i\omega t}\Big[\cos(gt)\,|e,0\rangle - i\sin(gt)\,|g,1\rangle\Big]$$

and the probability the system has emitted a photon into the cavity is

$$P_{\text{emit}}(t) = |c_g(t)|^2 = \sin^2(gt)$$

The probability is first maximized when $gt = \pi/2$, or

$$t_{\max} = \frac{\pi}{2g}$$

At this time, $|\psi(t_{\max})\rangle = |g,1\rangle$. The excitation has been transferred from the two-level system to the cavity field. Because the ideal cavity is lossless, the photon will subsequently be reabsorbed, and the excitation will continue to oscillate between $|e,0\rangle$ and $|g,1\rangle$.

16.4 Continuum Limit

We now let $V = L^3 \to \infty$. Since the spacing between neighboring components of the wavevector is $\Delta k_j = 2\pi/L$, one allowed wavevector occupies a volume $\Delta^3 k = (2\pi)^3/V$ in wavevector space. In the continuum limit, we replace sums with integrals:

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3}\int d^3k,$$

and we rescale the ladder operators by introducing

$$a_\mu(\mathbf{k}) = \sqrt{\frac{V}{(2\pi)^3}}\; a_{\mathbf{k}\mu} \tag{16.81}$$

In the continuum limit, the discrete commutation relations are replaced by

$$\begin{aligned}\left[a_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] &= \delta_{\mu\mu'}\delta^{(3)}(\mathbf{k}-\mathbf{k}'),\\ \left[a_\mu(\mathbf{k}), a_{\mu'}(\mathbf{k}')\right] &= \left[a^\dagger_\mu(\mathbf{k}), a^\dagger_{\mu'}(\mathbf{k}')\right] = 0.\end{aligned} \tag{16.82}$$

The continuum electric-field operator is then

$$\boxed{\mathbf{E}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\hat{\mathbf{e}}_{\mathbf{k}\mu}a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.83}$$

Similarly, the magnetic-field operator becomes

$$\boxed{\mathbf{B}(\mathbf{r},t) = \frac{i}{(2\pi)^{3/2}}\sum_{\mu=1}^{2}\int d^3k\,\sqrt{\frac{\hbar\omega_\mathbf{k}}{2}}\,\left(\hat{\mathbf{k}}\times\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)a_\mu(\mathbf{k})e^{i(\mathbf{k}\cdot\mathbf{r}-\omega_\mathbf{k}t)} + \text{h.c.}} \tag{16.84}$$

Lastly, the continuum field Hamiltonian is

$$\boxed{H_\text{field} = \sum_{\mu=1}^{2}\int d^3k\,\hbar\omega_\mathbf{k}a^\dagger_\mu(\mathbf{k})a_\mu(\mathbf{k}) + \frac{V}{2}\sum_{\mu=1}^{2}\int\frac{d^3k}{(2\pi)^3}\,\hbar\omega_\mathbf{k}.} \tag{16.85}$$

Notice that the vacuum energy becomes infinite for two conceptually distinct reasons. First, the total vacuum energy is proportional to the volume V, as expected for a uniform energy density, and therefore becomes infinite when the spatial volume is taken to infinity. Second, the vacuum-energy density is itself divergent because the integral includes modes with arbitrarily large wavevectors. This latter divergence is called an *ultraviolet divergence*.

Chapter 17

Time-Dependent Perturbation Theory

17.1 Minimal Coupling

In this section, we derive the atom-field Hamiltonian in the electric-dipole approximation and motivate the need for time-dependent perturbation theory. It can be skipped if you're short on time.

In Chapter 16, we studied the quantum theory of a source-free electromagnetic field. We now want to study an atom interacting with that field. The photon operators already introduced describe the radiation field; the new question is how those operators affect the dynamics of a charged particle. It is useful to express the electromagnetic field in terms of potentials:

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

The potentials are not unique: different choices can produce the same electric and magnetic fields. In the *Coulomb gauge*, $\nabla \cdot \mathbf{A} = 0$, the free radiation field has $\phi = 0$, so

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \qquad \mathbf{B} = \nabla \times \mathbf{A}.$$

For the quantized free field in a periodic box, we can write

$$\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k},\mu} \sqrt{\frac{\hbar}{2V\omega_k}}\, \hat{\mathbf{e}}_{\mathbf{k}\mu} a_{\mathbf{k}\mu} e^{i(\mathbf{k}\cdot\mathbf{r} - \omega_k t)} + \text{h.c.}$$

This gives the electric-field operator in Chapter 16 by taking $-\partial_t \mathbf{A}$. The free-field energy is

$$H_{\text{field}} = \frac{1}{2}\int_V d^3r \left[\left(\frac{\partial \mathbf{A}}{\partial t}\right)^2 + (\nabla \times \mathbf{A})^2\right].$$

In the Coulomb gauge, once charges are present, ϕ is determined by the Poisson equation $\nabla^2\phi = -\rho$. However, we'll include the Coulomb interaction in the atomic Hamiltonian so it does not contribute another independent radiation mode.

For a single particle of charge q and mass m interacting with the radiation field, the *minimal coupling* rule gives

$$H_{\text{particle}} = \frac{[\mathbf{p} - q\mathbf{A}]^2}{2m} + q\phi + U$$

where U denotes any additional potential energy. The Hamiltonian has this form because it is invariant under a gauge transformation and it gives the classical Lorentz force equation. For hydrogen, take the proton to be fixed and use the Coulomb potential

$$U_C = -\frac{e^2}{4\pi\epsilon_0 r}$$

The total Hamiltonian describing the atom, the field, and their interaction is

$$H = H_{\text{field}} + \frac{[\mathbf{p} + e\mathbf{A}]^2}{2m_e} + U_C.$$

Include electron spin by simply adding its magnetic interaction with the field via $-\boldsymbol{\mu}_s \cdot \mathbf{B}$.

Expanding the Hamiltonian separates it into a part with no matter–field interaction and interaction terms:

$$H = H_0 + H_1 + H_2,$$

where, omitting the constant zero-point energy of the field, and using $\lambda = (\mathbf{k}, \mu)$ to label radiation modes,

$$H_0 = \frac{\mathbf{p}^2}{2m_e} + U_C(\mathbf{r}) + \sum_\lambda \hbar\omega_k \, a_\lambda^\dagger a_\lambda,$$

$$H_1 = \frac{e}{2m_e}\left[\mathbf{p}\cdot\mathbf{A}(\mathbf{r}) + \mathbf{A}(\mathbf{r})\cdot\mathbf{p}\right] + \frac{e}{m_e}\mathbf{S}\cdot\mathbf{B}(\mathbf{r}),$$

$$H_2 = \frac{e^2}{2m_e}\mathbf{A}^2(\mathbf{r}).$$

Because $\nabla\cdot\mathbf{A} = 0$, the symmetrized orbital term simplifies to $\mathbf{p}\cdot\mathbf{A} = \mathbf{A}\cdot\mathbf{p}$, so its contribution to H_1 can also be written $(e/m_e)\mathbf{p}\cdot\mathbf{A}(\mathbf{r})$. For an atom small compared with the radiation wavelength, $kr \ll 1$, expand the vector potential about the atomic center. Using the summation convention,

$$A_i(\mathbf{r}) \approx A_i(0) + r_j \partial_j A_i(0) + \cdots$$

Starting in the Coulomb gauge with $\nabla\cdot\mathbf{A} = 0$ and $\phi = 0$, define $\mathbf{A}_0(t) = \mathbf{A}(0,t)$ and choose

$$\chi(\mathbf{r}, t) = -\mathbf{r}\cdot\mathbf{A}_0(t)$$

as the "gauge function" such that under the gauge transformation

$$\mathbf{A}' = \mathbf{A} - \nabla\chi, \qquad \phi' = \phi - \partial_t\chi,$$

we have

$$\mathbf{A}' = \mathbf{A}(\mathbf{r}, t) - \mathbf{A}_0(t), \qquad \phi' = -\mathbf{r} \cdot \mathbf{E}(0, t)$$

Because $\mathbf{A}'$ is spatially constant, we are still in the Coulomb gauge, only with $\phi' \neq 0$. We can see that the transformed scalar potential gives the electric-dipole interaction

$$q\phi' = -q\,\mathbf{r} \cdot \mathbf{E}(\mathbf{0}, t) = -\mathbf{d} \cdot \mathbf{E}(\mathbf{0}, t), \qquad \mathbf{d} = q\mathbf{r}.$$

The spatial variation of $\mathbf{A}'$ contains the magnetic-dipole and higher multipole interactions. In this gauge, the constant part of $\mathbf{A}$ has been removed, leaving its spatial variation:

$$A_i'(\mathbf{r}, t) = A_i(\mathbf{r}, t) - A_i(\mathbf{0}, t) \approx r_j\, \partial_j A_i(\mathbf{0}, t).$$

Since $\mathbf{A}' = 0$, the part of the Hamiltonian that is linear in this residual vector potential is

$$H_{\text{orb}}^{(1)} = -\frac{q}{m}\, p_i r_j\, \partial_j A_i(\mathbf{0}, t).$$

Split the gradient into antisymmetric and symmetric parts:

$$\partial_j A_i = \frac{1}{2}(\partial_j A_i - \partial_i A_j) + \frac{1}{2}(\partial_j A_i + \partial_i A_j).$$

The antisymmetric part is fixed by the magnetic field:

$$\partial_j A_i - \partial_i A_j = -\epsilon_{ijk} B_k.$$

Its contribution satisfies

$$p_i r_j\, \frac{\partial_j A_i - \partial_i A_j}{2} = \frac{1}{2}\, \mathbf{L} \cdot \mathbf{B}, \qquad \mathbf{L} = \mathbf{r} \times \mathbf{p}.$$

Therefore the orbital magnetic-dipole term is

$$H_{\text{orb}} = -\frac{q}{2m}\, \mathbf{L} \cdot \mathbf{B}(\mathbf{0}) = -\boldsymbol{\mu}_L \cdot \mathbf{B}(\mathbf{0}), \qquad \boldsymbol{\mu}_L = \frac{q}{2m}\mathbf{L}.$$

Including spin adds its magnetic-moment coupling:

$$H_{\text{B-dipole}} = -(\boldsymbol{\mu}_L + \boldsymbol{\mu}_S) \cdot \mathbf{B}(\mathbf{0}).$$

For an electron with $g \simeq 2$, $\boldsymbol{\mu}_S = -e\mathbf{S}/m_e$, so

$$H_{\text{B-dipole}} = \frac{e}{2m_e}(\mathbf{L} + 2\mathbf{S}) \cdot \mathbf{B}(\mathbf{0}).$$

The symmetric part of $\partial_j A_i$ gives the electric-quadrupole contribution in the multipole expansion. These terms depend on the spatial variation of $\mathbf{E}$, and in the dipole approximation, these terms

are usually neglected. The only consistent way to justify the approximation is to neglect the magnetic dipole interaction as well. Indeed, the electric-dipole interaction usually dominates *when the electric-dipole transition is allowed.*

Let a be the size of the atom and $k = \omega/c$. For an allowed electric-dipole transition, the typical interaction energy is

$$H_{\mathrm{E1}} \sim e\, a\, E.$$

The orbital magnetic-dipole energy is roughly

$$H_{\mathrm{M1}} \sim \frac{e\hbar}{m_e} B \sim \frac{e\hbar}{m_e c} E,$$

using $B \sim E/c$ for radiation. Thus

$$\frac{H_{\mathrm{M1}}}{H_{\mathrm{E1}}} \sim \frac{\hbar}{m_e c a}.$$

For an atom with size $a \sim a_0$, this ratio is on the order of the fine-structure constant,

$$\frac{\hbar}{m_e c a_0} = \alpha \ll 1.$$

The quadrupole interaction is roughly

$$H_{\mathrm{E2}} \sim e\, a^2\, |\nabla E| \sim e\, a^2 k E, \qquad \frac{H_{\mathrm{E2}}}{H_{\mathrm{E1}}} \sim ka \ll 1.$$

For ordinary atomic transitions, ka is also typically of order α. So both M1 and E2 amplitudes are generally smaller than an allowed E1 amplitude. The qualification is that an E1 matrix element can vanish by selection rules. Then an M1 or E2 transition can be the leading contribution, even though it is smaller in the long-wavelength scaling. In that case, compare the allowed M1 and E2 matrix elements for the specific transition; neither automatically dominates.

We finally arrive at a fully quantum-mechanical description of the electric-dipole interaction. For hydrogen with a fixed proton, the field-free atomic Hamiltonian is

$$H_{\mathrm{atom}} = \frac{\mathbf{p}^2}{2m_e} - \frac{e^2}{4\pi\epsilon_0 r}.$$

The free radiation field is described by

$$H_{\mathrm{field}} = \sum_{\lambda} \hbar\omega_k\, a_\lambda^\dagger a_\lambda, \qquad \lambda = (\mathbf{k}, \mu),$$

where the zero-point energy has been omitted. We take the unperturbed Hamiltonian to be

$$H_0 = H_{\mathrm{atom}} + H_{\mathrm{field}}.$$

Restricting attention to the electric-dipole interaction, the interaction Hamiltonian is

$$H_1 = -\mathbf{d}\cdot\mathbf{E}(\mathbf{0}),$$

where $\mathbf{d} = -e\mathbf{r}$ is the electron's dipole operator, and $\mathbf{E}(\mathbf{0})$ is the external radiation field evaluated at the atomic center. Thus, for hydrogen,

$$H_1 = e\,\mathbf{r}\cdot\mathbf{E}(\mathbf{0}).$$

Using real polarization vectors, the Schrödinger-picture field operator at the atomic center is

$$\mathbf{E}(0) = i\sum_{\mathbf{k},\mu}\sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}}\,\hat{\mathbf{e}}_{\mathbf{k}\mu}\left(a_{\mathbf{k}\mu} - a^\dagger_{\mathbf{k}\mu}\right),$$

so that

$$H_1 = ie\sum_{\mathbf{k},\mu}\sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}}\left(\mathbf{r}\cdot\hat{\mathbf{e}}_{\mathbf{k}\mu}\right)\left(a_{\mathbf{k}\mu} - a^\dagger_{\mathbf{k}\mu}\right).$$

We neglect the quadratic interaction term for the first-order electric-dipole treatment considered here. The eigenstates of H_0 are products of atomic states and photon occupation-number states:

$$|A\rangle\,|\{n_\lambda\}\rangle \equiv |A;\{n_\lambda\}\rangle\,.$$

Here $|A\rangle$ is an atomic energy eigenstate, and n_λ is the number of photons in mode λ. Their energies are

$$E_A + \sum_\lambda n_\lambda\hbar\omega_k.$$

The unperturbed Hamiltonian $H_0 = H_{\text{atom}} + H_{\text{field}}$ is exactly solvable: its eigenstates are products $|A;\{n_\lambda\}\rangle$ of atomic states and photon-number states. The interaction $H_1 = -\mathbf{d}\cdot\mathbf{E}(\mathbf{0})$, however, couples these states, allowing the atomic states and photon numbers to change. Expand the atom–field state in the product eigenstates of H_0:

$$|\Psi(t)\rangle = \sum_\alpha c_\alpha(t)|\alpha\rangle, \qquad H_0|\alpha\rangle = E_\alpha|\alpha\rangle,$$

where α labels both an atomic state and a photon occupation-number state. Substitution into the Schrödinger equation gives

$$i\hbar\dot{c}_\alpha(t) = E_\alpha c_\alpha(t) + \sum_\beta\langle\alpha|H_1|\beta\rangle c_\beta(t).$$

The interaction couples each state to many others, and there are infinitely many photon modes and atomic states. Solving all these coupled equations exactly is generally impractical. Since the atom–field interaction is weak in the regime of interest, we instead solve for the coefficients

approximately, treating H_1 as a perturbation. We need a *time-dependent perturbation theory.*

17.2 The Interaction Picture

Consider a quantum-mechanical system with a Hamiltonian

$$H(t) = H_0 + V(t) \tag{17.1}$$

where H_0 is a time-independent Hamiltonian that can be easily diagonalized, i.e., we know its eigenstates and corresponding energies; and V is a generally time-dependent perturbation. In the context of time-dependent perturbation theory, we wish to calculate *transition amplitudes*

$$\text{amplitude for } |i\rangle \to |f\rangle \equiv \langle f|U(t)|i\rangle \tag{17.2}$$

where $|i\rangle, |f\rangle$ are eigenstates of H_0 and $U(t)$ is the exact time-evolution operator (we assume $t_0 = 0$). Recall from Chapter #2 that $U(t)$ satifies a differential equation of the form

$$i\hbar\frac{\partial U(t)}{\partial t} = H(t)U(t) \tag{17.3}$$

In order to calculate transition amplitudes, we must solve the differential equation for $U(t)$. If $V(t)$ may be treated as a perturbation, we can use an approximation scheme.

Start by defining the "unperturbed time-evolution operator"

$$U_0(t) \equiv \exp\left(-iH_0t/\hbar\right) \tag{17.4}$$

Here, $U_0(t)$ describes the time evolution of the system only in the case $V(t) = 0$. Using $U_0(t)$, we can define the *interaction picture*, which is similar to the Heisenberg picture but only in terms of $U_0(t)$, not the exact time evolution operator. Think of the interaction picture as if we are factoring out the portion of the time evolution induced by H_0. Define an interaction-picture state vector in terms of the corresponding Schrödinger-picture vector as

$$|\psi_I(t)\rangle = U_0^\dagger(t)\,|\psi_S(t)\rangle \tag{17.5}$$

The vectors coincide at $t = 0$, i.e., $|\psi_I(0)\rangle = |\psi_S(0)\rangle$. For a Schrödinger picture operator A_S, define the interaction-picture operator

$$A_I = U_0^\dagger(t)A_SU_0(t) \tag{17.6}$$

Observe that expectation values are preserved in the two pictures, just as in the Heisenberg picture:

$$\langle\psi_I|A_I|\psi_I\rangle = \langle\psi_S|A_S|\psi_S\rangle$$

Since $|\psi_S(t)\rangle = U(t)\,|\psi_S(0)\rangle = U(t)\,|\psi_I(0)\rangle$, it follows that $|\psi_I(t)\rangle = U_0^\dagger(t)U(t)\,|\psi_I(0)\rangle$. Evidently,

the time-evolution operator in the interaction picture is

$$W(t) \equiv U_0^\dagger(t)U(t). \tag{17.7}$$

Remarkably, the operator $W(t)$ satisfies a differential equation of the same form as $U(t)$ where $V_I(t)$ takes the place of the Hamiltonian. In other words,

$$i\hbar\frac{\partial W}{\partial t} = V_I(t)W(t) \tag{17.8}$$

where $V_I(t)$ is the interaction picture perturbation. Applying the differential equation to the initial state $|\psi_I(0)\rangle$, we find a Schrödinger equation for $|\psi_I(t)\rangle$ in the interaction picture:

$$i\hbar\frac{\partial |\psi_I(t)\rangle}{\partial t} = V_I(t)\,|\psi_I(t)\rangle\,. \tag{17.9}$$

The approximation scheme we employ begins with a formal integration of the differential equation for $W(t)$ from 0 to t. Integrating, we find

$$W(t) = \mathbb{1} + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t')W(t'), \quad W(0) = \mathbb{1} \tag{17.10}$$

This expression is exact, and we notice that the portion which involves $W(t)$ on the right-hand side is proportional to the perturbation $V(t')$. Even though the expression is exact, it's an implicit equation for $W(t)$ since it appears on both the left- and right-hand sides. We can effectively push $W(t)$ to second order in the perturbation by substituting the expression into itself... Iterating the implicit equation, we find

$$W(t) = \mathbb{1} + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t') + \frac{1}{(i\hbar)^2}\int_0^t dt' \int_0^{t'} dt''\, V_I(t')V_I(t'')W(t'') \tag{17.11}$$

Continuing this procedure, we obtain the *Dyson series* for $W(t)$. If you like, you could introduce λ as $H = H_0 + \lambda V(t)$ to keep track of the order of each term. We can approximate the exact time evolution by truncating the series. For example, the first-order correction is

$$W^{(1)}(t) = \frac{1}{i\hbar}\int_0^t dt'\, V_I(t') \tag{17.12}$$

It is convenient to formulate the approximation in terms of expansion coefficients. Assume the initial state satisfies $H_0\,|i\rangle = E_i\,|i\rangle$ and $|\psi_I(0)\rangle = |i\rangle\,$. Using $|\psi_I(t)\rangle = W(t)\,|i\rangle$, we find

$$|\psi_I(t)\rangle = |i\rangle + \frac{1}{i\hbar}\int_0^t dt'\, V_I(t')\,|i\rangle + \frac{1}{(i\hbar)^2}(\cdots) \tag{17.13}$$

Expand the state vector in the unperturbed eigenstates:

$$|\psi_I(t)\rangle = \sum_n c_n(t)\,|n\rangle \tag{17.14}$$

Assuming the unperturbed eigenstates are orthonormal,

$$c_n(t) = \langle n|W(t)|i\rangle = e^{-iE_n t/\hbar}\,\langle n|U(t)|i\rangle \tag{17.15}$$

The first term represents the "rapid evolution" due to H_0 and the second term is the transition amplitude from $|i\rangle$ to $|n\rangle$. Using the approximation for $W(t)$, we can write $c_n(t)$ as a series

$$c_n(t) = \delta_{ni} + c_n^{(1)}(t) + c_n^{(2)}(t) + \cdots \tag{17.16}$$

where

$$c_n^{(1)}(t) = \frac{1}{i\hbar}\int_0^t dt'\ \langle n|\,V_I(t')\,|i\rangle \tag{17.17}$$

$$c_n^{(2)}(t) = \frac{1}{(i\hbar)^2}\int_0^t dt' \int_0^{t'} dt''\ \langle n|\,V_I(t')V_I(t'')\,|i\rangle \tag{17.18}$$

We can always switch back to the Schrödinger picture. For instance, the first-order correction is

$$\boxed{\begin{aligned} c_n^{(1)}(t) &= \frac{1}{i\hbar}\int_0^t dt'\ \langle n|\,U_0^\dagger(t')V(t')U_0(t')\,|i\rangle \\ &= \frac{1}{i\hbar}\int_0^t dt'\, e^{i\omega_{ni}t'}\,\langle n|\,V(t')\,|i\rangle \end{aligned}} \tag{17.19}$$

where ω_{ni} are the Bohr frequencies

$$\omega_{ni} = \frac{E_n - E_i}{\hbar} \tag{17.20}$$

Using a resolution of the identity, namely $\sum_k |k\rangle\langle k| = \mathbb{1}$, the second-order correction becomes

$$c_n^{(2)}(t) = \frac{1}{(i\hbar)^2}\int_0^t dt' \int_0^{t'} dt''\ \sum_k e^{i\omega_{nk}t' + i\omega_{ki}t''}\,\langle n|V(t')|k\rangle\ \langle k|V(t'')|i\rangle \tag{17.21}$$

Some people like to add colorful language to the expressions above. The first-order correction corresponds to a "direct transition" from $|i\rangle$ to $|f\rangle$, driven by the perturbation, whereas the second-order correction involves a sum over transitions through all "intermediate states" $|k\rangle$. We will mostly focus on the first-order transition amplitudes, and since we are typically interested in the transition to a state other than the one we start with, when $n = f \neq i$, the transition probability is

$$P_f(t) = |\,\langle f|U(t)|i\rangle\,|^2 = |c_f(t)|^2 \approx |c_f^{(1)}(t)|^2 \tag{17.22}$$

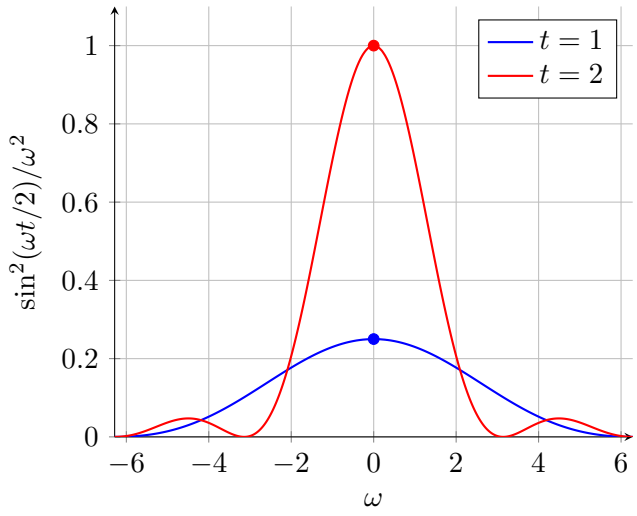


Figure 17.1: Time-dependent factor in the transition probability for $t = 1$ and $t = 2$. Longer interaction times produce a taller, narrower peak centered at $\omega = 0$.

Notice that the first-order correction has the familiar form of a Fourier transform, so we interpret the result as the Fourier coefficient (in frequency space) of the perturbation at the Bohr frequency ω_{fi}. If the perturbation $V(t)$ has an appreciable component at the Bohr frequency ω_{fi} associated with the energy difference between the initial state $|i\rangle$ and some other state $|f\rangle$, then the probability of the system making a transition from $|i\rangle$ to $|f\rangle$ is large.

17.3 Time-Independent Perturbations

Even a time-independent perturbation can drive transitions between different unperturbed states. Taking V to be time-independent, the transition amplitude depends on the integral

$$\int_0^t dt' e^{i\omega t'} = 2e^{i\omega t/2}\,\frac{\sin(\omega t/2)}{\omega} \tag{17.23}$$

where $\omega = \omega_{fi} = (E_f - E_i)/\hbar$. The transition probability is

$$P_f(t) = \frac{4}{\hbar^2}\frac{\sin^2(\omega t/2)}{\omega^2}\left|V_{fi}\right|^2 \tag{17.24}$$

where $V_{fi} = \langle f|V|i\rangle$ is the corresponding matrix element of the perturbation. The time-dependent term depends on the the unperturbed energies and the matrix element of the perturbation depends on the unperturbed states and all properties that come with them (momentum, spin, etc.)

Figure 17.1 shows the time-dependent factor in the transition probability. It is symmetric about $\omega = 0$, where its limiting value is $t^2/4$: $1/4$ for $t = 1$ and 1 for $t = 2$. As the interaction time increases, the central peak grows taller. Its first zeros are at $\omega = \pm 2\pi/t$, so they move closer to the origin as t increases. This shows that, for longer interaction times, the time-dependent factor

becomes sharply concentrated around $\omega = 0$, corresponding to transitions where E_f is close to E_i.

In a distributional sense, this concentration is expressed by

$$\delta(\omega) = \lim_{t\to\infty} \frac{2}{\pi t} \frac{\sin^2(\omega t/2)}{\omega^2}. \tag{17.25}$$

This identity explains how energy conservation emerges in the long-time limit: transitions with nonzero energy difference contribute less, while the contribution becomes concentrated at $E_f = E_i$. For finite interaction times, transitions are possible over a range of energies around E_i. At sufficiently long times, the transition probability can become large, and first-order perturbation theory then ceases to be reliable. Defining

$$\Delta_t(\omega) = \frac{2}{\pi t} \frac{\sin^2(\omega t/2)}{\omega^2} \tag{17.26}$$

we can express the transition probability as

$$P_f(t) = \frac{2\pi t}{\hbar^2} \left|V_{fi}\right|^2 \Delta_t(\omega) \tag{17.27}$$

In the long-time limit, the linear time dependence allows us to define a **transition rate** as the probability per unit time:

$$\boxed{R_f = \frac{d}{dt} P_f(t \to \infty) = \frac{2\pi}{\hbar^2} \left|V_{fi}\right|^2 \delta(\omega)} \tag{17.28}$$

In many practical applications, there is a spread of final energy states rather than a single discrete quantum state. In those cases, the relevant transition rate is a sum over the rates to all accessible states. We assume that these rates are incoherent so that we can add the rates (i.e., probabilities) rather than the amplitudes.

Example 17.1: A Spin-1/2 Example

Consider

$$H_0 = \frac{\hbar\omega_0}{2}\sigma_z, \qquad V = \frac{\hbar g}{2}\sigma_x,$$

where V is constant in time. The unperturbed states are $|\uparrow\rangle$ and $|\downarrow\rangle$, with energy difference $\hbar\omega_0$. Since σ_x flips these states,

$$V_{fi} = \langle\downarrow |V| \uparrow\rangle = \frac{\hbar g}{2}.$$

If the system starts in $|\uparrow\rangle$, first-order time-dependent perturbation theory gives

$$P^{(1)}_{\text{flip}}(t) = \frac{g^2}{\omega_0^2} \sin^2\left(\frac{\omega_0 t}{2}\right).$$

Although V is constant, the transition probability depends on time. The exact result is

$$P_{\text{flip}}(t) = \frac{g^2}{\omega_0^2 + g^2} \sin^2\left(\frac{\sqrt{\omega_0^2 + g^2}\, t}{2}\right).$$

When $g \ll |\omega_0|$, the prefactor becomes g^2/ω_0^2, and the oscillation frequency becomes $|\omega_0|$. Thus, the perturbative result is the leading weak-coupling approximation.
In the degenerate case $\omega_0 = 0$, the exact result is

$$P_{\text{flip}}(t) = \sin^2\left(\frac{gt}{2}\right).$$

For $gt \ll 1$, this reduces to $g^2t^2/4$, the same as the perturbative result. At longer times, the exact probability oscillates, while the t^2 approximation eventually fails.

17.4 Harmonic Perturbations

The study of resonant interactions is the most important application of time-dependent perturbation theory. This is because the most efficient way to make a transition is to impose a sinusoidal perturbation *at the transition frequency*. At $t = 0$ we turn on a time-dependent perturbation Hamiltonian that has separate space and time parts:

$$V'(t) = 2V\cos(\omega_0 t) = V\left(e^{-i\omega_0 t} + e^{i\omega_0 t}\right) \tag{17.29}$$

where V is a Hermitian operator so that the perturbation V' is Hermitian. The probability amplitude for making a transition from $|i\rangle$ to $|f\rangle$ under the harmonic perturbation is

$$c_f(t) = \frac{2}{i\hbar} V_{fi}\left[e^{i(\omega-\omega_0)t/2}\,\frac{\sin(\omega-\omega_0)t/2}{\omega-\omega_0} + e^{i(\omega+\omega_0)t/2}\,\frac{\sin(\omega+\omega_0)t/2}{\omega+\omega_0}\right] \tag{17.30}$$

To find the probability, we square this amplitude, which leads to cross terms and a complicated expression. This is what we have to do if the two terms inside the square brackets are of comparable size, which happens if the frequency is far from a resonance. However, if the resonance condition is satisfied or nearly satisfied, then one of the two terms inside the square brackets dominates because the denominator approaches zero. Which term dominates depends on the sign of the energy difference.

If the final state has higher energy than the initial state, then $\omega = (E_f - E_i)/\hbar > 0$ and the first term dominates when $\omega_0 = \omega$. In this case, the dominant probability amplitude is for the transition from a lower state to an upper state, which we call **absorption**; we say that the system absorbs energy from the perturbation. If the final state has *lower* energy than the initial state, then $\omega = (E_f - E_i)/\hbar < 0$ and the second term dominates when $\omega_0 = -\omega$. The dominant probability amplitude is for the transition from a higher state to a lower state, which we call **emission**; we

say that the system emits energy to the perturbation. Since the emission is "stimulated" by the perturbation, we call this **stimulated emission**.

For now, we consider the absorption case. The probability of measuring the system in the final state is

$$P_f(t) = \frac{2\pi t}{\hbar^2}\left|V_{fi}\right|^2 \Delta_t(\omega - \omega_0) \tag{17.31}$$

and the corresponding transition rate in the long-time limit is given by

$$\boxed{R_f = \frac{2\pi}{\hbar^2}\left|V_{fi}\right|^2 \delta(\omega - \omega_0)} \tag{17.32}$$

The delta function effectively enforces energy conservation: the quantum of energy causing the transition must match the energy difference between the two states.

Before we tackle the fully quantum-mechanical problem of photon emission, let us begin with a semiclassical model. We will treat the atom quantum mechanically, but represent the electromagnetic field as a classical wave. This simpler model lets us see how an oscillating electric field drives transitions between atomic energy levels, and why the transition is strongest when the field frequency matches the energy difference between the states. We will first consider absorption, in which the atom gains energy from the field, and then stimulated emission, in which the atom transfers energy to the field. Later, we will consider the quantized field.

The interaction Hamiltonian between an atom and an applied electromagnetic field is the same as we used for the Stark effect, except that the electric field is now time dependent. We neglect the interaction of the atom with the magnetic component of the electromagnetic field because it is smaller by a factor of the fine structure constant. The electric dipole Hamiltonian is

$$V'(t) = -\mathbf{d} \cdot \mathbf{E}(t) \tag{17.33}$$

where $\mathbf{d}$ is the atomic dipole moment operator and the electric field is

$$\mathbf{E}(t) = 2\mathcal{E}_0 \cos(\omega_0 t)\, \hat{\mathbf{e}} \tag{17.34}$$

which has the same form as the generic harmonic perturbation considered earlier. In the case of hydrogen, the perturbation can be written as

$$V'(t) = e\mathcal{E}_0\, \hat{\mathbf{e}} \cdot \mathbf{r}\, \left(e^{-i\omega_0 t} + e^{i\omega_0 t}\right) \tag{17.35}$$

Since $\mathbf{r}$ is the only term in the matrix element that depends on the atomic states, we can immediately write down the transition rate as

$$R_f = \frac{2\pi e^2 \mathcal{E}_0^2}{\hbar^2}\left|\, \hat{\mathbf{e}} \cdot \langle f|\mathbf{r}|i\rangle \,\right|^2 \delta(\omega - \omega_0) \tag{17.36}$$

The delta function is not physical, so we need to learn how to apply this to a real situation. In order

to make the situation a bit more realistic, we consider a *broadband light source* that is isotropic and unpolarized. A broadband source is composed of many frequency components. To model it, let k label a plane-wave component, including its frequency ω_k, polarization $\hat{\mathbf{e}}_k$, and amplitude $\mathcal{E}_k$. The electric field of each component is

$$\mathbf{E}_k(t) = 2\mathcal{E}_k \hat{\mathbf{e}}_k \cos(\omega_k t + \phi_k). \tag{17.37}$$

We assume the phases of different components are uncorrelated, so their transition probabilities add when averaged over time or over the source. For one component, the electric-dipole perturbation is

$$V'_k(t) = e\mathcal{E}_k \, \hat{\mathbf{e}}_k \cdot \mathbf{r} \, \left(e^{-i\omega_k t} + e^{i\omega_k t}\right). \tag{17.38}$$

Its contribution to the absorption transition rate is

$$R^{(k)}_{i\to f} = \frac{2\pi e^2 \mathcal{E}_k^2}{\hbar^2} \left| \, \hat{\mathbf{e}}_k \cdot \, \langle f|\mathbf{r}|i\rangle \, \right|^2 \delta(\omega - \omega_k), \tag{17.39}$$

We now describe how much field energy lies in a frequency interval. Recall from Phys 152/162, the time-averaged total energy density of one plane-wave component is

$$u_k = 2\epsilon_0 \mathcal{E}_k^2 \tag{17.40}$$

This differs from the familiar expression because we've chosen $2\mathcal{E}_k$ to be the amplitude of a single plane-wave component. The total energy density is then a sum over all plane-wave contributions

$$u = 2\epsilon_0 \sum_k E_k^2 \tag{17.41}$$

Now define $\rho(\omega')$ to be the time-averaged energy density per unit angular frequency interval. By definition,

$$u = \int \rho(\omega') d\omega' \tag{17.42}$$

For discrete components, it follows that

$$\rho(\omega') = 2\epsilon_0 \sum_k E_k^2 \delta(\omega' - \omega_k) \tag{17.43}$$

since integrating this expression over a frequency interval returns the energy density of the components in that interval. A real broadband source has a dense set of components, so $\rho(\omega')$ becomes a smooth spectrum.

Summing the individual transition rates gives

$$\sum_k R^{(k)}_{i\to f} = \frac{2\pi e^2}{\hbar^2} \sum_k \left| \, \hat{\mathbf{e}}_k \cdot \, \langle f|\mathbf{r}|i\rangle \, \right|^2 \mathcal{E}_k^2 \delta(\omega - \omega_k) \tag{17.44}$$

The radiation is isotropic and unpolarized, meaning the polarization vector is randomly oriented, so we average over all possible directions of the polarization vector. The three-dimensional spatial average of $|\hat{\mathbf{e}}_k \cdot \hat{\mathbf{r}}|^2$ is 1/3. Therefore,

$$\begin{aligned} R_{i\to f} &= \frac{2\pi e^2}{3\hbar^2} \big| \langle f|\mathbf{r}|i\rangle \big|^2 \sum_k \mathcal{E}_k^2 \delta(\omega - \omega_k) \\ &= \frac{\pi e^2}{3\epsilon_0 \hbar^2} \big| \langle f|\mathbf{r}|i\rangle \big|^2 \rho(\omega) \end{aligned} \tag{17.45}$$

There are two contributions to the transition rate; there is an atomic factor determined by the states and a field factor determined by the radiation.

The atomic factor is the squared dipole matrix element. It measures how strongly the initial and final states are coupled by the position operator. If this matrix element vanishes, the transition is forbidden in the electric-dipole approximation, regardless of the intensity of the radiation. Its nonzero values give the electric-dipole selection rules, including $\Delta\ell = \pm 1$ and $\Delta m = 0, \pm 1$. The field factor, $\rho(\omega)$, is the radiation energy per unit volume per unit angular-frequency interval at the transition frequency $\omega = (E_f - E_i)/\hbar$. Only the part of the broadband radiation near this frequency contributes appreciably to the transition. Increasing the radiation energy density at ω increases the transition rate proportionally.

The atomic factor is defined as the **Einstein *B* coefficient** such that

$$R_{i\to f} = \frac{\pi e^2}{3\epsilon_0 \hbar^2} \big| \langle f|\mathbf{r}|i\rangle \big|^2 \rho(\omega) = B\,\rho(\omega) \tag{17.46}$$

The Einstein B coefficient is the same if we swap initial and final states, as is the electromagnetic energy density evaluated at the transition frequency, so the rates of emission and absorption are the same. This would suggest that a collection of two-level atoms in thermal equilibrium with a radiation field would have equal populations of its two energy levels. From statistical mechanics, the Boltzmann distribution law tells us that the populations of levels decrease as the energy level increases. There must be a third process—**spontaneous emission**—connecting the two levels in order for thermal equilibrium to be maintained. This is essentially the starting point for Einstein's derivation of the transition rate for stimulated emission, which we call the **Einstein *A* coefficient**. As expected, the spontaneous emission rate is independent of the applied field and is found to be

$$A = \frac{\hbar\omega^3}{\pi^2 c^3} B = \frac{e^2 \omega^3}{3\pi\epsilon_0 \hbar c^3} \big| \langle f|\mathbf{r}|i\rangle \big|^2 \tag{17.47}$$

The decay of a state caused by spontaneous emission implies that excited states are not stationary states. Excited states have an inherent finite lifetime due to spontaneous emission. If we have a system of atoms in the excited state, then the rate equation for the upper level is

$$\frac{dN_{\text{upper}}}{dt} = -N_{\text{upper}} A \tag{17.48}$$

Solving this differential equation yields the time dependence of the upper level population

$$N_{\text{upper}}(t) = N_{\text{upper}}(0)e^{-At} = N_{\text{upper}}(0)e^{-t/\tau} \tag{17.49}$$

The upper level population decays exponentially with a lifetime τ given by

$$\tau = \frac{1}{A} \tag{17.50}$$

We will not pursue Einstein's derivation here; rather, we will move on to calculating the Einstein A coefficient from first principles by treating both the atom and the field quantum mechanically.

Selection Rules

Electric dipole transitions obey **selection rules**, which we derive in this section. If the matrix element in Eq. (17.47) vanishes, the excited state will not decay as a result of the electric dipole interaction. This doesn't mean the excited state is completely stable because there can still be other decay channels, e.g., magnetic dipole transitions and second-order field effects. These channels typically have much longer lifetimes since the interactions involved are relatively weak.

The position operator acts on the spatial part of the atomic state and does not act on spin. Consequently, the initial and final states must have the same spin and spin projection:

$$\Delta s = 0, \qquad \Delta m_s = 0.$$

We now neglect spin and label the atomic states by $|n, l, m\rangle$. First, consider the z-component of the position operator. Since $[L_z, z] = 0$, we find

$$\langle n', l', m' | [L_z, z] | n, l, m \rangle = \hbar(m' - m) \langle n', l', m' | z | n, l, m \rangle = 0.$$

Thus, a transition due to the z-component of the electric dipole operator can occur only when

$$\Delta m = 0.$$

For the components perpendicular to z, the angular momentum commutators are

$$[L_z, x \pm iy] = \pm\hbar(x \pm iy).$$

Taking matrix elements gives

$$\hbar(m' - m) \langle n', l', m' | x \pm iy | n, l, m \rangle = \pm\hbar \langle n', l', m' | x \pm iy | n, l, m \rangle .$$

Therefore, transitions due to the components perpendicular to z require

$$\Delta m = \pm 1.$$

To find the allowed changes in l, we use the operator identity

$$[L^2, [L^2, r_i]] = 2\hbar^2(r_i L^2 + L^2 r_i),$$

which holds for each Cartesian component r_i of $\mathbf{r}$. Taking its matrix element between $|n, l, m\rangle$ and $|n', l', m'\rangle$, and using

$$L^2 \,|n, l, m\rangle = \hbar^2 l(l+1)\,|n, l, m\rangle\,,$$

we obtain

$$\hbar^4\big[l'(l'+1) - l(l+1)\big]^2 \,\langle n', l', m'|r_i|n, l, m\rangle = 2\hbar^4\big[l'(l'+1) + l(l+1)\big]\,\langle n', l', m'|r_i|n, l, m\rangle\,.$$

Rearranging and factoring gives

$$(l + l')(l + l' + 2)\big[(l - l')^2 - 1\big]\,\langle n', l', m'|r_i|n, l, m\rangle = 0.$$

For $l + l' > 0$, this matrix element can be nonzero only if

$$l - l' = \pm 1.$$

The case $l = l' = 0$, for which the prefactor vanishes, is also forbidden: an s-state has even parity, whereas $\mathbf{r}$ changes sign under parity. Thus the orbital angular momentum selection rule is

$$\Delta l = \pm 1.$$

In summary, the **selection rules for electric dipole transitions** are

$$\boxed{\Delta l = \pm 1, \qquad \Delta m = 0, \pm 1} \tag{17.51}$$

For example, the $2p \to 1s$ transition is allowed by the electric dipole interaction, but $2s \to 1s$ is forbidden by the rule $\Delta\ell = \pm 1$.

17.5 Spontaneous Emission

Consider a single atom interacting with an electromagnetic field confined to a finite periodic box. Each quantized mode has a spatial dependence $e^{i\mathbf{k}\cdot\mathbf{r}}$, so the field variation across the atom is measured by ka where a is the characteristic size of the atom. If $ka \ll 1$, the field is effectively uniform over the atom and the leading interaction is the electric-dipole term. For example, for spontaneous emission of hydrogen as it transitions from $2p \to 1s$, energy conservation selects photon modes with $\omega_k = \omega_{21}$ where $\omega_{21} = (E_{2p} - E_{1s})/\hbar$. The corresponding wavenumber is

$k = \omega_{21}/c$, and the relevant small parameter is

$$ka = \frac{\omega_{21} a}{c} = \frac{2\pi a}{\lambda_{21}} \tag{17.52}$$

For hydrogen's $2p \to 1s$ transition, $\lambda_{21} \approx 121.6\,\text{nm}$, while the orbital dimensions are on the scale of the Bohr radius. This gives $ka_{\text{atom}} \ll 1$, so the field's spatial variation across the atom is small. The dipole approximation is therefore well justified for this transition.

Take the position of the atom to be $\mathbf{r} = 0$. In the electric dipole approximation,

$$V = -\mathbf{d} \cdot \mathbf{E}(0) \tag{17.53}$$

In a quantization volume V, using real polarization vectors, the electric-field operator is

$$\mathbf{E}(0) = i \sum_{\lambda} \sqrt{\frac{\hbar \omega_k}{2\epsilon_0 V}}\, \hat{\mathbf{e}}_\lambda \left(a_\lambda - a_\lambda^\dagger \right). \tag{17.54}$$

where $\lambda = (\mathbf{k}, \mu)$ denotes a photon mode with wavevector $\mathbf{k}$ and polarization μ. In the study of spontaneous emission, we are interested in the transition

$$|i\rangle = |2\rangle_{\text{atom}} |0\rangle_{\text{field}} \to |f\rangle = |1\rangle_{\text{atom}} |1_\lambda\rangle_{\text{field}} \tag{17.55}$$

We consider tensor product states of the form $|\text{atom}\rangle\,|\text{field}\rangle$, where the field is described in terms of photon-number states. The atom begins in a higher-energy state $|2\rangle$ while the field occupies the vacuum state. We would like to calculate the transition rate for the atom to move into the lower-energy state $|1\rangle$ with the corresponding emission of a photon in the λ mode. Remarkably, even though the field is in the vacuum state, we will find a nonzero transition rate!

Using first-order perturbation theory, we essentially have a time-independent perturbation, so the transition rate for each mode is given by Eq. (17.28)

$$R_f^{(\lambda)} = \frac{2\pi}{\hbar^2} \left| V_{fi} \right|^2 \delta(\omega_1 + \omega_k - \omega_2) = \frac{2\pi}{\hbar^2} \left| V_{fi} \right|^2 \delta(\omega_k - \omega_{21}) \tag{17.56}$$

There is no way of knowing which mode the photon will enter upon emission. The magnitude of k is constrained by energy conservation via the delta function, but the direction of propagation and polarization are not constrained. Assuming the modes are independent of each other, we can calculate the total transition rate for spontaneous emission as an incoherent sum of transition rates into each mode as

$$R_f = \frac{2\pi}{\hbar^2} \sum_{\lambda} \left| V_{fi} \right|^2 \delta(\omega_k - \omega_{21}) \tag{17.57}$$

Before we consider the sum over photon modes, we focus our attention on the matrix element V_{fi}. With $\mathbf{d} = -e\mathbf{r}$, the matrix elements factorize as

$$\langle f | V | i \rangle = e \, \langle 1 | \mathbf{r} | 2 \rangle \cdot \langle 1_\lambda | \mathbf{E} | 0 \rangle \tag{17.58}$$

The annihilation operator gives zero on the vacuum, but the creation operator does not:

$$a_\lambda^\dagger \left|0\right\rangle = \left|1_\lambda\right\rangle .$$

Thus, the matrix element for emission into mode λ is, up to an irrelevant phase,

$$\langle f|V|i\rangle \;\propto\; e\sqrt{\frac{\hbar\omega_k}{2\epsilon_0 V}}\,\langle 1|\mathbf{r}|2\rangle\cdot\hat{\mathbf{e}}_\lambda. \tag{17.59}$$

The squared magnitude is

$$\left|V_{fi}\right|^2 = \frac{e^2\hbar\omega_k}{2\epsilon_0 V}\left|\,\hat{\mathbf{e}}_\lambda\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2 \tag{17.60}$$

Substituting into the transition rate, Eq. (17.57), we find

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar V}\sum_\lambda \omega_k\left|\,\hat{\mathbf{e}}_\lambda\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2\delta(\omega_k-\omega_{21}) \tag{17.61}$$

Using periodic boundary conditions in a cubic box with $V = L^3$, the spacing between allowed k-values is $2\pi/L$. Each wavevector therefore occupies a volume $(2\pi)^3/V$ in k-space. In a large quantization volume, the spacing between photon states becomes small, approaching a continuum as $V\to\infty$. In the continuum limit, replace sums with integrals:

$$\sum_k \to \frac{V}{(2\pi)^3}\int d^3k$$

Hence, in the continuum limit, the transition rate is independent of the quantization volume:

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar(2\pi)^3}\int d^3k\sum_\mu \omega_k\left|\,\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\,\langle 1|\mathbf{r}|2\rangle\,\right|^2\delta(\omega_k-\omega_{21}) \tag{17.62}$$

Using spherical coordinates for $\mathbf{k}$, and choosing the two real polarization vectors to be $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$, we can separate the integral into a radial k-integral and an angular integral:

$$R_f = \frac{\pi e^2}{\epsilon_0\hbar(2\pi)^3}\left[\int_0^\infty k^2\,dk\,(ck)\,\delta(ck-\omega_{21})\right]\left[\int d\Omega\sum_\mu\left|\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\mathbf{r}_{12}\right|^2\right]. \tag{17.63}$$

where $\mathbf{r}_{12} = \langle 1|\mathbf{r}|2\rangle$ and $d\Omega = \sin\theta\,d\theta\,d\phi$. After some tedious calculations, we find

$$\int d\Omega\sum_\mu\left|\hat{\mathbf{e}}_{\mathbf{k}\mu}\cdot\mathbf{r}_{12}\right|^2 = \frac{8\pi}{3}\left|\mathbf{r}_{12}\right|^2 \tag{17.64}$$

As for the radial integral, we use a delta-function identity to write

$$\delta(ck-\omega_{21}) = \frac{1}{c}\,\delta\left(k-\frac{\omega_{21}}{c}\right) \tag{17.65}$$

and thus,

$$\int_0^\infty k^2\,dk\,ck\,\delta(ck-\omega_{21}) = \int_0^\infty k^2\,dk\,ck\,\frac{1}{c}\delta\Big(k-\frac{\omega_{21}}{c}\Big) = \Big(\frac{\omega_{21}}{c}\Big)^3 \tag{17.66}$$

Putting the radial and angular results into the original expression,

$$\boxed{R_f = \frac{\pi e^2}{\epsilon_0\hbar(2\pi)^3}\left(\frac{\omega_{21}^3}{c^3}\right)\left(\frac{8\pi}{3}\right)|\mathbf{r}_{12}|^2 = \frac{e^2\omega_{21}^3}{3\pi\epsilon_0\hbar c^3}\big|\,\langle 1|\mathbf{r}|2\rangle\,\big|^2.} \tag{17.67}$$

We have arrived at a remarkable result: *the atom can decay even when the electromagnetic field begins in its vacuum state.* The vacuum has no photons, but that does not mean that the electric-field operator vanishes. Its expectation value is zero, while its matrix element for creating a photon is nonzero. The creation-operator part of the field couples the initial state $|2;0\rangle$ to the final state $|1;1_{\mathbf{k}\mu}\rangle$. The atom loses energy $\hbar\omega_{21}$, and that energy becomes the emitted photon's energy. This is **spontaneous emission**: it does not require an incoming classical field or an initially occupied photon mode. The rate we derived is the Einstein A coefficient we discussed earlier. If we have described the atom's only decay channel, its excited-state lifetime is $\tau = 1/A$.

Consider the $2p \to 1s$ transition of hydrogen. Since there are three states in the $2p$ energy level, we must calculate the matrix element $\langle 1|\mathbf{r}|2\rangle$ for all three and then take an average. This is because we assume the atom is initially unpolarized and each of the three m_ℓ values occur with equal probability. Luckily, the matrix elements are identical, and the square magnitude is given by

$$\big|\,\langle 1|\mathbf{r}|2\rangle\,\big|^2 = \frac{2^{15}}{3^{10}}a_0^2. \tag{17.68}$$

Substituting this into the spontaneous emission rate gives

$$A = \frac{2^{15}}{3^{11}\pi}\frac{e^2a_0^2\omega_{21}^3}{\epsilon_0\hbar c^3} \tag{17.69}$$

Using $e^2/(4\pi\epsilon_0) = \alpha\hbar c$, this can also be written as

$$A = \frac{2^{17}}{3^{11}}\,\alpha\,\omega_{21}\left(\frac{a_0\omega_{21}}{c}\right)^2. \tag{17.70}$$

and since $\omega_{21} = (E_{2p}-E_{1s})/\hbar = 3\alpha c/(8a_0)$ for hydrogen,

$$A = \frac{2^8}{3^8}\,\alpha^5\frac{m_ec^2}{\hbar} \approx 6.27\times10^8\ \mathrm{s}^{-1} \tag{17.71}$$

This corresponds to a lifetime of

$$\tau_{2p} = \frac{1}{A} = \frac{3^8}{2^8\alpha^5}\frac{\hbar}{m_ec^2} \approx 1.60\times10^{-9}\ \mathrm{s}. \tag{17.72}$$

So the hydrogen $2p$ lifetime is approximately 1.60 ns. The agreement with experiment is excellent, and our ability to calculate this lifetime is one of the great triumphs of quantum mechanics!